Spatial Competition and Resistance Control in Adaptive Cancer Therapy

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Abstract

Drug resistance remains a major obstacle to lasting cancer treatment because ecological and spatial interactions can allow resistant clones to expand when therapy removes their sensitive competitors. Adaptive therapy seeks to preserve this competition, yet many implementations neglect the spatial organization in which resistance evolves. This dissertation develops an integrated experimental framework to investigate how spatial organization shapes resistance evolution and how this knowledge can guide adaptive therapy.

To enable controlled competition assays, I established traceable human melanoma populations with defined drug-response phenotypes and used them to develop a spatially structured two-dimensional colony assay. There, I found local crowding to constrain resistant proliferation and loss of spatial confinement to be associated with accelerated resistant outgrowth and loss of control. Adaptive therapy strategies that preserved sufficient sensitive-cell-mediated competition delayed escape and resistant progression while reducing treatment exposure compared to continuous therapy *in vitro*, with a spatially explicit *in silico* model reproducing these dynamics through local physical interactions.

To connect spatial competition to treatment design, a separate non-spatial model was designed to capture the relevant dynamics. Before escape, global population measures retained sufficient information about local competitive conditions to support treatment optimization. Reinforcement learning within this effective model favoured isolated treatment pulses, from which I derived an interpretable burden-threshold strategy with performance comparable to the learned policies while limiting treatment exposure.

To examine how structural integrity affects resistance control in three dimensional space, I extended the framework to spheroids, where loss of cohesion was associated with faster resistant expansion. Stable collagen embedding enabled long-term adaptive therapy comparisons, with trends toward more persistent restraint of resistant growth than under continuous treatment.

Together, these results identify crowding, confinement, and escape as linked determinants of resistance control, while showing how aggregate measurements can guide treatment rules that exploit spatial competition. These findings connect drug resistance in cancer to broader principles of evolutionary ecology and biophysics, showing how spatial organization shapes population dynamics and treatment responses.

Zusammenfassung

Arzneimittelresistenz bleibt ein zentrales Hindernis für langfristig wirksame Krebsbehandlungen. Wenn eine Therapie sensitive Zellen eliminiert, können ökologische und räumliche Wechselwirkungen die Expansion resistenter Klone begünstigen. Adaptive Therapien sollen diese Konkurrenz erhalten, doch viele Modelle vernachlässigen die räumliche Organisation, in der sich Resistenzen entwickeln. Diese Dissertation entwickelt einen integrierten experimentellen Ansatz, um zu untersuchen, wie räumliche Organisation die Resistenzentwicklung prägt und wie sich dieses Wissen für adaptive Therapien nutzen lässt.

Ich etablierte nachverfolgbare humane Melanompopulationen mit definierten Therapieantworten und entwickelte einen räumlich strukturierten 2D-Kolonieassay. Lokale Packungsdichte hemmte resistentes Wachstum, während der Verlust räumlicher Begrenzung mit beschleunigter Expansion und Kontrollverlust einherging. Adaptive Therapieschemata verzögerten das Entkommen und Wachstum resistenter Zellen bei geringerer Therapieexposition als kontinuierliche Behandlung *in vitro*. Ein räumlich explizites *in silico* Modell reproduzierte diese Dynamik durch lokale physikalische Wechselwirkungen.

Ein separates, nicht-räumliches Modell erfasste die Dynamik vor dem Entkommen resistenter Zellen. In dieser Phase enthielten globale Populationsmessgrößen genügend Information über lokale Konkurrenzbedingungen, um Therapien zu optimieren. Reinforcement Learning bevorzugte in diesem Modell vereinzelte Therapieimpulse. Daraus leitete ich eine interpretierbare Schwellenwertstrategie auf Basis der Gesamtzelllast ab, die bei begrenzter Therapieexposition ähnlich gut abschnitt wie die erlernten Strategien.

Zur Untersuchung der Resistenzkontrolle im dreidimensionalen Raum übertrug ich den Ansatz auf Sphäroide, in denen ein Verlust des Zellzusammenhalts mit schnellerer Expansion resistenter Zellen verbunden war. Eine stabile Kollageneinbettung ermöglichte langfristige Therapievergleiche, die tendenziell eine länger anhaltende Begrenzung resistenten Wachstums unter adaptiver als unter kontinuierlicher Behandlung zeigten.

Insgesamt identifizieren diese Ergebnisse lokale Packungsdichte, räumliche Begrenzung und das Entkommen resistenter Zellen als verknüpfte Faktoren der Resistenzkontrolle und zeigen, wie aggregierte Messgrößen Therapiestrategien unterstützen, die räumliche Konkurrenz nutzen. Damit verbinden sie Therapieresistenzen mit grundlegenden Prinzipien der Evolutionsökologie und Biophysik, indem sie den Einfluss räumlicher Organisation auf Populationsdynamik und Therapieantworten verdeutlichen.

List of Publications

The research carried out during my PhD, along with further work published during this time, resulted in the following publications and preprints:

[1] M. Eiche*, E. F. Fischer*, M. Milas*, T. Citak, S. Aif, A.-S. Smith, and J. Kayser.
Crowding governs resistance confinement and guides the improvement of adaptive therapy in a 2D melanoma model system.
bioRxiv (2026), preprint
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[2] S. Aif, M. Eiche, N. Appold, E. Fischer, T. Citak, and J. Kayser.
Reinforcement Failing guides the discovery of emergent physical dynamics in adaptive tumor therapy.
npj Artificial Intelligence (2026)
https://doi.org/10.1038/s44387-026-00157-4

[3] M. Eiche and J. Kayser.
Wie mikrobielle Modellsysteme helfen, Tumorevolution zu entschlüsseln.
Biospektrum (2022), 28:250–252
https://doi.org/10.1007/s12268-022-1745-2

[4] N. Bauer, B. Oswald, M. Eiche, L. Schiller, E. Langguth, C. Schantz, A. Osterlehner, A. Shen, S. Misaghi, J. Stingele, and S. Ausländer.
An arrayed CRISPR screen reveals Myc depletion to increase productivity of difficult-to-express complex antibodies in CHO cells.
Synthetic Biology (2022), 7(1):ysac026
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"What I cannot create, I do not understand."

— Richard P. Feynman

"Seek simplicity and distrust it."

— Alfred North Whitehead

"One must imagine Sisyphus happy."

— Albert Camus

Introduction

Cancer remains a leading cause of death worldwide, accounting for millions of deaths annually despite substantial advances in early detection, targeted therapy, and immunotherapy [5]. While modern treatments can induce profound initial responses, durable control remains limited in advanced cancers [6]. A central obstacle to long-term success is the evolutionary capacity of tumor cells to adapt under therapeutic pressure [7]. Through genetic diversification, phenotypic plasticity, and ecological interactions within the tumor microenvironment, resistant subpopulations emerge that evade the effects of treatment [8–10]. This capacity for adaptation transforms cancer therapy from a purely pharmacological problem into one of evolutionary and ecological dynamics, where selective pressures imposed by therapy reshape the composition and behavior of the tumor ecosystem.

Conventional treatment paradigms commonly seek maximal tumor reduction [11]. However, eliminating sensitive cells may release resistant clones from local ecological competition, accelerating their outgrowth and driving relapse [12]. Adaptive therapy (AT) instead modulates treatment intensity or timing to preserve sensitive populations that competitively suppress resistant subclones [12]. By exploiting this intra-tumoral competition, AT aims to delay progression and sustain control rather than pursue eradication [13].

Preclinical studies have shown that AT can prolong tumor control with progressively reduced dosing after initial stabilization [13], and an early clinical pilot reported delayed progression with reduced drug exposure relative to CT [14]. Yet, many implementations remain heuristic, relying on empirically chosen thresholds rather than mechanistic insight. Without a quantitative understanding of how treatment schedules interact with tumor ecology, it remains difficult to predict when adaptive strategies will succeed [15, 16], and failures may result in substantial clinical consequences. Identifying the ecological and evolutionary conditions that enable sustained tumor control, particularly the role

of spatial competition, is therefore essential to establishing AT as a broadly applicable treatment principle [15, 17]. Experimental systems capable of isolating and measuring these ecological interactions are critical to bridge this gap.

Bulk or well-mixed two-dimensional (2D) cultures capture overall growth but provide limited access to spatial interactions shaping clonal competition [18] and AT outcomes [19]. Tumor spheroids and *in vivo* models offer greater physiological relevance, but nutrient, oxygen, and drug gradients and limited single-cell imaging make resource limitation, intrinsic fitness differences, and spatial confinement difficult to disentangle. Bridging this gap requires a framework that retains quantitative control while allowing spatial structure to emerge under uniform environmental conditions. Tracking sensitive and resistant populations over time would then enable the ecological mechanisms underlying AT to be observed and systematically dissected.

This thesis therefore asks how local spatial ecological interactions between sensitive and resistant tumor cells shape AT control, and how these interactions can be made experimentally observable, mechanistically interpretable, and actionable for treatment design. Rather than attempting to reproduce complete tumor complexity in a single model, I developed a modular framework of controlled experimental analogues of tumor microregions that links genetic engineering, spatially structured *in vitro* systems, mathematical modelling, and data-driven treatment optimization.

Against this backdrop, the first part of this thesis in chapter 2 focuses explicitly on methods development. I established a genetic engineering toolbox for generating melanoma cell lines that differ in their drug-resistance status and fluorescent identity. Stable integration into the adeno-associated virus integration site 1 (AAVS1) safe-harbor locus provided a reproducible framework in which resistant and sensitive populations could be distinguished, tracked, and placed into direct competition under identical environmental conditions. Together with a well-defined Geneticin (G418) selection system, this created the validated foundation for the ecological competition experiments presented in chapters 3 and 4. The platform was further extended by the design of an Inducible Resistance Module (IRM), intended to enable controlled emergence of resistance within initially homogeneous populations in future iterations.

To examine AT under spatially structured yet experimentally controlled conditions, I used the stable reporter-line platform in a 2D competition assay with uniform environmental conditions and full optical accessibility. Defined initial cell ratios and densities enabled

reproducible comparisons of CT and AT while tracking population growth and the spatial organization of resistant clones. The investigations presented in Chapter 3 asked how local crowding and clone position influence resistant expansion, and how treatment schedules affect spatial confinement. Long-term imaging and quantitative analyses were combined to examine the conditions under which sensitive cells can constrain resistant growth and support sustained control under AT.

Experimental observations were complemented by a spatially explicit dissipative particle dynamics (DPD) model representing cells as mechanically interacting agents. This approach tested how local interactions contribute to competition and escape. A separate effective generalized Lotka–Volterra (GLV) model examined whether dynamics before escape could be captured through population-level variables without explicitly resolving spatial structure. Within this effective model, reinforcement learning (RL) was used to explore treatment schedules based on aggregate tumor burden and recent treatment history. The resulting decision logic informed an interpretable treatment strategy, which was evaluated experimentally and in spatial simulations. Together, these approaches addressed how mechanistic understanding of spatial competition could guide adaptive dosing rules based on accessible population measurements.

While the 2D assay provided a mechanistically transparent system for identifying density-dependent competition and spatial escape as key drivers of therapy failure, its planar geometry necessarily abstracts from the structural complexity of real tumors. To address this limitation, in chapter 4, I extended the framework into three-dimensional (3D) systems, where spatial organization and diffusion constraints become increasingly important. Initial 3D studies indicated that the structural integrity of the spheroid environment was associated with resistance dynamics. Loss of cohesion in semi-stable systems accompanied faster resistant expansion, whereas stable embedding enabled long-term comparisons in which AT-associated trends were compatible with more persistent restraint of resistant expansion. Together, these advances provide a first step toward translating the ecological principles identified in 2D into more physiologically relevant tumor architectures.

Beyond the core 2D and 3D systems, this thesis explores how the platform could be extended toward greater biological and physical complexity. These exploratory directions include mechanical tuning of the microenvironment, *in vivo* tracking of resistant populations in zebrafish xenografts, immune–tumor measurements, and marker-based readouts of population burden. Although preliminary, these studies outline possible routes for expanding the framework beyond *in vitro* competition assays and are presented in chapter 5.

At its core, this thesis asks how spatial organization shapes competition between drug-sensitive and resistant tumor cells, and whether this competition can be exploited to improve adaptive therapy. Addressing this question requires connecting local cellular interactions to population-level dynamics and treatment response. To this end, the following chapters combine genetically engineered melanoma reporter lines, controlled 2D and 3D experimental systems, and computational approaches ranging from mechanistic modeling to reinforcement learning. Together, these complementary approaches aim to clarify under which conditions spatial competition can constrain resistant growth and how therapeutic interventions might preserve or exploit this constraint. The overall research framework is illustrated in Figure I.

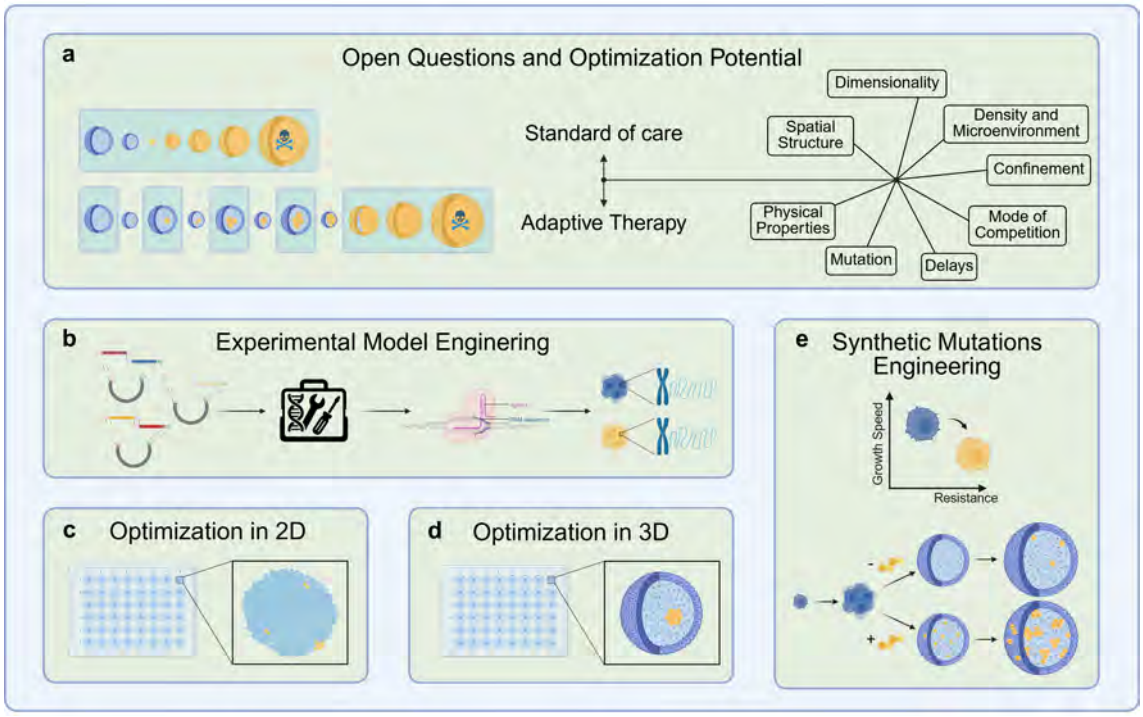


Figure I: Research framework for investigating spatial competition in adaptive therapy (a) introduces the ecological rationale of AT, contrasting standard-of-care treatment with adaptive dosing and highlighting open questions. (b) summarizes the engineering of fluorescent melanoma reporter lines with defined resistance phenotypes. These populations underpin the 2D and 3D frameworks in (c) and (d), investigating treatment scheduling under spatial crowding and increasing structural complexity. (e) illustrates a proposed resistance switch, designed to generate resistant descendants from initially homogeneous populations for investigations of evolutionary dynamics. Created with BioRender.com [20]

1 Background

This chapter introduces the biological, physical, computational, and experimental concepts required to interpret evolutionary therapies as a spatial ecological control problem. It first develops an eco-evolutionary and biophysical view of tumor growth, then introduces adaptive therapy, model-guided treatment design, and the experimental systems needed to test these principles under controlled and observable conditions.

1.1 Eco-Physical Foundations of Tumor Growth

1.1.1 Cancer as an Eco-Evolutionary and Physical System

Cancer remains one of the leading causes of mortality worldwide, characterized by the uncontrolled proliferation of cells that escape the regulatory architecture of normal tissues [5]. Despite remarkable advances in molecular profiling and targeted therapies, durable remission remains rare in advanced cancers, largely because tumor cells continue to adapt and evolve under therapeutic selective pressures [7, 12].

This adaptive capacity reflects that cancer is, at its core, an evolutionary process unfolding within the constraints of a host tissue [21]. Somatic mutation and epigenetic plasticity continually generate variation, while selection acts not only through therapy but also through competition for resources, immune surveillance, and the physical properties of the microenvironment [22–25].

These evolutionary pressures give rise to the cellular capabilities that characterize cancer, allowing malignant cells to outcompete normal tissues while navigating biochemical and biophysical constraints. The ”hallmarks” framework distilled these capabilities and, in its 2022 update, explicitly emphasized phenotypic plasticity and non-mutational epigenetic reprogramming as central features of tumor progression [26, 27]. This is directly relevant to adaptive control: if cells can shift states without new mutations, therapy faces a moving target [28]. Many hallmarks also manifest through physical processes, as angiogenesis, invasion, and immune evasion depend on force generation, adhesion remodeling, and movement through crowded, pressurized tissues, highlighting that molecular traits and physical context must be interpreted jointly [26, 29].

These phenotypic capabilities arise within a genetically heterogeneous background, as articulated in Nowell's clonal evolution model [21]. Genetic instability seeds subclones, and environmental or therapeutic pressures select fitter lineages. Multi-region sequencing revealed extensive branched evolution within tumors, showing that any single biopsy can under-sample the genetic landscape [30]. Some cancers also show periods consistent with effectively neutral ”Big Bang”–like expansion, where subclone abundance is set largely by mutation timing rather than ongoing selective sweeps [31, 32]. Yet the evolutionary fates of these lineages are shaped not only by genetic alteration, but also by the mechanical, metabolic, and spatial constraints imposed by the surrounding tissue [29].

Integrating these evolutionary trajectories with their physical setting provides a useful perspective in which tumors become ecosystems embedded in space: Growth can be limited by local nutrient accessibility, dispersal proceeds along permissive tissue structures, and the effective drug dose experienced by cells is filtered by transport mechanisms such as perfusion, diffusion, and interstitial pressure [33–35]. This eco-physical framing extends the classical molecular “hallmarks of cancer” perspective by emphasizing that clonal behavior in tissue is shaped not only by intrinsic cellular capabilities, but also by evolutionary dynamics, mechanical constraints, and transport limitations [21, 36, 37].

At this physical scale, cancer cells inhabit a mechanically active microenvironment. Re-

modeling of the extracellular matrix (ECM) by tumor and stromal cells stiffens and anisotropically organizes the tissue scaffold, while continued expansion generates solid stress that can compress blood and lymphatic vessels and restrict drug and oxygen transport [37, 38]. These forces are not merely by-products of growth: low-perfusion or high-compression zones can favor slow-cycling, stress-tolerant phenotypes, whereas aligned or softened tracks facilitate invasion [38–41]. Mechanotransduction pathways—integrin–focal adhesion kinase (FAK) signaling and the Hippo/YAP–TAZ network—translate stiffness and confinement into transcriptional programs that can reinforce aggressive and therapy-resistant states [37, 42, 43].

By shaping local competition, dispersal, treatment exposure, and adaptation, spatial gradients determine where and how selection acts and motivate ecological therapies that treat tumors as spatially structured systems rather than homogeneous targets.

1.1.2 Spatial Organization in Tumor Evolution

Against this broader backdrop, spatial organization emerges as a defining property of solid tumors rather than a secondary detail [44]. As visualized in Figure 1.1, tumor masses expand within constrained tissue space, cell lineages segregate into distinct clonal sectors and interact across physical interfaces [31, 45]. Local crowding, nutrient and oxygen gradients, and uneven drug penetration establish spatially varying selective pressures that determine which variants persist and where they do so [10, 46]. These spatial patterns shape ecological interactions by determining which cells compete or cooperate. At the same time, they influence therapeutic exposure by controlling which regions receive sufficient drug concentrations to exert therapeutic pressure.

Microbial range-expansion experiments provide a minimal physical baseline for understanding how space, crowding, and frontier geometry can shape evolution in compact, proliferating cell populations. In radially growing microbial colonies, proliferation is largely restricted to a narrow layer at the advancing edge, causing initially mixed populations to demix into persistent monoclonal sectors [45]. Lineages that by chance occupy small portions of the frontier can ”surf” the expanding edge and rise to high frequency, whereas interior cells become physically confined and no longer contribute descendants to the advancing frontier. Similarly, mutations arising behind the active frontier may fail to contribute to long-term expansion [47]. These systems help distinguish three contribu-

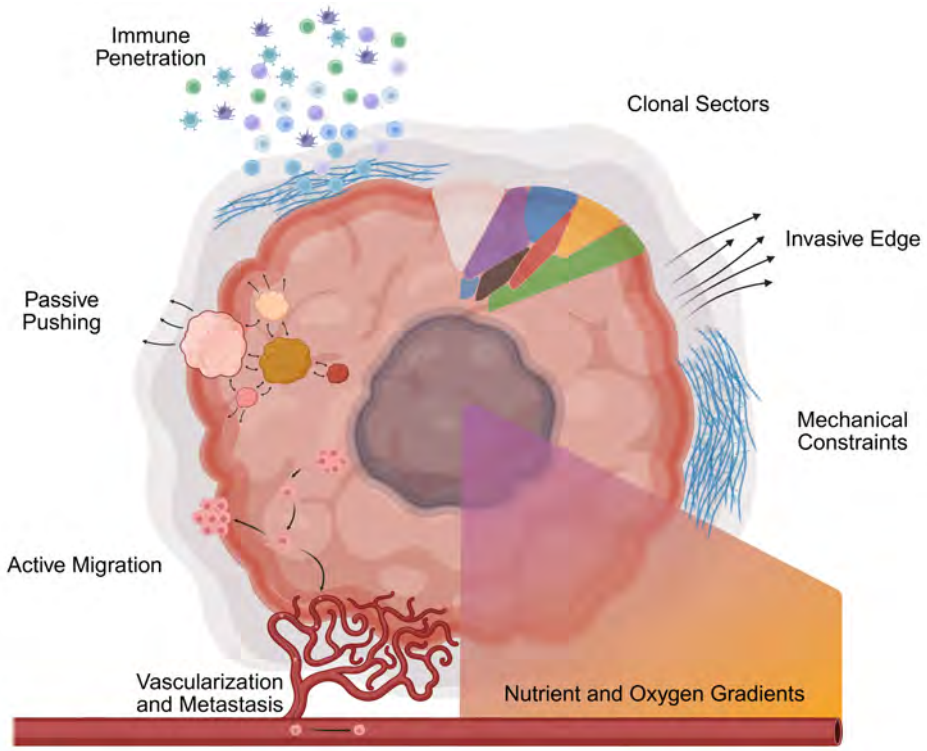


Figure 1.1: Spatial structure in solid tumors shapes ecological dynamics This schematic illustrates the physical and biological mechanisms shaping the spatial fitness landscape of solid tumors. Nutrient and oxygen gradients toward the necrotic core favor stress-tolerant phenotypes, while proliferative and invasive fronts exert mechanical pressure against surrounding tissue. Distinct clonal sectors compete for space and resources and interact with infiltrating immune cells, whose access is constrained by tissue architecture. Passive pushing and active migration drive cellular rearrangements, linking local mechanics to vascular invasion and metastatic spread. Created with BioRender.com [20]

tions to clonal expansion: stochastic access to the frontier, differences in expansion speed, and mechanical coupling between neighboring cells.

For a neutral sector at an approximately flat front, a minimal model can treat its two boundaries as independent diffusive random walks [48]. Over a small forward displacement Δr of the front, the sector-width increment ΔX then satisfies

$$\langle \Delta X \rangle = 0, \qquad \left\langle (\Delta X)^2 \right\rangle = 4 D_X \Delta r, \tag{1.1}$$

where angular brackets denote an ensemble average and D_X quantifies the wandering of one boundary per unit front advance, with units of length. The factor of four accounts for the independent fluctuations of both boundaries. When the boundaries meet, the enclosed lineage loses access to the frontier, allowing genetic drift to eliminate sectors

without a fitness disadvantage. This diffusive approximation neglects front roughness, which can produce superdiffusive boundary wandering. Boundary wandering also connects a lineage's persistence at the advancing frontier to the size of its final clone. Lineages that lose frontier access become enclosed, whereas those that persist can generate exceptionally large "jackpot" clones, as demonstrated by spatial Luria–Delbrück assays [49].

Fusco et al. describe these clone-size distributions using a scaling framework that accommodates both flat and rough fronts. The probability $\Pi(x)$ that a mutation produces a clone occupying at least a fraction x of the final population takes the general form [49]:

$$\begin{aligned} \Pi(x) &= \Pi_c \, \chi\left(\frac{x}{x_c}\right), \\ \chi(\xi) &\sim \begin{cases} \xi^{-\alpha}, & \xi \ll 1, \\ \xi^{-\beta}, & \xi \gg 1. \end{cases} \end{aligned} \tag{1.2}$$

Here, $\xi = x/x_c$, χ is a dimensionless scaling function, x_c marks the crossover between predominantly enclosed clones ("bubbles") and frontier-persistent sectors, and Π_c sets the corresponding probability scale. The exponents α and β depend on dimensionality and the statistics of boundary wandering.

For the two-dimensional rough-front model of Fusco et al., $\alpha = 2/5$ and $\beta = 4$, giving cumulative clone-size distributions proportional to $x^{-2/5}$ below and x^{-4} above the crossover. This example includes the effect of front roughness neglected in the preceding diffusive model. Stochastic loss or persistence of frontier access can therefore generate highly unequal clone sizes even without fitness differences. Although derived for neutral growth without cell death, these scaling laws provide a physical baseline for understanding how spatial enclosure and frontier access can influence resistant clonal growth.

The preceding examples concern neutral lineages. Differences in expansion speed add a systematic bias to clonal dynamics, which can be isolated using the equal-time construction of Korolev et al. [50]. For an established sector of a faster-expanding lineage, assume constant, isotropic front speeds $v_1 > v_2 > 0$ and no invasion of already occupied territory. Equal time increments for radial advance of the slower population and advance along the sector boundary give

$$\frac{\mathrm{d}r}{v_2} = \frac{\sqrt{(\mathrm{d}r)^2 + r^2(\mathrm{d}\phi)^2}}{v_1}, \tag{1.3}$$

where (r, ϕ) are polar coordinates centered on the colony. Integration yields logarithmic

spiral boundaries,

$$\phi(r) = \pm\sqrt{\frac{v_1^2}{v_2^2} - 1}\ln\left(\frac{r}{R_0}\right), \tag{1.4}$$

where R_0 is the initial colony radius and angles are measured from the sector origin. The two signs describe the opposing boundaries. Thus, a difference in expansion speeds generates a systematic increase in the faster lineage's angular share of the frontier. This deterministic description neglects stochastic boundary fluctuations.

Mechanical coupling can modify this geometric prediction. In dense expanding populations, growth-driven cell–cell pushing causes neighboring lineages to move collectively, reducing the relative motion through which selection acts [51, 52]. Kayser et al. described this behavior using a phenomenological model in which the local normal front speed depends on curvature [51]:

$$v = v_{\text{type}}\left(1 + T\kappa\right), \tag{1.5}$$

where v_{type} is the flat-front expansion speed of the local cell type, κ is the front curvature, and $T > 0$ is an effective surface-tension parameter with units of length. Curvature is positive for inward indentations and negative for outward bulges. Indentations therefore advance faster, smoothing the front through growth-driven pushing. Together with sharp clonal boundaries that move normal to the front, this smoothing aligns neighboring cell trajectories and delays the exclusion of small, slower-growing clones [51]. Thus, dense packing and frontier access jointly shape which lineages persist in expanding populations.

Geometric inflation provides an additional influence on the fate of mechanically coupled clonal sectors during radial expansion. In compact microbial colonies and spatial tumor simulations, this widening can counteract selection against slower-growing lineages [53]. For a slower-growing sector, the deterministic change in its width can be expressed as an inflation–selection balance:

$$\partial_r w_{\text{tot}} = \partial_r w_{\text{inf}} - \partial_r w_{\text{sel}}, \tag{1.6}$$

where w is the sector's arc length along the frontier and r is the colony radius. The terms $\partial_r w_{\text{tot}}$, $\partial_r w_{\text{inf}}$, and $\partial_r w_{\text{sel}}$ represent the total width change, widening due to inflation, and narrowing due to selection per unit radial advance, respectively. The inflation contribution is $\partial_r w_{\text{inf}}(w, r) = w/r$, while the selection contribution can depend on clone width. At a given radius, a balance is stable if slightly narrower clones widen and slightly wider clones shrink. Mechanical screening can produce this restoring behavior by sufficiently weakening selection at small clone widths. This can stabilize narrow resistant sectors

over extended growth intervals, although stochastic fluctuations can still drive individual clones to extinction.

Beyond their influence on frontier geometry, collective mechanical interactions may also affect competition by regulating cellular growth itself. In nutrient-rich *E. coli* colonies, Wittmann et al. observed reduced cell elongation in crowded regions and reproduced this behavior using a statistical-mechanical model of mechanically coupled growth. Their extension to competing strains further predicts that differences in mechanical responsiveness can influence population composition [54].

These microbial systems do not reproduce the full biological complexity of tumors, but they reveal general biophysical constraints—dense packing, limited dispersal, growth-driven pushing, and frontier-restricted expansion—that provide a useful conceptual foundation for interpreting spatial genealogies in more complex tissues [45, 51, 53, 55].

In many compact tumor regions, proliferation is enriched near accessible growth fronts or perivascular niches, whereas poorly supplied or compressed interiors frequently become quiescent [37, 41]. Under such conditions, clonal expansion is limited not only by intrinsic fitness but also by the ability of a lineage to physically occupy, maintain access to, or reach permissive growth regions. Spatial tumor models and compact-population experiments indicate that limited dispersal and local crowding can alter sweep probabilities and generate apparently near-neutral patterns [51, 56]. However, direct demonstration of mechanical crowding as a determinant of clonal fitness in tumors remains limited [44, 53, 57]. Thus, rather than acting through fitness differences alone, selection in compact tumors is likely shaped by the spatial and mechanical opportunities available to each lineage.

Within tumors, access to growth-permissive regions is spatially heterogeneous, reflecting mesoscale domains shaped by vascular architecture, oxygen availability, ECM organization, mechanical stress, and drug accessibility [58–60]. Within such regions, cells experience partially shared ecological conditions. Poorly perfused or hypoxic interiors may select for slow-cycling, stress-tolerant phenotypes, whereas perivascular zones are more exposed to oxygen and therapeutics and can sustain higher turnover [61]. Matrix organization further structures these domains by guiding migration along aligned ECM tracks and by locally modifying cell shape, adhesion, and mechanical feedback [62]. Rather than arising from generic frontier physics alone, tumor microregions reflect tumor-specific interactions among vascular remodeling, ECM architecture, tissue mechanics, and treatment exposure [25, 63]. These microregions therefore shape not only the opportunities

for clonal expansion and phenotypic adaptation but also the local exposure of tumor cells to therapy.

Hypoxic and poorly perfused tumor regions can experience reduced exposure to cytotoxic agents due to heterogeneous vascular perfusion and impaired drug transport [35, 64]. Mechanically compressed regions can likewise exhibit reduced perfusion, while compression itself imposes substantial mechanical stress that can restrict cell growth and proliferation [41, 64]. In parallel, features of the tumor microenvironment can spatially exclude cytotoxic T cells and thereby create regions of relative immune privilege [65]. These microenvironments are dynamic and continually reshaped by tumor growth, vessel remodeling, stromal interaction, and treatment-induced changes in tissue structure [25]. The same structural and vascular heterogeneity generates biophysical barriers to therapy: accumulated solid stress can compress intratumoral vessels, while elevated interstitial fluid pressure can reduce the pressure gradients driving transvascular fluid transport [37, 38]. Even under spatially uniform systemic dosing, these mechanical and transport constraints produce steep concentration gradients from perfused regions toward compressed or necrotic interiors, resulting in substantial dose heterogeneity within the tumor mass [46].

Because spatial organization shapes both ecological competition and drug exposure, therapeutic schedules may alter the spatial fitness landscape rather than merely change the total amount of drug delivered [66]. Strategies that sustain or normalize perfusion, through vascular remodeling or metronomic exposure, aim to flatten gradients in oxygen and therapeutic concentration. Experimental studies demonstrate that improving vascular function and alleviating hypoxia can enhance therapy efficacy, while modeling work predicts that metronomic schedules may support normalization of perfusion and drug distribution [67, 68]. By reducing the steepness of drug and oxygen gradients, such approaches may diminish protected refugia and alter the spatial biases that favor clones located in permissive or weakly exposed regions.

Beyond strategies that modify spatial drug exposure, understanding resistance evolution requires identifying the local constraints that govern clonal expansion and persistence. Spatially defined tumor microregions provide a suitable scale for this investigation, as crowding, matrix architecture, and growth-layer thickness can produce local differences in the effective strength of selection [10, 53]. The *in vitro* systems developed in this thesis are designed to approximate such microregional environments in a controlled manner. They do not aim to reproduce the full tumor, but to isolate experimentally tractable

domains in which crowding, confinement, structural integrity, and therapy response can be measured quantitatively [37, 60, 69].

1.1.3 Cell Mechanics and Tissue Physics

The spatial organization of tumors is not only reflected in gradients of oxygen, nutrients, and therapeutic exposure, but also in the physical forces experienced by individual cells and multicellular assemblies. The same mechanical constraints that distort transport within tumors act directly on the cells that generate and remodel the tissue [63, 70]. At smaller scales, the interplay between cell mechanics, crowding, and active motility governs how cellular assemblies proliferate, migrate, and compete [71]. Whether a population behaves as a solid-like, jammed aggregate or as a fluid-like, expanding front depends on mechanical compliance and collective rearrangements within the tissue [72, 73]. Geometry and transport continue to filter the effective dose each cell experiences, while local organization and topology at interfaces guide motion, extrusion, and confinement [46, 74].

A minimal tissue-level description of mechanical growth regulation introduces a homeostatic pressure p_h, at which cell division balances apoptosis. Near this state, pressure and net proliferation can be expanded to first order around the homeostatic density ρ_h [75]:

$$\begin{aligned} p &= \chi^{-1}(\rho - \rho_h) + p_h, \\ k_d - k_a &= -\kappa(\rho - \rho_h). \end{aligned} \tag{1.7}$$

Here, ρ is the cell number density, p is the mechanical tissue pressure, and k_d and k_a are the per-cell division and apoptosis rates. The coefficient $\chi > 0$ describes the density response to pressure, while $\kappa > 0$ measures how strongly net proliferation decreases with increasing density. This local approximation describes growth below, and net cell loss above, the homeostatic pressure under otherwise fixed conditions.

Complementary experimental approaches characterize the mechanical phenotype of individual cells. Advances in single-cell biophysics—including high-throughput methods such as real-time deformability cytometry (RT-DC)—enable quantitative mapping of mechanical phenotypes across thousands of cells [76]. Such approaches complement optical stretcher and atomic force microscopy studies showing that malignant cells frequently adopt softer, more fluid mechanical states that ease pore transit and neighbor exchange

[77, 78]. At the tissue scale, collective behavior is tuned by density, adhesion, and active forces: jammed monolayers can behave solid-like, whereas unjamming transitions can permit coordinated flows and accelerated front advance [79, 80]. In spheroids, solid stress can inhibit net growth [70].

Cells sense and convert mechanical cues such as stiffness, shape, and tension into biochemical signals that regulate proliferation and survival [81]. This mechanotransduction mainly relies on integrin–FAK/Src signaling and the Hippo–YAP/TAZ pathway, whose nuclear activation increases on stiff or well-spread substrates [42, 82]. Through these routes, tissue mechanics directly influence gene expression, promoting growth, migration, and therapy tolerance [83, 84]. Consistent with these mechanisms, elevated ECM stiffness connects to increased proliferation and chemotherapeutic resistance in cancer cells, whereas soft substrates can restore drug sensitivity in some contexts [85].

Higher-order spatial organization adds an additional layer of mechanical regulation. Defect-driven flows and extrusion events in epithelial sheets exemplify how mechanical pathways may operate during collective invasion in cancer [74]. In patterned monolayers, boundary geometry can concentrate traction stresses and proliferation [86], while compression can suppress growth in tumor spheroids [41]. These mechanical and topological factors can influence local carrying capacity, cellular rearrangement, and access to expanding edges, thereby modulating which cells are able to compete within growing tissues [51].

Together, stiffness, adhesion, curvature, and confinement define a mechanical fitness landscape that governs how tumor cells compete and adapt under therapy. However, mechanically mediated fitness dynamics have only rarely been demonstrated directly in cancer, and even fewer experimental systems explicitly connect mechanical interactions to quantitative fitness differences and evolutionary outcomes. This gap motivates controlled model systems in which density and tissue structure can be perturbed and measured directly.

1.1.4 Melanoma Biology and Resistance

The mechanobiological interactions outlined above are not unique to any single cancer type but recur across solid tumors, influencing growth patterns and therapeutic response [29]. To investigate these processes within a defined and experimentally tractable framework, this work focuses on melanoma—a malignancy in which mechanical, spatial, and molecular determinants intersect and can be systematically manipulated *in vitro*.

In melanoma, oncogenic activation of the mitogen-activated protein kinase (MAPK) pathway, most commonly through $BRAF^{V600E}$ mutations, remains a central therapeutic target in *BRAF*-mutant disease [87]. Targeted inhibitors and immune-checkpoint blockade together define the current clinical standard, although resistance and relapse remain major clinical challenges [87]. Disease progression often arises from MAPK reactivation, pathway rewiring, and phenotype switching toward invasive or stress-tolerant states [28, 88]. These molecular routes intersect with spatial ecology: the location of resistant cells within the tumor, their motility, and the mechanical or hypoxic character of their niche may influence how effectively treatment constrains them [37, 44, 89].

Canonical resistance mechanisms against BRAF-inihibitor therapies include secondary mutations, *BRAF* amplification or splice variants, alterations in mitogen-activated protein kinase kinases 1 and 2 (MEK1/2), and receptor tyrosine kinase (RTK) upregulation feeding RAS–RAF–MEK signaling. Additional escape routes include bypass through phosphoinositide 3-kinase (PI3K)–AKT signaling, metabolic adaptation, and transcriptional state changes [28, 90–94]. Genomic and transcriptomic analyses further demonstrate that melanoma exhibits extensive intra- and inter-patient heterogeneity [95, 96], with multiple resistance mechanisms that can emerge in parallel across metastases during BRAF inhibitor therapy [96]. Although combined BRAF/MEK inhibition produces deeper and more durable responses than monotherapy [97], current treatment algorithms are not routinely individualized using quantitative measures of intratumoral spatial competition. Spatial models nevertheless predict that such structure can alter resistance and regrowth dynamics [44, 98].

Additionally, melanoma cells traverse a spectrum between $\text{MITF}^{\text{high}}/\text{AXL}^{\text{low}}$, more differentiated/proliferative states and $\text{MITF}^{\text{low}}/\text{AXL}^{\text{high}}$, mesenchymal/invasive, stress-tolerant states called the MITF rheostat [99–101]. This switching, which can be driven by stress signaling and microenvironmental cues, underlies tolerance and acquired resistance to MAPK blockade and influences immunotherapy responsiveness [102]. Mechanotransduction intersects with phenotype switching: actin remodeling and YAP/TAZ activation promote invasive reprogramming and BRAF inhibitor resistance, while experimental dampening of cytoskeletal tension or YAP/TAZ signaling can resensitize subsets of resistant melanoma cells [43]. Emerging evidence suggests that more differentiated $\text{MITF}^{\text{high}}$ melanoma subpopulations may be more effectively targeted by immunotherapy, whereas AXL^{high} cells may be comparatively refractory [103]. This supports the broader view that phenotype composition can influence therapy response, potentially in interaction with

local microenvironmental conditions.

In parallel to targeted therapy, immunotherapy has transformed the clinical landscape of advanced melanoma. Antibodies blocking programmed cell death protein 1 (PD-1), programmed death-ligand 1 (PD-L1), or cytotoxic T-lymphocyte–associated protein 4 (CTLA-4) can produce durable remissions in a subset of patients [104, 105]. Yet immune-checkpoint responses depend on pre-existing antitumor immunity and tumor antigenicity, and may emerge only after delayed immune engagement [106]. In rapidly progressive disease, this difference in response kinetics can be clinically relevant, because immunotherapy may not achieve effective disease control quickly enough in all patients [107]. To mitigate this risk, clinical strategies have explored combining or sequencing targeted MAPK inhibition with immune checkpoint blockade, using the former to achieve rapid tumor debulking and enhance antigen release before or during immunotherapy [87, 108].

DREAMseq demonstrated a survival advantage for an immunotherapy-first sequence in *BRAF*-mutant metastatic melanoma, with supportive long-term findings from the non-comparative phase II SECOMBIT trial [107, 109]. However, treatment sequencing remains clinically and biologically nuanced. In some settings, targeted therapy is still used first to achieve rapid disease control [87, 107]. This strategy may come with a potential biological trade-off: switching to immunotherapy only after targeted-therapy failure may occur in an altered tumor context, where MAPK-inhibitor-resistant disease can be associated with immune-evasive microenvironmental changes [110]. From an evolutionary perspective, such patterns do not invalidate targeted-first approaches, but they suggest that their potential benefit depends strongly on timing. Once resistance is fixed, the ecological leverage for immune engagement may be reduced. A deeper understanding of how population composition and spatial structure change under alternating selective pressures may therefore support treatment schedules that preserve immune control rather than attempt to recover it after ecological escape.

1.2 Adaptive Therapy: Principles, Models, and Experimental Frameworks

1.2.1 Principles and Scope of Adaptive Therapy

Adaptive therapy refers to evolutionary treatment strategies that seek to control, rather than eradicate, cancer by exploiting competition between drug-sensitive and drug-resistant cells [12]. It departs from CT approaches, which aim for maximal elimination but often select for resistant clones that drive relapse (see Figure 1.2). Instead, AT treats tumors as evolving populations: rather than maximizing immediate kill, it maintains sensitive cells that can suppress resistant lineages through ecological competition [12, 111]. The central trade-off is explicit. Continuous high-intensity dosing can yield rapid regression but removes competitors and accelerates resistant outgrowth, whereas adaptive exposure can extend control by preserving competition [14]. In practice, AT is implemented through adaptive feedback rules that modulate treatment intensity in response to measured changes in tumor burden, growth, or treatment response [14].

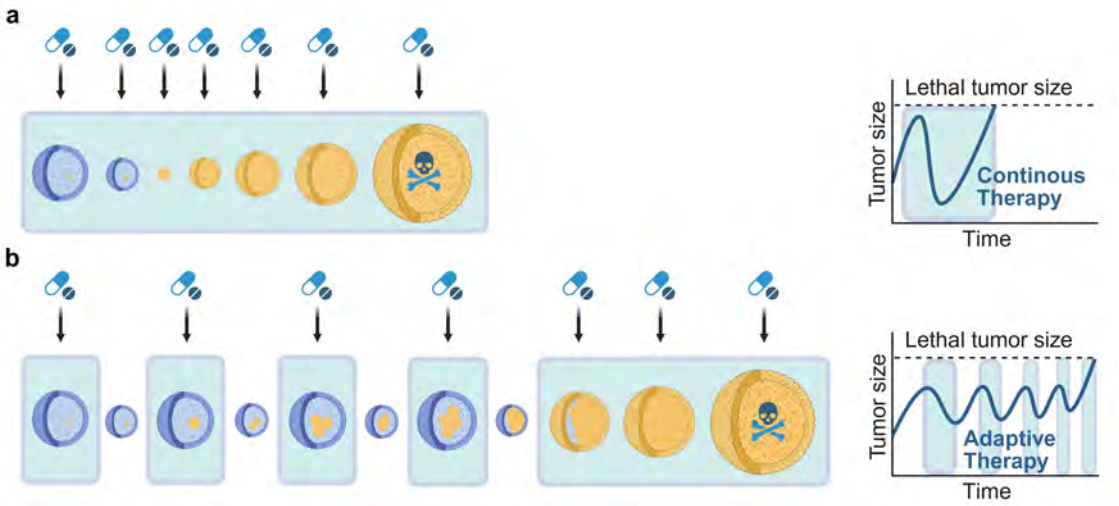


Figure 1.2: Conceptual principle of AT compared to CT Under CT (a), high-dose drug exposure eradicates most sensitive (blue) tumor cells but spares pre-existing or newly emerging resistant cells (yellow), which subsequently expand unchecked and drive rapid relapse. AT (b) instead modulates treatment intensity over time, alternating between drug-on and drug-off phases to maintain a population of sensitive competitors that suppress resistant growth through resource competition. Pale-blue blocks represent drug-on intervals. Created with BioRender.com [20]

AT should be distinguished from predefined intermittent treatment schedules. Fixed-schedule intermittent therapy (IT) uses predefined, calendar-based drug holidays with the aim of reducing treatment exposure and delaying resistance [112]. AT builds on the same rationale but differs fundamentally in how treatment modulation is determined: instead of following fixed on/off cycles, it introduces real-time feedback and adjusts treatment intensity according to the tumor's measured response, aiming to maintain ecological balance between sensitive and resistant populations [12]. Intermittent schedules have been clinically explored for decades, but often failed to prevent or reverse resistance, potentially because pauses were mistimed relative to tumor dynamics or because substantial overtreatment had already occurred before modulation began [14, 112–114]. These outcomes do not diminish the ecological rationale behind evolutionary control. Rather, they highlight the need to adapt treatment in real time based on measurable ecological feedback while accounting for the spatial dynamics of resistance.

As discussed in section 1.1.2, spatial heterogeneity defines the ecological landscape in which AT operates in solid tumors. A cell's effective fitness is not purely intrinsic but emerges from its position within the microenvironment, where gradients of oxygen, nutrients, growth factors, mechanical stress, and matrix attachment generate locally distinct selective pressures. Lymphatic and interstitial flow can additionally reshape local immune conditions [10, 41, 115]. In structured cancer co-cultures and spatial tumor models, these differences can determine where resistant clones establish and how competition unfolds under AT [19, 116].

Under constrained dispersal, resistant clones may persist without rapidly sweeping because spatial confinement slows competitive exclusion and prolongs lineage coexistence, potentially maintaining the sensitive–resistant competition that AT seeks to exploit. Conversely, when spatial organization is disrupted or dispersal increases, resistant cells may gain access to new growth regions, weakening local containment [19, 44]. Experimental and theoretical studies therefore show that spatial bottlenecks arising from crowding and limited dispersal modulate the tempo of selection and can delay fixation of resistance [116, 117].

Similar containment concepts arise in other resistance-management systems. In antimicrobial therapy, alternating or cycling drugs to exploit collateral sensitivities can redirect evolution and delay multidrug resistance *in vitro* [118, 119]. In pest management, untreated refuges preserve sensitive populations and can delay resistance by reducing selection for resistant genotypes [120, 121]. In contrast, structured antiretroviral treatment

interruption produced worse clinical outcomes in trials, illustrating that drug holidays do not automatically create evolutionary control [122]. Together, these examples illustrate both the potential and limitations of evolution-informed treatment strategies across biological systems.

In oncology, however, the specific determinants of successful containment remain incompletely understood. Potential determinants include the fitness cost of resistance, the timing and amplitude of tumor regrowth, the accuracy of response monitoring, and the extent to which spatial or pharmacokinetic refuges protect resistant clones. Defining how these elements interact in tumors represents the central theoretical and experimental frontier of AT and motivates the framework developed in this work.

1.2.2 Computational Models for Spatial Tumor Dynamics

Quantitative *in silico* models offer a means to translate the complex spatial dynamics of tumors into tractable, testable descriptions. Rather than seeking exhaustive realism, the goal is to isolate the dominant processes that govern how cell populations grow, spread, interact, and respond to treatment. Early theoretical efforts described tumor growth with simple laws linking resource limitation to slowing proliferation [123]. These descriptions were later extended by continuum, agent-based, and hybrid spatial frameworks that captured how clonal populations expand through space and interact with their local environment [15, 124–126]. Only in recent years have data-driven and machine-learning approaches begun to integrate such spatial and kinetic descriptions into AT design, formalizing how therapy might respond to evolving tumor dynamics [19, 127]. Used together, these approaches try to form a minimal yet mechanistically transparent vocabulary linking experimental observations like cell number, density maps, interface motion and spatial confinement to the broader ecological logic of tumor control.

At the continuum level, these processes can be connected through a local cell-number balance [75]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = (k_\mathrm{d} - k_\mathrm{a})\rho, \tag{1.8}$$

where $\rho(\mathbf{x}, t)$ is the local cell number density, $\mathbf{v}(\mathbf{x}, t)$ the tissue velocity, and k_d and k_a the per-cell division and apoptosis rates. The divergence term describes redistribution through cell movement, while the right-hand side describes local cell production and loss. Thus, changes in local density reflect both population turnover and spatial transport.

This continuum approximation averages over regions containing many cells and assumes that tissue dynamics are slow compared with individual cellular rearrangements and mechanical relaxation [75].

For systems in which local crowding, mechanical rearrangement, and stochastic clonal escape are central, spatially explicit agent-based models provide a useful complement to continuum descriptions [128, 129]. Instead of treating the tumor as a smooth density field, these models represent individual cells or cell-like units whose growth, movement, division, and death are governed by local rules. This enables encoding interactions that are difficult to express in low-dimensional equations, including direct competition for space, stochastic extinction or expansion of small clones, and the emergence of tissue-scale structure from cell-level behavior [126, 128]. Such models are therefore well suited to experimental systems in which local neighborhood conditions, rather than only global population size, determine whether resistant cells remain confined or expand.

One established class of such models is based on DPD, a coarse-grained framework originally developed for soft-matter systems and later adapted to biological tissues [130–132]. In DPD-based tissue models, cells are represented by interacting particles or particle assemblies subject to conservative, dissipative, and stochastic forces.

The standard DPD equations of motion are [131]:

$$\begin{aligned} \dot{\mathbf{r}}_i &= \mathbf{v}_i, \\ m_i \dot{\mathbf{v}}_i &= -\frac{\partial V}{\partial \mathbf{r}_i} - \sum_j \gamma \omega^D(r_{ij})(\mathbf{v}_{ij} \cdot \mathbf{e}_{ij})\mathbf{e}_{ij} \\ &\quad + \sum_j \sigma \omega^R(r_{ij}) \frac{\mathrm{d}W_{ij}}{\mathrm{d}t} \mathbf{e}_{ij}. \end{aligned} \tag{1.9}$$

Here, $\mathbf{r}_i$, $\mathbf{v}_i$, and m_i denote particle position, velocity, and mass. V is the conservative interaction potential, γ the friction coefficient, and σ the noise amplitude. The weighting functions ω^D and ω^R set the distance dependence of friction and noise, and $\mathrm{d}W_{ij} = \mathrm{d}W_{ji}$ are symmetric pairwise Wiener increments representing random fluctuations. The interparticle distance is $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$, $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ is the relative velocity, and $\mathbf{e}_{ij} = (\mathbf{r}_i - \mathbf{r}_j)/r_{ij}$ is the interparticle unit vector. The sums run over $j \neq i$, excluding self-interactions. Growth and division require additional model rules. For example, in the two-particle tissue model of Basan et al., an intracellular repulsive force drives particle separation, and division occurs when this separation exceeds a prescribed threshold. Resistance from

neighboring cells can therefore slow expansion and delay division [130].

This allows tissue-scale behavior to emerge from local mechanical interactions, including repulsion, adhesion, growth, division, and apoptosis [132, 133]. Previous tissue-mechanics applications have used such formulations to study epithelial rheology, mechanical homeostasis, and competition under crowding, providing a physically grounded way to connect local cell mechanics to collective population dynamics [130, 132, 133].

In this thesis, a DPD-based spatial model is used to complement the 2D melanoma colony experiments in section 3.2 as a digital twin. The model represents sensitive and resistant cells as mechanically interacting agents whose proliferation, death, and treatment response depend on local conditions. Rather than aiming to reproduce the full molecular or physiological complexity of tumors, it provides a simplified framework for testing whether local mechanical interactions, crowding-regulated proliferation, and delayed treatment response can account for key features of spatial competition under therapy.

Beyond spatially explicit descriptions, GLV competition models describe density-dependent interactions through population sizes [134, 135]. In oncology, they have been used to study competitive release, coexistence under fitness costs, and suppression of resistant clones by persistent sensitive competitors [12, 14].

This framework can also incorporate delayed treatment responses. For example, the reduced model used in chapter 3 combines competition with delayed growth suppression and cell killing to describe pre-escape dynamics [1]. Its variables S and R denote sensitive and resistant cell numbers, normalized to the combined cell number at the first observation:

$$\begin{aligned} \frac{\mathrm{d}S}{\mathrm{d}t} &= \gamma_S g(t) S \left(1 - \frac{S+R}{K}\right) \\ &\quad - \delta S - D q(t) S - \Delta_{\min} \mathbb{I}_{q(t)>0}, \\ \frac{\mathrm{d}R}{\mathrm{d}t} &= \gamma_R R \left(1 - \frac{R+cS}{K}\right) - \delta R. \end{aligned} \tag{1.10}$$

Here, γ_S and γ_R are growth rates, K is the shared carrying capacity, δ is the baseline death rate, and c quantifies the competitive effect of sensitive cells on resistant cells.

The dimensionless functions $g(t)$ and $q(t)$ describe delayed growth suppression and treatment-induced death, respectively. Here, $g = 1$ denotes unsuppressed proliferation and $g = 0$ complete growth arrest, while $Dq(t)$ is the treatment-induced per-cell death rate, with $0 \leq q \leq 1$. The additional sensitive-population loss rate $\Delta_{\min}$ applies when $q(t) > 0$:

the indicator $\mathbb{I}_{q(t)>0}$ equals one in this case and zero otherwise. Population sizes are constrained to remain nonnegative. These responses depend on treatment history and can persist after treatment withdrawal. Such extensions connect ecological competition to the timing of treatment effects, providing a tractable framework for exploring feedback-based dosing strategies.

Modeling and experimentation together provide a quantitative language for AT. Image-derived maps of density and clonal composition can define the measurable states of the system while mathematical models describe how these states evolve under growth, dispersal, and treatment. AT can therefore be interpreted as a feedback-control problem applied to ecological population dynamics, where the tumor state s_t can be represented either by spatially resolved variables, such as local crowding and clone position, or by reduced quantities, such as sensitive and resistant population sizes. Therapy applies an action a_t, corresponding to dose intensity or treatment state, that modulates the model's growth and death terms. RL provides a computational framework to discover dosing policies $\pi(a_t \mid s_t)$ that balance tumor control and drug exposure [127]. Rather than functioning as autonomous decision-makers, such algorithms can help map which ecological variables such as density, growth rate, treatment history, or boundary dynamics carry predictive value for control. When combined with imaging-based measurements and mechanistic spatial models, these approaches provide a conceptual bridge between ecological insight and quantitative treatment design.

Together, spatial agent-based models, reduced GLV-type abstractions, and RL approaches provide complementary ways to study treatment response across different levels of resolution. Spatial models can represent how local interactions, crowding, and treatment-induced turnover shape population dynamics, whereas GLV-type models offer simplified population-level descriptions when key ecological effects can be captured by aggregate variables. RL can then be used to explore feedback-based treatment strategies within these reduced model environments and to identify general dosing principles under defined assumptions.

1.2.3 Experimental Model Systems

While theoretical and computational models clarify how ecological interactions shape treatment response, experimental systems of tumor growth provide the empirical foundation for testing these principles, from early spheroids and xenografts to modern organ-

otypic tumor spheroids [136–138]. Each system captures distinct facets of tumor ecology—nutrient and drug gradients, mechanical constraints, clonal heterogeneity, and immune interaction—while inevitably simplifying others [136, 139]. Because ecological interpretations depend on following the fates of defined subpopulations, many experimental formats also require stable lineage-tracing tools like fluorescent reporters, barcodes, or inducible switches to distinguish sensitive from resistant cells and quantify their relative abundance over time. Within this landscape, 2D cultures remain invaluable for optical accessibility and high-throughput perturbation, yet lack the architectural and transport complexity of living tissue [18]. 3D spheroids and organoids reproduce oxygen and drug gradients as well as aspects of ECM mechanics, though at the expense of geometric control and imaging depth [140]. *In vivo* models capture organism-level vascular, stromal, and pharmacokinetic influences on tumor growth and treatment response, but their physiological complexity and limited optical access can make local ecological interactions harder to isolate quantitatively [137]. In the context of AT, such models remain essential for testing whether containment mechanisms persist under systemic treatment conditions [13, 88]. For studying adaptive control, where ecological feedback between sensitive and resistant subpopulations is central, no single format suffices. The challenge is therefore not to replace existing models but to combine their strengths: to use systems that are structured enough to isolate key mechanisms, yet realistic enough to test how these mechanisms behave under physiological constraints.

Genetic Instrumentation for Observability and Control

Lineage tracing and genetic barcoding have become central tools for resolving clonal heterogeneity and therapy-induced selection in cancer models [141]. In adaptive-control experiments such identifiers provide the state information needed to distinguish sensitive from resistant cells, quantify their abundance, and follow clonal fates over time. Stable fluorescent reporters, deoxyribonucleic acid (DNA) barcodes, and color-labeling strategies have been used across *in vitro* and *in vivo* settings to connect lineage abundance to drug exposure, spatial context, and microenvironmental selection [141–143]. When combined with controlled genetic integration at safe-harbor loci such as AAVS1, these approaches support longitudinal readouts with reduced position effects and reporter drift [144–146].

Beyond passive tracing, inducible genetic switches create controlled perturbations that are directly relevant to AT [66]. Cre recombinase fused to a tamoxifen-inducible estrogen receptor mutant (CreERT2)-based systems, for example, enable time-locked recombination in response to tamoxifen-like small molecules, while clustered regularly inter-

spaced short palindromic repeats (CRISPR) interference or activation frameworks allow reversible modulation of gene expression without altering genomic sequence [147–149]. Selection-linked cassettes, such as antibiotic-resistance markers, can additionally create manipulable fitness differences that are optically traceable and experimentally selectable [150, 151]. Such synthetic selection systems do not reproduce the molecular specificity of clinical resistance mechanisms, but they provide a controlled way to isolate ecological variables such as fitness cost, competition, spatial position, and treatment-induced population turnover. In ecological therapy models, these tools convert abstract questions about competition, cost, confinement, and emergence into measurable experimental states. A practical caveat is that reporters, switches, and selection markers must remain genetically and phenotypically stable over the experimental horizon, so that measured ecological effects reflect space and treatment rather than barcode or construct biology.

2D Monolayers

2D model systems provide a controllable and optically accessible setting to study spatial growth and treatment response. Patterned or confined formats including stencil-based arrays, ring geometries, and tubular confinement have long been used to investigate collective migration, boundary formation, and clonal organization under well-defined environmental conditions [116, 152, 153]. Because diffusion across a monolayer is rapid and the surrounding medium is well mixed, cells experience nearly uniform exposure to drugs and nutrients. This decouples local interactions from global gradients and allows geometry, crowding, and mechanical feedback to be varied independently. Structured 2D systems, as illustrated in figure 1.3a, thus serve as quantitative testbeds for linking microscopic interactions to macroscopic growth patterns, enabling high-resolution imaging and longitudinal mapping of colony dynamics.

Recent advances in automated image analysis, artificial-intelligence-based image segmentation, and fluorescence-based tracking have expanded the scope of these assays, allowing direct measurement of lineage motion, interface shape, and spatial heterogeneity over time [154]. While such cultures omit matrix architecture, 3D transport, and stromal or immune components, they remain invaluable for building interpretable, mechanistically controlled systems that connect physical organization to therapeutic response.

3D Systems

3D tumor models introduce architectural and transport features that are absent in monolayers. Multicellular spheroids, matrix embeddings, and organoid systems recreate diffu-

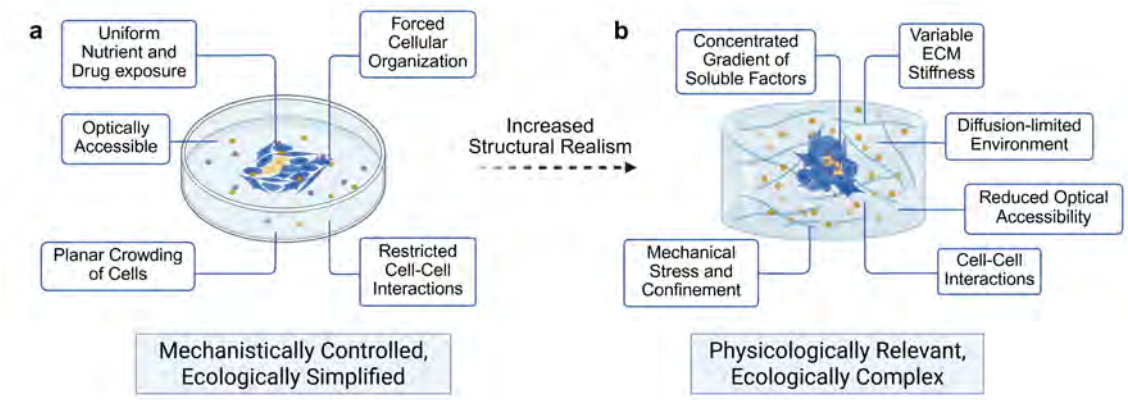


Figure 1.3: Comparison of 2D and 3D tumor models This schematic contrasts the ecological and physical environments of tumors modeled in 2D versus 3D. In 2D systems (a), tumor cells grow on a planar substrate under uniform nutrient and drug exposure, with forced organization and high optical accessibility but reduced architectural complexity and predominantly planar cell interactions. In contrast, 3D ECM models (b) introduce diffusion gradients, variable stiffness, and mechanical confinement, creating a physiologically more realistic yet experimentally less tractable environment. Purple dots indicate soluble secreted factors, and yellow dots represent nutrients and drugs, which are evenly distributed in 2D but form gradients in 3D. Created with BioRender.com [20]

sion limits, mechanical confinement, and ECM organization that shape growth, invasion, and therapy response [136, 140]. In 3D systems, as depicted in figure 1.3b, gradients of oxygen, nutrients, and therapeutic compounds naturally develop during growth, giving rise to proliferative outer rims and quiescent or necrotic cores that recapitulate aspects of *in vivo* tumor physiology [136, 155].

For a spherical, avascular spheroid with uniform tissue density, constant oxygen diffusivity, and constant consumption throughout the oxygenated tissue, the steady-state diffusion–consumption balance is [155]:

$$a(r) = \frac{D}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial C}{\partial r}\right), \tag{1.11}$$

where r is radial distance, D the oxygen diffusion coefficient, C the oxygen volume per unit tissue mass, and $a(r)$ the local oxygen consumption rate in the same units per time, taken to be constant, $a(r) = a$, throughout the oxygenated tissue. For a spheroid of radius r_o, the boundary conditions are $C(r_o) = C_o$ and $C'(0) = 0$. Surface oxygen availability can also be expressed as a partial pressure $p_o = \Omega C_o$, where Ω converts oxygen volume per unit tissue mass into partial pressure using Henry's law. The diffusion-limit radius r_l,

at which central oxygen first reaches zero, is [155]:

$$r_l = \sqrt{\frac{6Dp_o}{a\Omega}}. \tag{1.12}$$

This critical radius increases with oxygen availability and diffusivity but decreases with cellular consumption. For DLD1 tumor spheroids, Grimes et al. estimated a critical radius of approximately 232 ± 22 µm, corresponding to a diameter of approximately 460 µm under their experimental conditions [155]. Importantly, reduced proliferation and quiescence can emerge before complete oxygen depletion, while spheroids exceeding the critical radius may develop an anoxic centre and subsequently a necrotic core. Thus, increasing spheroid size can progressively restrict proliferation to an outer, better-oxygenated region, generating spatially heterogeneous growth even under otherwise uniform culture conditions.

Beyond diffusion-driven heterogeneity, matrix-embedded 3D models allow stiffness and viscoelasticity to be tuned *in vitro* to investigate how mechanical cues regulate cell migration and mechanotransduction [156, 157].

Limitations remain in optical throughput, segmentation complexity, and matrix variability, and many workflows retain substantial operator-dependent steps that introduce variability in spheroid formation, embedding, or imaging [139]. Achieving reliable long-term stability and cross-experiment reproducibility therefore remains non-trivial, especially as spheroid size, matrix stiffness, or nutrient conditions shift during culture. Yet 3D systems uniquely bridge simplified 2D cultures and the physiological context of *in vivo* tissue. For AT specifically, matrix architecture is not merely a realism-enhancing feature. It can determine whether sensitive and resistant populations remain spatially coupled long enough for competitive suppression to operate. Stable 3D systems therefore provide a versatile setting to examine how geometry, diffusion, mechanics, and structural integrity interact to shape treatment response under controlled but realistic constraints.

In Vivo Context and Comparative Considerations

In vivo models incorporate vascular delivery, host pharmacokinetics, and stromal context that are absent from standard culture systems. Immunocompetent or humanized models are additionally required when immune selection is part of the question [13, 88, 158]. These systems remain the benchmark for testing containment and relapse under realistic physiological constraints, yet they limit quantitative control and observability.

In the context of AT, animal models indicate whether mechanisms identified in simplified systems—such as competition or rebound under modulation—persist when gradients, migration, and immunity are physiologically coupled. They also reveal failure modes that *in vitro* assays rarely capture, including immune escape, metastasis, and protected niches. *In vivo* studies thus anchor ecological insights from 2D and 3D models within the systemic realities of the organism.

Ultimately, progress in ecological cancer therapy relies on the dialogue between models rather than any single format. Mechanistic hypotheses generated in 2D can be stress-tested in 3D architectures, then evaluated in *in vivo* tissues where immune, vascular, and stromal interactions complete the ecological landscape. This iterative use of models with increasing complexity allows both precision in mechanism and realism in validation, while also aligning with the Replace, Reduce, Refine (3R) principle—using simplified systems wherever possible, and reserving animal studies for questions that require full physiological context. Such synergy is indispensable for translating adaptive-control principles into durable therapeutic strategies.

1.2.4 Preclinical and Clinical Applications

AT has advanced from theory to preclinical and clinical testing, demonstrating that modulating treatment to preserve sensitive cells can delay resistance and reduce drug use, yet its implementation remains fragmented and limited by the lack of mechanistic, spatially resolved models.

One of the first preclinical demonstrations of AT established its feasibility in controlled xenograft models. In a mouse study of breast cancer, dose modulation based on tumor volume produced prolonged disease control compared with standard high-dose treatment [13]. Related evolutionary treatment strategies have also been explored in melanoma and ovarian systems, including intermittent targeted therapy and adaptive chemotherapy approaches [88, 159].

In *in vitro* experiments, elements of adaptive control have been reproduced, but typically under conditions that limit their clinical interpretability. Well-mixed cultures cannot resolve the local competition and spatial refuges that may shape containment in tumors, whereas structured assays can capture spatial ecological interactions but often evaluate predefined dosing schedules rather than adaptive feedback control [160]. As a result,

these experiments illuminate key mechanisms but do not yet provide the spatially resolved feedback loops required to guide patient dosing. Complementary computational work has started to define the theoretical boundaries of containment and underscored that spatial organization is central to adaptive success [19]. Yet experimental systems that combine spatially resolved clonal tracking with adaptive treatment feedback remain limited.

Early clinical applications have translated adaptive principles into pragmatic dose-adjustment protocols. A prominent early clinical example is metastatic prostate cancer, where treatment pauses are triggered by prostate-specific antigen (PSA) responses and therapy resumes when PSA returns to its pretreatment level [14]. This trial achieved prolonged control, demonstrating the potential of containment strategies in patients. Ongoing clinical studies are now exploring AT in melanoma and ovarian cancer [161, 162].

Despite these promising outcomes, implementation remains limited by the coarse observability of disease state. Biomarkers such as PSA or circulating tumor DNA (ctDNA) provide indirect readouts of tumor burden and composition, while the frequency and resolution of clinical measurements can constrain feedback-based treatment control [15]. Moreover, the fitness costs, dispersal rates, and ecological coupling that underpin adaptive success are rarely measurable in clinical settings [15, 163]. Translational progress therefore depends on model systems that can disentangle these factors under controlled, observable conditions [16].

1.2.5 Determinants of Adaptive Therapy Success

Insights from preclinical and clinical studies reveal that AT is neither inherently fragile nor universally effective, but succeeds or fails depending on how biological, physical, and operational factors interact [15]. Each determinant represents both a potential obstacle and a possible lever for control. Containment requires that sensitive cells persist, compete locally with resistant lineages, and be measurable and targetable with sufficient precision to inform dosing. Factors such as spatial refuges, cost-free resistance, rapid dispersal, or poor observability can undermine these conditions, whereas fitness costs, mechanical confinement, or drug addiction in resistant clones can be exploited to enhance them [88, 164, 165]. As adaptive concepts expand toward multidrug and combination regimens, the relevant space of interactions broadens further, introducing new possibilities for reciprocal sensitivity and ecological steering [166]. In the following, determinants are organized into

three domains: initial geometry and composition, dynamic and environmental filters, and pharmacologic and operational constraints.

The initial makeup and spatial arrangement of a tumor strongly influence whether adaptive containment is possible. A low fraction of resistant cells increases the likelihood that sensitive neighbors can suppress their expansion when treatment pressure is eased, whereas high initial resistance limits the scope for ecological control [164]. Geometry adds a second layer of influence: clustered resistant cells in the interior experience restricted access and stronger local competition, while those at the frontier can exploit free space and resource gradients [2]. The width and connectivity of these interfaces determine how efficiently competition transmits through tissue.

A related geometric consideration is how dimensionality affects the number of sensitive cells required to surround a growing resistant population. For a circular colony of radius r, the circumference $P = 2\pi r$ grows with the square root of its area $A = \pi r^2$. For a spherical aggregate, the surface area $O = 4\pi r^2$ grows with the two-thirds power of its volume $V = 4\pi r^3/3$. Assuming constant cell size and packing, the number of cells N_∂ required to cover the boundary therefore scales with the interior cell number N_R as

$$N_\partial \propto \begin{cases} N_\mathrm{R}^{1/2}, & \text{2D}, \\ N_\mathrm{R}^{2/3}, & \text{3D}. \end{cases} \tag{1.13}$$

To illustrate the consequence for spatial enclosure, consider an exponentially growing resistant interior and a surrounding sensitive population, with $N_i(t) = N_i(0)2^{\nu_i t}$ for $i \in \{\mathrm{R}, \mathrm{S}\}$. Here, ν_i denotes a constant doubling frequency, and the available sensitive population is assumed to remain at the boundary with approximately fixed coverage per cell.

Substituting $N_\mathrm{R}(t)$ into equation 1.13 gives a boundary coverage requirement growing as $2^{\nu_\mathrm{R} t/2}$ in 2D and $2^{2\nu_\mathrm{R} t/3}$ in 3D, compared with $2^{\nu_\mathrm{S} t}$ for the available sensitive population. For an initially covered boundary, sensitive cell numbers can keep pace with the increasing coverage requirement if

$$\nu_\mathrm{S} \geq \begin{cases} \frac{1}{2}\nu_\mathrm{R}, & \text{2D}, \\ \frac{2}{3}\nu_\mathrm{R}, & \text{3D}. \end{cases} \tag{1.14}$$

Under these idealized assumptions, maintaining spatial enclosure is geometrically more demanding in 3D than in 2D, as the sensitive population requires a higher relative growth rate to sustain boundary coverage. This scaling illustrates a potential additional challenge

for adaptive containment in three-dimensional systems, although actual outcomes also depend on mechanical interactions, local proliferation, and tissue organization.

Cellular movement introduces another important constraint on spatial enclosure. High cell motility, collective invasion, or stromal remodeling can erode spatial segregation and seed resistant cells into new territories, whereas limited dispersal and structural confinement preserve local competition [116, 117]. These same properties may also offer intervention points, as proliferation and migration are intrinsically coupled processes that shape both tumor expansion and therapeutic response [117]. In controlled experimental systems, cell composition and spatial organization can be manipulated to examine how spatial competition shapes treatment response [116]. Complementary spatial modelling shows that the initial distribution of resistant cells can substantially alter intra-tumor competition and the benefit of AT [19]. Together, these approaches motivate testing whether local interventions can first reconfigure unfavourable spatial arrangements before systemic adaptive schedules are applied. The emerging challenge is to understand not only how initial conditions constrain AT, but also how they might be intentionally shaped to enhance it.

The relative growth rates of sensitive and resistant populations define the competitive dynamics that shape outcomes under AT. When treatment is relaxed, faster regrowth of sensitive cells can restore local dominance. When resistant cells expand more quickly or carry little fitness cost, containment erodes [164]. Crowding and local carrying capacity modulate these effects by determining how strongly neighbors constrain one another: dense, near-jammed conditions can intensify competition, whereas open space might favor expansion [41, 116].

Phenotypic plasticity adds another layer of complexity. Transiently drug-tolerant states, lineage switching, or microenvironment-induced quiescence can blur the boundary between sensitive and resistant behavior, undermining predictable competition [9, 24, 167]. Yet these same transitions may become exploitable if drug withdrawal or environmental change induces re-sensitization, allowing cycling schedules to exploit reversibility [168]. Environmental filters like perfusion, diffusion limits, interstitial pressure, or solid stress create spatially variable selection, generating both refuges and vulnerabilities [35, 38]. Recognizing where these filters protect resistant pockets and where they expose them offers a path to refine dosing strategies and extend adaptive control.

Translating AT into practice depends on how closely treatment actions align with biological effects. Pharmacokinetic and pharmacodynamic delays can uncouple dosing from

response, introducing hysteresis that weakens feedback control [15]. Observability remains a central constraint: imaging such as magnetic resonance imaging (MRI) cannot be performed frequently enough to guide fine-grained adaptation [169, 170]. Progress therefore relies on developing better surrogate markers, such as ctDNA, and predictive simulations that infer internal tumor dynamics from sparse measurements [171].

Operational robustness also depends on adherence, logistical precision, and the patient's ability to tolerate uncertainty. Irregular dosing and imperfect adherence can alter the effective treatment exposure experienced by patients [172]. Delayed or noisy measurements and interpatient variability can further degrade feedback-based control, motivating schedules that are explicitly designed to remain effective under uncertainty [15, 173]. Metastatic spread adds further complexity, as new lesions often escape early detection and challenge the definition of tumor burden itself [98]. Accounting for such hidden compartments experimentally and *in silico* will be essential for AT to remain effective across spatially distributed disease.

Together, these determinants show that AT is not a fixed schedule but an ecological control problem shaped by spatial structure, plasticity, and observability. Progress will depend on linking biological insight to controllable model systems and predictive simulations so that dosing can be tuned to the evolving organization of the tumor.

1.3 Individualized Control and Translation

1.3.1 Model-Guided Adaptation in Oncology

As oncology moves toward increasingly individualized treatment strategies, the challenge lies not only in selecting a therapy but in determining how it should be adjusted over time. One-time molecular profiling provides valuable baseline information, yet it offers limited insight into the evolving dynamics that govern treatment response [16, 174]. In current clinical practice, substantial modifications to dosing or scheduling generally require dedicated trials that compare predefined treatment arms, rather than drawing on mechanistically grounded models capable of evaluating alternative strategies *in silico* [175, 176]. Quantitative approaches such as pharmacokinetic–pharmacodynamic analysis or radiotherapy dose–fractionation modeling demonstrate the value of mechanistic reasoning but remain restricted in scope [177, 178]. Classical response categories and biomarker

thresholds provide only coarse snapshots of disease state and lack the temporal resolution needed for feedback-based adaptation [179]. These limitations motivate model-guided control frameworks that can synthesize heterogeneous measurements, interpret dynamic changes, and forecast short-term responses to alternative interventions.

Historically, quantitative oncology has drawn on tumor-growth laws, pharmacokinetic–pharmacodynamic models, spatial descriptions, and ecological competition systems to describe how tumors expand and respond to treatment [123–125, 180]. More recently, advances in high-throughput measurement, computational inference, and systems-level modeling have expanded these approaches toward hybrid frameworks that combine mechanistic structure with patient- or experiment-specific data [181]. These developments provide the methodological basis for individualized treatment modeling, in which model parameters, state estimates, or treatment forecasts are updated from empirical observations rather than fixed once at baseline.

Clinical digital twins represent one ambitious vision for this line of development. In the strict sense, a digital twin denotes a patient-specific computational representation that is repeatedly synchronized with empirical measurements and used to forecast disease trajectories or compare candidate interventions [182, 183]. Unlike static individualized models, such frameworks typically require longitudinal data assimilation, uncertainty quantification, and a sufficiently accurate mapping between observable measurements and hidden tumor states. Representative examples include spatial models calibrated to serial MRI [184] for glioma growth forecasting and model-informed precision-dosing frameworks that personalize systemic therapy using individualized pharmacokinetic models [185, 186]. RL and optimal-control approaches extend this logic by exploring treatment strategies that may remain effective under uncertainty [2, 187, 188].

The present thesis does not construct a patient-specific clinical digital twin in the strict sense. Instead, it develops a preclinical experimental and computational framework that captures key elements of digital-twin logic by combining longitudinal measurements with mechanistic modeling to evaluate adaptive treatment strategies [127, 168]. Rather than reproducing the full complexity of individual patients, this framework provides a controlled setting in which the principles of adaptive control can be systematically investigated.

This distinction is important for translation. Prospective digital-twin implementation in oncology remains challenging because clinical measurements are sparse, tumor states are only partially observable, and model structure and parameters remain uncertain [171, 183].

Preclinical model–experiment loops can address a more limited but essential task: defining which variables must be observed, and at what resolution, for feedback-guided therapy to remain mechanistically meaningful. In the context of AT, this includes determining when global observables such as tumor burden are sufficient, when spatial information such as local density or clonal position is required, and how uncertainty in these measurements affects treatment decisions. Iterative cycles between experiment, simulation, and policy evaluation therefore provide a practical testbed for ecological dosing strategies, while also clarifying the components required before similar principles can be transferred into patient-specific digital-twin frameworks.

1.3.2 Controversies and Boundaries

As AT concepts advance toward quantitative and individualized implementation, the field is moving from broad conceptual promise toward explicit, testable boundary conditions for ecological control. Key questions concern how spatial organization, dosing depth, monitoring frequency, and algorithmic control interact with biological variability and clinical responsibility [15]. These boundaries define both the scientific scope and the translational context of AT.

At the biological level, tumors differ in the extent to which their evolution is well mixed or spatially structured [31, 89]. This makes containment inherently context dependent: AT is most plausible when sensitive and resistant populations interact strongly enough for competition to matter, but least reliable when resistant cells are spatially protected, rapidly dispersing, or only weakly constrained by sensitive competitors. In practice, limited access to high-resolution information on tumor architecture, clonal composition, and spatial heterogeneity restricts the personalization of AT beyond indirect or population-level proxies [15].

At the therapeutic level, reduced-intensity dosing may limit selection but also risks rebound growth or renewed expansion if treatment is withdrawn too deeply or for too long [15]. Plasticity and tolerance can further erode containment by blurring the distinction between sensitive and resistant states [28, 189], and clinical experience with structured antiretroviral treatment interruption shows that treatment holidays can increase clinical risks [190]. In settings of cost-free resistance, rapid dispersal, or poor observability, CT may therefore remain the safer control strategy [15]. This boundary becomes particularly

important when resistant cells may be present at low but nonzero frequencies. If resistance is truly absent and eradication is clinically achievable, CT may remain appropriate. If resistance is already seeded, even at low levels, AT may prolong control by maintaining sensitive competitors that suppress resistant expansion [15, 191]. The difficulty is that these scenarios are hard to distinguish clinically: when resistant cells are rare or undetectable, it is often unclear whether they are absent or merely below the detection threshold. This ambiguity complicates treatment choice and underscores the need for reliable markers of ecological state [171].

From a computational and operational perspective, optimization and learning-based controllers can uncover non-intuitive dosing strategies *in silico*, but their translational value depends on robustness to uncertainty and real-world variability [192]. RL policies may overfit simulator assumptions, while optimal-control solutions can remain sensitive to parameter errors, delays, and model misspecification [173]. These concerns motivate uncertainty quantification, constraint-aware design, interpretability, and human oversight as practical requirements for translation. Implementation further depends on reliable data integration and clinical workflows capable of executing adaptive regimens transparently. Patient adherence is an additional practical determinant of whether the intended treatment exposure is achieved, particularly for oral anticancer agents [172].

Finally, at the ethical and societal interface, these technical challenges intersect with patient perception and clinical decision-making. Treatment holidays may be difficult to communicate within clinical cultures oriented toward continuous tumor reduction, and algorithmic optimization makes explicit value judgments about how survival, toxicity, tumor burden, and quality of life are weighted [15, 193, 194]. AT therefore remains not only a biological and computational problem but also a clinical and ethical one, requiring transparency, shared decision-making, and responsible automation.

These issues represent only a subset of the obstacles facing AT. Progress will depend on sustained efforts across modeling, experimentation, and clinical practice to identify when ecological control is justified, measurable, and safer than eradication-oriented treatment. Many of the constraints discussed above, spatial heterogeneity, phenotypic plasticity, ecological interactions, and limited observability, also define the variables that future adaptive strategies must learn to measure, model, and control.

1.3.3 Concluding Remarks

The background of this thesis establishes cancer as an evolving eco-physical system shaped by spatial, mechanical, and transport constraints. Solid tumors cannot generally be treated as well-mixed populations at the spatial and temporal scales relevant to resistance evolution. Instead, geometry, crowding, tissue mechanics, and environmental gradients influence how malignant cells proliferate, compete, persist, or escape therapy. Within this framework, AT emerges as a strategy of ecological control: instead of aiming solely for maximal tumor reduction, it seeks to modulate selective pressure so that sensitive cells can continue to constrain resistant populations. Spatial agent-based models, reduced ecological descriptions, and RL approaches provide complementary tools for linking these local mechanisms to measurable population-level behavior and for exploring how treatment schedules might exploit them.

Translating these principles into experimentally testable systems requires the simultaneous observation and control of the ecological mechanisms that govern AT. *In vivo* models reproduce physiological complexity, but local competition, spatial confinement, and clonal state are difficult to measure directly. Classical cell-culture systems provide control and optical access, but often lack the spatial structure and physical constraints that make containment meaningful. Bridging this gap requires experimental platforms that are both mechanistically transparent and ecologically structured: systems in which spatial organization, density, confinement, and drug exposure can be manipulated while the fates of sensitive and resistant populations remain observable over time. Stable genetic instrumentation and fluorescence-based tracking provide the technical foundation for such systems by enabling lineage-resolved measurements and controlled perturbation of resistance. Paired with mechanistic *in silico* models, these platforms can serve as preclinical testbeds for simulation–experiment cycles in which ecological dosing strategies are developed, evaluated, and refined under controlled conditions.

The following chapters build directly on these conceptual and methodological foundations. They first describe the construction of a modular genetic toolbox for engineering cancer cell populations with defined sensitivities and optical labels, establishing a reproducible platform for studying AT *in vitro*. These cell lines were then used to develop structured 2D and 3D melanoma assays that test how crowding, confinement, spatial escape, and ecological feedback shape therapeutic control. In this way, the theoretical principles introduced in the background are translated into experimentally accessible and quantitatively

interpretable model systems.

2

Experimental Model Engineering

To study adaptive therapy *in vitro*, I established a genetic toolbox that enables fluorescently distinguishable melanoma cell populations with defined drug-response phenotypes to be placed into direct competition under controlled conditions. Stable fluorescent labels allow the sensitive and resistant populations to be tracked longitudinally, while defined resistance markers permit treatment to be applied as a controlled ecological perturbation.

The principal validated output of this toolbox is a matched Mel Juso pair comprising the G418-sensitive line $\alpha 2$ labelled with enhanced green fluorescent protein (EGFP) and the G418-resistant line $\alpha 4$ labelled with mScarlet-H (hereafter mScarlet). These lines form the operational basis of the 2D and 3D competition experiments presented in Chapters 3 and 4.

Additional reporter and resistance constructs broaden the platform. The design and assembly of a first-generation inducible resistance module provided a molecular starting point for future development, with functional establishment in melanoma cells remaining an open step.

2.1 Design and Assembly of Plasmid Constructs

To achieve stable genomic integration at a defined locus, I chose a CRISPR/Cas9 system. Viral and transposon-based methods can yield stable expression but integrate at non-predefined sites, potentially causing variable expression or disruption of endogenous loci [195]. In contrast, CRISPR/Cas9 enables programmable targeting of a defined locus [196]. A single-guide RNA (sgRNA) directs the Cas9 nuclease to create a double-strand break, which can be repaired by homology-directed repair (HDR) using a co-delivered donor plasmid with matching homology arms.

As the genomic landing pad, the AAVS1 locus was selected, located in intron 1 of the *PPP1R12C* gene on chromosome 19. This locus is widely recognized as a "safe harbor". Genomic integration here impairs the endogenous protein but yields stable and robust expression of transgenes without interfering with crucial cell function [197]. The AAVS1 locus has been extensively validated in therapeutic and experimental contexts, making it an ideal choice for the engineered cancer cell model [145].

A further advantage of the AAVS1 donor system developed by Oceguera-Yanez et al. lies in its ability to couple targeted genomic integration to selectable marker expression [145]. In the donor design used here, the puromycin resistance cassette lacks its own promoter and is designed to become expressed through splicing into the endogenous *PPP1R12C* transcript after integration into intron 1. This arrangement provides a practical means of enriching for cells with targeted integration during antibiotic selection, facilitating the establishment of stable reporter populations [145].

The system comprised an AAVS1-targeting Cas9/sgRNA plasmid, an EGFP donor for establishing the sensitive reporter line, and an empty donor backbone for generating custom reporter–resistance combinations. These constructs are described below using Roman identifiers and functional names; their corresponding identifiers and full source identities are listed in table 7.10. The engineered donors and IRM constructs are mapped to their identifiers in table 7.11.

I. AAVS1_Cas9

This plasmid provides the Cas9 endonuclease and an sgRNA targeting the AAVS1-T2 site. The enhanced specificity Cas9 variant eSpCas9(1.1) was chosen to minimize off-target activity while maintaining efficient cleavage [144, 198].

II. AAVS1_EGFP

This donor combines AAVS1 homology arms and the promoterless puromycin selection cassette with EGFP expression under the strong cytomegalovirus (CMV) early enhancer/chicken β-actin (CAG) promoter [145]. It enabled initial testing of the integration strategy and established the EGFP-labelled, G418-sensitive α2 cell line used in the subsequent competition experiments.

III. AAVS1_empty

This backbone contains the same AAVS1 homology arms and puromycin selection cassette but no fluorescent reporter or additional resistance payload [145]. It served as the starting point for my custom donor constructs, allowing alternative fluorophores and resistance markers to be combined within a shared targeting framework.

2.1.1 Choice of Fluorescent Reporters

Tracking the spatial dynamics of competing sensitive and resistant populations over time required fluorescent reporters that were spectrally distinct, photostable, and suitable for repeated live-cell imaging. I therefore selected four monomeric fluorophores covering complementary regions of the visible spectrum:

EGFP: the enhanced green fluorescent protein, a widely used green-fluorescent protein derivative with robust expression and well-characterized excitation/emission properties [199]. EGFP served as a benchmark fluorophore in initial construct testing and provided compatibility with existing laboratory imaging workflows.

mNeonGreen (Neon): a bright green fluorescent protein with high molecular brightness compared to classical GFP derivatives, making it particularly suitable for long-term live-cell imaging [200].

mScarlet: a monomeric red fluorescent protein with excellent brightness and maturation kinetics, serving as the red counterpart to the green reporters and enabling two-color competition assays [143].

mCarmine: a synthetic far-red fluorophore engineered for deep tissue penetration and reduced autofluorescence, intended to enable tracking of an additional cell population in multi-color experiments [201].

2.1.2 Reporter–Resistance Constructs

Starting from AAVS1_empty, I generated two donor constructs combining mScarlet with either *NeoR* for G418 resistance or a $BRAF^{V600E}$ expression cassette for investigating a clinically motivated resistance mechanism. Both donors retained puromycin selection for cell-line engineering.

The donors were assembled from polymerase chain reaction (PCR)-amplified fragments using high-fidelity DNA assembly and verified by sequencing, following the workflow described in chapter 7.

Reporter	**None**	***NeoR***	***BRAF*^{V600E}**	***MGMT***	***HygroR***
None	III	–	–	–	–
EGFP	II	–	–	–	–
mScarlet	–	IV, IX_{NLS}	V	–	–
mCarmine	–	VII	–	VI	–
Neon	–	–	–	–	VIII_{NLS}

Table 2.1: Reporter and resistance combinations. Donor constructs individually described in this section, grouped by reporter and additional resistance. Roman identifiers refer to the numbered construct descriptions; NLS subscripts indicate nuclear localization. "None" denotes the absence of an additional reporter-associated resistance payload. Backbone selection markers are specified in Methods. Dashes indicate combinations not represented among the donors shown.

IV. AAVS1_mScarlet_NeoR

This donor combines mScarlet with the neomycin phosphotransferase II gene (*NeoR*), which confers resistance to G418 [143, 202]. It was used to generate the red-fluorescent, G418-resistant Mel Juso line $\alpha 4$ for competition assays. Whereas puromycin served primarily for cell-line engineering, G418 provided the defining drug-response difference between the sensitive $\alpha 2$ and resistant $\alpha 4$ populations in subsequent ecological experiments.

V. AAVS1_mScarlet_$BRAF^{V600E}$

This construct combines mScarlet with a $BRAF^{V600E}$ expression cassette intended to explore a clinically motivated resistance mechanism [143, 203]. $BRAF^{V600E}$ was chosen because activating *BRAF* mutations are common drivers in melanoma and define a clinically

important therapeutic target. Acquired resistance to BRAF inhibitors can arise through copy-number gain of $BRAF^{V600E}$, which restores MAPK signaling despite drug exposure [94]. *BRAF* amplification has also been identified in progressing melanomas following combined BRAF/MEK inhibition [97, 204]. By introducing an additional $BRAF^{V600E}$ cassette, this donor was designed to explore whether increased mutant BRAF abundance could confer MAPK-inhibitor resistance in the respective parental background. Mel Juso (α), which lacks $BRAF^{V600E}$, was unsuitable for modeling this resistance mechanism. SK-Mel-28 (β), which already harbors $BRAF^{V600E}$, was therefore the more appropriate background for testing the construct [205].

To provide an alternative selection route, I also generated a hygromycin B-selectable version of AAVS1_empty. This provided a backup if puromycin selection proved suboptimal and enabled independent selection of different genetic modules. From this backbone, I generated derivatives combining mNeonGreen with puromycin resistance [206], mScarlet with NeoR, or mScarlet with $BRAF^{V600E}$. The latter two retained the reporter–resistance combinations of the donors described above while providing hygromycin B selection, thereby extending the available combinations of reporters, resistance determinants, and engineering-selection markers.

2.1.3 Far-Red and Nuclear-Localized Reporters

To extend competition assays to additional cell populations and improve automated image segmentation, I developed far-red reporter constructs and nuclear-localized variants of the green and red reporters. I explored mCarmine as a far-red reporter to distinguish an additional cell population in multi-color competition assays, potentially including immune cells [201]. Far-red fluorescence was selected for its favorable imaging properties: improved tissue penetration, reduced interference from cellular autofluorescence, and spectral separation from the green and red channels [207, 208].

VI. AAVS1_mCarmine_MGMT

This plasmid combined mCarmine with the *MGMT* gene. *MGMT* encodes a DNA repair enzyme that counteracts the cytotoxic DNA alkylation induced by dacarbazine [201, 209]. Elevated MGMT expression is a well-established mechanism of resistance to dacarbazine and related alkylating agents, as it restores DNA integrity by removing O-6-methylguanine

adducts [210, 211]. Thus, this donor was designed not only to expand the spectral palette but also to introduce a resistance mechanism directly relevant to clinical chemotherapy.

VII. AAVS1_mCarmine_NeoR

This construct paired mCarmine with *NeoR*, providing a selectable far-red counterpart to the mScarlet-based donors [201, 202].

To improve automated segmentation, I also generated reporter variants carrying a nuclear localization signal (NLS). Nuclear localization was particularly attractive because nuclear signals are more uniform in intensity, less influenced by changes in cell shape, and usually facilitate easier segmentation in automated microscopy pipelines [212].

VIII. AAVS1_NLS-Neon_HygroR

This donor combined nuclear-localized Neon with a hygromycin B resistance cassette, providing a green reporter designed for nuclear imaging and segmentation [206]. Additional puromycin-resistance and cytosolic variants provided alternative selection and localization options. Their identities and construction relationships are listed in table 7.11.

IX. AAVS1_NLS-mScarlet_NeoR

This construct was generated by adding an NLS to AAVS1_mScarlet_NeoR, yielding a nuclear-localized mScarlet reporter under puromycin selection. The rationale was to facilitate nuclear imaging and more reliable quantification of individual resistant cells.

Summary

Starting from the empty AAVS1 donor backbone, I established a modular library of reporter–resistance constructs for competition experiments (table 2.1). Alternative selection markers increased flexibility in cell-line engineering, while far-red and nuclear-localized reporters expanded the available designs for population tracking and image segmentation. Additional resistance cassettes provided starting points for investigating clinically motivated resistance mechanisms.

The EGFP and mScarlet_NeoR donors were used to establish the functionally validated $\alpha 2/\alpha 4$ cell-line pair selected for the core spatial competition experiments of this thesis. The remaining constructs provide a molecular basis for future extensions of the platform. The following section describes the generation and validation of these reporter cell lines through CRISPR/Cas9-mediated knock-in.

2.2 Engineering and Validation of Melanoma Reporter Cell Lines

With the donor plasmid library established, I generated stable melanoma reporter lines by co-transfecting AAVS1_Cas9 and the respective donor vector for CRISPR/Cas9-mediated targeting of the AAVS1 safe-harbor locus. Although recombinant Cas9 with synthetic sgRNAs was evaluated, the plasmid-based system was ultimately used for all core experiments due to its consistency in melanoma cells.

Melanoma was selected as a model system because of its aggressive clinical behavior, high mutational burden, and well-characterized resistance pathways [213, 214]. Two commonly used human melanoma lines were obtained through collaborations in Erlangen: Mel Juso (referred to as α), derived from a primary tumor, and SK-Mel-28 (referred to as β), originating from a metastatic lesion. The experimental work subsequently focused on α cells, which provided a broader therapeutic window and more homogeneous growth behavior than β cells. Nevertheless, the overarching goal was to derive mechanistic insights into resistance management that extend beyond melanoma to spatially structured cancers more broadly.

2.2.1 Establishing Stable Reporter Expression

Before applying my custom donor constructs, I carried out proof-of-concept transfections to establish whether CRISPR/Cas9-mediated knock-in was feasible in melanoma cells and to optimize the integration workflow. These initial experiments used the unmodified AAVS1_EGFP donor, allowing direct monitoring of EGFP expression and survival under puromycin as readouts for successful integration. A detailed protocol for transfection and selection can be found in chapter 7.

Transient EGFP expression was already visible 24 hours post-transfection in Cas9$^+$ donor conditions, while no signal was observed in negative controls. Under puromycin selection, most cells in control wells died rapidly, whereas colonies of EGFP-positive cells began to appear in Cas9$^+$ donor wells. By day 6, distinct resistant colonies were clearly visible, confirming that the combined delivery of Cas9 and donor plasmid could generate stably integrated knock-ins in melanoma cells. SK-Mel-28 cells expanded more slowly than Mel Juso cells, but by day 18 both lines had established confluent EGFP-positive populations.

To evaluate whether linearizing the donor plasmid would improve knock-in efficiency, I digested the donor with the restriction enzyme NotI prior to transfection. In this experiment, however, EGFP-positive colonies also appeared in donor-only control wells after puromycin treatment, indicating that linearized donors were prone to random integration. This highlighted the necessity of including $Cas9^+$ donor together with Cas9-only and donor-only controls in every experiment, to discriminate true homology-directed knock-ins from random insertions.

These experiments established that CRISPR/Cas9 knock-in was feasible in melanoma cells. Donor linearization was not pursued further because it did not prevent random integration. Careful titration of antibiotic selection was also critical, as overly harsh early selection could eliminate correctly targeted clones, whereas insufficient dosing allowed survival of non-edited cells. Based on these results, I established a standardized protocol that I subsequently applied to my custom AAVS1_empty-derived donors.

2.2.2 CRISPR/Cas9 Knock-In of the Donor Plasmids

Each construct CRISPR-integrated in this chapter contained specific combinations of fluorescent reporters and resistance elements tailored for AT experiments, as described in section 2.1. Transfections followed the standardized protocol described in chapter 7, with only minor adjustments when required by plasmid-specific or cell line–specific factors. An overview of successful genomic integrations for representative constructs is provided in Figure 2.1.

Most of the donor constructs described above were used in CRISPR/Cas9 integration experiments. These efforts generated a series of Mel Juso and SK-Mel-28 reporter lines, of which only a subset was used in subsequent experiments. The Mel Juso-derived $\alpha 2/\alpha 4$ pair was selected as the central model for this thesis owing to its more suitable growth kinetics and phenotypic consistency.

The core validated system comprised $\alpha 2$, generated using AAVS1_EGFP and serving as the EGFP-labelled, G418-sensitive population, and $\alpha 4$, generated using AAVS1_mScarlet_NeoR and serving as the mScarlet-labelled, G418-resistant population. Together, these lines formed the operational sensitive–resistant pair throughout this dissertation. The correspondence between cell-line identifiers and the plasmids used to generate them is listed in table 7.13.

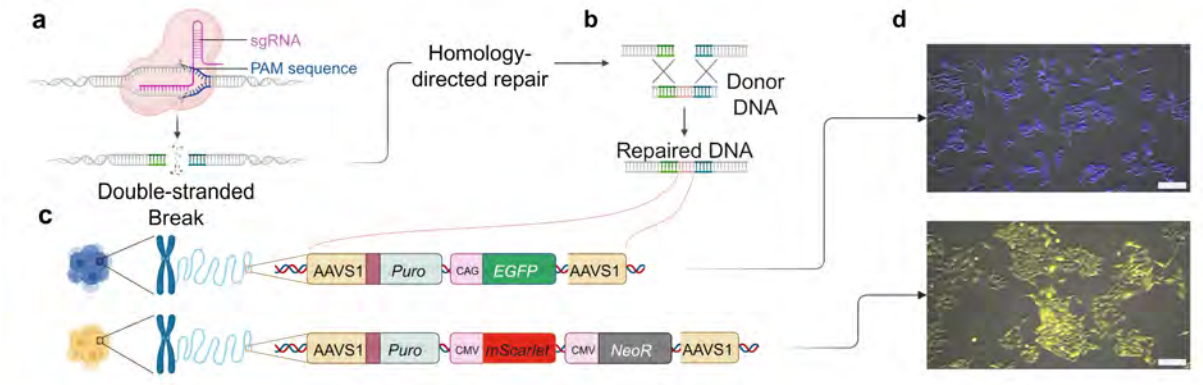


Figure 2.1: Engineering of matched melanoma model cell lines This schematic illustrates the conceptual workflow used to create fluorescently labeled tumor cell lines differing in drug response. The CRISPR/Cas9 system (a) uses a sgRNA to guide Cas9 to the AAVS1 target sequence adjacent to a protospacer adjacent motif (PAM) site, where Cas9 introduces a double-strand break. During HDR (b), a donor DNA template carrying the desired transgene is integrated into the genome. The donor constructs (c) encode a selection cassette with EGFP for the sensitive (blue) population and mScarlet_NeoR for the resistant (yellow) population. The resulting melanoma lines (d), derived from a shared parental background, display distinct fluorescence and resistance profiles. Scale bar: 30 µm. Created with BioRender.com [20].

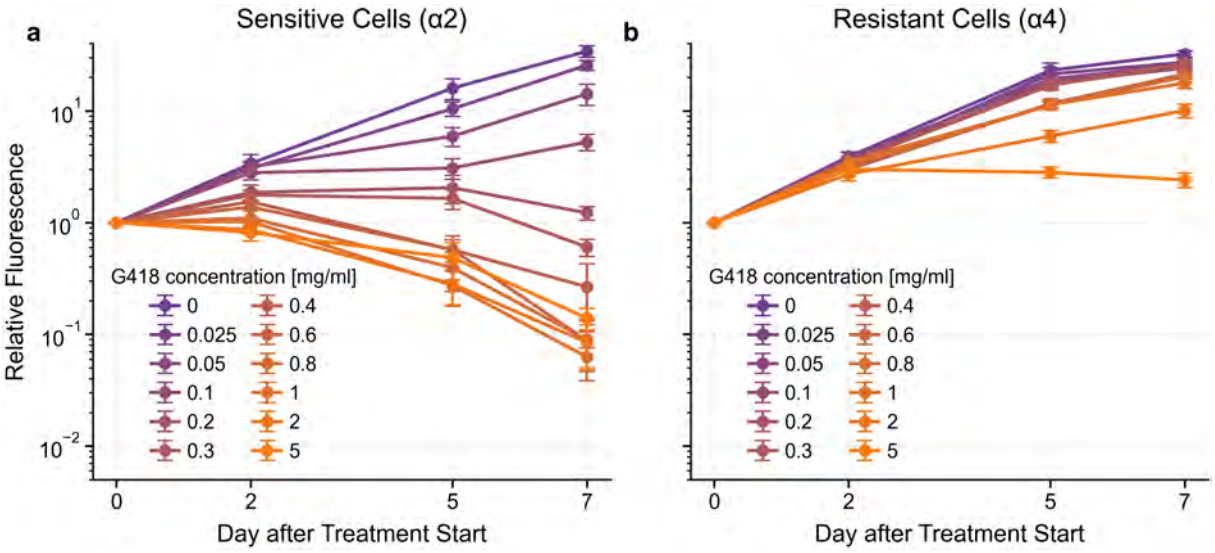


Figure 2.2: Differential drug-responses of sensitive and resistant cells. Panels (a) and (b) show AlamarBlue fluorescence of sensitive $\alpha 2$ and resistant $\alpha 4$ cells, respectively, over seven days, plotted on a logarithmic scale. Fluorescence reflects cellular metabolic activity and was background-corrected, calibrated against Rhodamine B, and normalized to each well's pretreatment measurement (day 0 = 1), taken two days after seeding immediately before treatment initiation (section 7.2.2). Points represent means ± SD of eight technical replicates per condition.

Drug sensitivity was quantified to functionally validate both lines. Using the AlamarBlue assay described in chapter 7, G418 concentrations were tested to identify conditions under which resistant cells survived while sensitive cells were eliminated. As shown in figure 2.2, sensitive cells were efficiently killed between 0.2 and 1.0 g/l G418, whereas resistant cells remained viable.

2.3 Design of an Inducible Resistance Module

The reporter-cell platform established above enables controlled competition experiments between pre-existing sensitive and resistant populations. These experiments begin with predefined resistance phenotypes and therefore focus on ecological interactions between established competitors. To extend the framework toward evolutionary dynamics, I sought to implement a system in which individual cells could transition from a sensitive to a resistant state during population growth.

In biological tumors, the probability of new resistant variants emerging generally increases with total proliferative population size [215], and the interplay between competitive inhibition and continual introduction of new variants is inherently nonlinear [15]. Existing *in vitro* models rarely capture this balance. Most rely on fixed mixtures of resistant and sensitive cells or on bulk selection protocols that lack spatial and temporal precision [216]. Controlled, stochastic, and temporally adjustable resistance induction would allow this interplay to be investigated within growing populations.

To realize this functionality, I designed an inducible resistance module (IRM) based on Cre/loxP recombination. The system uses CreERT2 recombinase, which remains largely inactive in the cytoplasm until exposure to 4-hydroxytamoxifen (4-OHT). Upon induction, CreERT2 translocates to the nucleus and catalyzes recombination between flanking loxP sites, permanently converting the genetic payload. This established mechanism provides a basis for temporally controlled recombination in mammalian systems [217, 218]. I adapted it to couple a change in fluorescent barcode to a transition from a hygromycin B-sensitive to a hygromycin B-resistant state. The IRM was thereby designed to introduce optically traceable resistant cells within previously homogeneous populations. A conceptual overview is shown in figure 2.3.

2.3.1 Molecular Architecture of the IRM

The construct comprises four functional components.

- First, a CRISPR/Cas9-compatible backbone provides defined homology arms for targeted integration into the AAVS1 safe-harbor locus and retains the puromycin cassette for cell-line engineering.
- Second, a CreERT2 recombinase module under a weak promoter enables ligand-dependent recombination and thereby control over recombination frequency.
- Third, a loxP-flanked expression cassette contains the *mKate2* and *NeoR* genes, marking the unswitched state and enabling its maintenance under G418 selection while leaving cells sensitive to hygromycin B.
- Fourth, a downstream promoterless cassette encodes EGFP and a hygromycin B resistance, which become transcriptionally active only after recombination and thereby define the hygromycin B-resistant recombined state.

This modular layout couples fluorescent color and drug resistance in a mutually exclusive manner, allowing visual discrimination and pharmacological selection of each phenotype. The design also ensures compatibility with future modifications, including alternative promoters or fluorophores, without altering the overall recombination logic.

The *CreERT2* gene is expressed from an enhancer-free simian virus 40 (SV40) promoter to limit basal recombinase activity. Downstream, an EF1α promoter drives the initial loxP-flanked cassette, consisting of *mKate2*, an SV40 polyadenylation signal, and a CMV-driven *NeoR* gene. mKate2 was selected because its red emission, brightness, and photostability support imaging in optically dense samples [219, 220]. Although the red channel was conceptually intended to mark rare resistant cells, in this first construct it labels the initial sensitive state for technical reasons.

Upon 4-OHT-induced CreERT2 activation, the loxP-flanked *mKate2–NeoR* cassette is excised, placing the EF1α promoter upstream of a previously promoterless *EGFP*–T2A–*HygroR* cassette. This recombination event switches cells from an mKate2-positive, G418-resistant initial state to an EGFP-positive, hygromycin B-resistant recombined state. The T2A sequence enables production of EGFP and a hygromycin B resistance protein from a single transcript, which is terminated by an SV40 polyadenylation signal. All coding sequences were codon-optimized for human expression.

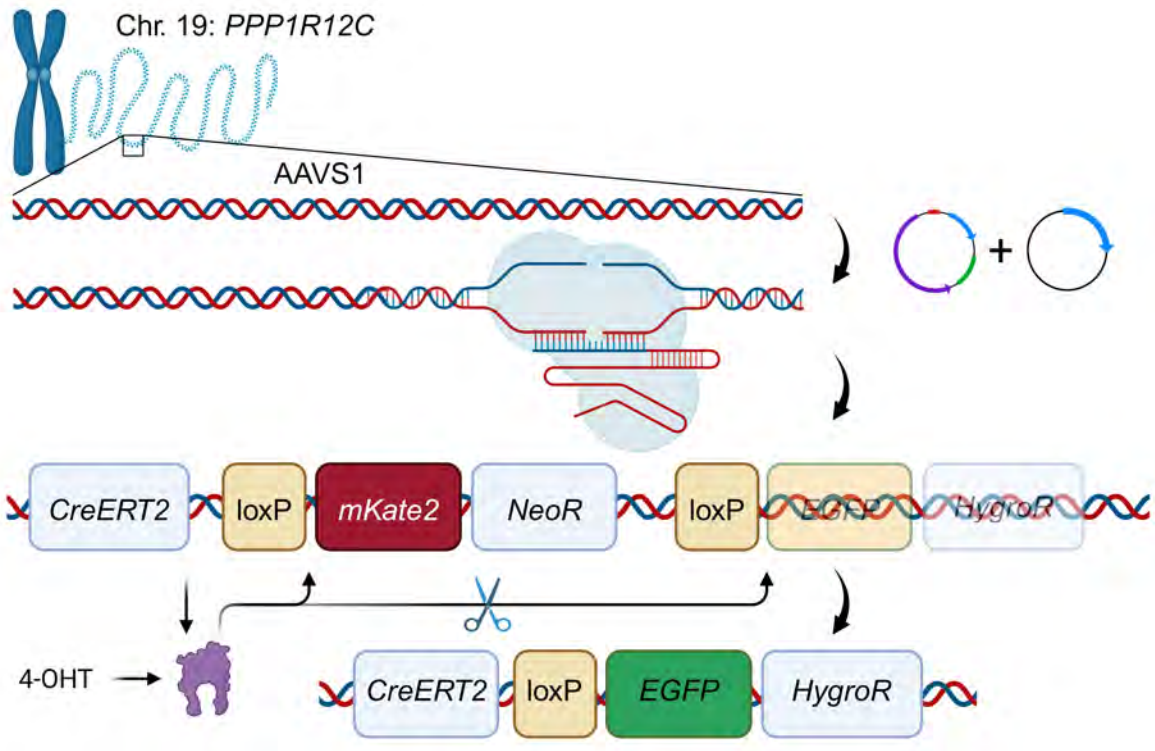


Figure 2.3: Conceptual workflow for introducing synthetic mutations Schematic overview of the proposed IRM for modelling synthetic resistance events in melanoma cells. The cassette was designed for CRISPR/Cas9-mediated knock-in at the AAVS1 safe-harbor locus and contains an mKate2-marked initial state, CreERT2, loxP sites, and a downstream EGFP-linked resistance cassette. Upon 4-OHT exposure, CreERT2 is intended to excise the loxP-flanked sequence, switching cells to an EGFP-labelled resistant state. Created with BioRender.com [20]

2.3.2 Construction and Initial Evaluation of the IRM

To implement the IRM design, I adapted AAVS1_empty for modular assembly and obtained the cassette as six synthetic DNA modules. Alternative promoter and resistance modules were obtained so that expression control and resistance phenotype could be adapted within the modular design (table 2.2). Only one version of each functional element was used in a given assembly. The first-generation design described above used the hygromycin B resistance option. Backbone preparation and assembly procedures are described in section 7.1.10. The modified backbone, intermediate plasmids, and final donor clones are identified in table 7.11.

Initial one-step Golden Gate and Gibson Assembly approaches did not yield a correctly assembled full-length construct. Two sequence-verified intermediates containing comple-

Module	Alternative 1	Alternative 2
SV40 promoter	Ubc promoter	
CreERT2		
EF1α–mKate2		
EF1α promoter	CMV promoter	cBh promoter
NeoR–EGFP–T2A		
$BRAF^{V600E}$	hygromycin B resistance	

Table 2.2: DNA modules and design alternatives for IRM construction. The ordered synthetic DNA modules included alternative promoter and resistance-cassette components to support adaptation of expression control and resistance phenotype.

mentary portions of the cassette were therefore generated. When direct joining also proved unsuccessful, the assembly was redesigned around four overlapping PCR fragments to reduce individual amplification lengths while limiting the number of assembly junctions. This Gibson strategy yielded two full-length first-generation IRM donor clones.

Whole-plasmid sequencing identified a nonsynonymous mutation in the puromycin resistance gene in both clones and an additional amino acid substitution in *CreERT2* in one. Full-length assembly was thus achieved, with the remaining sequence deviations requiring correction or functional assessment.

Subsequent transfection into melanoma cells did not establish a functionally validated IRM reporter line. The sequence deviations may have impaired antibiotic selection or recombination, while the large donor size may also have limited Lipofectamine3000-mediated delivery. The experiments did not resolve the relative contributions of these possible limitations. The next subsection outlines design and delivery changes intended to address them and enable functional testing of the proposed resistance switch.

2.3.3 Future Improvements for IRM Design

The first-generation IRM donor provides a molecular starting point for a sequence-corrected redesign. Full-length plasmid synthesis would allow the identified sequence deviations to be corrected and the following design changes to be incorporated directly:

- Reporter configuration: Replace the current EGFP and mKate2 reporters with

mGreenLantern and mScarlet or a far-red fluorophore, assigning green to the initial sensitive state and red or far-red to the recombined resistant state. This aims to improve brightness, spectral separation, and detection in 3D spheroids.

- Expression architecture: Link each reporter to its resistance gene via a T2A sequence and remove redundant or inverted polyadenylation sites to reduce cassette size and improve transcriptional reliability.
- Two-cassette configuration: Place the switching and selection modules on separate donor constructs to reduce individual donor size and allow independent modification. Integration into different alleles would require verification to ensure that both constructs do not occupy the same allele.
- Delivery: Test electroporation in place of Lipofectamine3000 to determine whether it improves delivery of the large donor constructs and whether delivery limitations contributed to the unsuccessful integrations [221].

The redesigned IRM would then require functional evaluation of fluorescence readout, recombination dynamics, and selection fidelity.

The validated reporter-cell pair established in this chapter provides the basis for the competition experiments that follow. Functional establishment of the IRM will extend this framework toward controlled resistance emergence within spatially structured populations in the future.

3

Biophysical Mechanisms of Resistance Control

Treatment-induced depletion of sensitive tumor cells can weaken local competition and permit resistant populations to expand. AT aims to exploit these competitive interactions by preserving sensitive populations that constrain resistant growth [12]. To investigate how these principles operate in practice, this chapter presents a 2D *in vitro* framework that enables long-term tracking of mixed melanoma populations under controlled, dynamically modulated treatment schedules.

Building on the genetic toolbox established in Chapter 2, the assay combines sensitive and resistant reporter lines in a spatially structured colony system. The platform enables clone-resolved observation of population dynamics, allowing local competitive interactions and spatial constraints on resistant expansion to be quantified over time. Using this framework, the chapter investigates how treatment-induced changes in crowding influence the growth and spatial organization of resistant clones, and under which conditions competitive suppression can be maintained or lost. Understanding these density-dependent

spatial dynamics is critical for identifying strategies that sustain long-term treatment control. Both experimentally defined treatment schedules and model-guided optimization approaches are therefore evaluated, linking treatment decisions to their spatial and population-level consequences.

The preliminary assay development and optimization presented in Section 3.1 draw substantially on work performed by Elias Fischer during his master's thesis under my supervision, beginning in 2023. The experimental work included both experiments we conducted together and experiments Elias Fischer performed under my supervision. Results from Elias Fischer's master's thesis are presented alongside additional assay-development work to document the establishment of the experimental platform, with selected figures, tables, and passages reproduced or adapted from his thesis with his permission and are duly acknowledged. After continuing on the project as a PhD student in 2024, Elias Fischer established the two-step seeding protocol and further improved its reliability. The contributions to the subsequent publication-based investigations are described separately at the beginning of Section 3.2.

3.1 Development and Optimization of the 2D Colony Assay

Most existing *in vitro* cancer models used to study AT or the evolution of resistance under therapy omit well-controlled spatial structure. Conventional 2D monolayers provide a homogeneous environment without confinement, while 3D spheroids introduce gradients of oxygen and nutrients that complicate interpretation [18, 160, 216]. Neither approach allows systematic observation of how local crowding and spatial confinement influence the dynamics of sensitive and resistant subpopulations under (adaptive) therapy. To overcome these limitations, a mechanistically transparent 2D colony system was developed that preserves spatial organization while remaining optically accessible. This design provides an intermediate platform between abstract ecological models and complex 3D tumor systems by retaining spatial confinement and local competition while minimizing confounding microenvironmental gradients, thereby enabling direct experimental analysis of the ecological mechanisms underlying AT.

The guiding principle was to generate dense but planar colonies that mimic the crowding and boundary conditions of tumor growth while limiting the formation of strong nutrient

and oxygen gradients inherent to spheroids. These colonies are compatible with widefield microscopy, enabling segmentation and quantification at the level of individual resistant cells. Furthermore, the system remains stable over months and supports dynamic, population-informed treatment decisions in addition to fixed dosing regimens. Colony densities reach levels sufficient to induce strong crowding effects, allowing controlled investigation of density-dependent competition.

3.1.1 First Colony Seeding Assay

The first step toward establishing a spatially structured assay was the development of a colony-seeding protocol. For this purpose, $\alpha 2$ (sensitive) and $\alpha 4$ (resistant) cell lines were selected based on their pronounced therapeutic window and their bright, homogeneous, and stable fluorescence. A droplet-based seeding method was adopted and refined, in which approximately 9,000 cells in a 3 µl suspension were pipetted into the center of wells of a 48-well plate. Resistant cells were mixed with sensitive cells at low initial frequencies (0.025–0.1%), such that most colonies contained only 1-5 resistant cells at the start. To minimize evaporation, the peripheral wells were filled with phosphate-buffered saline (PBS). After overnight adhesion, 1 ml of medium was carefully added to each well to avoid displacing the forming colonies. This procedure reliably generated circular colonies with a well-defined boundary and internal spatial structure.

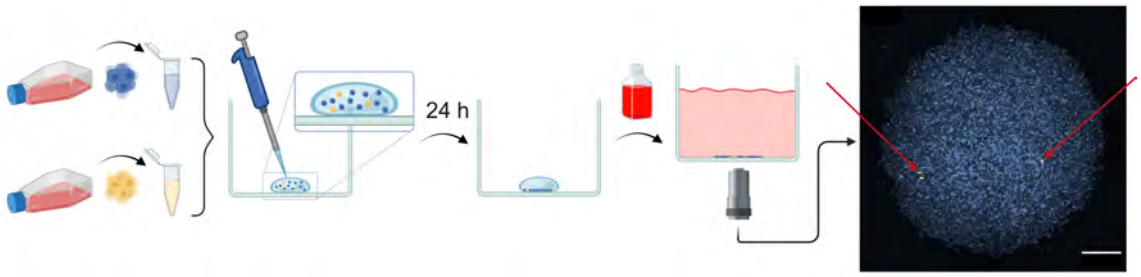


Figure 3.1: Illustration of the initial seeding workflow This schematic shows the standardized workflow used for generating 2D melanoma colonies. Sensitive (blue) and resistant (yellow) cell lines were detached from culture, counted, and mixed at defined ratios before seeding as 3 µl droplets of 3×10^6 cells/ml onto the center of a well. After a 24 h adhesion phase, Roswell Park Memorial Institute (RPMI) medium supplemented with 10% fetal bovine serum (FBS) was added to cover the well, yielding stable, radially confined colonies ready for imaging. Red arrows indicate single resistant cells. Scale bar: 500 µm. Created with BioRender.com [20]

While this initial setup provided proof-of-concept that spatially structured colonies could

be generated and maintained, several limitations became apparent. In densely populated regions, fluorescence signals were prone to overexposure, and medium exchange procedures occasionally introduced artifacts or displaced cells. Moreover, the spatial distribution of resistant cells was suboptimal for AT experiments, as many resistant clones emerged near the expanding front, effectively precluding confinement. This observation highlights that initial spatial positioning critically influences the ability of surrounding sensitive cells to exert competitive suppression. These limitations underscored the need to refine seeding, imaging, and handling protocols before the assay could provide reproducible insights into spatial competition under AT.

3.1.2 Assay Optimization

A first line of optimization addressed the instability of colonies during medium exchanges. I hypothesized that covering colonies with a thin gel layer could both stabilize their position and reduce artifacts caused by convection currents during handling. Low-melting point agarose overlays were tested in concentrations ranging from 0.5% to 2%. Overlays solidified well at 37 °C, but applying them at 42 °C caused immediate cell death, underlining the sensitivity of colonies to even modest thermal stress. At concentrations $\geq 1\%$, agarose overlays produced pronounced central cell loss, consistent with the emergence of radial viability gradients or impaired transport within the colonies. Very low agarose concentrations (0.5%) initially appeared promising for stabilization. Over extended culture, no pronounced radial viability differences were apparent before treatment, but treatment cycles revealed clear death gradients across the colonies. Because these radial viability patterns would have introduced unwanted ecological complexity into the system, agarose overlays were ultimately discarded as a stabilization method. Collagen overlays between 0.6 and 2.4 mg/ml in contrast provided more stable coverage and eliminated visible viability gradients, but markedly altered cell morphology by inducing elongated, fiber-aligned growth. These tests showed that overlays could eliminate medium-exchange artifacts but at the cost of either creating undesirable gradients or altering cell morphology as seen in figure 3.2. As a result, overlays were abandoned for the experiments in this thesis.

Since overlays did not provide a sustainable solution, attention shifted to optimizing medium exchange directly. Complete aspiration of wells frequently disrupted colonies and caused resistant cells to be transferred between wells. Several strategies were compared, including using Pasteur pipettes, serological pipettes, and multipipettes. The most robust

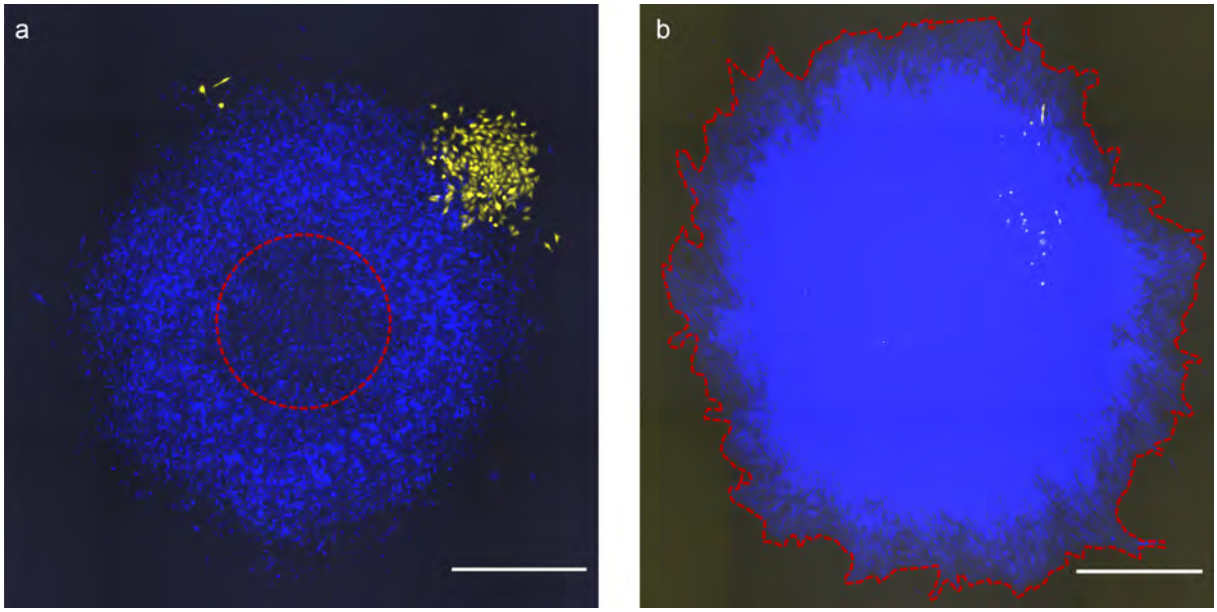


Figure 3.2: Effects of matrix overlays on 2D melanoma colonies Fluorescence microscopy images show 2D melanoma colonies overlaid with different matrices to test their stability during treatment handling. (a) Colonies covered with 0.5% agarose and treated with G418 developed a pronounced radial pattern of cell death, with faster loss of viability in the center (indicated with a red circle). (b) Collagen-overlaid colonies without treatment exhibited pronounced peripheral protrusions (highlighted with a dotted red line) and altered tissue morphology compared to standard conditions. Scale bar: 300 µm.

solution was to remove only approximately 800 µl of the 1 ml medium per well, leaving a thin fluid layer above the colonies. This adjustment stabilized colonies during handling and reduced artifacts, while also preventing cross-contamination. Medium addition with a multipipette on the lowest speed setting further minimized disturbances.

Another recurring problem was increased cell death in certain wells immediately after seeding, which was traced back to local evaporation effects. Increased cell death was repeatedly observed when plates were stacked in the incubator, consistent with uneven evaporation and drying of the colony droplets. Avoiding plate stacking and supporting plates individually during incubation markedly improved colony survival and reproducibility. Additionally, before seeding warm PBS was added to the spaces around the wells and after seeding, plates were placed inside a wet chamber within the incubator to keep humidity high and avoid evaporation issues.

Unexpectedly, the sensitivity of $\alpha 2$ cells to G418 was found to depend strongly on their passage history after thawing as quantified in figure 3.3. Colonies established directly

after thawing died quickly at 0.2 mg/ml G418 in the first treatment cycle but survived subsequent cycles at the same dose, suggesting cycle-dependent tolerance. To eliminate this variability, a standardized protocol was adopted. Resistant and sensitive cells were thawed, expanded, and only used for seeding after two weeks of passaging. Colonies seeded at this point displayed consistent responses to drug treatment across experiments, underlining the importance of growth-state equilibration after thawing.

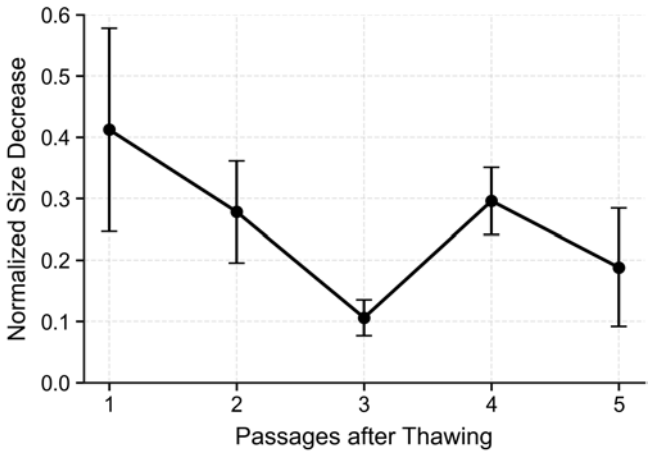


Figure 3.3: Treatment response depends on passage number The plot quantifies the decrease of normalized cell number from the initial value at day 0 to the post-treatment minimum following a 2-day G418 pulse at 0.2 mg/ml and subsequent regrowth, as a function of the number of passages after thawing. Although no strict linear relationship was observed, cells tested immediately after thawing displayed greater sensitivity to treatment, whereas cultures stabilized after two to three passages. Data represent the mean ± SD from $n = 10$ independent colonies per averaged data point.

Colony stability during drug exposure also depended on the time cells adhered before treatment. Experiments revealed that colonies required at least two days of adhesion after seeding before drug application produced reproducible results. To compensate for the higher cell numbers resulting from the longer growth period, the initial seeding density was reduced from 9,000 to approximately 6,000 cells per colony. This adjustment ensured that colonies reached the desired confluency by the onset of treatment without becoming overcrowded.

Reliable segmentation of individual cells was a central requirement for tracking resistant clones over time. While the $\alpha 2$ line provided bright cytosolic EGFP that already enabled good segmentation, I explored whether the $\alpha 3$ line carrying a nuclear mNeonGreen protein could further improve segmentation accuracy. However, $\alpha 3$ exhibited two major issues: strong spectral crosstalk with the red channel and unstable fluorescent intensity during

timelapse imaging. Colonies frequently developed non-fluorescent clones, complicating clone identification. Consequently, the $\alpha 2$ line was selected for all subsequent experiments. Attempts to use Hoechst 33342 as an additional nuclear marker were discontinued, as the dye persisted in dead cells and interfered with reliable viability measurements and segmentation.

Additionally, untreated colonies often developed dome-shaped multilayered structures over time, exceeding the focal depth of widefield imaging and rendering segmentation unreliable. To approximate total cell numbers in such cases, the area of overexposed pixels was multiplied by the average maximum density observed in still-segmentable colony areas and added to the Cellpose-derived counts. This correction mainly, but not exclusively, affected untreated wells, as colonies under CT or adaptive cycles typically remained mostly planar and segmentable.

Together, these iterative improvements defined the operational boundaries of the 2D colony system. The challenges encountered along the way clarified the practical constraints of spatially structured culture and ultimately informed the reproducible protocol used for the subsequent therapy experiments.

3.1.3 Towards Functional Therapy Experiments

After establishing a stable and reproducible colony assay, drug treatments were introduced to assess whether AT schedules could be meaningfully tested. The central variable at this stage was the treatment depth, defined by the extent to which sensitive cells were eliminated at a given dose of G418. Identifying the correct treatment depth was essential. Doses that were too high would eradicate the sensitive population too rapidly and prevent competition, while doses that were too low would fail to sustain selective pressure on sensitive cells.

In the AlamarBlue assay described in section 7.2.2, an initial effective G418 window of 0.2–1.0 mg/ml was identified. However, because treatment response depends strongly on spatial structure and local confluency, this range needed to be re-evaluated directly in the colony assay. Initial experiments, therefore, examined the response of colonies to a range of G418 concentrations delivered as a single pulse for two days. This is quantified and visualized in figure A.1, with primary image data shown in figure 3.4 for CT and in

figure 3.5 for a single-pulse treatment. Increasing drug concentrations produced progressively deeper treatment effects: At ≥0.7 mg/ml, most sensitive cells were eradicated, and colonies recovered only after prolonged lag phases, whereas at ≤0.2 mg/ml treatment was shallow, affecting primarily peripheral cells. At intermediate doses around 0.4 mg/ml, colonies displayed partial reduction of sensitive cells with delayed but measurable recovery.

AT schedules were first tested across different G418 concentrations to identify a treatment depth compatible with ecological competition. In this work, AT50, AT80, and AT100 denote dosing rules in which treatment is initiated once the total cell number exceeds its baseline value and withdrawn again when it falls below 50%, 80%, or 100% of baseline, respectively. eAT100 is an adapted version of this strategy that accounts for the delayed treatment-induced cell death by initiating treatment above 100% of baseline and withdrawing it once the cell number falls below 137%.

At 0.7 mg/ml, the first treatment cycle eliminated nearly all sensitive cells, leaving resistant clones as the dominant population (see figure 3.5C). As a consequence, AT50, AT80, and AT100 converged toward similar dosing patterns, and no advantage over CT was detectable. At 0.4 mg/ml, limited cycling of sensitive cells was observed, indicating that some ecological interactions were preserved. However, resistant growth appeared modestly reduced at this concentration, consistent with the small decrease in relative growth observed for $\alpha 4$ cells in the AlamarBlue assay (see figure 2.2). In this initial assay configuration, 0.4 mg/ml therefore still impaired resistant growth too strongly to cleanly isolate competition effects. By contrast, 0.2 mg/ml enabled repeated cycles of sensitive-cell depletion and regrowth under AT50, AT100 and eAT100. Exemplary image sequences are shown for CT and AT50 in figure 3.6 and figure 3.7, respectively.

Under these shallow treatment conditions, net resistant growth was negatively associated with sensitive-cell abundance across the pooled longitudinal data (see figure 3.8). In a generalized estimating equation (GEE) that accounted for repeated measurements within colonies and adjusted for treatment strategy and time, each additional 1,000 sensitive cells was associated with an average reduction of 112 resistant cells between consecutive observations (95% confidence interval (CI): −153 to −71; $p = 7.2 \times 10^{-8}$; 28 colonies). The magnitude of this association differed between treatment strategies, indicating that treatment context modulated the relationship (see figure A.2). These early experiments suggested that sensitive-cell abundance and spatial context influence resistant expansion.

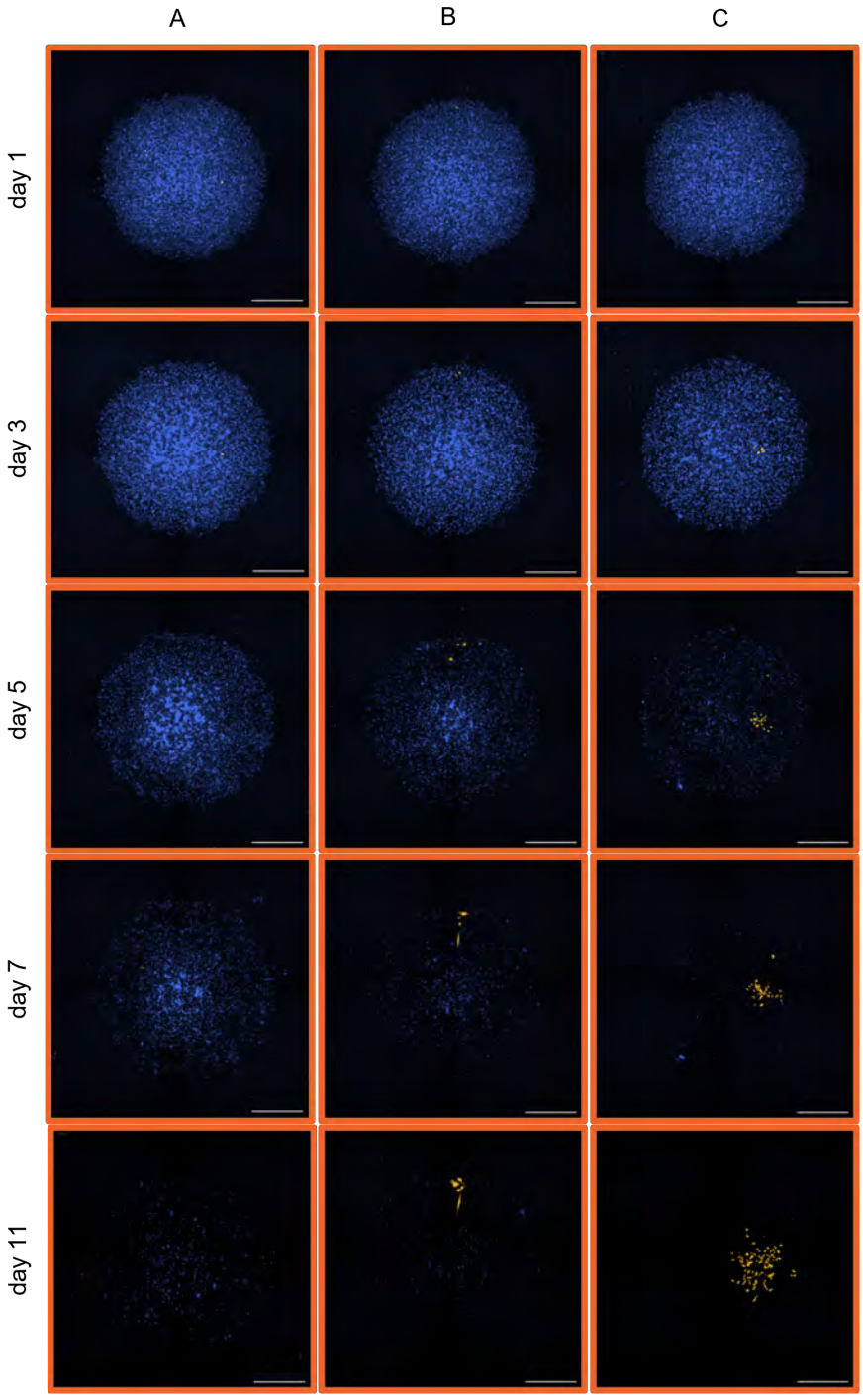


Figure 3.4: Increasing G418 doses accelerate the death of sensitive cells in CT Time-lapse fluorescence images of sensitive $\alpha 2$ (blue) and resistant $\alpha 4$ (yellow) cells in a colony exposed to three G418 concentrations (A: 0.20 mg/ml, B: 0.40 mg/ml, C: 0.70 mg/ml). Higher G418 doses lead to more rapid depletion of the sensitive population and earlier emergence of resistant sectors. Orange frames indicate treatment periods. Scale bar: 1 mm. Figure and text adapted from Elias Fischer's master's thesis.

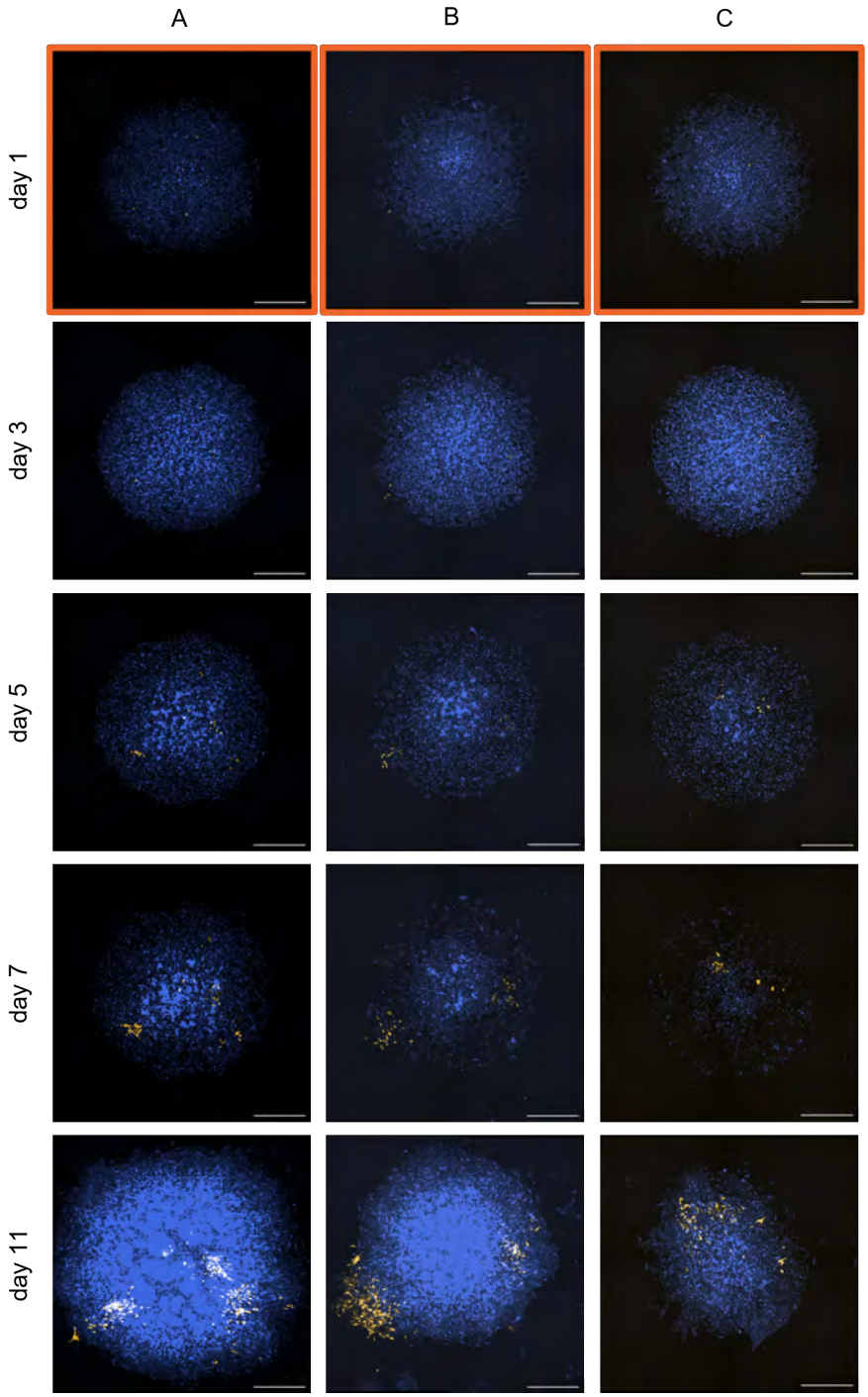


Figure 3.5: Higher G418 doses induce stronger population collapse under pulse treatment Time-lapse fluorescence images of sensitive α2 (blue) and resistant α4 (yellow) cells in a colony subjected to a single treatment pulse at three G418 concentrations (A: 0.20 mg/ml, B: 0.40 mg/ml, C: 0.70 mg/ml). Increasing G418 dose leads to more pronounced declines in total cell number after treatment and earlier emergence of resistant sectors. Orange frames indicate treatment periods. Scale bar: 1 mm. Figure and text adapted from Elias Fischer's master's thesis.

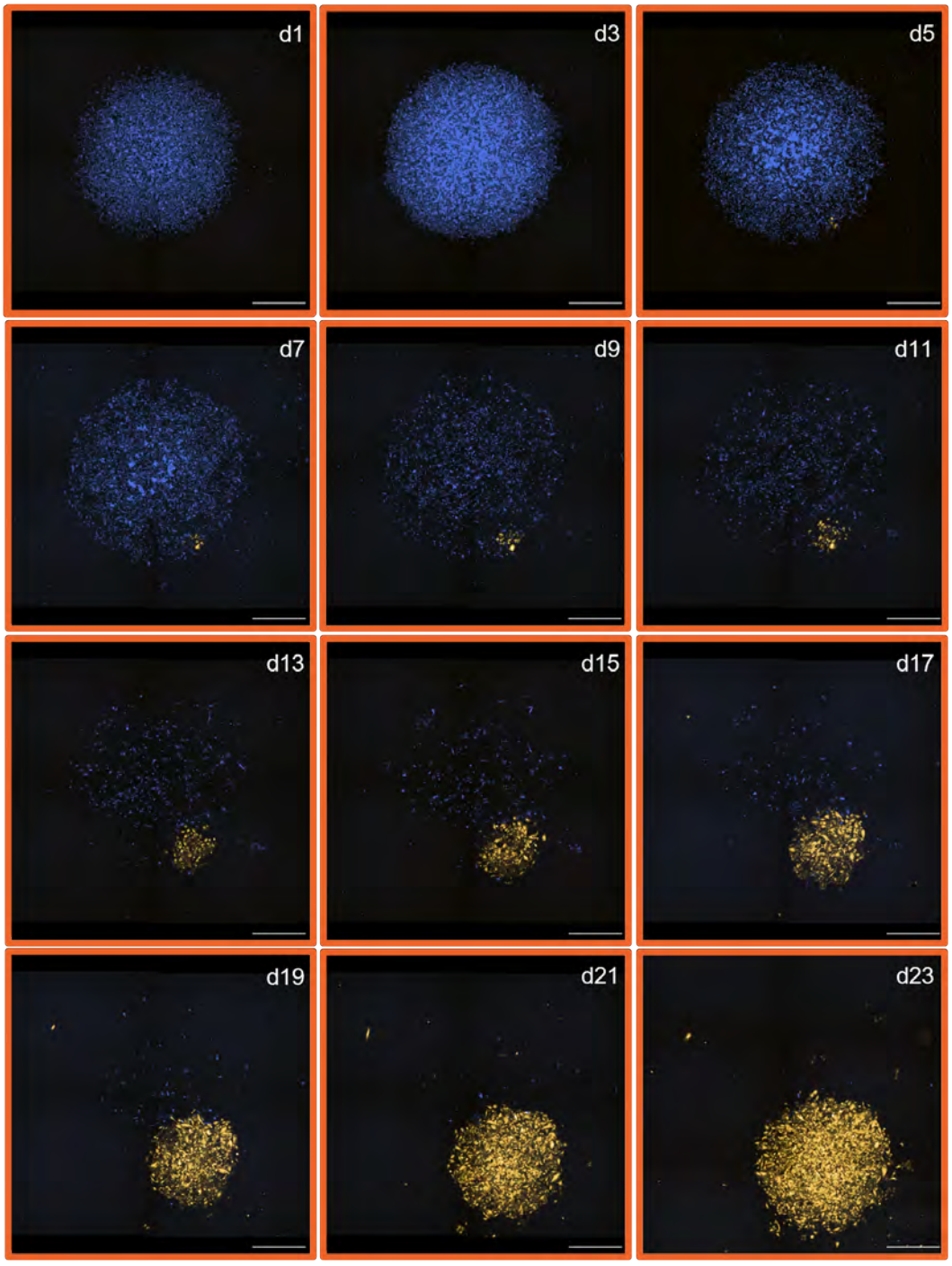


Figure 3.6: Gradual depletion of sensitive cells under CT Time-lapse fluorescence imaging of an example $\alpha 2$–$\alpha 4$ colony under CT with 0.20 mg/ml G418. Sensitive cells are shown in blue and resistant cells in yellow. Orange frames indicate treatment periods. Scale bar: 1 mm. Figure and text adapted from Elias Fischer's master's thesis.

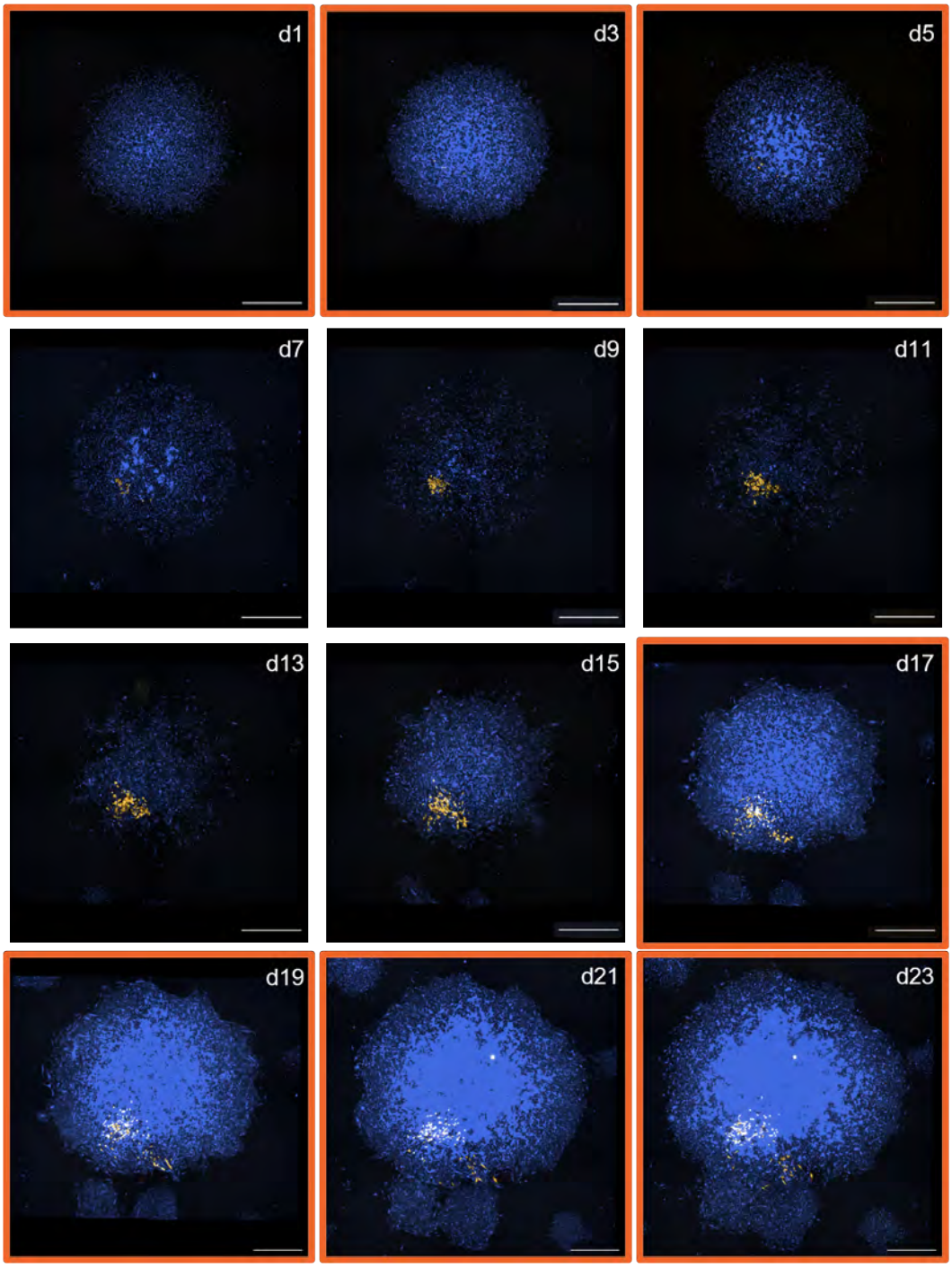


Figure 3.7: Preliminary cycling behavior under AT50 Time-lapse fluorescence imaging of an $\alpha 2$–$\alpha 4$ colony during AT50 treatment with 0.20 mg/ml G418. Sensitive cells are shown in blue and resistant cells in yellow. Orange frames indicate treatment periods. Scale bar: 1 mm. Figure and text adapted from Elias Fischer's master's thesis.

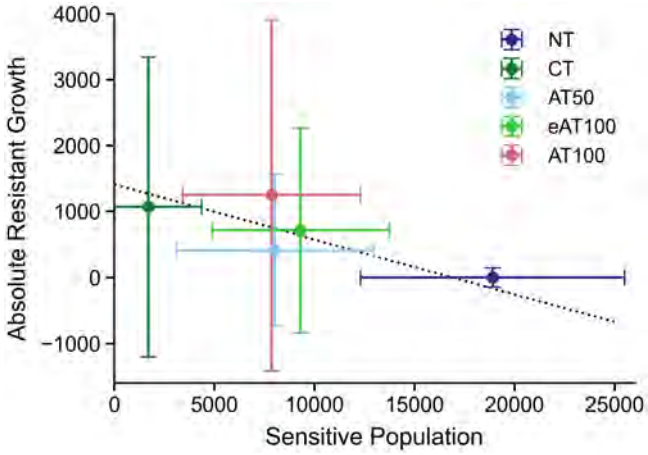


Figure 3.8: Sensitive-cell abundance is associated with lower net resistant growth Net resistant growth, defined as the change in resistant cell number between consecutive observations, is plotted against the corresponding sensitive-cell abundance for each treatment strategy (No treatment (NT), CT, AT50, AT100, and eAT100). Large markers indicate the mean per strategy, with error bars representing the SD. The dashed line shows a pooled descriptive linear fit across all observations. Across 954 longitudinal observations from 28 colonies, a GEE accounting for repeated measurements within colonies and adjusting for treatment strategy and time showed a negative population-average association, with each additional 1,000 sensitive cells associated with 112 fewer resistant cells between observations (95% CI: -153 to -71; $p = 7.2 \times 10^{-8}$). Figure and text adapted from Elias Fischer's master's thesis.

While shallow treatment conditions using 0.2 mg/ml G418 revealed the ecological interactions of interest, the spatial distribution of resistant clones emerged as a key determinant of experimental outcome. Colonies in which resistant cells were located near the boundary frequently showed rapid outward expansion via escape, whereas centrally positioned clones remained confined for longer, emphasizing that spatial placement strongly influences how resistance unfolds in structured populations. Because the initial seeding procedure often placed resistant cells close to the expanding front—allowing them to escape confinement during the first shallow treatment cycle—this outward growth pattern highlighted the need to refine the seeding strategy to position resistant clones more centrally.

To address this, after his master's thesis, a sequential seeding protocol was established by Elias Fischer in which resistant and sensitive populations were plated in two steps. A 3 µl droplet containing resistant cells at 1,333 cells/ml was first deposited in the center of a 24-well plate and allowed to adhere for 3 h. A 20 µl droplet containing sensitive cells at 1×10^6 cells/ml was then added to the same location and allowed to adhere for a further 18 h before medium was added. This approach generated colonies in which a sparse resistant core was enclosed by a dense rim of sensitive cells.



Figure 3.9: Illustration of the sequential seeding workflow Resistant cells (yellow) were seeded centrally as a 3 µl droplet at 1,333 cells/ml and allowed to adhere for 3 h. A 20 µl droplet of sensitive cells (blue; 1×10^6 cells/ml) was then added to the same location and allowed to adhere for a further 18 h. RPMI medium supplemented with 10% FBS was subsequently added to cover the well. Created with BioRender.com. [20]

This refinement introduced two key improvements to the assay: precise spatial placement of resistant cells and increased long-term stability of growing colonies. By positioning resistant clones centrally and surrounding them with a dense population of sensitive cells, the sequential seeding protocol established a well-defined spatial configuration in which AT schedules could be meaningfully compared.

Following the transition to the larger two-step core–rim assay and standardized post-thaw handling, treatment depth was reassessed using a single-pulse experiment. In this final configuration, with colonies larger in size then previous colonies, 0.4 mg/ml G418 provided reproducible sensitive-cell reduction without the near-complete collapse observed at higher doses.

The resulting assay thus provided a reproducible, spatially structured, and mechanistically tractable *in vitro* platform, forming the methodological foundation for the experiments described in section 3.2.

3.1.4 Core Conclusions from System Development

The development of the 2D colony system was a strongly iterative process, with many technical obstacles that initially limited reproducibility but ultimately clarified the constraints under which AT can be studied *in vitro*. Several overarching lessons emerged from this process:

1. Colony stability during media exchange is critical. Overlay approaches with agarose or collagen were explored, but either produced radial viability patterns or altered cell

morphology. The decisive improvement was optimizing medium exchange through partial aspiration and careful handling, which made additional overlays unnecessary.

2. Environmental stability must be controlled during seeding. Colony viability was highly sensitive to evaporation-related instability during seeding, particularly when plates were stacked in the incubator. Reliable survival required consistent incubation conditions, with each plate supported individually.

3. Cellular state influenced drug response: sensitive cells exhibited vulnerability to G418 immediately after thawing. Standardizing a post-thaw recovery period of approximately two weeks ensured a more consistent response profile.

4. Timing of first treatment application matters. Colonies required at least two days of adhesion before drug exposure to produce stable responses. Adjusting the initial seeding confluency downward prevented overgrowth during this period.

5. Imaging and segmentation require careful design. Expression stability, fluorophore brightness, and filter choice determined the feasibility of long-term clone tracking. Abandoning the $\alpha 3$ line in favor of $\alpha 2$, and avoiding Hoechst as a viability marker, were critical steps.

6. Treatment depth defines experimental utility. In the final two-step assay configuration, 0.4 mg/ml G418 provided a workable shallow treatment regime, whereas higher doses caused excessive sensitive-cell loss and obscured schedule-dependent ecological effects.

7. Spatial arrangement of resistant cells is decisive. Random initial distributions often resulted in colonies without resistant cells or with resistant clones located close to the periphery. The sequential seeding protocol, in which resistant cells were deliberately placed at the colony center before adding sensitive cells, enabled consistent generation of colonies with centrally positioned resistant clones suitable for AT experiments.

Together, these insights produced a reproducible and tractable 2D assay. The final system was designed to preserve spatial crowding while minimizing externally imposed gradients and supporting long-term imaging under AT regimens.

3.2 Crowding-Mediated Resistance Control and Model-Guided Therapy Optimization

Building on the stabilized assay described above, the sequential two-step seeding workflow provided the standardized experimental foundation for the main investigations presented in this chapter. This section and the corresponding discussion later in this chapter draw on results, analyses, figures, interpretations, and text from the joint study by Eiche *et al.* [1]:

Crowding governs resistance confinement and guides the improvement of adaptive therapy in a 2D melanoma model system

The material has been revised and expanded for this dissertation, with selected text passages reproduced verbatim and figures reproduced or adapted from the publication.

The study combined sustained experimental work with complementary theoretical expertise. Experiments lasting up to 96 days required Elias Fischer and me to share responsibility for their organization, execution, and experimental decisions under jointly established protocols.

My individual contributions for this study comprised generating the CRISPR/Cas9-engineered, fluorescently labelled melanoma cell lines and performing the confinement and escape-margin analyses. The shared contributions included study conception and design together with Elias Fischer and Dr. Jona Kayser, and the long-term experiments conducted jointly with Elias Fischer. Together with the co-authors, I also interpreted the results, prepared figures, and wrote and revised the manuscript.

Elias Fischer developed the two-step seeding protocol and 2D competition assay, established the image-analysis pipeline for closed-loop treatment decisions, and performed the crowding and local-density analyses.

Maja Milas carried out and analyzed the spatial DPD simulations, calibrating the model against the experimental data. Selected simulation results are presented here to complement the experiments and support their mechanistic interpretation. A comprehensive investigation of the digital twin, including its broader implications, detailed computational analyses, further simulation studies, and model refinement, lies beyond the scope of this dissertation and is discussed in the study [1]. Dr. Serhii Aif fitted the effective

generalized Lotka–Volterra model, constructed the reinforcement-learning training environment together with Timon Citak, and trained the reinforcement-learning agents. Maja Milas additionally performed the policy-behaviour analysis of the trained agents. These computational contributions informed the treatment strategies evaluated experimentally by Elias Fischer and me.

The complete author contribution statement is provided in the publication and the corresponding co-author declaration form accompanying this dissertation. Detailed experimental procedures, model specifications, and additional analytical information are provided in the publication.

3.2.1 An Integrated Experimental and Computational Framework for Studying Adaptive Therapy

To investigate how global therapy dynamics and local competition mechanisms are connected across scales under AT (Fig. 3.10a), we combined the established experimental system with mechanistic modelling of interactions between individual cells. This integrated *in vitro/in silico* framework links long-term tracking of sensitive and resistant populations at whole-colony and single-cell resolution to the local mechanical interactions underlying spatial competition.

The experiments used the CRISPR-engineered Mel Juso-derived cell lines $\alpha 2$ and $\alpha 4$, generated using the plasmid constructs described in Section 2.1 and validated in Section 2.2.2. As established above, $\alpha 2$ cells are G418-sensitive and EGFP-labelled (pseudocoloured in blue), whereas $\alpha 4$ cells are G418-resistant and mScarlet-labelled (pseudocoloured in yellow; Fig. 3.10b, c). Using the sequential two-step seeding protocol described in Section 3.1.3, sparse resistant cells were surrounded by a large sensitive population, forming densely packed colonies with a defined boundary (Fig. 3.10d). This configuration enabled the expansion of initially rare resistant clones to be followed under controlled spatial conditions. Longitudinal therapy-mimicry experiments were conducted in parallel for up to 96 days, with whole-colony images acquired at single-cell resolution three times per week.

For closed-loop implementation of AT schedules, same-day image analysis was performed using a custom automated analysis pipeline (Fig. 3.10e). Fluorescence images were used for segmentation of individual cells, extracting the population identity and centre-of-mass position of each cell [154, 222, 223]. These data were used to compute population-specific

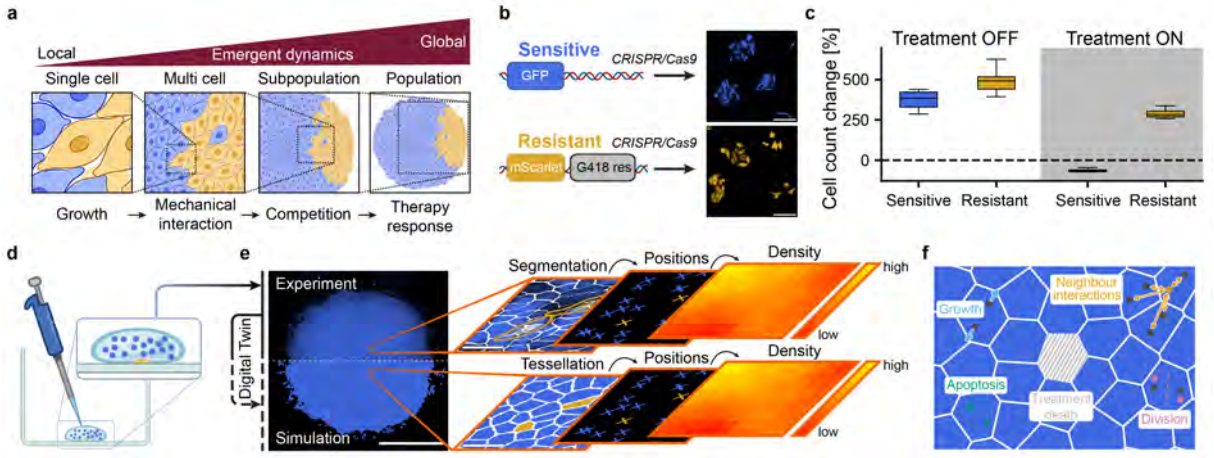


Figure 3.10: An integrated experiment-simulation system to study spatial competition (a) Schematic of how system-level phenomena emerge from local interactions. (b) Genotype schematic and example images of synthetically engineered G418-resistant (yellow) and sensitive (blue) melanoma cells. Scale bar, 100 µm. (c) Type-specific treatment effects as measured by change in cell count after 3 days. Boxes show median and interquartile range. Whiskers indicate 1.5× IQR, $n = 8$ replicate wells per cell line and condition. (d) Illustration of colony seeding. (e) Example of whole-colony, single-cell resolution fluorescence image (scale bar 2 mm, gamma = 0.7) with schematic segmentation, position-map and generation of density maps *in vitro*. Scale bar, 50 µm. Bottom row: Dissipative particle dynamics simulations serve as digital twins that are initiated from experimentally derived cell positions, following the same analysis pipeline. (f) Schematic of the particle-based simulation framework. Adapted from [1].

cell counts η_{sen} and η_{res} and spatial density maps. Total cell counts η_{total} informed adaptive treatment decisions for each colony, while spatial density maps enabled analysis of the local competitive environment.

To resolve and understand the mechanobiological constraints governing resistant cell expansion, we complemented experiments with DPD simulations [130]. Acting as digital twins, these simulations have been shown to quantitatively capture tissue development [130, 133]. In this mechanically explicit framework, each cell is represented by two particles and incorporates cell growth, interactions with neighbouring cells [130], a division rate regulated by cell size [133, 224], lifetime-dependent stochastic apoptosis, and therapy-induced death determined by cell sensitivity and drug absorption capacity (Fig. 3.10f). Importantly, key population-level phenomena, such as competition and therapy response, emerge intrinsically from the underlying local dynamics. Following parameterization, each digital twin was initialized from an experiment-derived colony configuration and subjected to the same treatment strategies, based on simulated cell counts (Fig. 3.10e, f).

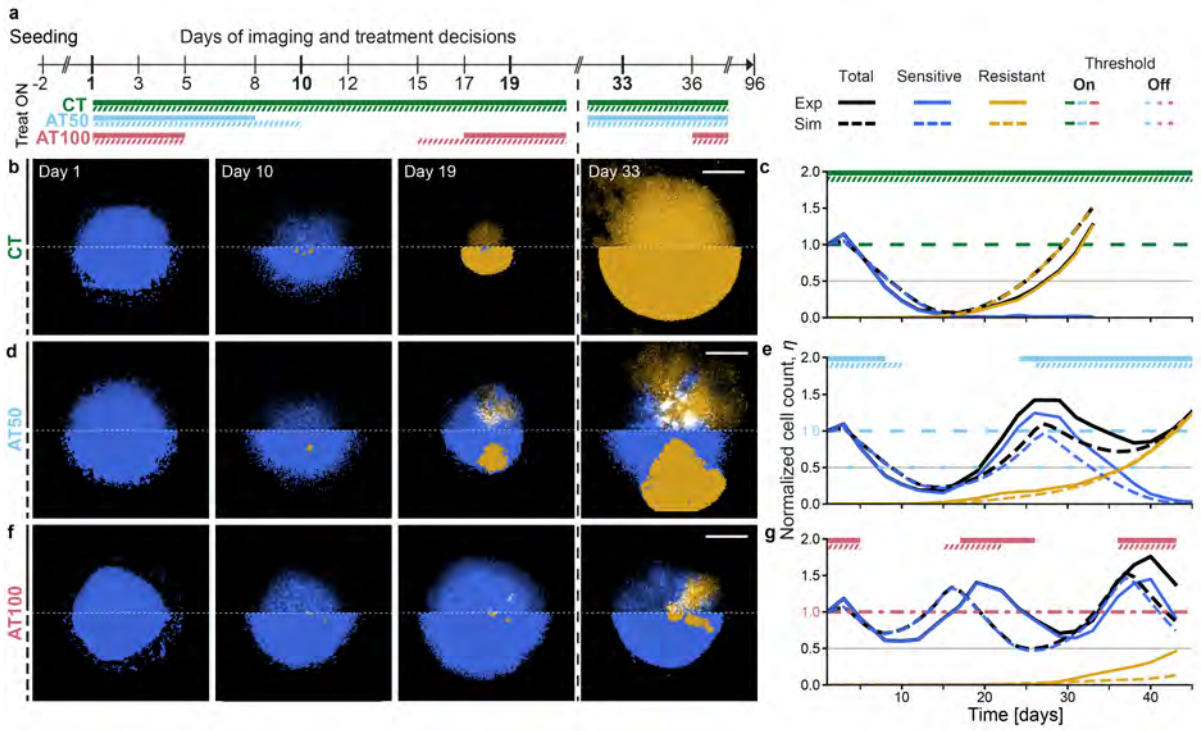


Figure 3.11: Matched experimental and simulated colony dynamics under varying therapy (a) Timeline indicating imaging, medium exchange and treatment decision days. Bold timepoints correspond to images below. Bars show periods of treatment for CT (green), AT50 (blue) and AT100 (red) replicates *in vitro* (solid) and *in silico* (striped). (b, d, f) Example images of colonies under CT (b), AT50 (d) and AT100 (f) *in vitro* and matched *in silico*. Scale bar, 2 mm, blue channel: gamma = 0.1, yellow channel: gamma = 0.2. Blue and yellow indicate sensitive and resistant cells, respectively, while white denotes overlap. (c, e, g) Corresponding total (black), sensitive (blue) and resistant (yellow) cell count η (normalized to initial total) of CT (c), AT50 (e) and AT100 (g) colonies *in vitro* (solid) and *in silico* (dashed). Horizontal lines give the treatment thresholds of each strategy; bars above give treatment periods. Adapted from [1].

Using this platform, we conducted matched *in vitro*/*in silico* experiments to compare continuous therapy (CT) and adaptive therapies (AT) (Fig. 3.11a). For continuous therapy, a constant drug dose of 0.4 mg/ml G418 was maintained over the entire course of the experiment. For adaptive strategies, treatment was paused when the total cell count dropped either below 50% (AT50) or 100% (AT100) of the initial cell count and resumed upon regrowth above baseline.

In both *in vitro* and *in silico* the CT baseline strategy yields complete eradication of the sensitive population, followed by rapid growth of resistant cells (Fig. 3.11b, c). For AT50, we find a delayed rebound of sensitive cells that, while slowing resistance growth, is typ-

ically insufficient to maintain resistance confinement for more than one treatment cycle (Fig. 3.11d, e). Only AT100 exhibits a more pronounced effect on resistance suppression, often maintaining control over several treatment cycles (Fig. 3.11f, g). Extended trajectories of representative *in vitro* colonies are depicted in figure 3.12 for CT, figure 3.13 for AT50 and figure 3.14 and 3.15 for AT100.

Comparing population-level dynamics by longitudinal measurements of type-specific cell numbers we find that the DPD simulations using a single globally calibrated parameter set capture the experimental observations both qualitatively (Fig. 3.11b, d, f) and quantitatively (Fig. 3.11c, e, g).

However, the suppressing effects of sensitive cell regrowth during adaptive strategies remain limited, especially for AT50, warranting a closer investigation of how spatial population structure affects therapy response.

3.2.2 Spatial Escape of Resistant Clones Drives Loss of Therapeutic Control

As first noted during assay development (Section 3.1.3), resistant clones near the colony boundary can escape spatial confinement and expand into unoccupied space. At this boundary, outward growth is no longer constrained by a surrounding layer of sensitive cells, even if the total sensitive population subsequently recovers. This spatial competitive release, referred to as "escape" (Fig. 3.16a), may therefore limit sustained resistance control. We thus investigated whether delaying escape prolongs therapeutic control.

To study the dynamics of resistance escape more systematically, we quantify the degrees of confinement or confinement margin by the number of sensitive cells between the resistant clone and open space before escape, $\eta_{\text{sen}}^{\text{conf}}$, and the degree of escape by the number of resistant cells more forward than any sensitive cells, $\eta_{\text{res}}^{\text{esc}}$ (Fig. 3.16a, b).

The dynamics of the confining sensitive layer reveal a key difference between both adaptive strategies. For AT100, $\eta_{\text{sen}}^{\text{conf}}$ deteriorates during the initial round of treatment but rebounds before the second cycle. In contrast, AT50 shows only marginal rebound, despite both strategies recovering to the initial total population size. This decrease of confinement layer rebound can also be observed for AT100, but only at a later stage *in vitro* as well as *in silico*.

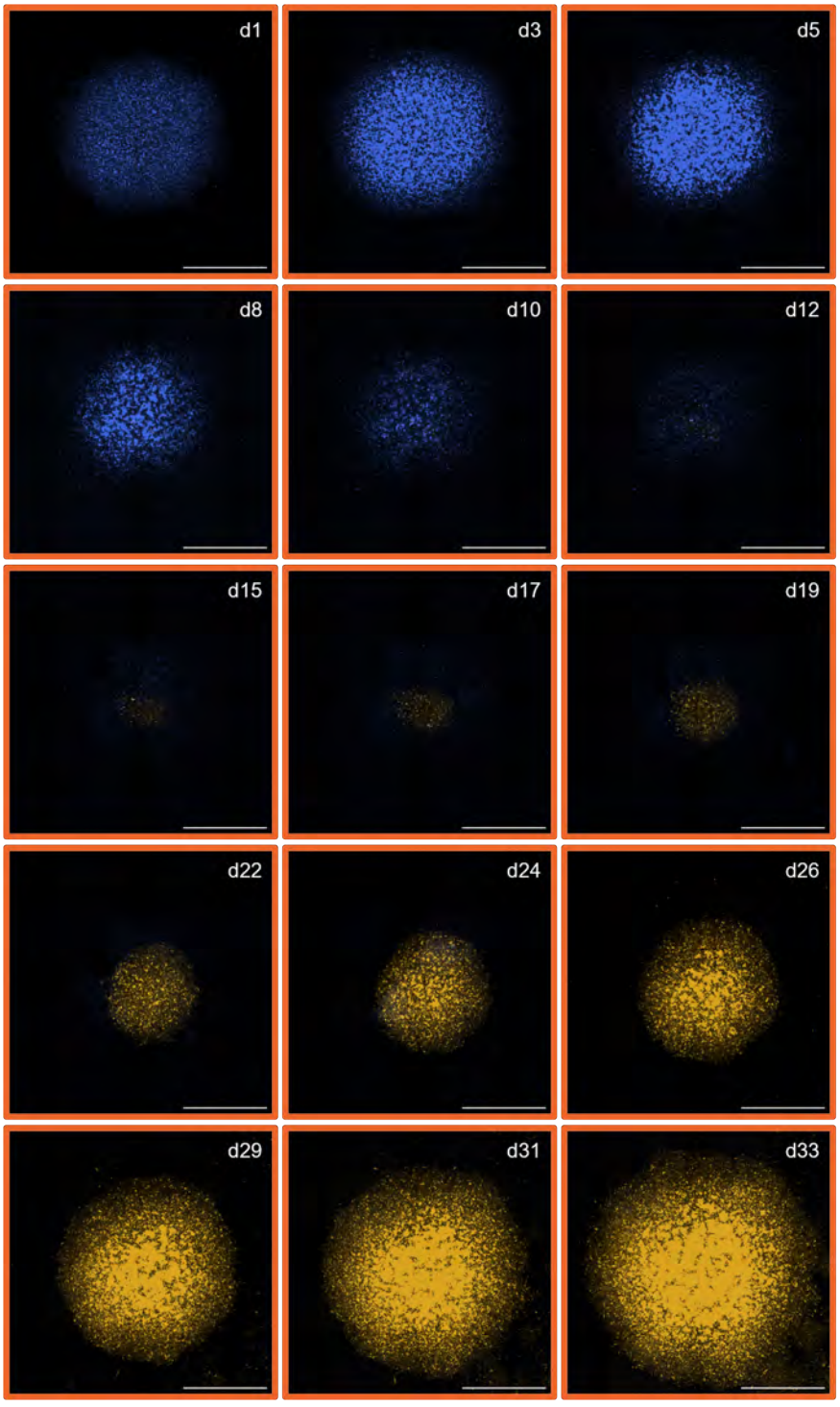


Figure 3.12: Rapid competitive takeover of resistant cells under CT Fluorescence time-lapse images of $\alpha 2$–$\alpha 4$ colonies subjected to CT. Sensitive cells are shown in blue and resistant cells in yellow. Under CT conditions, resistant clones emerge early and expand steadily, ultimately overtaking the colony. Orange frames indicate treatment periods. Scale bar: 2 mm.

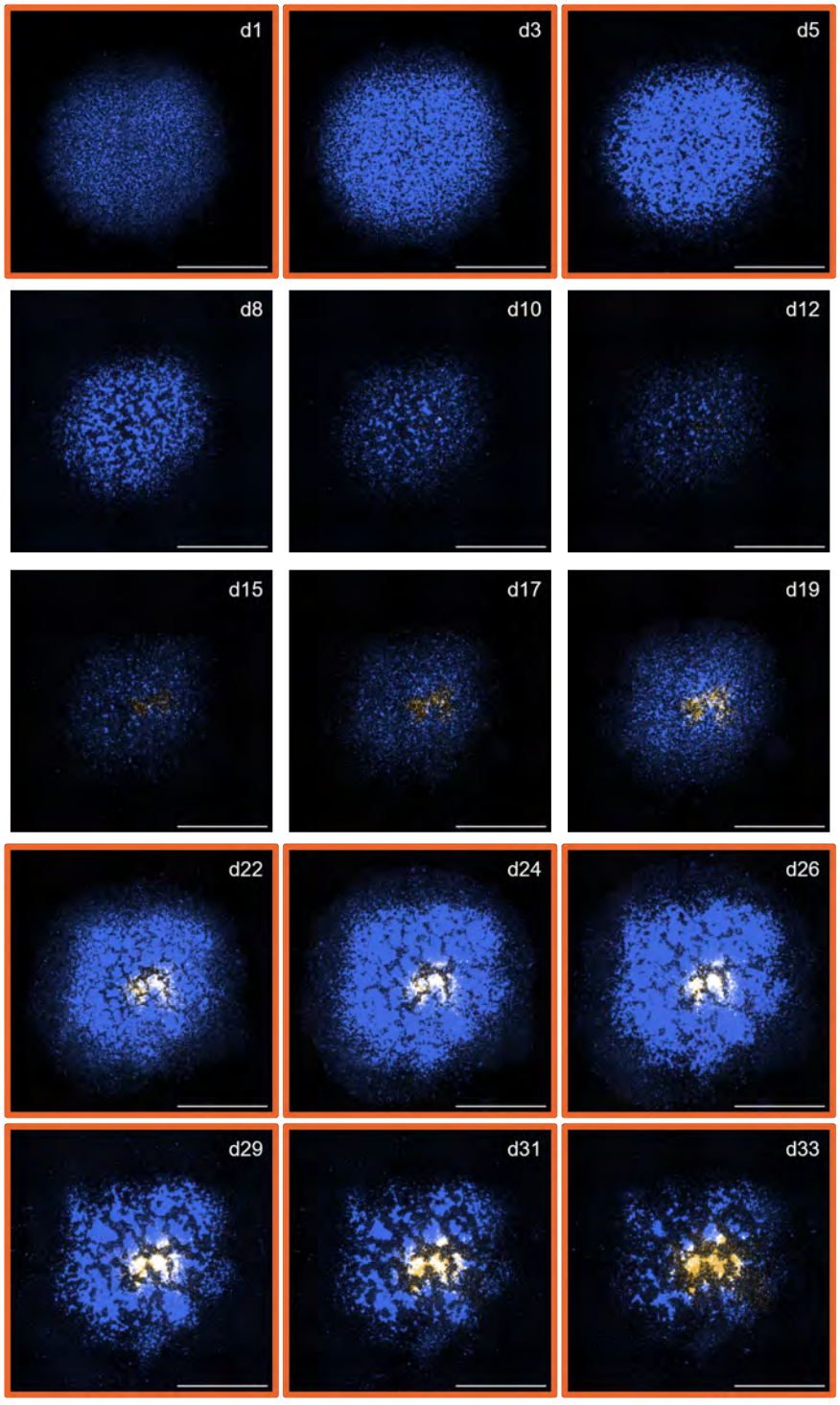


Figure 3.13: AT50 delays resistant takeover by maintaining competitive pressure Fluorescence time-lapse images of $\alpha 2$–$\alpha 4$ colonies subjected to AT50 treatment. Sensitive cells are shown in blue and resistant cells in yellow. Under AT50, resistant clones emerge later and repeatedly regress when sensitive populations recover during drug-free intervals, prolonging competitive suppression of the resistant subpopulation. Orange frames indicate treatment periods. Scale bar: 2 mm.

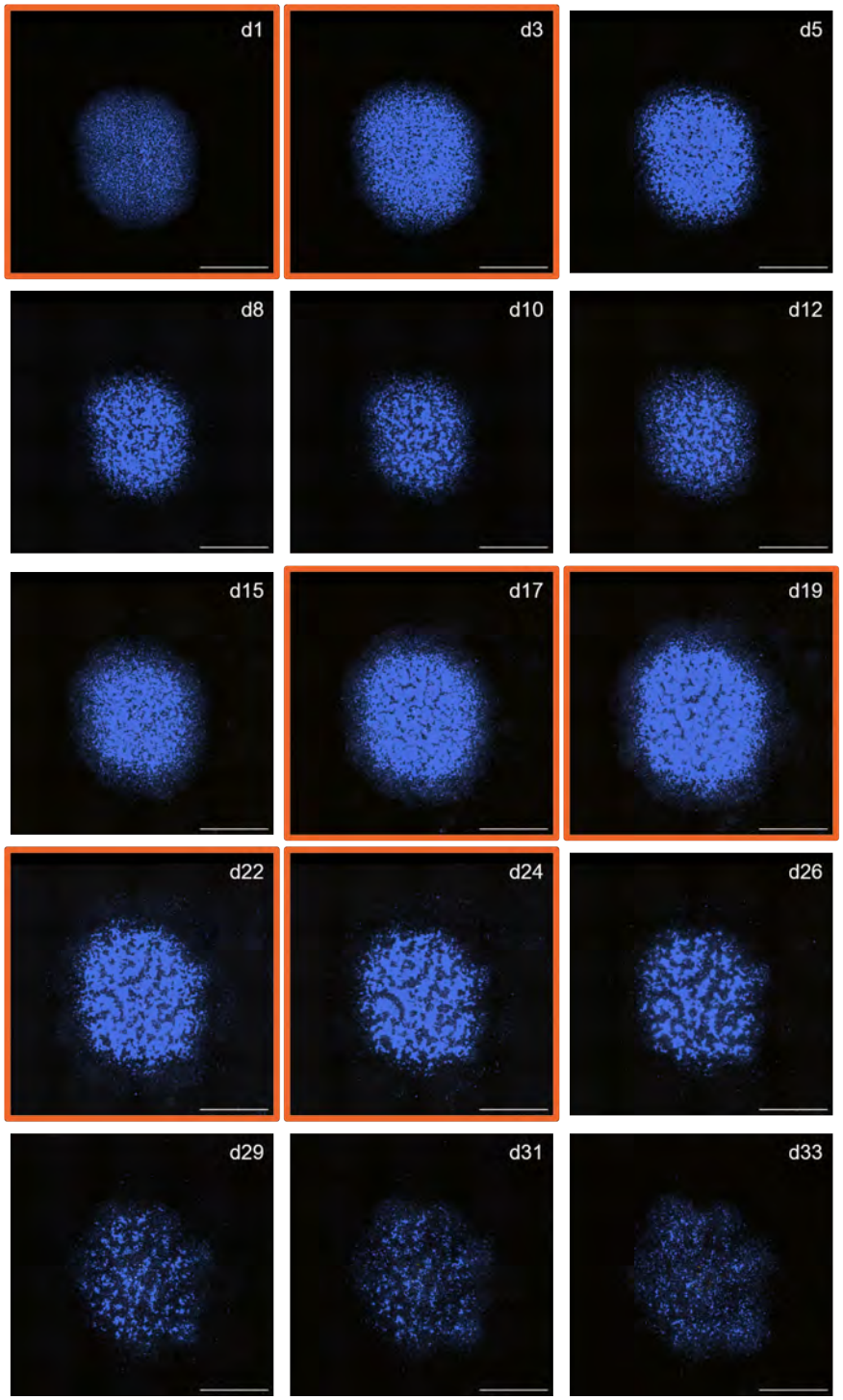


Figure 3.14: Early AT100 time-lapse of α2–α4 colonies (days 01–33) Fluorescence images of α2–α4 colonies under AT100 treatment from day 1 to day 33. Sensitive cells are shown in blue and resistant cells in yellow. Treatment periods are indicated by orange frames. Over this early phase of the experiment, the colony maintains its overall structure with slowly emerging resistant foci but no pronounced expansion. Scale bar: 2 mm.

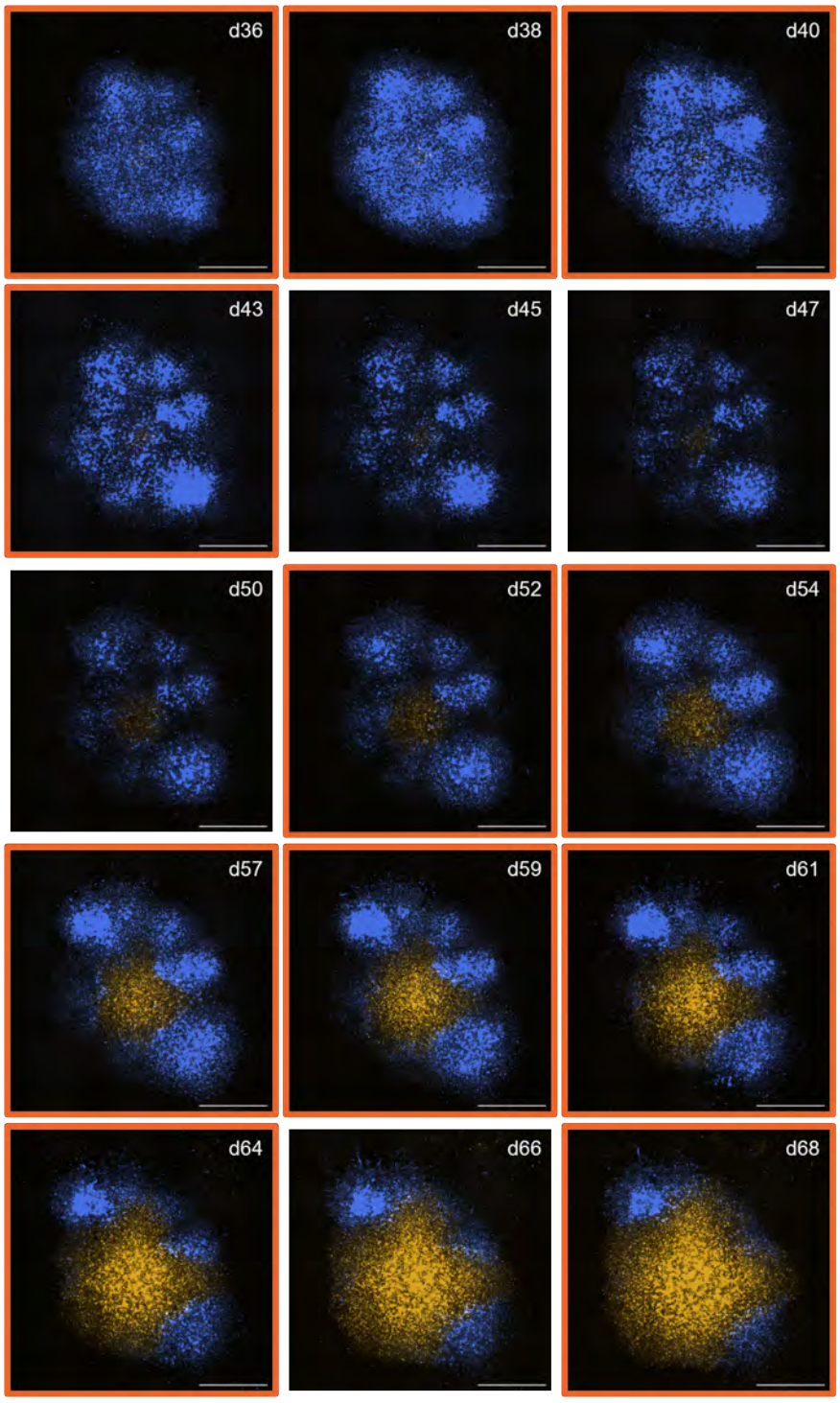


Figure 3.15: Continued AT100 time-lapse of α2–α4 colonies (days 36–68) Continuation of the AT100 time-lapse imaging shown in figure 3.14, covering days 36 to 68. Sensitive cells are shown in blue and resistant cells in yellow, with treatment periods indicated by orange frames. Throughout this extended timeframe, resistant foci arise only late and remain limited as recurrent regrowth of sensitive cells restores competitive pressure. Scale bar: 2 mm.

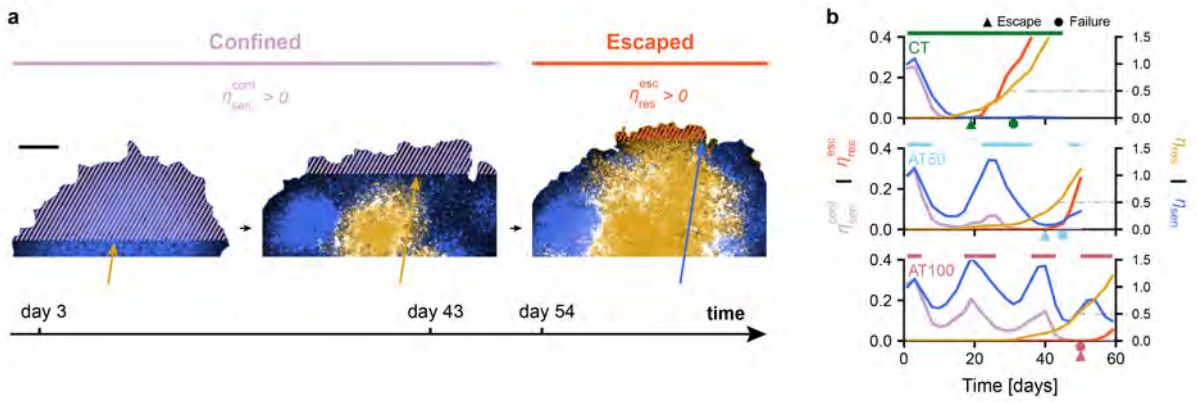


Figure 3.16: Spatial progression and quantification of resistant clone escape (a) Representative fluorescence images illustrating escape of a resistant clone over time. Striped areas indicate the region of confining sensitive cells ($\eta_{\text{sen}}^{\text{conf}}$, purple), and escaped resistant cells ($\eta_{\text{res}}^{\text{esc}}$, orange). Arrows indicate the most forward resistant or sensitive cell. Scale bar = 200 µm, gamma = 0.3. (b) Quantification of representative escape dynamics over time for CT, AT50, and AT100. Horizontal lines above indicate treatment. Gray lines indicate treatment failure threshold. Adapted from [1].

To probe how confinement dynamics relate to treatment success, we measured the time to escape (TTE, $\eta_{\text{res}}^{\text{esc}} > 0$; Fig. 3.17a) and the time to treatment failure of resistance control (TTF, $\eta_{\text{res}} \geq 0.5$; Fig. 3.17b), for each strategy in both *in vitro* and *in silico*. Compared with CT, AT100 significantly extended both TTE and TTF in experiments and simulations (Exp: ΔTTE = 23 days; 95% CI = 7–49 days; $p = 1.13 \times 10^{-5}$; ΔTTF = 19 days; 95% CI = 2–33 days; $p = 0.0138$; Sim: ΔTTE = 58 days; 95% CI = 42–75 days; $p = 5.29 \times 10^{-19}$; ΔTTF = 55 days; 95% CI = 47–67 days; $p = 2.31 \times 10^{-16}$). AT50 likewise significantly extended TTE and TTF relative to CT in both experiments and simulations, although the simulated delay in TTE was smaller than that observed for AT100 (Exp: ΔTTE = 16 days; 95% CI = 0–23 days; $p = 4.02 \times 10^{-4}$; ΔTTF = 14 days; 95% CI = 0–21 days; $p = 0.0357$; Sim: ΔTTE = 4 days; 95% CI = 2–14 days; $p = 9.07 \times 10^{-8}$; ΔTTF = 11 days; 95% CI = 8–17 days; $p = 1.75 \times 10^{-13}$).

Pooling CT, AT50, and AT100 replicates with both events observed, TTE was strongly correlated with TTF (Pearson's $r = 0.847$, 95% CI = 0.667–0.930, $p = 5.1 \times 10^{-10}$, $n = 33$ *in vitro*; $r = 0.979$, 95% CI = 0.954–0.991, $p = 4.1 \times 10^{-49}$, $n = 71$ *in silico*), corroborating that escape is an important driver of therapy failure without establishing causality (Fig. 3.17c).

Supported by high-resolution *in silico* data, these findings suggest that the observed dynamics of the confining sensitive cell layer result from a two-sided spatial ratchet:

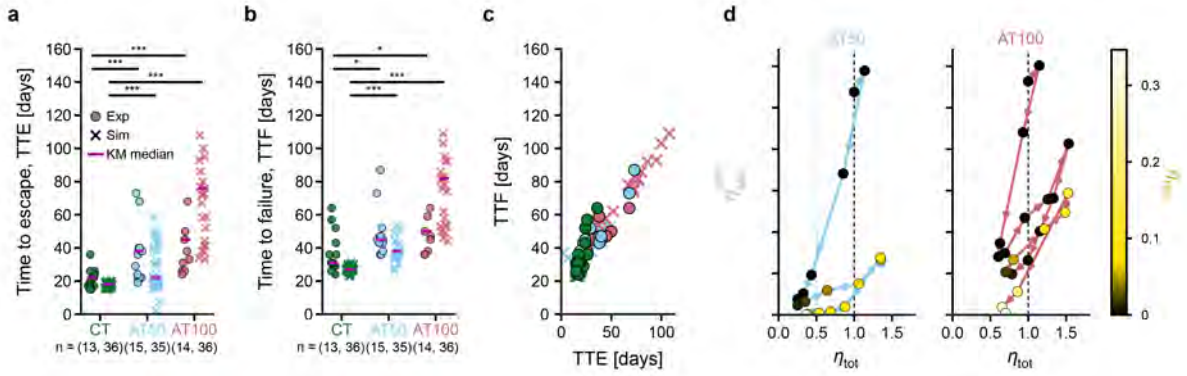


Figure 3.17: Control of resistant clones depends on clonal escape dynamics (a) Time to escape (TTE, $\eta_{\text{sen}}^{\text{conf}} = 0$) for experiments and simulations. Points are individual colonies, bars indicate median. Numbers below groups indicate respective n. Pairwise comparisons used unadjusted log-rank tests. Statistical results are reported in detail in the main text. (b) Time to treatment failure (TTF, $\eta_{\text{res}} \geq 0.5$), analogous to a. (c) Correlation between TTE and TTF (*in vitro*: Pearson's $r = 0.85$, $p = 5.1 \times 10^{-10}$; *in silico*: $r = 0.98$, $p = 4.1 \times 10^{-49}$). (d) Visualization of ratchet-like clonal escape trajectories. Marker colors indicate the corresponding η_{res}. Dashed lines indicate initial values. Dotted line denotes the AT50 treatment-off threshold. Adapted from [1].

Treatment deteriorates confinement through both peripheral killing of sensitive cells and triggering resistance expansion (Fig. 3.17d). Both processes together cause a decrease of $\eta_{\text{sen}}^{\text{conf}}$. However, while the decrease in sensitive cell number is reversible, resistance expansion is not. Since sensitive rebound is capped by the total burden threshold, this irreversible ratchet will gradually cause confinement deterioration and eventually escape.

3.2.3 Crowding as the Mechanistic Lever of Resistance Growth Inhibition

Going beyond the limits accessible experimentally, the simulations allow us to directly probe how the observed population dynamics emerge from interactions at the level of individual cells, where proliferation is modulated by the mechanical constraints exerted by neighboring cells (Fig. 3.18a). During the expansion phase of the cell cycle, each cell has to extend against these external forces, decreasing its expansion rate (λ) and delaying the entry into the division phase (Fig. 3.18b) [133]. Relating λ to the local total cell density ρ_{tot}, a measure also available in experiments, shows that higher densities are strongly associated with decreased cell expansion (Fig. 3.18c). This is consistent with previous work on mechanosensitivity of cell proliferation [130, 133, 224, 225]. We therefore

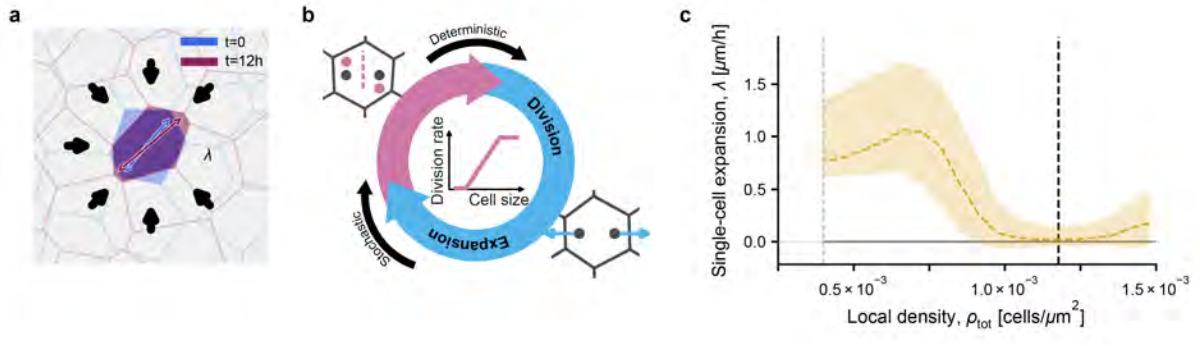


Figure 3.18: Cell expansion governs the suppression of resistant growth *in silico* (a) Schematic of mechanical forces countering cell expansion *in silico*. (b) Schematic illustrating the simulated cell cycle. (c) Single cell expansion λ as a function of total local density ρ_{tot} for resistant cells *in silico*. Dashed yellow curve shows median, the shaded region IQR. ρ values below $0.4 \times 10^{-3}\,\mathrm{cells/\mu m^2}$ correspond to non-confluent states and are excluded. Dashed black vertical line marks the homeostatic cell density. Adapted from [1].

use density as a proxy for crowding-mediated mechanical constraints on cell growth. It should be noted that the effects of density can also be cell type dependent, and that in our parameterization the relation between λ and ρ_{tot} for sensitive cells differs from that of resistant cells. However, we will here focus on density effects on resistance growth.

The mechanical effects on cell growth are clearly reflected in spatial expansion and contraction of resistant and sensitive subpopulations, which we quantify by computing spatially-resolved density maps $\rho_{\mathrm{res}}(x, y)$ and $\rho_{\mathrm{sen}}(x, y)$ alongside associated density-change maps $\Delta\rho_{\mathrm{res}}(x, y)$ for both simulations and experiments (Fig. 3.19a, b). Together, these reveal anti-correlated dynamics in which resistance expansion is arrested in high-density environments but can proceed when crowding is reduced due to treatment-induced depletion of sensitive cells (Fig. 3.19c, d). Despite substantial spatial and temporal heterogeneity across different treatment patterns, this relationship can be quantitatively summarized by a master curve $\Delta\tilde{\rho}_{\mathrm{res}}(\rho_{\mathrm{tot}})$ that collapses the spatial maps into a crowding-growth axis (Fig. 3.19e, f). Here, the tilde denotes normalization of the local resistant density change by the local resistant density at the preceding observation. Division by the elapsed time Δt gives the relative density-change rate plotted in the figure. Across all strategies, high-density regions were associated with resistant stasis, whereas low-density regions permitted growth. Although these density changes reflect cell production, loss, and spatial redistribution rather than division alone (Eq. (1.8)), the consistent relationship between density and resistant growth suggests that local density could serve as an effective control parameter for therapeutic decisions.

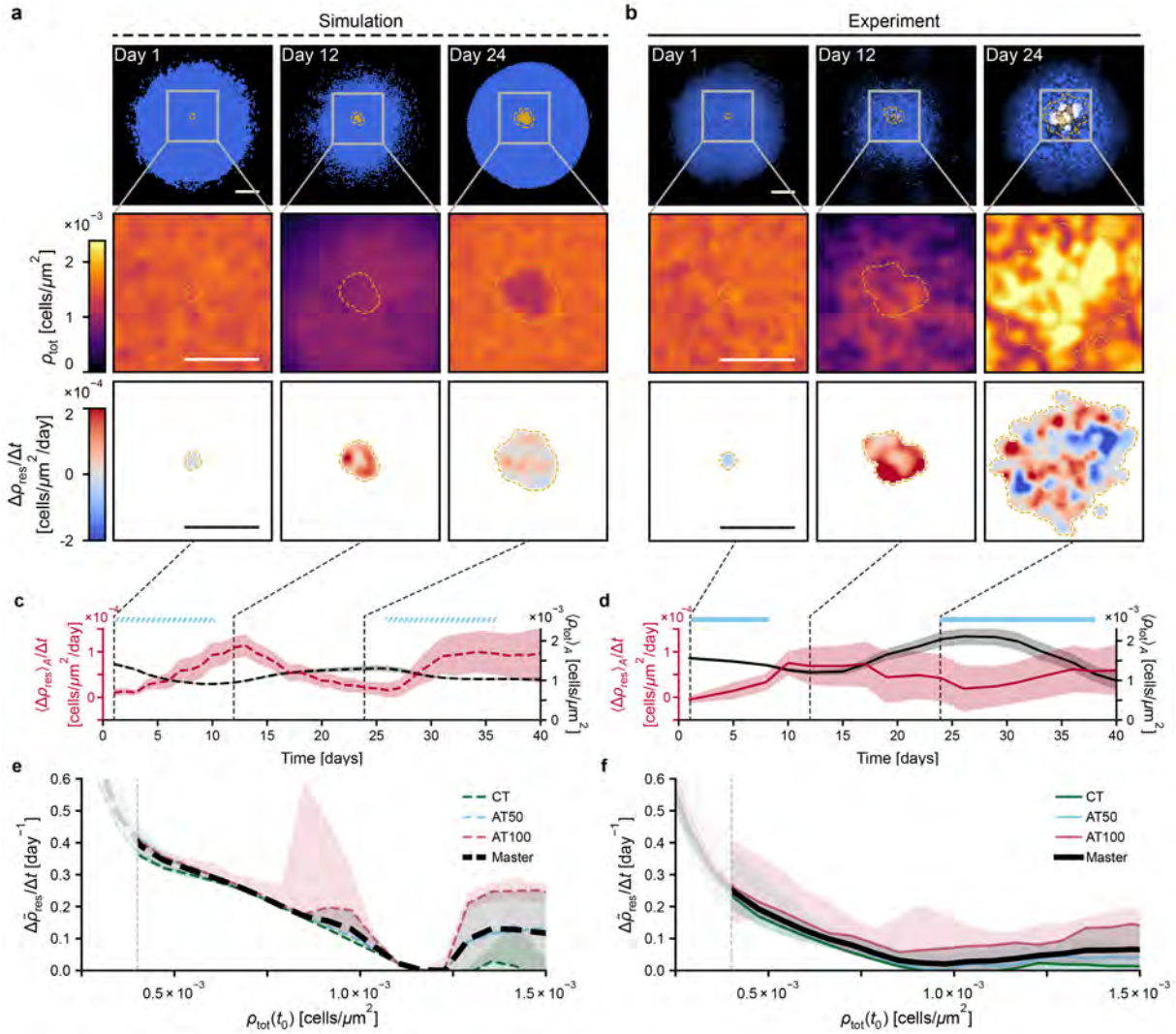


Figure 3.19: Local crowding governs the suppression of resistant growth (a, b) Representative *in silico* (a) and *in vitro* (b) colonies, coarse-grained total density ρ_{tot} and resistant density-change $\Delta\rho_{\mathrm{res}}/\Delta t$ maps for low and high density time points. Yellow dashed outlines indicate pixels with sufficient local resistant-cell density. Gamma = 0.45/0.2 (blue/yellow channel). (c, d) Corresponding area-averaged resistant density change $\langle\Delta\rho_{\mathrm{res}}\rangle_A/\Delta t$ (red, left axis) and total density $\langle\rho_{\mathrm{tot}}\rangle_A$ (black, right axis), running mean with SD (shaded). Bars above give treatment periods, dashed lines mark the days shown in a and b. (e, f) Binned relative resistant density change $\Delta\tilde{\rho}_{\mathrm{res}}/\Delta t$ as a function of ρ_{tot} *in silico* (e) and *in vitro* (f) for CT, AT50 and AT100, per-strategy median with IQR (shaded), and the mean over strategies (black) with SD (shaded). ρ values below $0.4\times10^{-3}\mathrm{cells}/\mu\mathrm{m}^2$ excluded as in Fig. 3.18c. Adapted from [1].

Using ρ_{tot} as an experimentally accessible proxy for the local mechanical environment further enables direct comparison between *in silico* and *in vitro* dynamics, which reveal the same qualitative relationship. Although mechanical forces cannot be measured directly in our experiments, resistant expansion predominantly occurs in regions where the local sensitive-cell density has decreased, whereas densely populated sensitive neighborhoods restrict resistant growth. The consistency between experimental observations and mechanically explicit simulations therefore supports local crowding as a key determinant of resistant confinement and release, with cell density providing an experimentally accessible readout of mechanically mediated competition. As expected, the *in vitro* relationship is considerably noisier, likely reflecting cell migration, segmentation artifacts, and intrinsic biological heterogeneity.

3.2.4 Local Competition Abstraction into a Population-Level Model for RL-Guided Therapy Optimization

Since spatially resolved measurements of crowding are rarely available for treatment decisions in a clinical context, we asked whether the outcome relevant dynamics could be further abstracted into a coarse-grained model based on more accessible population-level observables without losing predictive power on strategy optimality.

Because normalized total-cell number η_{tot} closely tracked total-cell density $\langle \rho_{\text{tot}} \rangle_A$ in experiments before escape (Fig. 3.20a), we used the experimental data (Fig. 3.20b) to construct a minimal non-spatial effective model based on an established two-population generalized Lotka–Volterra (GLV) system (Fig. 3.20c). This reduction follows the treatment-dependent competition structure introduced in Eq. (1.10), with the full calibrated specification given in [1]. The calibrated GLV model reproduced the qualitative pre-escape population dynamics for AT100. Importantly, AT100 consistently outperformed AT50 and CT across the GLV model, the spatial simulations and the experiments [1]. Thus, while the GLV model should not be interpreted as a faithful quantitative description of the complete spatial dynamics, it represents a control-relevant abstraction of the pre-escape regime in which competition can still be exploited. As a direct consequence, the GLV model may be used as a computationally efficient environment surrogate for treatment optimization beyond fixed-threshold strategies.

More specifically, this abstracted model can serve as the training environment for rein-

forcement learning (RL), which optimizes observation-based treatment decisions to maximize a predefined reward function, such as long-term outcome (Fig. 3.20d). This approach allows dosing principles to emerge from the model dynamics and reward function rather than from a predefined scheduling rule.

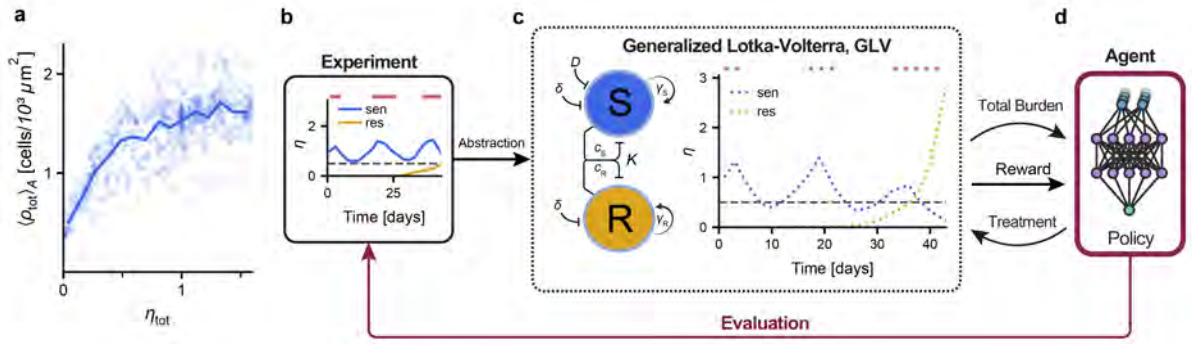


Figure 3.20: An effective population model enables reinforcement learning-based therapy optimization (a) Mean normalized total-cell number η_{tot} and area-averaged total-cell density in resistant clone region $\langle \rho_{\mathrm{tot}} \rangle_A$ before escape. (b) Example *in vitro* normalized cell count with dashed horizontal bars above indicating treatment. Horizontal gray line indicating treatment failure threshold. (c) Effective two-population GLV model of sensitive (S) and resistant (R) cells, with a representative GLV fit to the AT100 example in b. Dashed bars above indicate treatment periods. Horizontal gray line indicating treatment failure threshold. (d) Schematic of agent trained on the GLV environment in c. Adapted from [1].

The RL agent was trained in a fixed effective-model environment using proximal policy optimization and two sequential reward functions. At each decision point, the agent observed only the normalized total population burden η_{tot} at the four most recent measurements together with the corresponding treatment decisions and selected whether treatment should be applied during the next interval. It had no access to clonal composition, local density or spatial position. To assess the robustness of the learned decision logic to uncertainty in model calibration, we subsequently refitted the effective model to 20 stratified bootstrap datasets and trained one additional policy for each refit. The policy selected for experimental scheduling and validation is hereafter termed ATagent (Fig. 3.21a).

To identify which features of the observation history governed treatment decisions, we evaluated the trained policies across a predefined set of synthetic inputs (Fig. 3.21b). The fraction of deterministic treatment-on decisions could not be reduced to current burden alone. Across the input ensemble, treatment was selected more frequently when burden had risen over the preceding observations rather than depending only on its instantaneous value, and more frequently after a treatment-free interval than when therapy had already

been applied recently. We distilled this behaviour into a simplified decision heuristic, AT^+, which applies treatment only if it was withheld at the preceding decision point and burden exceeds the treatment threshold. For consistency with treatment initiation of ATagent, we defined the experimental AT^+ treatment-decision threshold as the total cell count on day 3. All reported normalized cell counts retained the day-1 total cell count as their denominator.

To test whether this treatment logic transferred beyond the effective model used for training, we applied ATagent and AT^+ to *in vitro* colonies and corresponding digital twins, recording population-level metrics as before. Both strategies adhered to the experimental measurement time points and used total burden as their only population-level input. Representative trajectories reveal the same qualitative pulse-pause dynamics in experiments and spatial simulations (Fig. 3.21c). Representative *in vitro* trajectories for ATagent and AT^+ are shown in Fig. 3.23, 3.24, 3.25 and 3.26. ATagent and AT^+ delivered short pulses separated by treatment-free intervals during which sensitive cells and total burden could recover. Importantly, corresponding $\eta_{\text{sen}}^{\text{conf}}$ trajectories remained more stable than under AT50 and AT100, delaying the eventual confinement collapse.

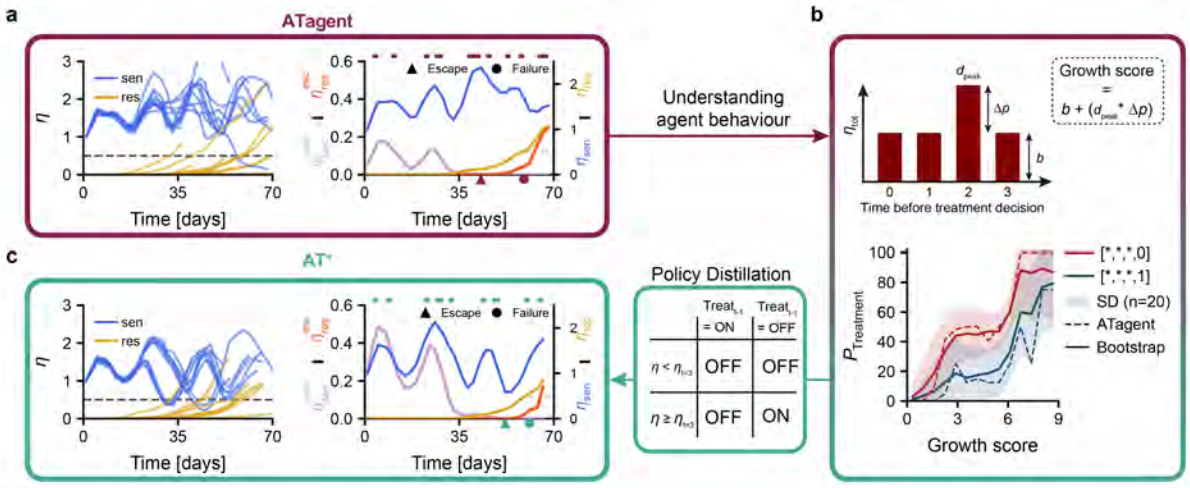


Figure 3.21: RL-guided optimization of adaptive therapy strategies (a) *In vitro* normalized cell count trajectories under ATagent, with a representative example showing confining sensitive $\eta_{\text{sen}}^{\text{conf}}$ and escaped resistant cells $\eta_{\text{res}}^{\text{esc}}$. Horizontal gray line indicating treatment failure threshold. (b) Retrospective policy-behaviour analysis of 20 PPO agents trained on bootstrap-refitted effective models. Shading indicates ± 1 sample SD. Synthetic four-observation burden histories are summarized by the growth score $G = b + (d_{\text{peak}} * \Delta p)$, with baseline burden b, peak magnitude Δp, and peak position $d_{\text{peak}} = 0, \ldots, 3$ from oldest to most recent. (c) Qualitative and quantitative depiction of the colony dynamics under the distilled AT^+ strategy as in a. Adapted from [1].

Although ATagent was never provided with density measurements, its pulse pattern - and consequently those of AT^+ - avoided prolonged low-burden states associated with reduced local density. Finally, comparison of all strategies at the population level confirmed this advantage in both experimental colonies and their digital twins (Fig. 3.22a and b). Kaplan–Meier curves showed delayed failure from CT through AT50 and AT100 to the model-derived strategies. Median time to failure increased from 31 days (CT) to 45/50 days (AT50/AT100) and 57/61 days (AT^+/ATagent) *in vitro*, with an even stronger separation *in silico* (27, 38, 82, 93, and 107 days, respectively). Only CT versus ATagent reached statistical significance *in vitro* ($p = 0.0052$). In contrast, *in silico*, all pairwise comparisons were significant except AT100 versus AT^+ and AT^+ versus ATagent (CT versus ATagent, $p = 5.9 \times 10^{-15}$; two-sided log-rank tests with Bonferroni correction). Extending the evaluation beyond disease control alone, we compared all strategies in an adversity space defined by the percentage of days under treatment and the normalized total burden η_{tot} (Fig. 3.22c). In this joint comparison, AT^+ achieved a more favorable balance than ATagent, with similar treatment exposure and a lower time-averaged normalized population burden (median burden difference, AT^+ minus ATagent: Exp, -0.44, 95% CI $[-0.51, -0.31]$, $p = 0.00012$; Sim, -0.74, 95% CI $[-0.77, -0.62]$, $p = 9.6 \times 10^{-11}$). Full statistical results, including all pairwise comparisons and additional analyses, are reported in the corresponding study [1].

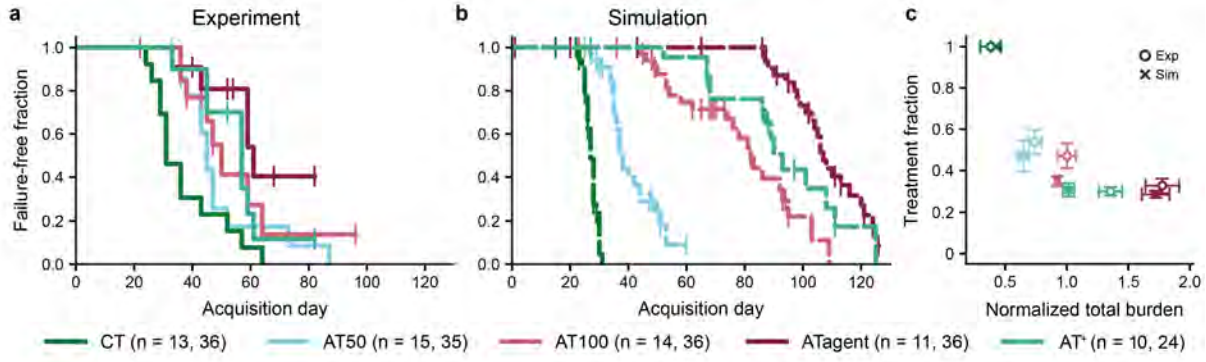


Figure 3.22: Improved adaptive therapy prolongs resistance control (a, b) Failure-free fraction per strategy *in vitro* (a) and *in silico* (b); ticks mark replicates censored due to artifacts arising from cell loss during washing. n = (*in vitro*, *in silico*). (c) Treatment fraction against time-averaged normalized total burden $\langle \eta_{\text{tot}} \rangle_t$ per strategy *in vitro* and *in silico*. Bars indicate SD. n = (*in vitro*, *in silico*). Adapted from [1].

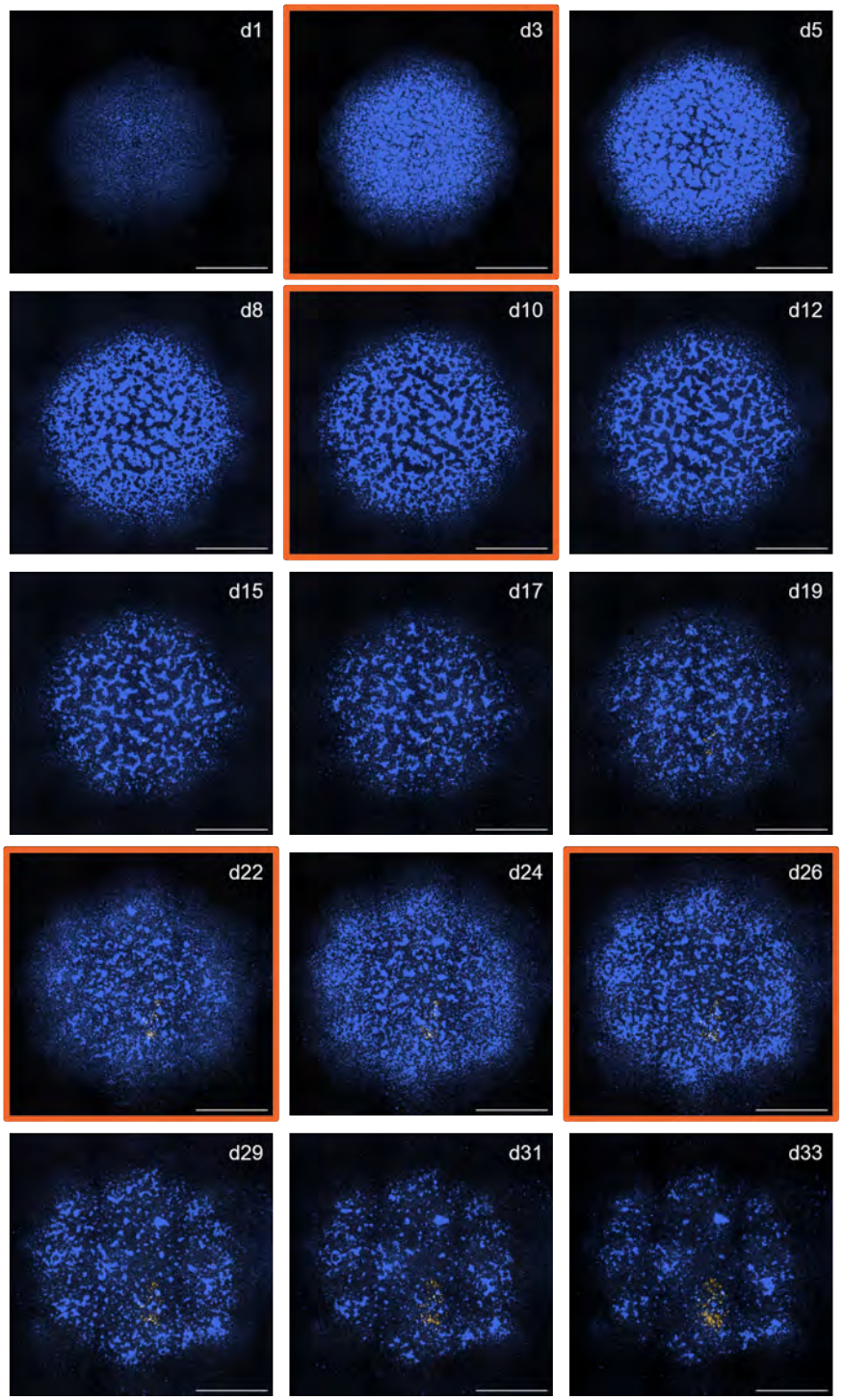


Figure 3.23: Early LV-based RL time-lapse of α2–α4 colonies (days 01-33) Fluorescence images of an α2–α4 colony treated according to an RL policy trained on the GLV model. Sensitive cells are shown in blue and resistant cells in yellow. Treatment periods are indicated by orange frames. During this early phase, the agent maintains a high density of sensitive cells, limiting resistant expansion and preserving spatial confinement. Scale bar: 2 mm.

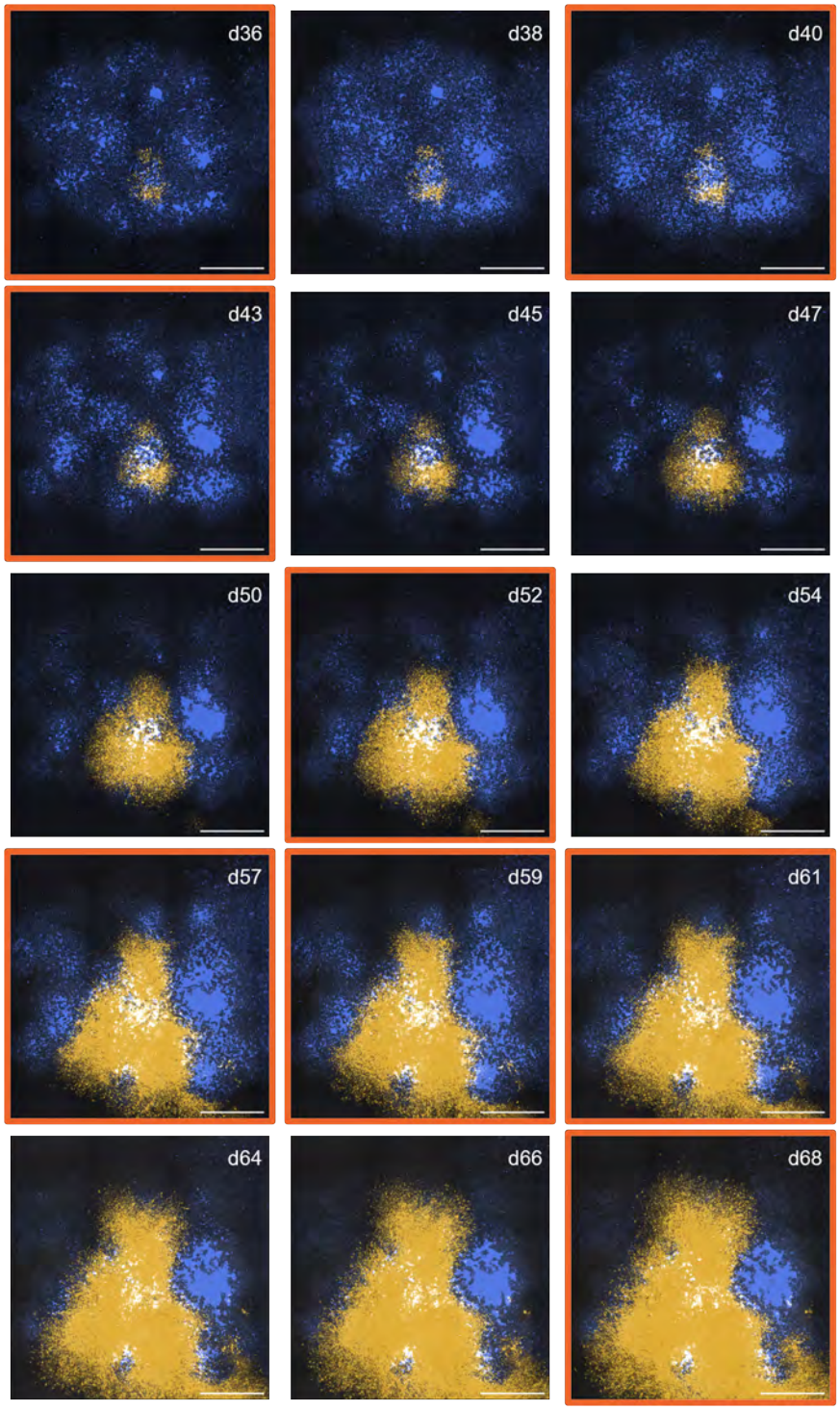


Figure 3.24: Continued GLV-based RL time-lapse of α2–α4 colonies (days 36-68) Continuation of the GLV-based RL time-lapse imaging shown in figure 3.23. Sensitive cells are shown in blue and resistant cells in yellow, with treatment periods indicated by orange frames. Throughout this extended timeframe, repeated recovery of the sensitive population maintains competitive suppression, restricting resistant expansion and delaying spatial escape until failure. Scale bar: 2 mm.

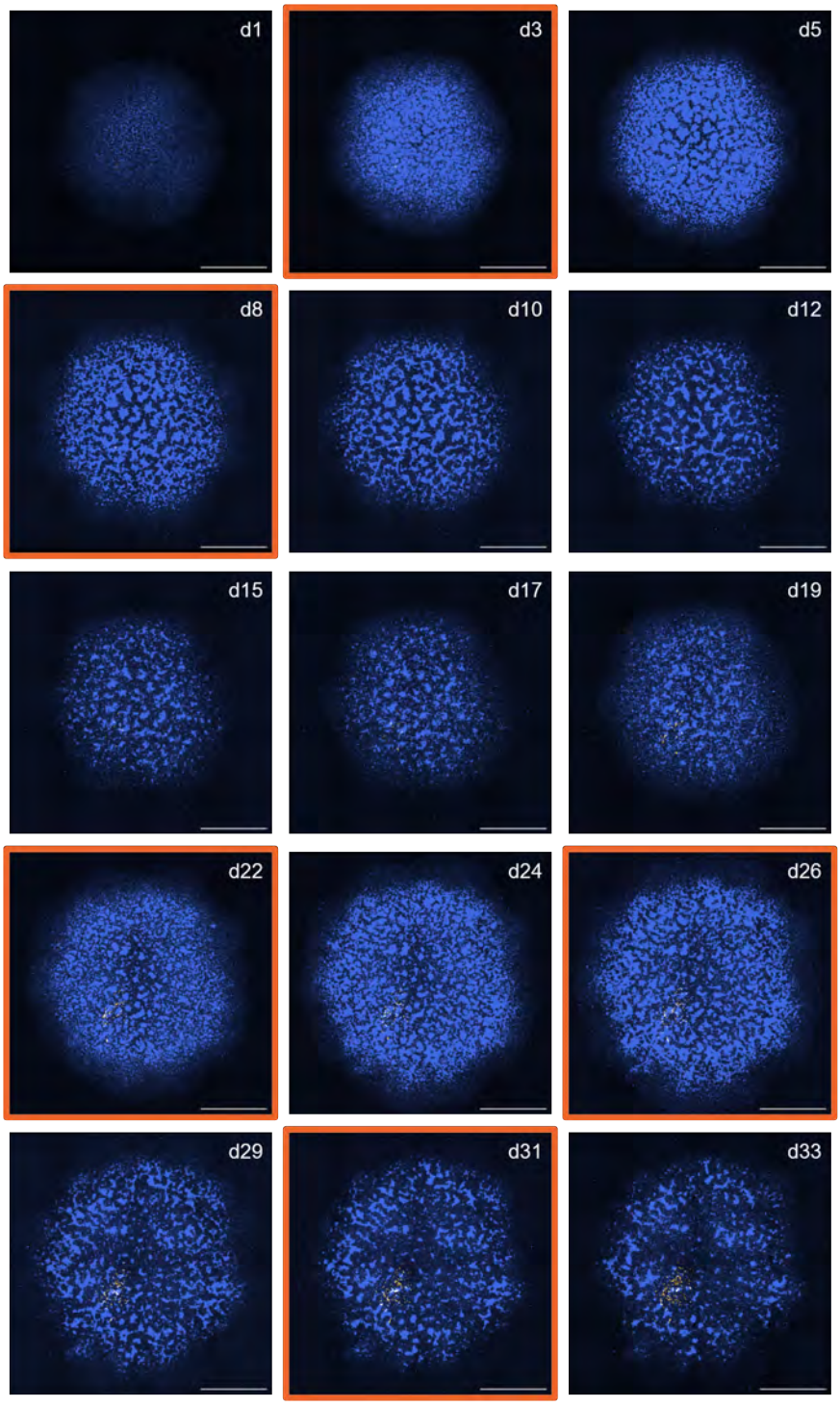


Figure 3.25: Early AT$^+$ time-lapse of α2–α4 colonies (days 01-33) Fluorescence images of an α2–α4 colony treated with AT$^+$. Sensitive cells are shown in blue and resistant cells in yellow. Treatment periods are indicated by orange frames. Over this early phase of the experiment, the colony maintains its overall structure while the sensitive population remains abundant, preserving competitive pressure and limiting resistant expansion. Scale bar: 2 mm.

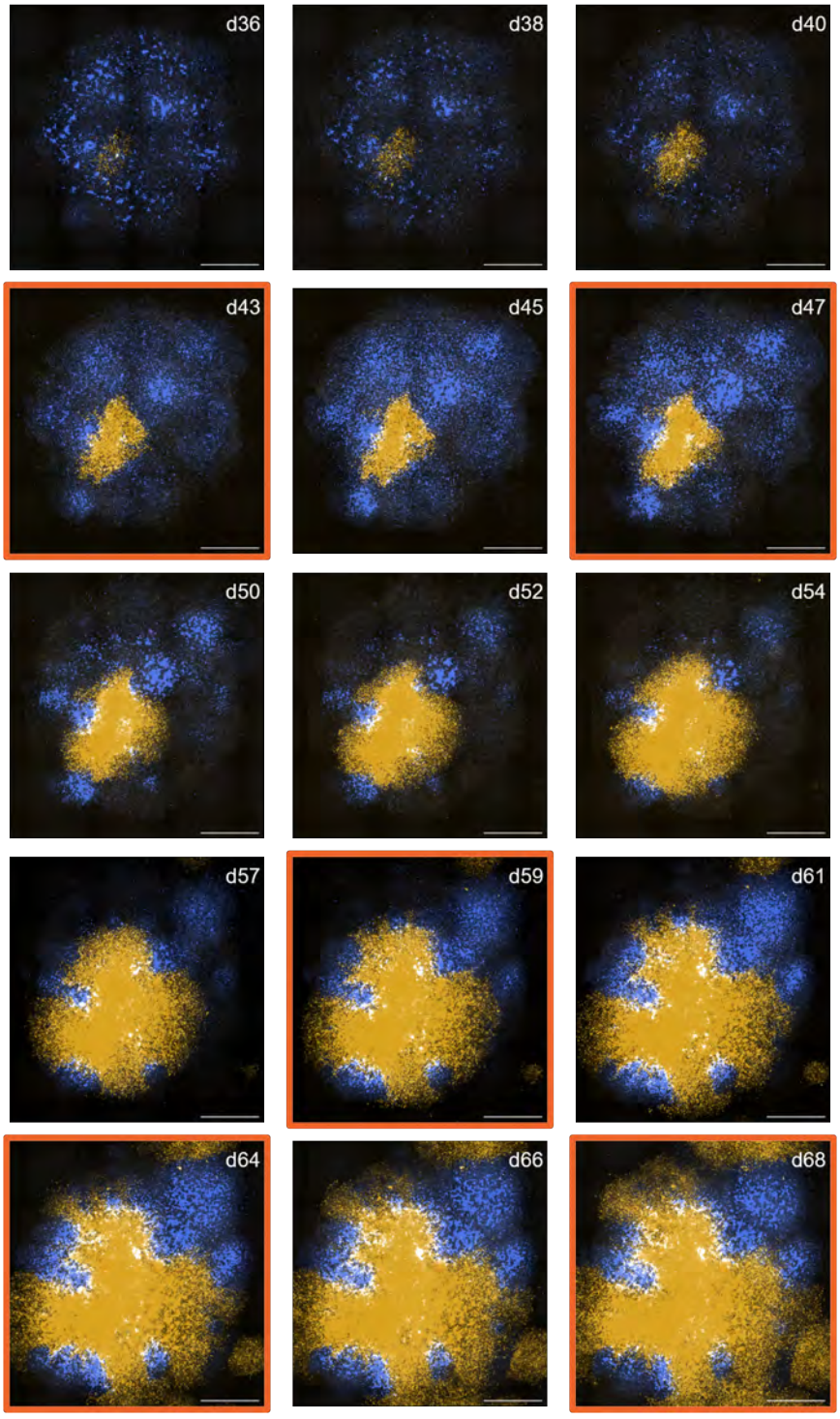


Figure 3.26: Continued AT$^+$ time-lapse of α2–α4 colonies (days 36-68) Continuation of the AT$^+$ time-lapse imaging shown in figure 3.25. Sensitive cells are shown in blue and resistant cells in yellow, with treatment periods indicated by orange frames. Throughout this extended timeframe, recurrent recovery of the sensitive population partially maintains competitive suppression, mirroring the behaviour of the RL agent. Scale bar: 2 mm.

3.3 Discussion and Outlook

The interpretation and synthesis in this discussion build on the results reported in Eiche *et al.* [1] and have been revised and expanded for the present dissertation.

3.3.1 Crowding as the Basis of Competition—Escape as Its Failure Mode

The first central insight of this chapter links tumor ecology to tissue biophysics: resistant cell growth is governed by local crowding and spatial competition. At high densities, resistant proliferation is suppressed, in some cases approaching complete inhibition - independent of treatment strategy. This interpretation connects the assay to tissue-mechanical studies of growth inhibition under crowding and compression [41, 73], as well as to statistical-mechanical models of growth inhibition in bacterial colonies [54].

AT exploits this dependence by balancing cytotoxic effects with the preservation of sensitive-cell density to maintain competitive pressure. Overtreatment disrupts this balance by creating low-density regions that permit resistant outgrowth. These observations provide experimental support for the view that spatial organization can amplify competition between populations and thereby shape the benefit of AT [19, 116].

Under permissive conditions, resistant clones expand in a ratchet-like manner. Treatment depletes the confining sensitive-cell layer while allowing resistant cells to advance into the resulting low-density regions. Subsequent recovery of the sensitive population does not restore the previous spatial configuration, allowing the confinement margin to erode across treatment cycles. Once resistant cells reach the colony boundary, outward expansion is released from surrounding sensitive-cell confinement. Figure 3.27 summarizes how population-level treatment feedback couples to these local processes. The accompanying *in silico* model shows that local mechanical interactions, crowding-dependent proliferation, and delayed treatment response are sufficient to reproduce the observed confinement, escape, and treatment-response dynamics.

This provides a physical explanation for crowding-mediated competition within this system and identifies escape as an emergent system-level transition in spatially structured tumor populations in an assay designed to minimize nutrient, oxygen, and drug gradients.

More broadly, this transition is consistent with work showing that, in compact populations, the fate of resistant lineages depends not only on intrinsic growth differences but also on their access to expanding spatial fronts [19, 53].

Escape, however, does not directly induce treatment failure. When escape occurs only shortly before TTF, substantial resistant expansion may already take place within the colony or along a still-restricted escape front. TTE is therefore likely to be particularly informative for treatment schedules that create a pronounced temporal separation between the loss of spatial confinement and resistant failure, which may become especially relevant in more complex spatial settings.

From a tumor-ecological perspective, the relationship between density and resistant growth suggests the existence of an operational window in which moderate density reductions have limited impact on resistant expansion, whereas stronger reductions lead to rapid growth bursts. This reflects the principle of competitive release in AT, whereby excessive treatment alleviates suppression of resistant populations [12, 13].

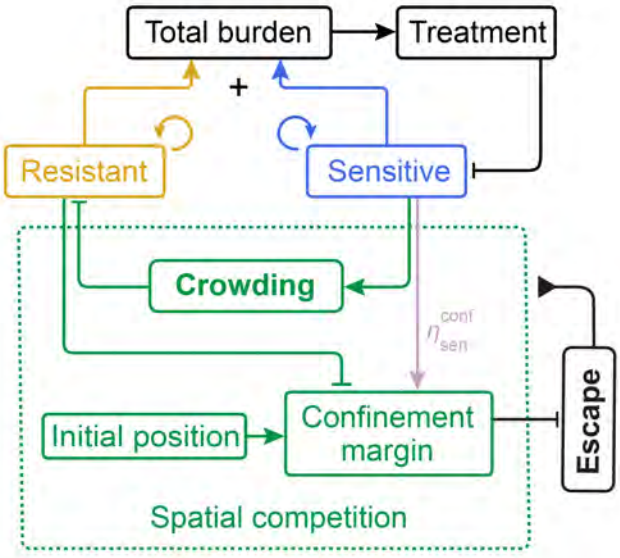


Figure 3.27: Interplay of spatial competition and treatment dynamics in controlling resistant escape Schematic illustrating the interplay between population-level treatment dynamics and local spatial competition. Treatment decisions depend on the total burden, while cellular crowding, initial clone position, and the confining sensitive-cell population ($\eta_{\text{sen}}^{\text{conf}}$) govern resistant clone confinement and escape. Arrows indicate positive interactions. Blunt-ended lines indicate inhibitory effects.

3.3.2 Global Control of Local Dynamics

The second central insight concerns the relationship between local mechanisms and global control. Resistant growth is fundamentally governed by local interactions, whereas treatment is applied and monitored at the scale of the entire tumor. This creates an inherent mismatch between the scale of control and the scale of the dynamics that determine resistant growth. Treatment strategies guided by global measurements therefore depend on the assumption that these observables adequately reflect the local competitive conditions relevant to resistance control, a correspondence that cannot be taken for granted. In the present system, the assay design aimed to minimize strong spatial gradients, potentially reducing this mismatch. Under these conditions, global observables, such as total tumor size or sensitive-cell abundance, remain informative of local crowding, allowing bulk measurements to serve as useful proxies for spatial competition before escape occurs. This defines a regime in which effective, non-spatial models can capture relevant aspects of system behavior, even if they are not fully predictive of the underlying local dynamics.

This observation has potential implications for clinical translation. While detailed spatial information on tumor architecture is difficult to obtain in patients, treatment decisions are routinely based on bulk measurements. The present results suggest that such measurements may nonetheless capture relevant aspects of the underlying ecological dynamics. However, this assumption is likely to break down in more heterogeneous or strongly structured environments. Determining when such inference remains reliable under heterogeneous clinical conditions remains a central translational challenge for AT [15].

Building on this reduced description, an RL agent trained on the GLV-like model identified treatment strategies that prolonged control while reducing treatment exposure. Importantly, analysis of the learned policy enabled the derivation of an interpretable treatment rule (AT^+), demonstrating that black-box optimization can reveal transparent decision principles. Experimentally, AT^+ was implemented in a later, larger colony state, whereas the matched *in silico* comparison isolated the scheduling rule more directly. This positions RL primarily as a hypothesis-generation tool for exploring effective treatment logic in model systems, rather than as a directly deployable clinical controller, in line with broader model-guided learning approaches to treatment scheduling [2, 127].

Together, these results establish crowding-mediated growth inhibition as a tractable ecological control mechanism and demonstrate that combining mechanistic modeling with data-driven optimization provides a powerful framework for designing AT strategies. As

model systems increase in complexity, for example through the inclusion of spatial gradients or evolutionary dynamics, such approaches will become increasingly important for identifying robust and clinically relevant treatment protocols.

3.3.3 Therapeutic Implications and Ecological Efficiency

Beyond the mechanistic interpretation, these experiments provide direct experimental evidence in a spatially structured *in vitro* system that AT can produce a measurable advantage over CT. The AT100 schedule significantly prolonged TTF relative to CT, demonstrating that the conceptual principles of AT translate into observable benefit when spatial context and competition are preserved, aligning with preclinical evidence [13, 226].

Importantly, this benefit did not require increased treatment intensity. Instead, treatment holidays allowed local crowding to recover between treatment phases, prolonging control relative to CT. All adaptive strategies also reduced treatment days compared to CT, with RL-derived approaches further limiting exposure. This reduction was accompanied by higher median population burdens within the assay, reflecting the central trade-off of adaptive control: prolonged containment relies on preserving sufficient sensitive-cell competition rather than minimizing total cell number. Although drug use *in vitro* does not directly translate to clinical toxicity, it provides a practical proxy for treatment burden. Conversely, the higher maintained burden under successful adaptive schedules could pose clinical constraints related to symptoms, organ function, metastatic risk, and patient preferences. Defining an acceptable balance between treatment and disease burden lies beyond the present model and requires future integration of tumor biology, pharmacology, clinical outcomes, and ethical perspectives into adaptive-treatment decisions.

3.3.4 Methodological Considerations and Limitations

Several experimental and analytical limitations define the scope of the conclusions drawn from this system. All experiments were performed using G418 as a drug within a shallow, experimentally selected dosing window. The observed balance between sensitive-cell recovery and resistant expansion may therefore differ for drugs with other pharmacodynamic profiles, narrower therapeutic windows, or indirect modes of action, including immunotherapies. Likewise, the central initial placement of resistant cells was necessary to quantify confinement and escape reproducibly, but it represents one favourable spatial

configuration and does not capture resistant cells arising near the colony boundary or in multiple dispersed locations. Although the colonies were treated as quasi-2D systems, residual vertical structure may also affect local-density estimates. In real tumors, preserving a large sensitive population may also increase the supply of new resistant variants, creating an evolutionary trade-off that this system does not capture.

A limitation of the experimental AT^+ implementation is that treatment began on day 3 and used the day-3 total cell count as its treatment-decision threshold, while reported cell counts remained normalized to day 1. Consequently, the observed improvement may reflect both the modified scheduling rule and the larger colony size throughout the experiment, whereas the corresponding *in silico* simulations isolate the scheduling effect under matched initial conditions.

By aiming to minimize nutrient, oxygen, and drug gradients, the assay provided a tractable setting for studying spatial competition. Their importance in solid tumors, however, motivates extending the model to include controlled environmental heterogeneity. Introducing controlled gradients - potentially through collagen overlays described in section 3.1 - into otherwise tractable 2D systems would allow direct testing of how heterogeneous local environments reshape the relation between crowding, confinement, and population-level control. Gradients need not necessarily reduce the scope for AT: hypoxic or necrotic regions can constrain proliferation and local carrying capacity, potentially imposing additional spatial limits on resistant expansion [19]. Whether these constraints strengthen or weaken containment will depend on how gradients jointly affect local growth, dispersal, drug exposure, and the relative fitness of sensitive and resistant populations [10].

Beyond environmental gradients, the mechanistic basis of crowding-mediated resistant-growth suppression remains uncertain. As local mechanical pressure was not measured *in vitro*, other factors, including biochemical signalling, contact-mediated effects, and cell-cycle regulation, may also contribute. Nevertheless, the spatial simulation showed that pressure-modulated proliferation, local mechanical interactions, and delayed treatment response suffice to reproduce the main confinement, escape, and treatment-response dynamics. Thus, the model provides a plausible minimal mechanistic explanation rather than a unique representation of the underlying biology.

Finally, the effective GLV-like model and RL framework were deliberately restricted to the early pre-escape regime, when global measurements still retained information about the underlying local competitive environment. This enabled systematic exploration of

treatment schedules but does not imply that an equally reduced controller would remain sufficient after resistant escape or in more heterogeneous systems. In this regime, the effective GLV-based model was not fully predictive of local density dynamics, but remained useful for treatment optimization. Likewise, the RL framework was trained using a simplified progression objective tailored to the experimental model system. Future controller development should extend this framework toward multi-objective optimization that jointly considers tumor burden, resistant-cell dynamics, treatment exposure, toxicity constraints, measurement uncertainty, and practical limitations on dose modulation.

3.3.5 Outlook

This chapter identified local crowding as a key regulator of resistant growth. Treatment-induced density reductions promote resistant expansion, whereas treatment holidays restore competitive suppression. Sustained resistant growth eventually erodes the surrounding sensitive layer, leading to escape and loss of AT control. Building on these findings, RL trained on a reduced GLV-like model yielded treatment strategies that prolonged TTF while reducing drug exposure *in vitro*.

Rather than marking the completion of the 2D system, these results establish it as a productive experimental platform in which ecological interactions can be isolated, manipulated, and quantitatively analyzed. The next steps involve extending this framework along two complementary directions.

First, introducing controlled nutrient or oxygen gradients within 2D systems—for example through matrix overlays—would allow environmental heterogeneity to be studied while maintaining full optical accessibility. Second, transferring these ecological principles into 3D systems will enable their evaluation under more physiologically relevant conditions, where gradients, mechanical constraints, and spatial heterogeneity play a central role.

The mechanisms identified here—treatment-induced space creation, re-crowding during recovery, and density-dependent suppression of resistant expansion—provide concrete hypotheses for how AT operates in structured environments. Testing these principles in 3D systems represents a critical next step and forms the conceptual bridge to the subsequent chapter.

4

Resistance Dynamics and Adaptive Therapy in Melanoma Spheroids

Although the 2D model system provides essential insights into the ecological interactions and growth dynamics of dense human cell populations, its limitations were evident from the outset. Tumors *in vivo* are three-dimensional structures in which geometry, diffusion, and mechanical constraints fundamentally alter cell behavior [29, 227]. In 3D structures, the surface-to-volume ratio limits diffusion, leading to gradients of oxygen and nutrients as well as the formation of necrotic cores—features that can only be partially reproduced in planar culture [155, 228]. To investigate these aspects, I developed a complementary 3D model system in parallel to the 2D assay.

The aim was to establish a stable, reproducible, and experimentally accessible platform for long-term AT studies under controlled 3D conditions. Achieving this required overcoming several technical challenges. First, reproducible methods were needed to generate uniform spheroids from 2D cultures. Second, spheroids had to be immobilized during imaging while still permitting limited mechanical expansion and allowing comparison with free-floating

conditions. Third, long-term stability had to be ensured so that spheroids neither disintegrated nor induced excessive matrix fissuring or nutrient depletion. Finally, imaging had to be optimized to enable repeated visualization at cellular resolution without compromising viability. The following sections describe the iterative solutions developed to meet these requirements and establish a robust platform for studying resistance evolution under AT in 3D.

4.1 Development of 3D Spheroid Embedding

To initiate 3D culture, cells were seeded in low-adhesion plates. Conventional tissue-culture plastic is treated to promote cell adhesion, leading most cell types to form monolayers on the stiff surface [229]. In contrast, low-adhesion well plates are engineered to prevent surface attachment through specialized coatings, forcing cells to interact with each other rather than the substrate [230]. Under these conditions, cells aggregate and compact into 3D spheroids, with formation kinetics and stability depending on cell type and culture conditions [231, 232].

Using 96-well low-adhesion plates, $\alpha 2$ cells aggregated into spheroids of varying sizes depending on the initial seeding count. These spheroids remained intact for several days, but their overall structure was unstable and prone to disintegration over longer periods. While many melanoma cell lines are known to form spheroids under appropriate culture conditions, the engineered cells used here showed limited structural cohesion and reproducibility [233, 234].

To improve spheroid stability, I collaborated with the group of Prof. Ben Fabry, with direct technical support from Dr. Christoph Mark. Following his suggestion, Mel Juso and SK-Mel-28 cells were cultured with CollagenR supplementation, a low-concentration, non-polymerized collagen supplement distinct from the collagen hydrogel used for subsequent embedding. Without supplementation, spheroids remained viable but gradually lost structural integrity, consistent with earlier observations.

Based on these observations, the final workflow involved seeding approximately 500 cells in 100 µl of medium per well, followed after 24 hours by addition of an equal volume of medium containing 4 µg/ml CollagenR. After a further seven days of culture, spheroids were used for subsequent embedding and imaging experiments.

Although free-floating spheroid cultures are widely used and were also explored later in this work, they posed practical constraints for the treatment–imaging workflow. Media exchanges between therapy cycles frequently perturbed spheroid architecture, and cytotoxic exposure often led to cohesion loss and fragmentation. Moreover, longitudinal tracking of individual cells or defined regions requires a fixed spatial orientation that is difficult to maintain in suspension. For these reasons, matrix embedding was required to provide mechanical stabilization and positional consistency, thereby enabling controlled dosing and repeated high-resolution imaging. In addition, the use of a matrix introduced an extra level of experimental control, as physical and transport parameters such as stiffness, elasticity, pore size, cell density, and nutrient accessibility could be systematically varied to probe their influence on spheroid behavior.

4.1.1 Mebiol Embedding

To implement matrix embedding under well-controlled and reproducible conditions, a synthetic polymer, Mebiol, was selected as the initial matrix material. Unlike biologically derived matrices such as Matrigel or collagen, Mebiol does not replicate specific cell–matrix interactions and therefore provides a simplified but highly defined environment. This reduction in biological complexity was advantageous at this stage, as the goal was to incrementally construct a controllable experimental system in a bottom-up manner before introducing additional sources of variability [235]. Mebiol offers consistent physical properties and reliable handling, making it well suited for methodological optimization [236]. A further practical advantage is its reversible thermoresponsive behavior. The polymer solidifies at 37 °C and liquefies at room temperature or below, allowing straightforward embedding and recovery of spheroids without mechanical disruption.

Bulk Mebiol Embedding

To establish a reliable embedding procedure for melanoma spheroids, I first tested a single-layer approach. The embedding was performed according to the following protocol:

Spheroid and Plate Preparation

- Day 0: Seed 500 α2 cells per well in a low-adhesion 96-well plate.

- Day 1: Add CollagenR at a 4 µg/ml dilution to promote spheroid cohesion.
- Day 8: Use mature spheroids for embedding.

Embedding procedure

- Prepare a Falcon tube containing Mebiol polymer solution and keep it on ice.
- Prepare two prechilled microtubes per spheroid—one intermediate tube for transfer and one final tube for embedding.
- Transfer preformed spheroids ($\approx$ 50 µl suspension) into the intermediate tube to allow better visibility during subsequent pipetting. Keep the tubes on ice.
- Using a 20 µl pipette with a 5 µl volume, gently transfer individual spheroids from the intermediate to the final tube.
- Add 150 µl of cold Mebiol to each final tube using a 200 µl pipette and reverse pipetting technique. Before use, cool the pipette tip in the Mebiol Falcon until the tip temperature equilibrates. Ensure that all tubes remain on ice throughout.
- Carefully aspirate 125 µl of the spheroid–Mebiol mixture with the spheroid positioned near the tip opening, and dispense it into a 96-well plate. Avoid introducing air bubbles.
- Place the plate in a 37 °C incubator for 2 minutes to induce gelation of Mebiol.
- After gelation, gently add 100 µl of 37°C culture medium to each well.

Although this procedure reliably produced gels containing intact spheroids, embedding depth remained difficult to control, and air bubbles frequently formed during pipetting. Spheroids positioned too close to the well bottom attached and spread across the surface, whereas those placed too high fell outside the imaging range or protruded into the medium. Even untreated control spheroids occasionally lost fluorescence and ceased growing. As daily imaging showed no apparent phototoxicity, impaired nutrient access was considered a more likely explanation. Variations in embedding depth and matrix thickness, potentially increasing diffusion distances and limiting access to fresh medium, thus emerged as major sources of variability affecting spheroid viability.

Droplet-Based Mebiol Embedding

To improve positional control, several iterations of a drop-embedding strategy were developed and tested. Preliminary experiments explored multiple droplet configurations

to identify conditions that yielded stable and centrally positioned spheroids. Different combinations of droplet size and spheroid number were evaluated to balance handling convenience and spatial control. While all variants allowed embedding in principle, positional reproducibility varied substantially. Handling was improved by combining precooled pipette tips, which minimized premature polymerization, with a prewarmed plate that reduced spreading and rapid polymerization of the Mebiol–spheroid mixture. Rapid aspiration followed by slow, controlled dispensing further aided vertical alignment of the viscous material. Nevertheless, although some spheroids were embedded at suitable imaging depths, most ultimately sank to the bottom of the well, underscoring the need for a more refined protocol.

Building on these observations, additional experiments were performed in which the Mebiol droplet was kept liquid with the plate on ice to allow spheroids to sink passively before controlled re-polymerization. After partial sinking, plates were inverted for a defined period to reposition spheroids closer to mid-height within the gel. This approach provided initial indications that the viscosity and temperature balance could be used to fine-tune spheroid height. However, spheroids that had once contacted the well bottom remained adhered and could not be repositioned. Gentle centrifugation to assist redistribution was tested but proved unsuitable, as even minimal acceleration disrupted the Mebiol-spheroid-droplets and destroyed their shape.

More reliable and reproducible placement was achieved when the spheroid was aspirated first and deposited at the end of the pipetting process, ensuring that it remained near the upper portion of the droplet prior to gelation. This modification significantly improved spatial control.

Spheroid and Plate Preparation

- Day 0: Seed 500 α2 cells per well in a low-adhesion 96-well plate.
- Day 1: Add CollagenR at a 4 µg/ml dilution to promote spheroid cohesion.
- Day 8: Use mature spheroids for embedding.
- Place a 48-well plate in a vacuum chamber overnight with 50 µl of 1H,1H,2H,2H-Perfluorodecyl-triethoxysilane (Silane) to hydrophobize the surface.
- Disinfect the plates with ultraviolet (UV) light prior to use.

Embedding procedure

- Resuspend the spheroid in 1 µl of medium in a 0.5 ml microtube placed in a cooled metal holder.
- Using a 20 µl pipette, aspirate 20 µl of Mebiol containing red calibration beads by inverse pipetting. Ensure the pipette tip is fully cooled before use.
- Add the Mebiol to the spheroid.
- Aspirate the spheroid first so that it is positioned at the top of the 20 µl Mebiol column inside the pipette tip—this step is critical for precise placement.
- Dispense the mixture very slowly through the holes of the lid of the silanized plate, keeping motion steady to avoid air bubbles. A thin Mebiol filament upon tip withdrawal indicates optimal viscosity and temperature.
- Overlay with 37 °C culture medium containing green calibration beads to stabilize the droplet.

This configuration provided the first reproducible placement of spheroids within the optical imaging range and was highly effective for positional control. Having largely stabilized spatial positioning, the main limitation now shifted toward nutrient accessibility. Spheroids embedded in thick Mebiol layers exhibited reduced long-term viability, suggesting diffusion constraints between the gel interior and the surrounding medium. To address this, the next series of experiments aimed to retain the spatial stability achieved in the single-drop configuration while improving access to nutrients by reducing the overall matrix thickness and adjusting the polymerization sequence.

Two-Step Thermal Mebiol Embedding

To reduce diffusion distances while retaining mechanical stability, I developed a two-step thermal embedding approach. A 20 µl pre-polymerized Mebiol layer was prepared on a warm plate, overlaid with a 2 µl spheroid–medium droplet, briefly cooled on ice to soften the polymer, and reheated to 37 °C for re-polymerization. Although initial reproducibility was limited, several spheroids were embedded within the confocal imaging range. Fluorescent beads enabled height quantification, and individual spheroids could be tracked for up to 78 days. This configuration also produced proof-of-concept treatment responses in 3D culture. Sensitive cells were selectively eliminated, followed by delayed resistant

regrowth under continued imaging (see figure 4.1). Varying cooling times produced reproducible sinking behavior, with spheroid depth increasing approximately linearly with cooling duration (slope = 4.11 µm/min, $R^2 = 0.9998$, $n = 4$; see figure 4.1b). However, viability still declined in most spheroids after several days.

To improve nutrient accessibility, I next attempted to reduce the total Mebiol volume by using thinner top layers. However, smaller droplets dehydrated rapidly, impairing re-melting and preventing proper fusion of the two layers during thermal cycling. Adding medium to maintain hydration improved re-melting, but frequently caused cavities to form around the spheroids. Within these cavities, spheroids initially grew freely before becoming compressed against the surrounding matrix, creating nonuniform mechanical conditions. Adjustments to cooling times, medium volumes, and rehydration steps reduced cavity size but did not eliminate the problem. Replacing preformed spheroids with cell suspensions also failed, yielding dispersed cells rather than cohesive aggregates.

Repeating the two-phase embedding experiments at room temperature rather than on a heated plate eliminated dehydration, as Mebiol polymerized sufficiently without additional heat. Spheroid sinking occurred independently of rehydration, while droplet structure remained stable. The most reproducible configuration was adopted as the protocol summarized below, offering the best compromise between positional control and structural stability. Although embedding quality was reproducible, spheroid height still varied between wells. More critically, spheroids frequently showed reduced long-term viability, suggesting persistent limitations in nutrient and oxygen availability. Thus, biological performance rather than technical feasibility became the main constraint. While automation of pipetting and temperature control might improve reproducibility and scalability, the approach was ultimately discontinued due to its viability limitations under the tested conditions.

Spheroid and Plate Preparation

- Day 0: Seed 500 α2 cells per well in a low-adhesion 96-well plate.
- Day 1: Add CollagenR at a 4 µg/ml dilution to promote spheroid cohesion.
- Day 8: Use mature spheroids for embedding.
- Place a 48-well plate in a vacuum chamber overnight with 50 µl of silane. Afterwards, UV-disinfect the plates prior to use.

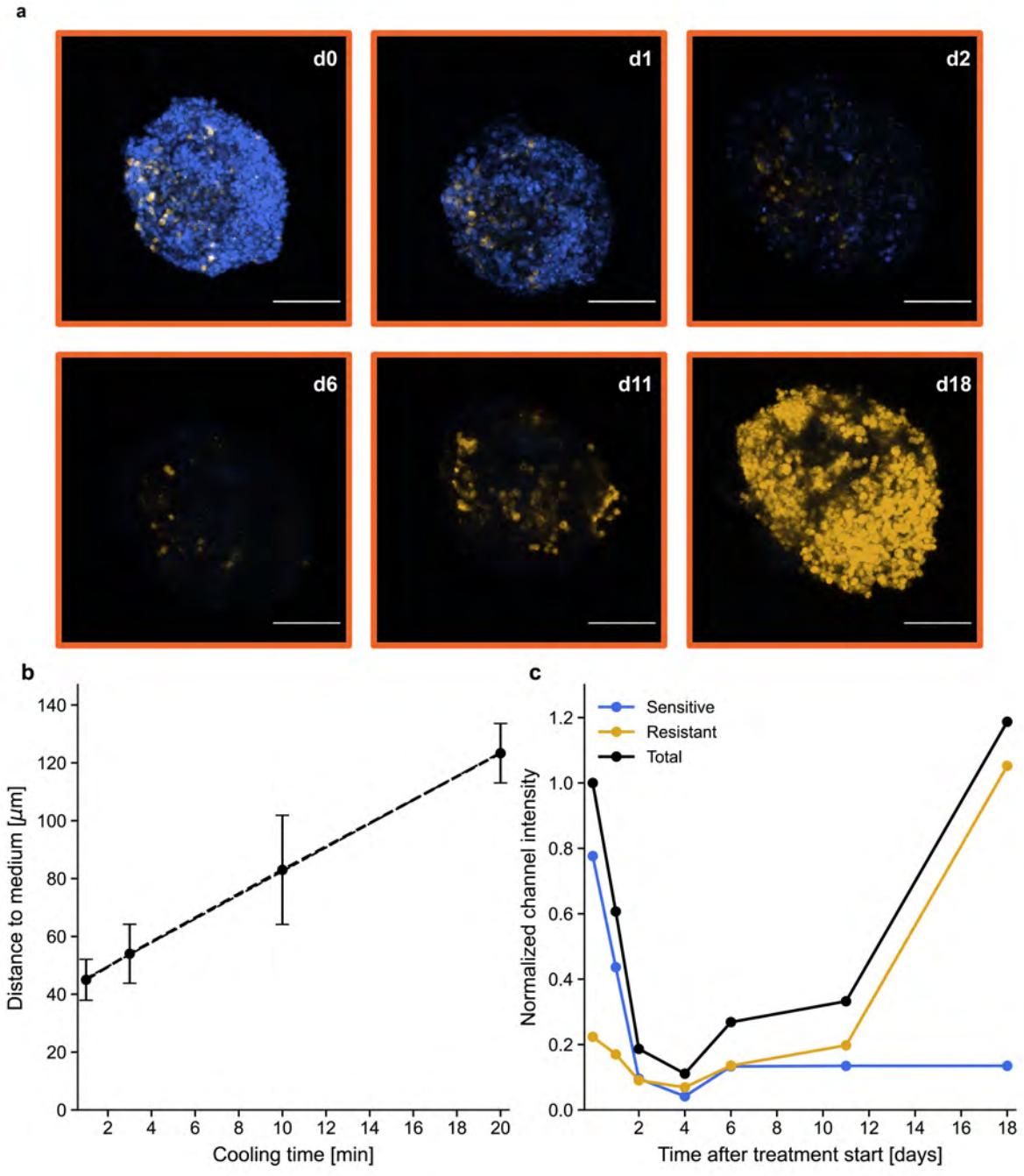


Figure 4.1: Spheroid sinking behaviour and initial CT response in embedded 3D models (a) Fluorescence time-lapse imaging of an $\alpha 2$–$\alpha 4$ spheroid under CT. Orange frames mark treatment periods. Maximum intensity projection (MIP) shown. Scale bar: 200 µm. (b) Spheroid sinking in Mebiol matrix. Points show mean immersion depth ± SD across $n = 4$ independent embeddings. The dashed line denotes the linear fit. Longer cooling increased immersion depth. (c) Proof-of-concept response to continuous G418 treatment. Fluorescence intensities are normalized to total fluorescence at day 0.

Embedding procedure

- Using a 20 µl pipette, aspirate 20 µl of Mebiol containing red calibration beads by inverse pipetting. Ensure that the pipette tip is thoroughly cooled before use. The plate should be kept at room temperature during this step.
- Resuspend the spheroid in 1 µl of culture medium and gently place it on top of the non- or partially polymerized Mebiol droplet.
- Cool the sample from above for 8 minutes using a metal block pre-cooled in the freezer to control polymerization and fix the spheroid at the desired height.
- Carefully overlay the droplet with 37 °C culture medium containing green calibration beads to maintain hydration and visualize the surface.

Miscellaneous Embedding

Beyond the configuration established above, several additional embedding strategies were explored to test the limits of Mebiol handling and structural control. These approaches were more experimental in nature but provided valuable insights into the mechanical and diffusion constraints of the polymer system. One variant, the layered Mebiol embedding approach, involved polymerizing a 200 µl base layer of Mebiol in a 48-well plate, placing a 2 µl spheroid–medium droplet on top, and then adding a 100–200 µl overlay of cold Mebiol. Controlled cooling (>5 min) allowed the two layers to fuse, preventing cell growth along the interface. While this setup achieved precise positioning and good structural integrity, long-term cell survival remained limited, likely due to diffusion constraints. Reagent consumption was also high, making the method cost-inefficient. Nevertheless, its reproducible layered configuration later inspired a similar collagen-based embedding approach aimed at overcoming the limitations in nutrient accessibility.

A three-drop variation, employing sequential additions of 20 µl, 1 µl, and 3 µl of Mebiol, was also tested to conserve material and shorten the diffusion distance between spheroids and the surrounding medium. Complete fusion between layers proved difficult to achieve, leading to growth along residual interfacial planes. This method was therefore discontinued. A rapid thermal polymerization approach, in which spheroid-containing Mebiol droplets were pipetted directly into warm medium to achieve immediate gelation, yielded inconsistent embedding depths and was abandoned after preliminary trials. Similarly, a direct injection embedding technique, in which spheroids were introduced into semi-solid

Mebiol using a distance-controlled pipette, provided clean individual samples but suffered from low reproducibility in manual handling. Together, these exploratory variants confirmed that Mebiol could be manipulated in multiple configurations, but none overcame the combined limitations of positioning, throughput, and long-term viability.

Summary

Across these iterations, Mebiol proved useful as a defined matrix for methodological optimization. The experiments established reproducible handling routines and clarified how temperature, hydration, viscosity, and matrix thickness influence spheroid positioning and stability. However, incomplete vertical standardization and limited long-term viability prevented its use for systematic AT experiments and indicated the need for a more permeable matrix system.

4.1.2 Collagen Embedding

Collagen was thus evaluated as a biologically active alternative with a porous fibrillar architecture that supports cell adhesion and molecular diffusion [237, 238]. Its type I fibers contain integrin-binding sites and provide greater physiological relevance, but also introduce increased variability and sensitivity to pH, ionic strength, and temperature. In addition, collagen polymerization is irreversible, making embedding and recovery less straightforward than in the thermoreversible Mebiol system. Despite these trade-offs, its permeability and biological relevance made collagen the more promising platform for the next stage of assay development.

In an initial experiment, collagen hydrogels were prepared according to the calculation-based and lot-calibrated polymerization protocol described in Section 7. Collagen concentration and gel volume were adjusted between experiments as required. Mixing was performed on ice and polymerization was initiated only after transfer into culture wells.

This collagen mixture was applied in a two-layer configuration. First, 25 or 50 µl of collagen were added to the well and polymerized at 37 °C for 10 min to form a base layer. Fully developed spheroids were collected from low-adhesion plates in 5 µl of medium, mixed with 50 µl of collagen solution, and dispensed onto the polymerized base layer to form the second layer.

Most spheroids in the initial two-layer collagen setup remained off-center or near the upper surface, as they failed to sink through the polymerizing gel. Consequently, many were outside the confocal working distance and could not be imaged reliably. Reducing the top-layer thickness offered little improvement, while lowering the collagen concentration facilitated sedimentation but eventually allowed cells to invade the interface between layers after approximately three weeks. Due to these structural limitations, the approach was initially set aside in favor of alternative embedding strategies.

The concept was later revisited together with Isabel Schneider, using collagen base layers of 50 or 80 µl in 96-well plates, followed by a 50 µl spheroid-containing collagen layer. Improved timing and more careful pipetting, ensuring that the spheroid-containing layer remained on the surface during application, substantially increased the proportion of spheroids settling within the optimal imaging plane. This revised configuration proved highly reproducible and was subsequently adopted in Isabel Schneider's master's project (see Section 4.3). The improved positional control and structural stability provided a promising foundation for further optimization.

Together, these trials demonstrated that collagen supported long-term spheroid culture more effectively than Mebiol, consistent with improved nutrient accessibility. While reliable positioning remained a technical challenge, the insights gained into collagen handling, polymerization, and spheroid settling enabled the development of a more robust two-layer embedding protocol. This configuration ultimately formed the basis for the standardized AT experiments developed in collaboration with Isabel Schneider.

4.2 Preliminary 3D Adaptive Therapy Experiments

The first AT experiments in 3D were conducted in parallel with ongoing optimization of the embedding methods. They served three complementary purposes: (i) to evaluate how well the current Mebiol- and collagen-based protocols supported reproducible treatment and imaging, (ii) to test whether ecological interactions observed in 2D—such as sensitive-cell-dependent restraint of resistant expansion—could also be detected in 3D spheroids, and (iii) to identify the experimental variables most critical for future assay design. In this sense, these early trials functioned as both biological and methodological tests, allowing the system itself to be characterized under therapy.

At this stage, collagen embedding enabled longitudinal imaging and treatment of spheroids, although positional consistency between samples remained limited. Variability in embedding height was the main unresolved technical challenge, as it affected both imaging conditions and experimental standardization. Nevertheless, these preliminary conditions were sufficient to initiate 3D treatment experiments, providing an essential first assessment of the feasibility of AT in this setting while revealing the biological and technical requirements for subsequent assay refinement.

To probe feasibility under these constraints, two complementary experimental strategies were pursued in parallel. First, AT schedules were tested in embedded spheroids using the best available Mebiol- (and later collagen-) based procedures, which offered geometric stability and optical access but suffered from imperfect control of vertical placement and diffusion. Second, free-floating spheroids in low-adhesion plates were used as an alternative that avoided embedding-related variability and allowed longer structural integrity, at the cost of reduced positional control. Together, these approaches provided a practical testbed to evaluate whether key ecological effects could be detected in 3D and to determine which technical parameters most strongly influenced treatment readouts.

Because longitudinal imaging generated repeated observations from the same spheroids over time, pooled point-level visualizations are interpreted descriptively. Where formal statistics are reported for free-floating spheroids, spheroid-aware analyses were used to account for within-spheroid dependence.

4.2.1 Embedded Spheroids

The embedded spheroid experiments served as the first direct attempt to implement AT in a 3D environment in this thesis. Mixed melanoma spheroids composed of 99.8% sensitive $\alpha 2$ and 0.2% resistant $\alpha 4$ cells were immobilized in silanized 48-well plates using a Mebiol-based embedding strategy in which a first Mebiol droplet was placed at the bottom of the well and a second droplet containing the spheroid was deposited on top. After one week of pre-growth in low-adhesion plates supplemented with CollagenR, spheroids were transferred into Mebiol droplets and cultured under daily confocal imaging. Adaptive schedules (AT50, AT80, AT100) were applied using G418 concentrations of 0.1–0.15 mg/ml, alongside continuous and intermittent dosing controls. These early trials aimed not only to test whether resistant outgrowth could be constrained by sensitive cells under adaptive

schedules, but also to assess how embedding geometry and diffusion conditions would affect spheroid stability and viability over extended observation periods.

As discussed before, the main technical limitation was variable embedding height. Spheroids near the well bottom attached and spread, whereas those close to the surface detached and those embedded too deeply showed impaired diffusion and viability. This well-to-well variability, occasionally compounded by air pockets that distorted spheroid morphology, reduced reproducibility and the number of usable samples. As a result, biological interpretation was often constrained by technical noise rather than treatment effects. In particular, spheroids subjected to CT frequently failed early due to embedding-related instability and could therefore not be analyzed reliably. In contrast, sufficient data could be obtained for no-treatment (NT) and AT conditions, allowing exploratory analysis of treatment-dependent trends. Together, these limitations emphasized the need for stricter control of embedding geometry and diffusion before large-scale AT assays could be performed.

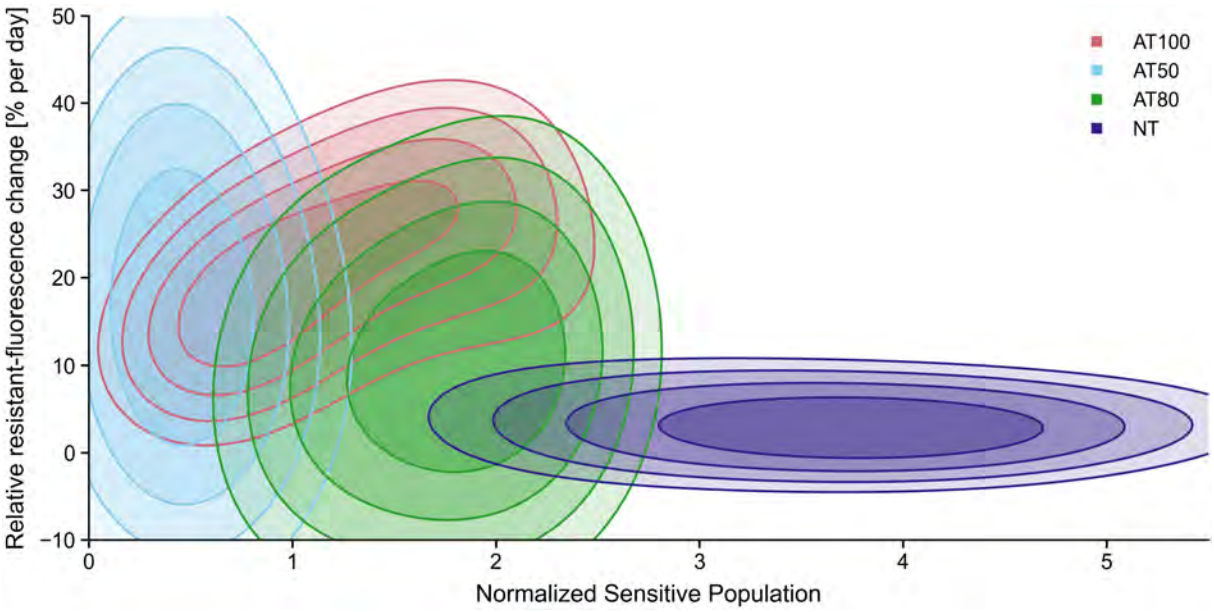


Figure 4.2: High sensitive burden coincides with low resistant growth For morphologically stable spheroids, percentage resistant-fluorescence change between successive retained observations was divided by elapsed days and plotted against interval-end sensitive fluorescence normalized to total fluorescence at treatment initiation. Each point represents one longitudinal observation. Filled contours show pooled two-dimensional Gaussian kernel-density estimates for each treatment strategy (bandwidth $= 0.7$), enclosing 10–40% cumulative probability mass. Darker regions indicate higher observation density.

Despite the limited reproducibility of the embedded assays, a subset of spheroids could be analyzed quantitatively, with examples shown in figure 4.3 and figure 4.4. To minimize the impact of absolute fluorescence intensity, which varied with imaging depth and spheroid orientation, I focused on relative changes over time. The cumulative red fluorescence signal was used as a proxy for resistant-cell burden, while the cumulative green signal represented the relative abundance of sensitive cells. From these signals, percentage resistant-fluorescence changes were divided by elapsed days to quantify relative growth per day. To approximate the global sensitive-cell state of each spheroid, the green-channel fluorescence intensity was normalized to the total fluorescence intensity (TFI) across the red and green channels at treatment initiation, providing a relative measure of sensitive-cell abundance within each spheroid. By plotting resistant growth rates against normalized sensitive population, I could assess whether resistant expansion was modulated by the presence of sensitive cells across different treatment schedules, independent of absolute intensity or positional artifacts (see figure 4.2).

In the pooled visualization, resistant growth remained close to zero at high sensitive-cell fluorescence, whereas lower sensitive fluorescence was associated with more variable resistant growth. This pattern was compatible with competition-driven dynamics, but owing to the exploratory design and repeated-measures structure, it was interpreted as a qualitative trend rather than an independently validated quantitative effect.

The pilot experiment also revealed substantial variation between wells, highlighting the influence of embedding height, cavity formation, and matrix diffusion on the measured dynamics. These technical limitations guided subsequent assay refinement and motivated the temporary transition to the simpler, geometry-independent free-floating spheroid format described in the next section.

4.2.2 Free-Floating Spheroids

This approach aimed to bypass embedding-related complexity while retaining a sufficiently stable 3D system for probing ecological interactions under therapy. Spheroids were generated from mixed populations of sensitive $\alpha 2$ and resistant $\alpha 4$ cells at different initial seeding sizes (1 000–10 000 cells). To enhance structural integrity, extracellular additives were systematically screened in parallel.

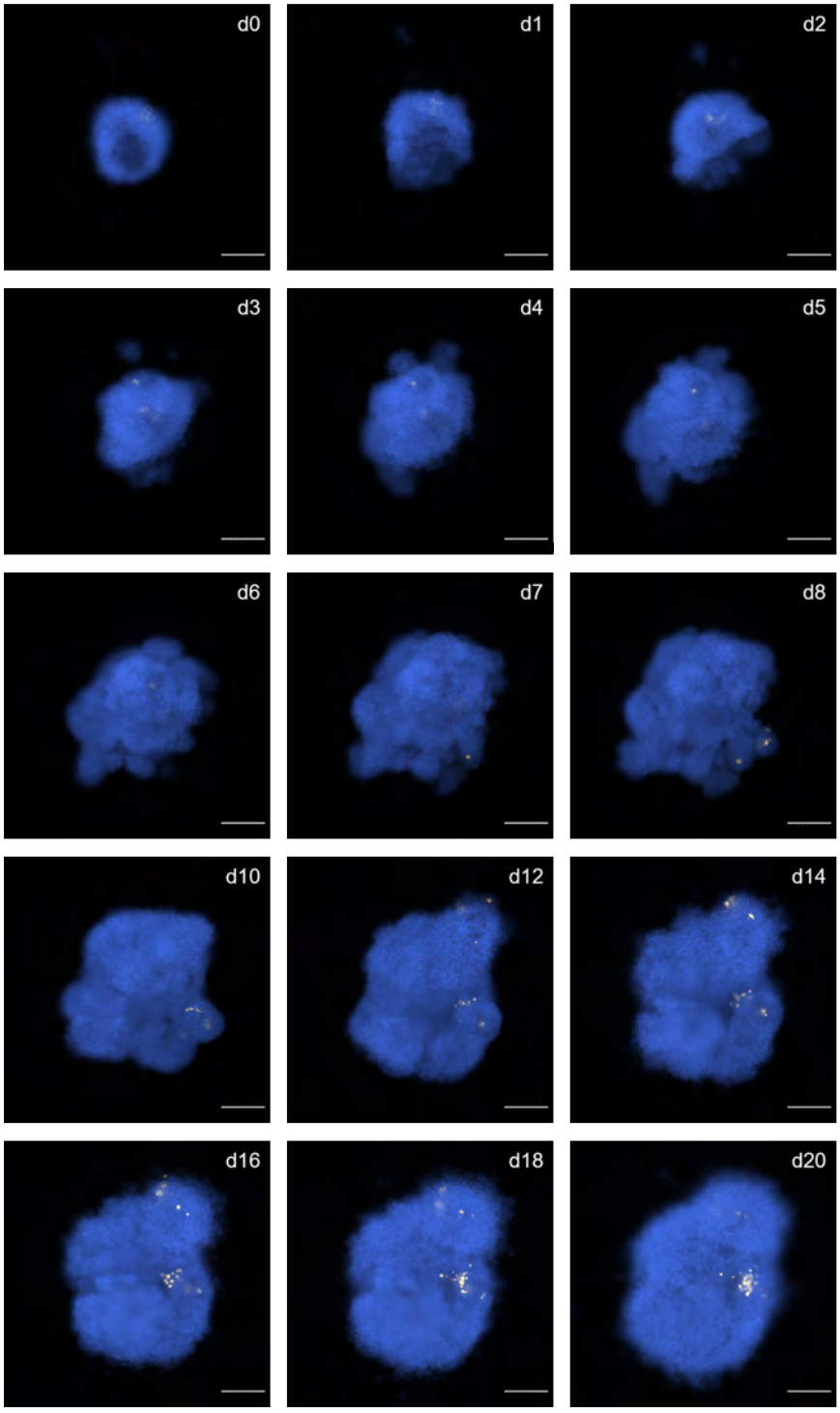


Figure 4.3: Growth of untreated spheroids with minimal resistant-cell emergence Fluorescence time-lapse images of an $\alpha 2$–$\alpha 4$ spheroid under NT conditions. Sensitive cells are shown in blue and resistant cells in yellow. The spheroid expands throughout the imaging period without substantial resistant outgrowth, consistent with resistance remaining rare in the absence of treatment-associated selection. Images are MIPs. Scale bar: 300 µm.

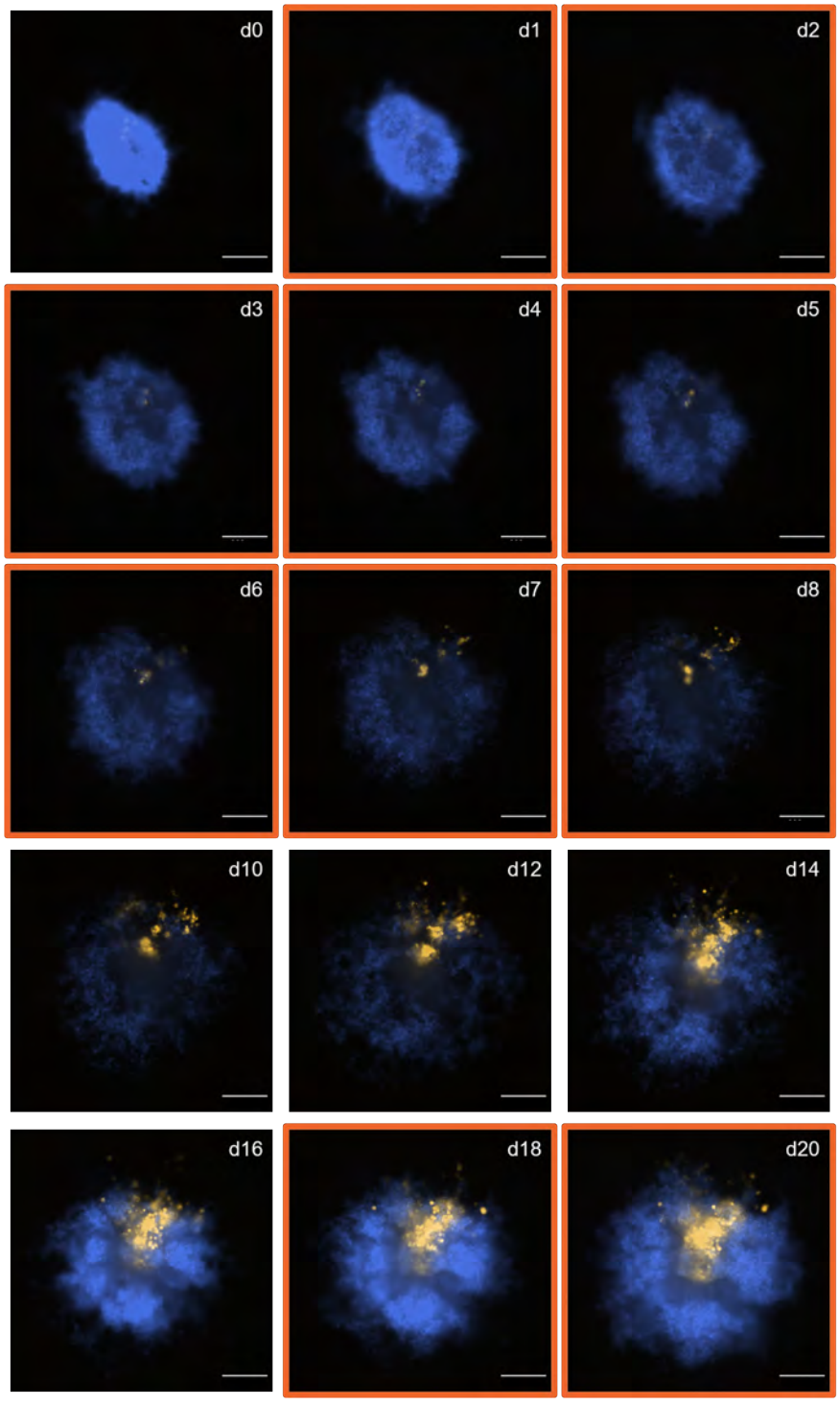


Figure 4.4: AT50 shows slow resistant expansion in α2–α4 spheroids Fluorescence time-lapse images of an α2–α4 spheroid subjected to AT50 treatment. Sensitive cells are shown in blue and resistant cells in yellow. Orange frames mark treatment periods. Resistant cells emerge but expand gradually, while sensitive cells remain a substantial fraction of the spheroid for much of the experiment. Images are MIPs. Scale bar: 300 µm.

To assess whether extracellular components could improve cohesion in suspension culture, CollagenR, RGD peptides, and hyaluronic acid (HA) were screened across a range of concentrations and dosing schemes using qualitative visual criteria. Spheroids initiated with approximately 3,000 cells appeared most suitable for longitudinal culture: 1,000-cell spheroids remained comparatively small, whereas 10,000-cell spheroids showed reduced structural stability. Regular supplementation with CollagenR improved apparent spheroid circularity and structural integrity under untreated and treated conditions, whereas spheroids cultured without CollagenR tended to disintegrate over time. Higher CollagenR concentrations (10 µg/ml) yielded no clear additional benefit while increasing medium viscosity and impairing handling. Accordingly, an intermediate concentration of approximately 5 µg/ml was selected for subsequent experiments.

RGD peptides (1–10 µg/ml) produced modest and variable improvements in cohesion. HA (0.1–1 mg/ml) also appeared to support spheroid stability, with continuous addition of 0.3 mg/ml providing a practical balance between structural integrity and handling. Overall, these exploratory observations indicated that continuous low-dose CollagenR was the most useful component for maintaining free-floating spheroids, whereas RGD peptides and HA provided less consistent additional benefits. Unless otherwise stated, subsequent free-floating assays were initiated with approximately 3,000 cells per spheroid and maintained with regular CollagenR supplementation (5 µg/ml).

Under these stabilized conditions, spheroids remained intact for extended periods—often several weeks—and could be subjected to a range of treatment regimens. Adaptive schedules (AT50, AT80, and AT100) were compared with CT, intermittent dosing, and pulse schedules, with representative image sequences shown in figures 4.6, figure 4.7, and figure 4.8. Treatments were applied at low G418 concentrations (0.2 mg/ml), and daily confocal imaging was performed to capture both green and red fluorescence signals. Analysis focused on three primary readouts: (i) time to treatment failure, dated at the second of two successive retained observations with relative total fluorescence > 5 or relative resistant fluorescence > 0.1, with spheroids annotated as lost or destroyed being right-censored unless threshold failure had already occurred, (ii) treatment burden, quantified as the fraction of observation days under therapy and (iii) the relative daily growth rate of resistant cells, analyzed both as a function of normalized sensitive-cell abundance and in relation to recent treatment exposure, including comparisons before and after spheroid disruption.

Kaplan–Meier curves (see figure 4.5a) suggested a tendency toward longer time to failure

under AT80 than under IT, although this comparison did not reach statistical significance ($p = 0.0625$; pairwise log-rank test). The comparison between AT80 and CT likewise did not establish a difference ($p = 0.284$). These results are therefore interpreted as a descriptive tendency rather than as evidence of a survival advantage. Treatment burden analysis revealed pronounced differences between therapy strategies. As expected, CT had a treatment fraction of 100%, whereas adaptive regimens reduced the fraction of retained post-baseline treatment-on observations by approximately 28–33% relative to CT, based on group medians (see figure 4.5b).

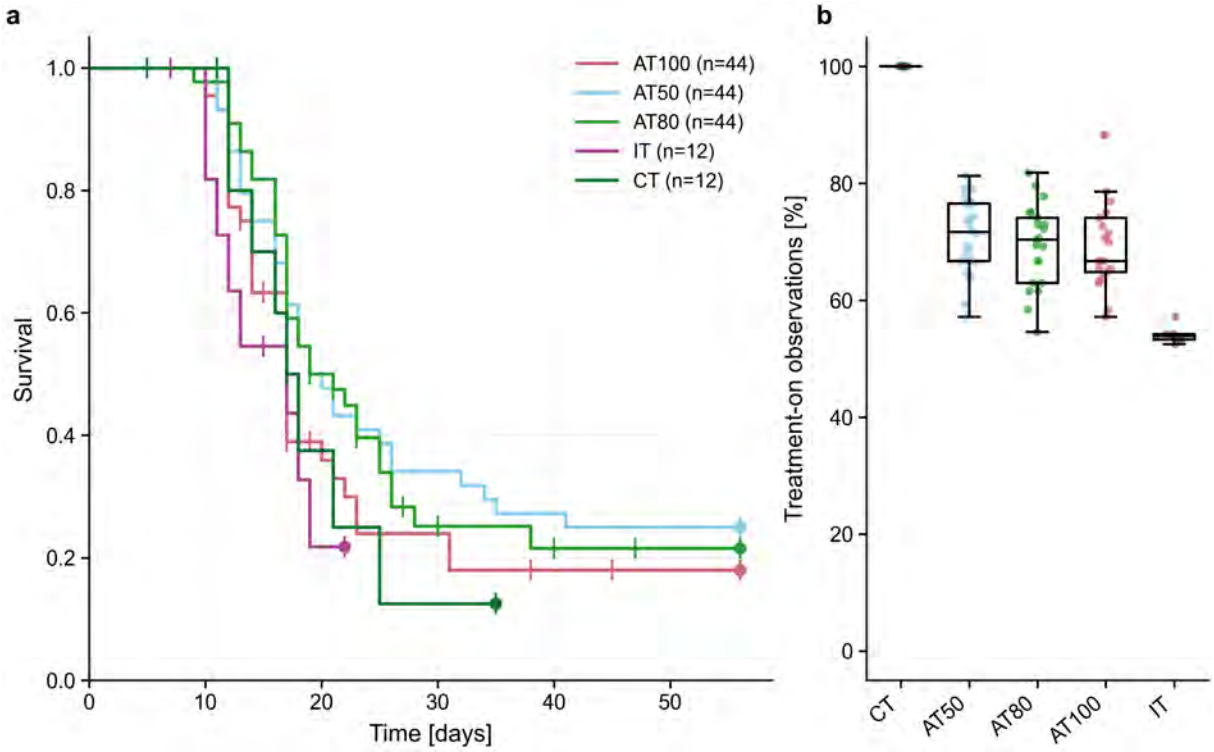


Figure 4.5: AT shows a tendency toward prolonged survival and reduced treatment frequency (a) Kaplan–Meier curves for non-embedded 3D spheroids. Failure was dated at the second of two successive retained observations with relative total fluorescence > 5, or relative resistant fluorescence > 0.1. Spheroids annotated as lost or destroyed were right-censored unless threshold failure had already occurred. Pairwise log-rank comparisons did not establish differences between AT80 and IT ($p = 0.0625$) or between AT80 and CT ($p = 0.284$). Sample sizes were $n = 44$ each for AT50, AT80, and AT100, and $n = 12$ each for IT and CT. (b) Fraction of retained post-baseline treatment-on observations for each strategy. Treatment fractions differed between regimens (Kruskal–Wallis $H = 39.9$, $p = 4.6 \times 10^{-8}$), with all AT schedules showing lower fractions than CT (all pairwise $p < 10^{-4}$). Sample sizes were CT: $n = 9$, AT50: $n = 28$, AT80: $n = 21$, AT100: $n = 21$, and IT: $n = 7$. Analysis excluded 70 spheroids marked as destroyed and calculated treatment fractions over available observation windows without time weighting.

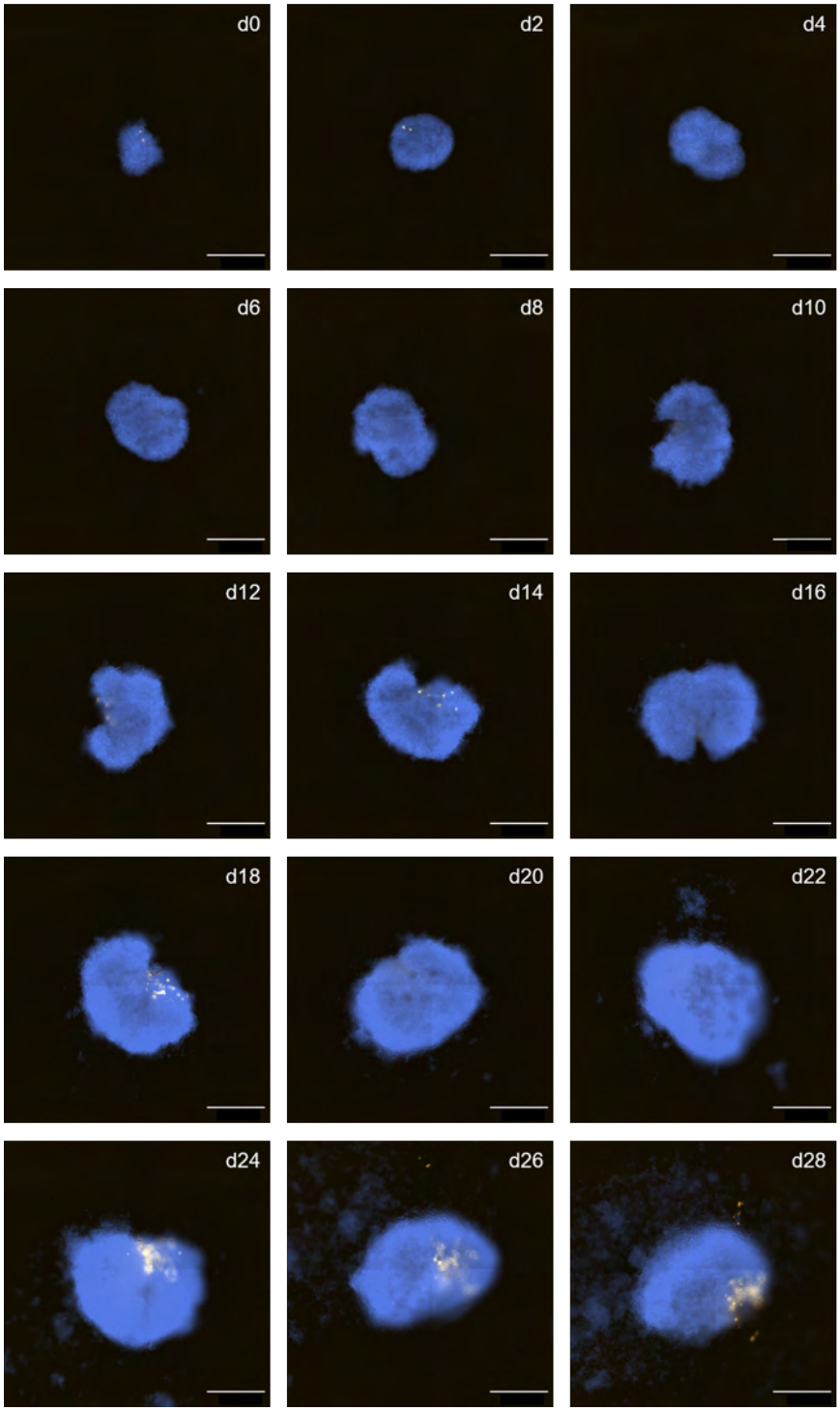


Figure 4.6: Growth of non-embedded α2–α4 spheroids under NT conditions Fluorescence time-lapse images of a non-embedded α2–α4 spheroid grown without treatment. Sensitive cells are shown in blue and resistant cells in yellow. The spheroid expands throughout the imaging period, with only minimal resistant-cell emergence, consistent with resistance remaining rare in the absence of treatment-associated selection. Images are MIPs. Scale bar: 500 µm.

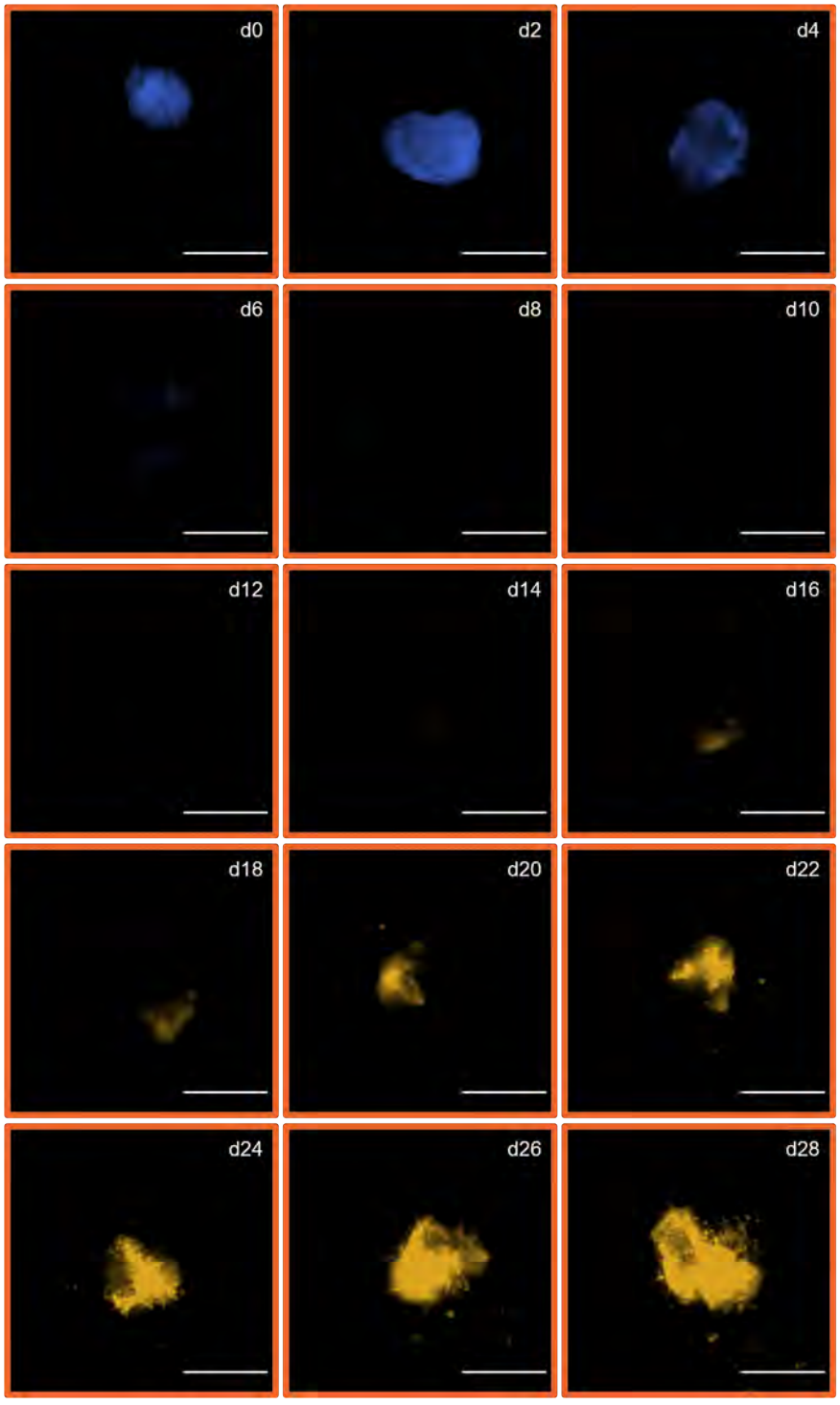


Figure 4.7: CT accelerates resistant takeover in non-embedded spheroids Fluorescence time-lapse images of a non-embedded $\alpha 2$–$\alpha 4$ spheroid under continuous G418 treatment. Sensitive cells are shown in blue and resistant cells in yellow. Orange frames mark treatment periods. Under CT, sensitive cells decline rapidly, while resistant clones emerge early, expand quickly, and ultimately dominate the spheroid. Images are MIPs. Scale bar: 500 µm.

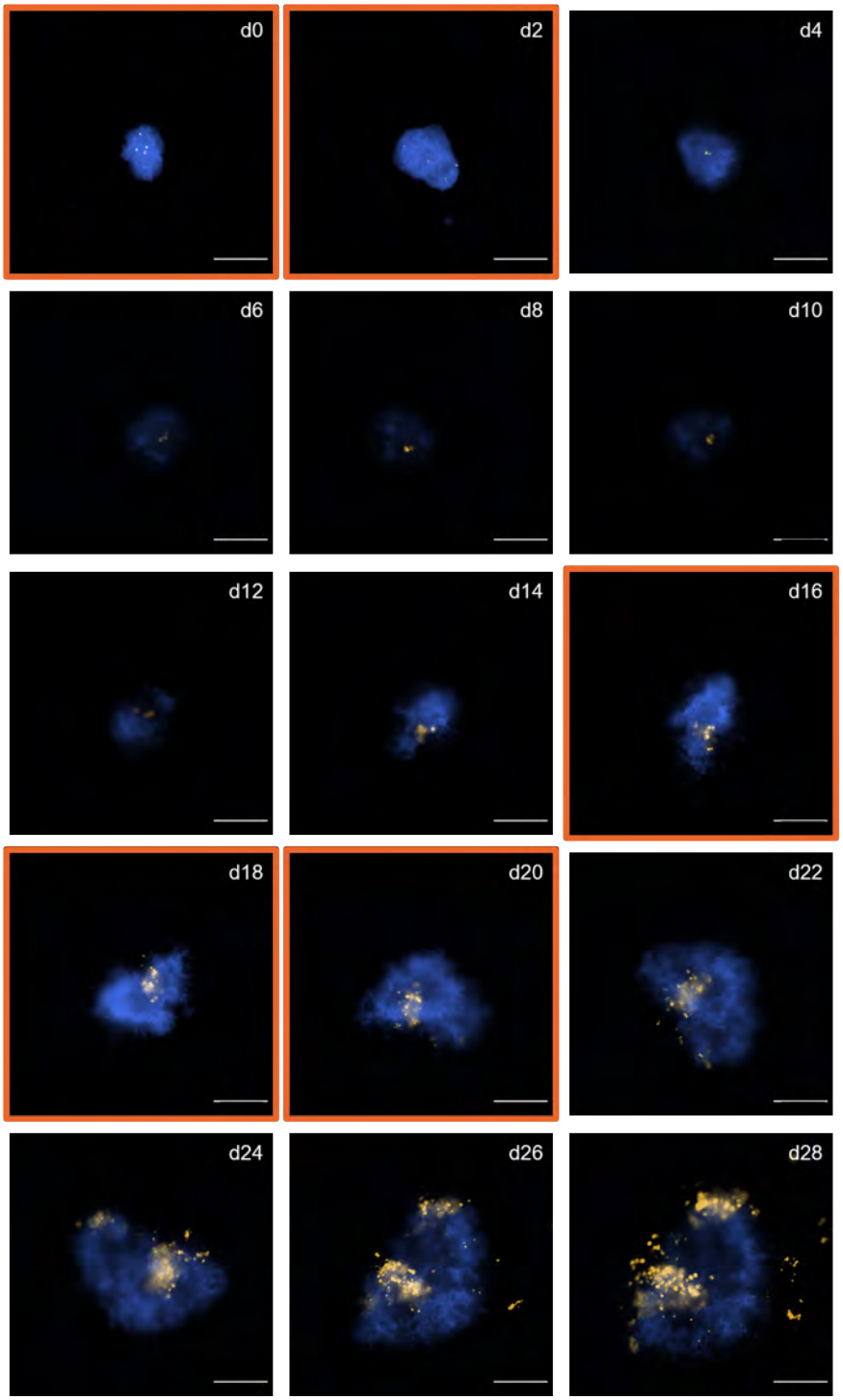


Figure 4.8: AT50 partially restrains resistant growth in non-embedded spheroids Fluorescence time-lapse images of a non-embedded $\alpha 2$–$\alpha 4$ spheroid subjected to AT50 treatment. Sensitive cells are shown in blue and resistant cells in yellow. Orange frames mark treatment periods. Small resistant foci emerge but remain relatively constrained for much of the experiment, while sensitive cells continue to comprise a substantial fraction of the spheroid. Images are MIPs. Scale bar: 500 µm.

Overall differences between strategies were highly significant (Kruskal–Wallis test: $H = 39.9$, $p = 4.6 \times 10^{-8}$). Pairwise comparisons confirmed that all AT schedules (AT50, AT80, AT100) had significantly lower treatment fractions than CT (all $p < 10^{-4}$), while no significant differences were observed between the individual adaptive strategies. Thus, adaptive regimens reduced treatment burden, with a descriptive tendency toward prolonged resistance control under AT80 relative to IT, but without an established time-to-failure advantage or evidence of equivalent system stability relative to CT.

Analysis of resistant-cell dynamics revealed a clear association with recent treatment exposure (see figure 4.9). Across 156 mixed spheroids and 4,787 observation intervals, relative resistant-fluorescence change was on average 4.86 percentage points per day higher following treated than untreated intervals (GEE clustered by spheroid; 95% CI: 3.70–6.03 percentage points per day; $p = 2.85 \times 10^{-16}$).

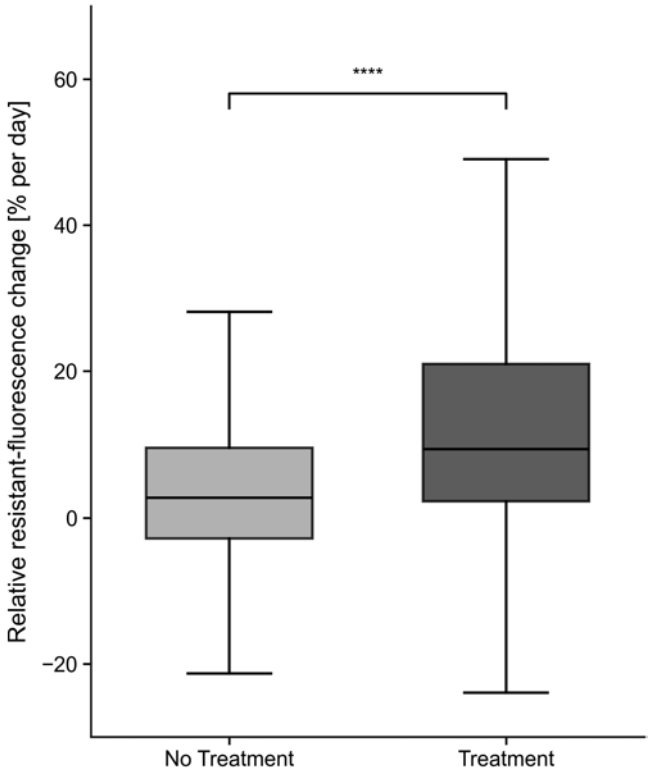


Figure 4.9: Recent treatment exposure is associated with increased resistant growth Resistant growth was quantified as the percentage change in resistant-channel fluorescence between successive retained observations, divided by elapsed days. Across mixed spheroids, resistant growth was higher following treatment-on than treatment-off intervals. A spheroid-aware longitudinal model estimated a mean increase of 4.86 percentage points per day (GEE clustered by spheroid; 95% CI: 3.70–6.03 percentage points per day; $p = 2.85 \times 10^{-16}$; 4,787 intervals from 156 spheroids). Included intervals spanned one day ($n = 4,512$) or two days ($n = 275$).

A paired within-spheroid analysis corroborated this pattern, consistent with ecological release following treatment-mediated depletion of sensitive cells. Although adaptive treatment decisions depended on prior spheroid state, the association supports this mechanistic interpretation without establishing causality.

Consistent with expectations from the 2D experiments, pooled resistant growth rates showed an inverse trend with normalized sensitive fluorescence (see figure 4.10). Spheroids with higher sensitive fluorescence generally exhibited lower resistant growth, whereas low-sensitive states showed greater variability and more frequent rapid expansion. Although the pooled data also reflect differences between spheroids, treatment strategies, and time points, this association supports a competition-based interpretation and aligns with the patterns observed in chapter 3 and the preliminary embedded-spheroid experiments described above.

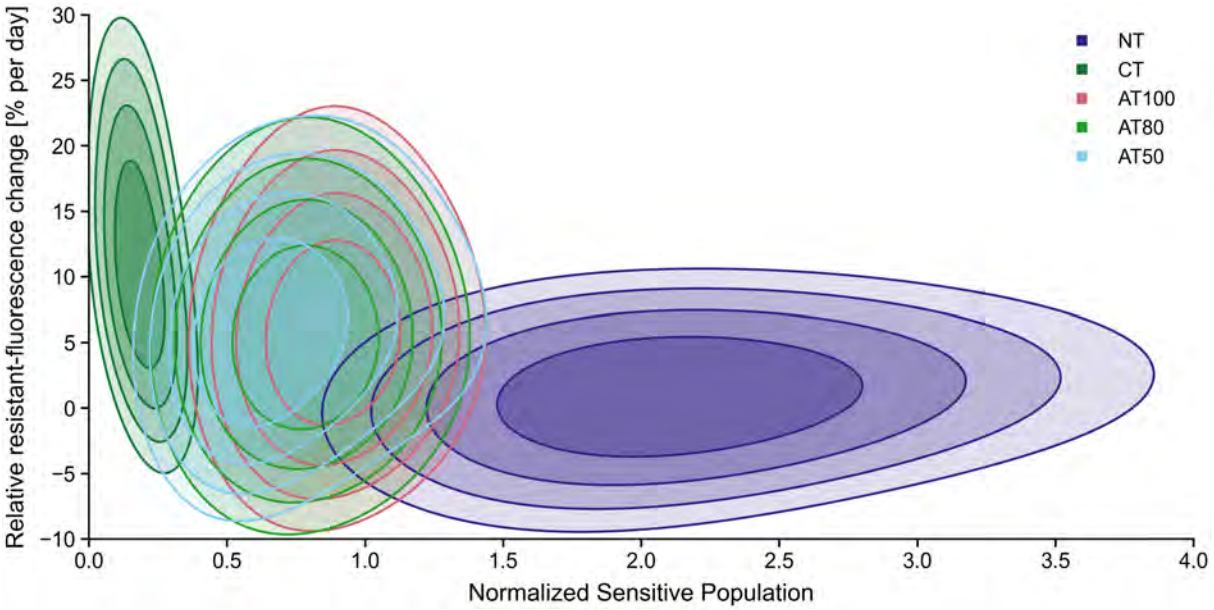


Figure 4.10: Competition dynamics in low-adhesion spheroids For non-embedded spheroids, percentage resistant-fluorescence change between successive retained observations, divided by elapsed days, was plotted against normalized sensitive fluorescence at the interval endpoint. Points represent longitudinal observations across NT, CT, AT50, AT80, and AT100 conditions. Nested filled contours show strategy-specific Gaussian kernel-density estimates, enclosing 10%, 20%, 30%, and 40% cumulative probability mass. Darker regions indicate higher local observation density.

To examine changes in resistance dynamics surrounding structural failure, time points were re-indexed so that "adjusted day = 0" represented the earliest visually annotated

loss of coherent spheroid structure, identified before growth filtering. In 141 collapse-annotated mixed spheroids, relative resistant-fluorescence change per day was higher in intervals ending on or after observed collapse than in intervals ending before it. A spheroid-aware longitudinal model estimated a mean increase of 8.40 percentage points per day (GEE clustered by spheroid; 95% CI: 6.33–10.47 percentage points per day; $p = 1.75 \times 10^{-15}$; figure 4.11). Thus, loss of structural integrity was associated with more rapid resistant expansion.

Taken together, these pilot experiments suggest that treatment exposure and structural stability strongly shaped resistant-cell dynamics in semi-stable 3D spheroids. Resistant-cell growth was higher after recently treated intervals.

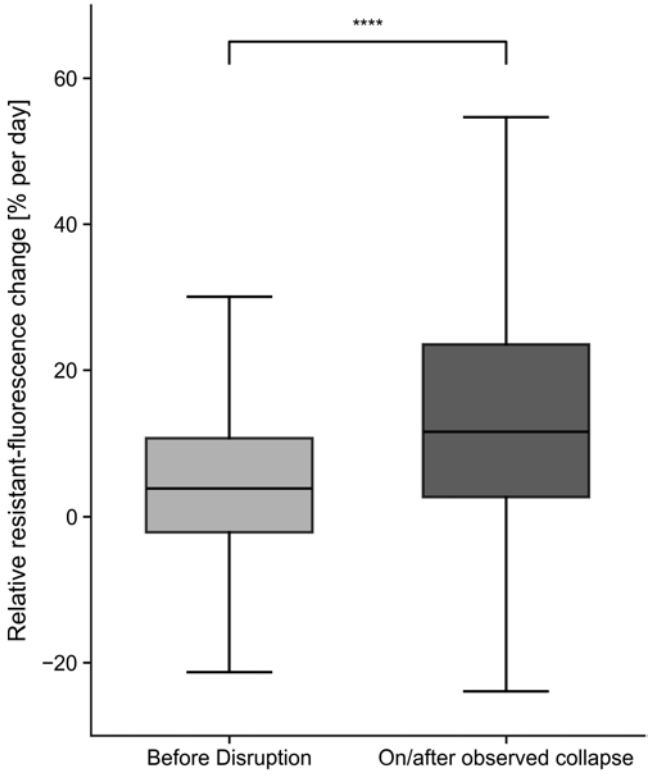


Figure 4.11: Spheroid disruption is associated with increased resistant growth Resistant growth was quantified as percentage resistant-fluorescence change between successive retained observations, divided by elapsed days. Collapse day was identified from the earliest visual annotation before growth filtering. Intervals ending on or after observed collapse were compared with those ending before collapse, including the interval spanning its first observation. A spheroid-aware longitudinal model estimated a mean increase of 8.40 percentage points per day (GEE clustered by spheroid; 95% CI: 6.33–10.47 percentage points per day; $p = 1.75 \times 10^{-15}$; 4,402 intervals from 141 spheroids).

The pooled fluorescence data also showed trends compatible with changing sensitive–resistant population relationships, consistent with the competition-driven behavior established in the 2D model. However, in these heterogeneous preliminary 3D experiments, sensitive-cell abundance alone could not be isolated as an independent quantitative determinant of resistant growth. The results therefore point toward two contrasting ecological conditions: where spheroid structure and mixed-population organization were maintained, resistant expansion appeared more constrained. Where structural integrity was lost, resistant outgrowth accelerated.

From a methodological perspective, these experiments underscored that reliable ecological inference in 3D systems requires the simultaneous maintenance of structural cohesion and adequate nutrient exchange. The stabilization screens using CollagenR, RGD peptides, and HA provided an initial framework for tuning these parameters and revealed how relatively small changes in matrix composition or diffusion could shift the ecological balance. At the same time, the limitations of semi-stable systems—such as Mebiol-based or free-floating spheroids—became apparent. These insights directly motivated the development of a fully stabilized 3D platform, which enabled systematic AT experiments under controlled and reproducible conditions and is described in the following section.

4.3 Adaptive Therapy in a Stable 3D Collagen Model

All work presented in this subsection was carried out in close collaboration with Isabel Schneider. The experimental design, execution, and analysis described here were jointly performed as part of her Master's thesis, which I supervised. Isabel contributed substantially to the experimental implementation, data acquisition, and quantitative analysis. The results and interpretations presented therefore reflect our joint effort. Selected figures, tables, and passages in this chapter have been reproduced or adapted from Isabel Schneider's master's thesis with her permission and are duly acknowledged here.

4.3.1 Stable Embedding Protocol

In parallel to the AT experiments described in section 4.2, the 3D culture system was further optimized with the goal of achieving a long-term stable embedding method. The

following collagen-based protocol proved reproducible and maintained structural and biological stability for several weeks without visible migration of spheroids into interface layers or signs of nutrient limitation. This procedure represents the final embedding strategy used for all subsequent 3D AT experiments in this thesis and is depicted in figure 4.12.

Spheroids were generated as described in Section 4.2, using low-adhesion well plates with a final concentration of 4 µg/ml CollagenR and cultured for eight days before embedding. Collagen was prepared at a final concentration of 2.4 mg/ml according to the standard protocol detailed in section 7.4.2.

- Collect mature spheroids in 150 µl of culture medium using 200 µl pipette tips and transfer them into microtubes.
- Transfer each spheroid into a smaller tube in 1 µl of medium using a 10 µl pipette tip.
- Dispense 50 µl of collagen solution into the center of a flat-bottom 96-well plate and polymerize at 37 °C for approximately 7 min to form the base layer.
- To prepare the upper embedding layer, draw 100 µl of collagen mix into a 200 µl pipette tip and dispense only 50 µl (first pressure point) onto the 1 µl spheroid suspension to mix both components.
- Re-aspirate the resulting 50 µl collagen–spheroid mixture, ensuring that the spheroid is aspirated last so that it remains near the upper portion of the mixture.
- Carefully pipette the collagen–spheroid mixture onto the pre-polymerized collagen base layer in the center of the well.
- After approximately 7 min of polymerization at 37 °C, add 185 µl of culture medium to cover the gel.
- Equilibrate embedded spheroids in the collagen matrix for three days before initiating experimental treatments.

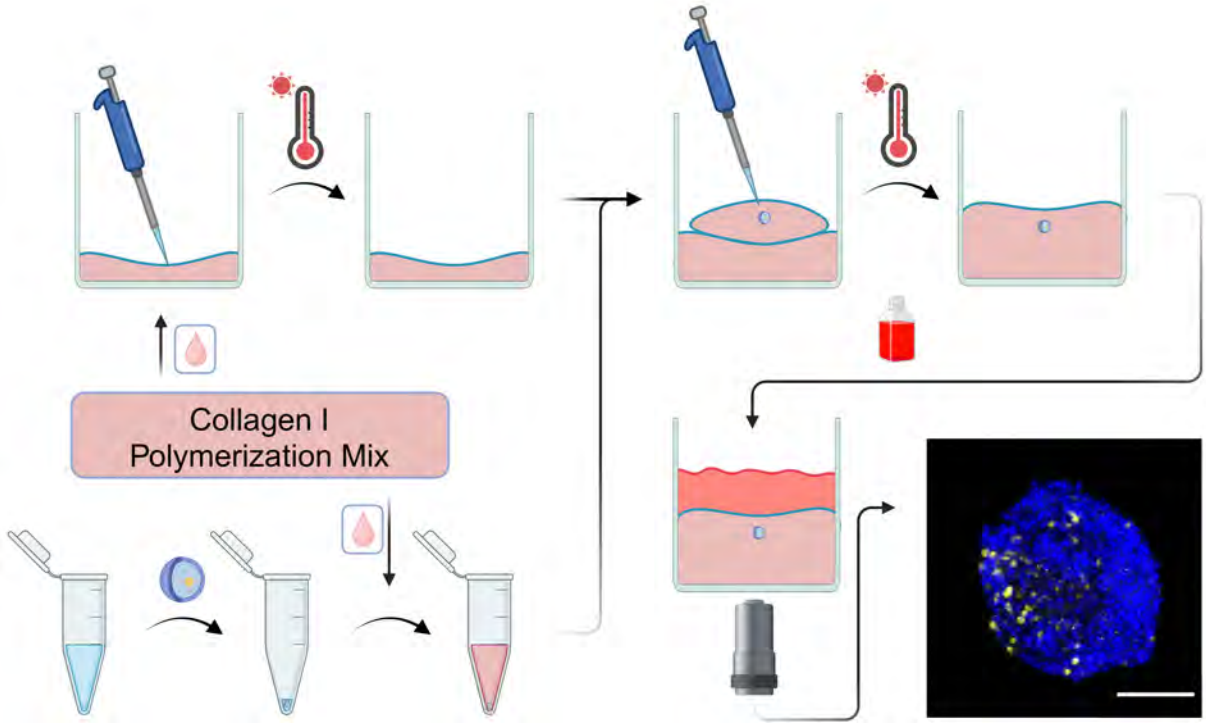


Figure 4.12: Protocol for long-term stable embedding of 3D spheroids Schematic of the optimized collagen-based workflow used to generate stable melanoma spheroids. A collagen base layer is added to flat-bottom wells and polymerized. Spheroids are pre-mixed in low-adhesion plates. Spheroids are transferred into a collagen polymerization mixture and embedded under controlled timing to ensure reproducible positioning and matrix solidification. Following polymerization, culture medium is added and spheroids are maintained under standard conditions. Example MIP of an embedded spheroid. Scale bar: 200 µm. A detailed procedure is provided in the main text.

4.3.2 Long-Term Resistance Dynamics in 3D

Building upon the protocol described above, a long-term 3D culture experiment was conducted to investigate resistance dynamics under spatial confinement. To this end, AT was compared directly with CT in collagen-embedded mixed spheroids composed of sensitive $\alpha 2$ and resistant $\alpha 4$ cells, with AT100 and CT serving as treatment strategies.

Spheroids were generated as described above, embedded in collagen (2.4 mg/ml), and equilibrated for three days before treatment initiation. The baseline G418 concentration was set to 0.15 mg/ml based on prior 3D titrations, in which 0.10 mg/ml did not fully eradicate $\alpha 2$ cells, whereas higher concentrations caused excessive cytotoxicity.

Therapy decisions were taken on Monday, Wednesday, and Friday for a total of 72 days.

Tumor burden was quantified from widefield images by calculating the TFI of the green and red fluorescent channels. Because G418 inhibits protein synthesis, fluorescence intensity reflects both cell abundance and translational activity rather than cell number alone.

Calibration spheroids composed of 500 $\alpha 2$ cells were imaged under conditions similar to those used for the experimental spheroids, and an equivalent area was used to calculate a correction factor for each fluorescence channel. In the adaptive-treatment regimen, G418 was withheld while TFI remained below baseline, applied at 0.15 mg/ml once TFI reached or exceeded baseline, and increased to 0.30 mg/ml when TFI reached or exceeded 150% of baseline. Treatment was withdrawn once TFI fell below baseline. Only spheroids in which resistant $\alpha 4$ cells were visibly detectable prior to embedding were included.

During maturation within the collagen matrix, red fluorescence from the core often diminished, likely due to optical attenuation and limited light penetration as well as cell loss. For 52% of all spheroids, no red signal was visible throughout the full experiment, raising the question of whether $\alpha 4$ cells persisted undetected in the core or had died within a necrotic region.

After CT was applied to all spheroids at the end of the study period, no re-emergence of resistant fluorescence was observed, suggesting that the inner $\alpha 4$ population had been lost in those cases. By contrast, 48% of spheroids developed detectable mScarlet-positive outgrowth during the experiment. These formed the datasets analyzed below, comprising $n = 8$ CT and $n = 12$ AT spheroids. All subsequent trajectory, time-to-failure, and distribution analyses are therefore conditional on detectable mScarlet-positive outgrowth and describe this analyzable subset rather than the full spheroid cohort or an overall estimate of treatment efficacy.

Over the 72-day observation period, treatment failure, defined as background-corrected calibrated mScarlet fluorescence reaching or exceeding 200% of the background-corrected initial total fluorescence, was observed in 5 of 8 spheroids under CT and in 5 of 12 spheroids under AT. Median final TFI reached approximately 330 % under CT, compared to about 200 % under AT, consistent with a trend toward slower overall expansion under adaptive dosing. The observed trajectories suggested slower expansion under AT, with a lower proportion of spheroids reaching the failure threshold and lower fluorescence accumulation over time. However, the log-rank comparison did not establish a difference in time to failure ($p = 0.277$; see figure 4.13).

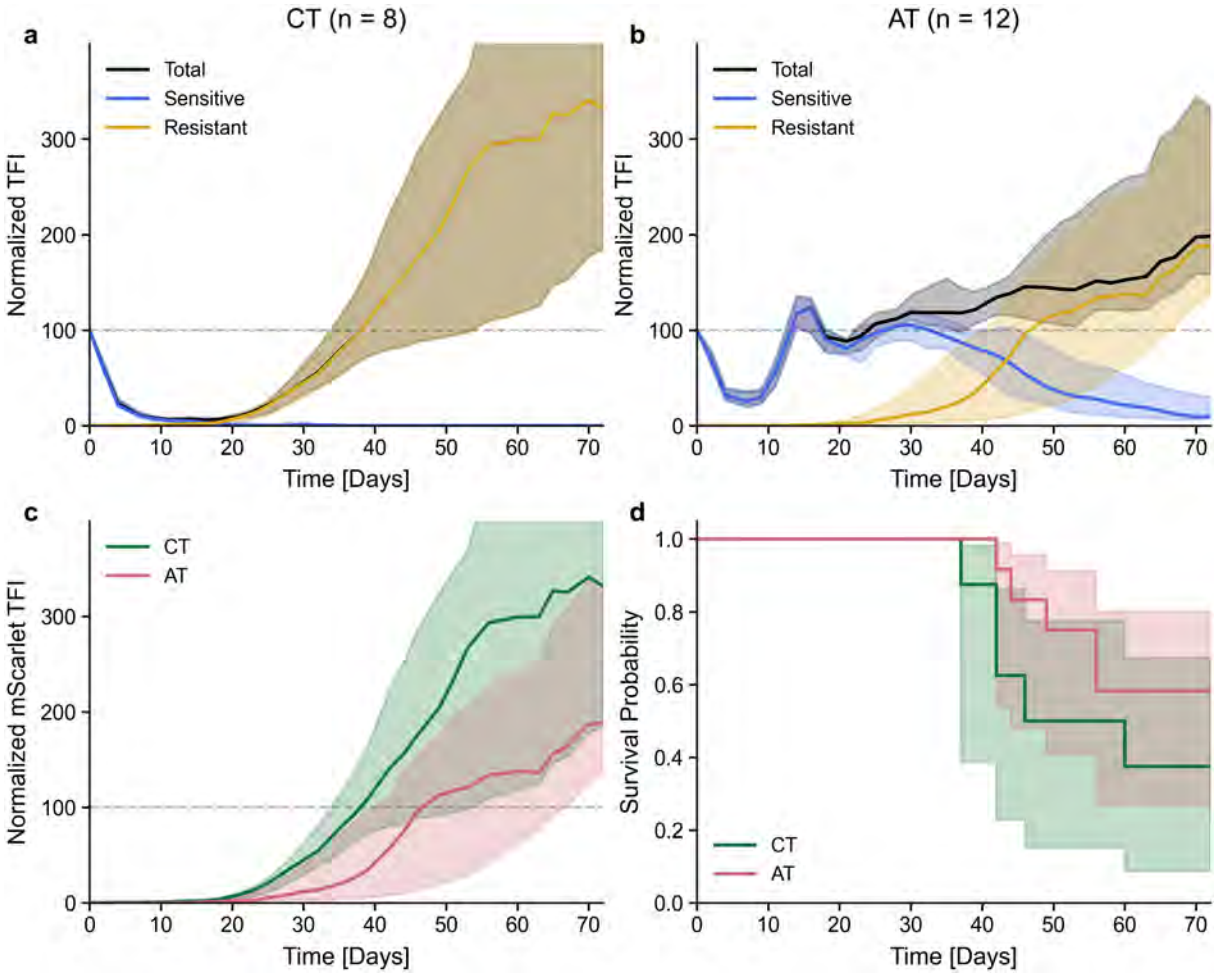


Figure 4.13: AT shows a trend toward delayed resistant progression in embedded spheroids (a, b) Median total, EGFP (sensitive), and mScarlet (resistant) fluorescence intensities over time for CT- and AT-treated collagen-embedded spheroids, respectively. Shaded regions indicate IQR. All plotted fluorescence intensities were background-corrected and normalized to the background-corrected initial total fluorescence of each spheroid. (c) Median normalized mScarlet fluorescence intensity, shown separately to visualize resistant-population dynamics under CT and AT. The dashed 100% line represents the initial total-fluorescence reference. (d) Kaplan–Meier analysis of time to treatment failure, defined as the first measured time point at which background-corrected calibrated mScarlet fluorescence reached or exceeded 200% of the background-corrected initial total fluorescence. Shaded regions indicate 95% confidence intervals. Sample sizes were $n = 8$ for CT and $n = 12$ for AT. Adapted from Isabel Schneider's master's thesis.

Several design constraints likely contributed to the uncertainty in quantitative outcome measures. First, G418 exposure differed between regimens. Spheroids under AT intermittently received escalated doses of 0.30 mg/ml during control phases, whereas CT spheroids were maintained at a constant 0.15 mg/ml. These fluctuations complicate direct compar-

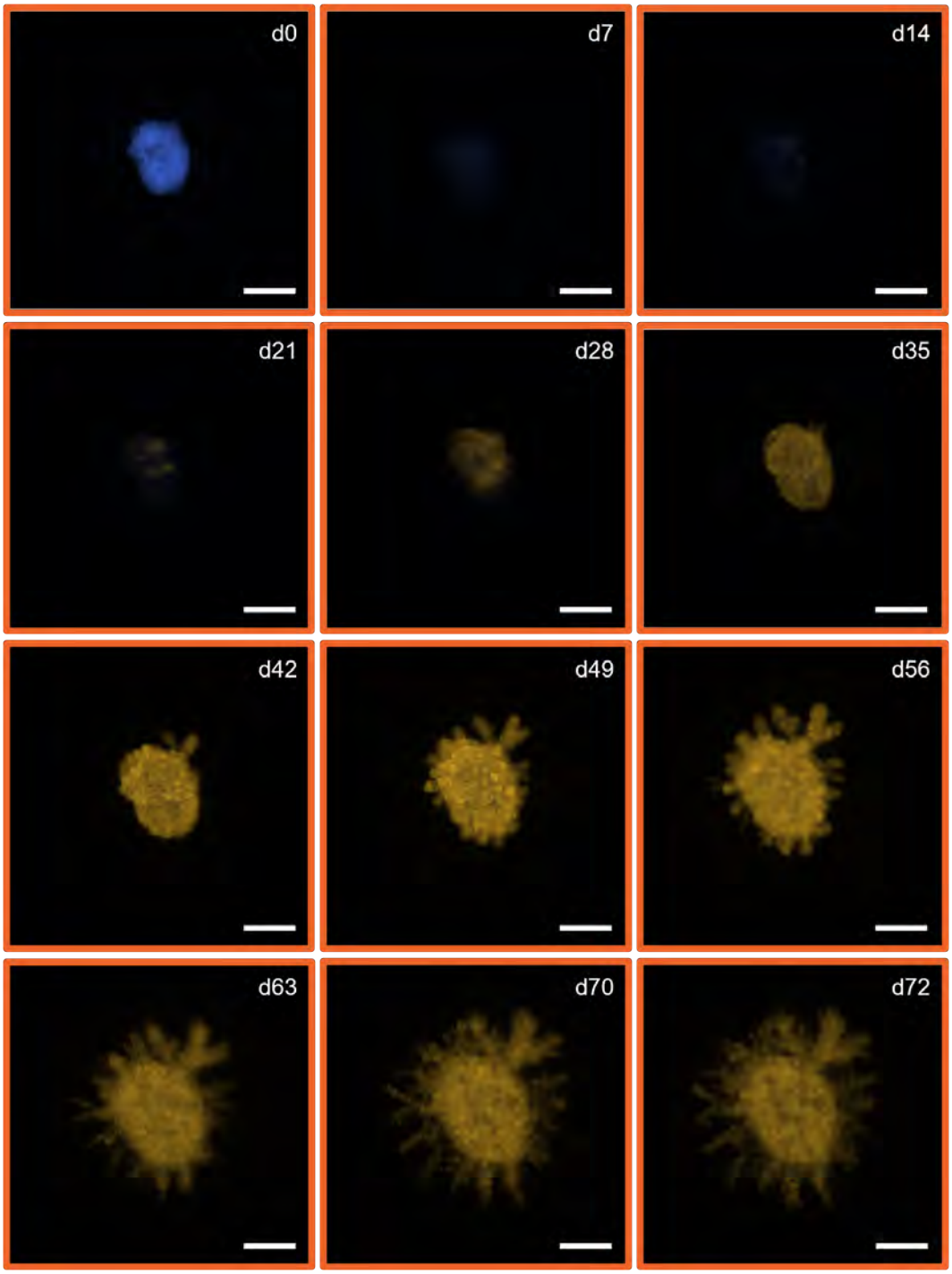


Figure 4.14: Extended time-lapse of an embedded spheroid under CT Representative epifluorescence time-lapse images of an embedded mixed $\alpha 2$–$\alpha 4$ spheroid under continuous G418 treatment. G418-sensitive cells are labeled with EGFP (blue), whereas resistant cells express mScarlet (yellow). Orange frames indicate treatment periods. Scale bar: 500 µm. Adapted from Isabel Schneider's master's thesis.

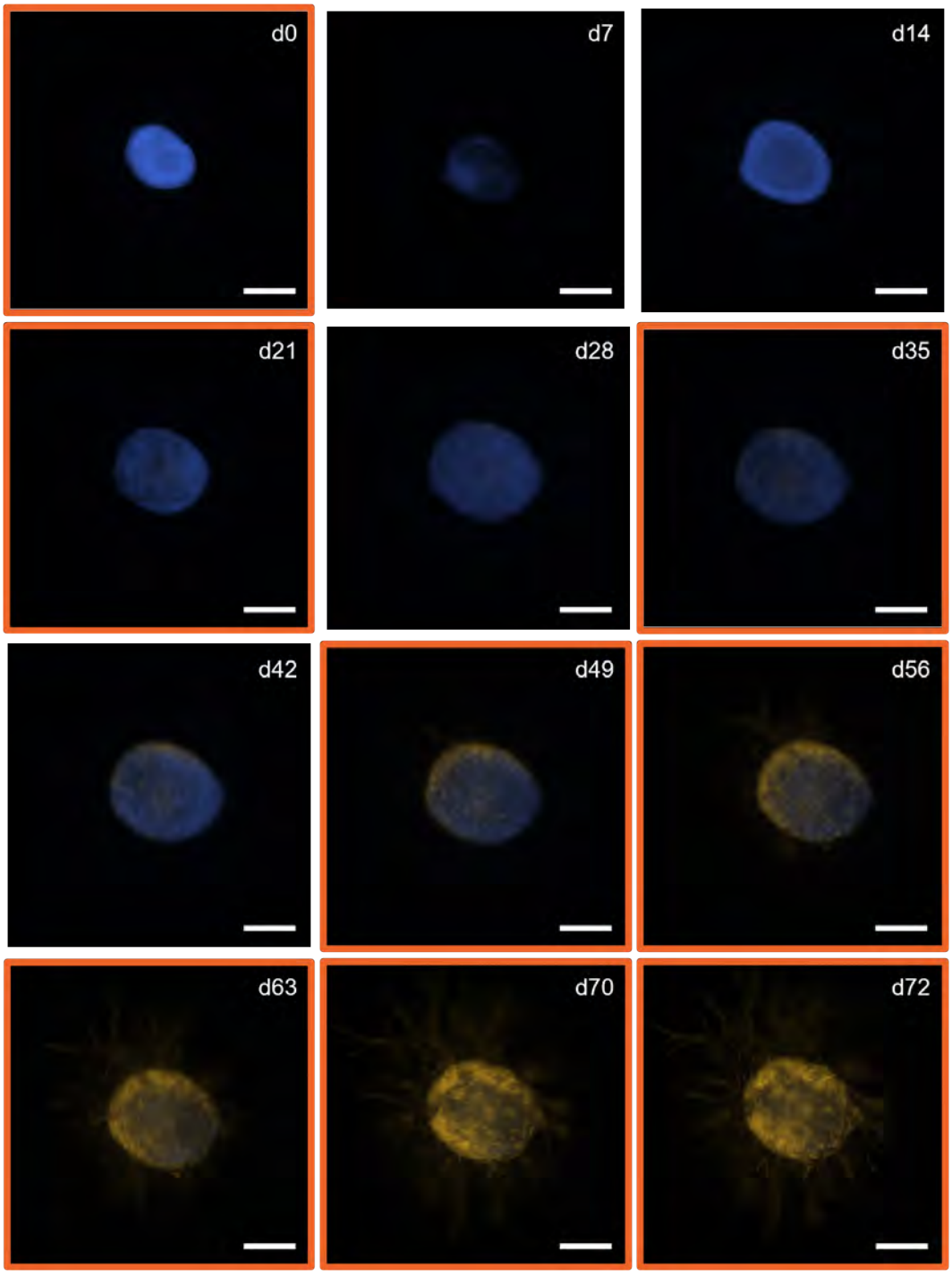


Figure 4.15: Extended time-lapse of an embedded spheroid under AT Epifluorescence time-lapse images of an embedded mixed $\alpha 2$–$\alpha 4$ spheroid under AT. G418-sensitive cells express EGFP (blue), whereas resistant cells express mScarlet (yellow). Orange frames indicate treatment periods. Scale bar: 500 µm. Adapted from Isabel Schneider's master's thesis.

isons of drug exposure and may have introduced additional inter-spheroid heterogeneity.

Second, tumor burden was estimated using TFI, which may not directly reflect absolute cell number following G418 addition. Because G418 inhibits protein synthesis, changes in fluorescence may reflect altered protein production rather than changes in cell viability alone. This effect is further amplified in 3D culture by optical attenuation within the collagen matrix, particularly for cells located in deeper layers of the spheroid.

Third, therapy was initiated three days after embedding, a time point that may have preceded full equilibration of the spheroid–matrix system. This is reflected in the pronounced early drop in TFI during the first AT cycle and likely imposed an additional stress on sensitive $\alpha 2$ cells. Such early depletion may have transiently reduced local competition, thereby facilitating partial release of resistant cells from ecological suppression.

Beyond differences in resistant growth dynamics, the spatial organization of resistant cells also differed markedly between treatment groups. This phenotype was evaluated semi-quantitatively using a 10-point visual distribution score assigned by an independent, treatment-blinded rater. Higher values indicated compact, bulk-like growth and lower values reflected diffuse, scattered distributions.

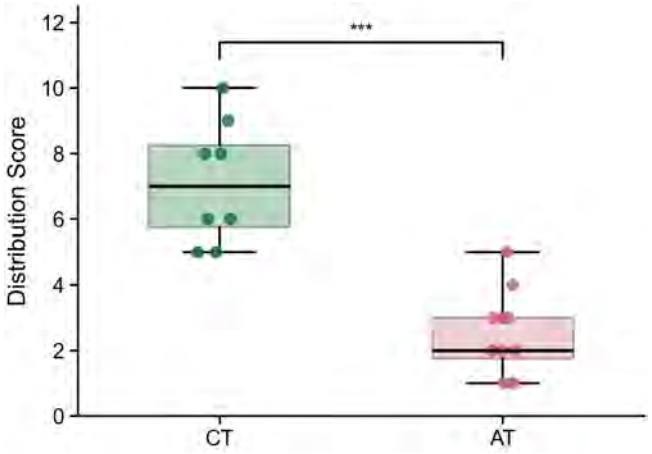


Figure 4.16: AT is associated with a more dispersed distribution of resistant cells 10-point distribution scores for resistant signal, assessed by a blinded rater for CT- and AT-treated spheroids. Higher scores indicate compact, bulk-like resistant outgrowth, lower scores indicate a diffuse distribution. Points represent individual spheroids. Boxes show the IQR, central lines the median, and whiskers the minimum and maximum. Groups were compared using an exhaustive two-sided label-permutation test based on pooled midranks, retaining ties ($p = 4.76 \times 10^{-5}$; $n_{\mathrm{CT}} = 8$, $n_{\mathrm{AT}} = 12$). Adapted from Isabel Schneider's master's thesis.

Using this metric, most spheroids under CT showed $\alpha 4$ cells forming compact, centrally

confined clusters, presumably due to the dominance of one or few resistant clones. In contrast, spheroids treated with AT exhibited more spatially dispersed mScarlet signal, suggesting either the persistence of multiple resistant clones or enhanced peripheral spreading (see figure 4.16 and Supplementary figure A.3). Despite the limited sample size, this difference was statistically significant (exhaustive two-sided midrank label-permutation test; $p = 4.76 \times 10^{-5}$; $n_{\mathrm{CT}} = 8$, $n_{\mathrm{AT}} = 12$). As this analysis is based on 2D projections of 3D spheroids, some orientation-dependent bias cannot be excluded. Future confocal imaging could provide a more detailed characterization of spatial organization in 3D.

4.4 Discussion and Outlook

A central outcome of this chapter is the establishment of a collagen-embedded melanoma spheroid assay for long-term 3D culture with preserved morphology and reproducible handling over periods exceeding two months. Achieving this required explicit control over two closely linked experimental constraints that repeatedly limited earlier approaches.

First, maintaining structural cohesion and positional stability of the entire spheroid to enable reliable longitudinal imaging and repeated medium exchange. Second, ensuring sufficient exchange of nutrients and oxygen through the matrix to sustain viable growth over extended periods. Under less permissive conditions, loss of viability and growth arrest were repeatedly observed, indicating that diffusion and media accessibility play an important role in sustaining viable growth at the level of the whole spheroid. Together, these optimizations highlight the need for structural integrity and sufficient environmental exchange in 3D ecological assays.

Before the fully stabilized collagen workflow was established, AT experiments were explored in semi-stable systems, including Mebiol embedding and spheroids maintained in suspension. Although these platforms were unsuitable for standardized long-term comparisons, they proved informative in two key respects. First, they demonstrated that longitudinal treatment and imaging in 3D are feasible and yielded operational readouts that remained interpretable across conditions. Second, they revealed a consistent and mechanistically relevant limitation: loss of spheroid integrity was associated with faster resistant expansion, highlighting the importance of structural stability for studying ecological control. These pilot experiments thus clarified the requirements for meaningful 3D

AT studies and motivated the emphasis on stable embedding and adequate environmental exchange in the final assay.

Using the stabilized collagen-embedded model developed together with Isabel Schneider, resistance dynamics could be examined under sustained spatial confinement, allowing treatment schedules to be compared without the confounding collapse events that dominated earlier pilot systems. Although inter-spheroid heterogeneity remained substantial, the direction of the outcome measures was consistent: relative to CT, AT tended to slow detectable resistant-cell expansion, as reflected by lower resistant-channel TFI. Importantly, the assay enabled these comparisons over extended time horizons, which is essential for studying resistance dynamics in 3D environments where diffusion limitations and spatial structure can alter selection pressures over time. A key implication is that treatment response in 3D cannot be inferred from bulk burden alone. Spatial structure imposes additional constraints—such as local crowding, nutrient access, and mechanical confinement—that shape how resistant clones expand. Adaptive dosing therefore influences resistance not only by altering total population size, but by modulating the competitive and physical context in which evolution unfolds.

Consistent with this view, the collagen assay revealed treatment-associated differences in the spatial organization of resistant cells. Under CT, resistant cells most often appeared as compact outgrowths in the projected fluorescence images. Under AT, resistant signal was more frequently distributed in a dispersed and fragmented pattern along the spheroid periphery. This indicates that the treatment schedule affects not only how much resistant biomass accumulates, but also how resistance is spatially organized within the spheroid and, consequently, how resistant cells access space and resources. While the visual scoring revealed a pronounced difference between treatment groups, future analyses could complement this assessment with automated quantitative measures of spatial dispersion.

One plausible interpretation is that CT-mediated depletion of sensitive cells reduces local competition and nutrient consumption, allowing nutrients to penetrate previously depleted regions and support contiguous resistant expansion. This is consistent with findings by Appold et al., who demonstrated in a spatial microbial model that treatment-induced suppression of sensitive growth can redistribute nutrients toward confined resistant clones, promoting their expansion [66]. Under AT, repeated recovery of sensitive cells may instead restore local resource competition, interrupt resistant expansion, and favor spatially distributed persistence rather than contiguous dominance. Although nutrient redistribution and internal 3D cell dynamics were not directly resolved here, this mechanism may

contribute to the observed spatial differences and suggests a possible interplay between treatment timing and spatial competition.

Several aspects of the experimental design complicate the interpretation of quantitative differences between CT and AT. Adaptive schedules involved transient dose escalation, which limits direct comparison of cumulative exposure and may have introduced additional pharmacodynamic variability. In addition, tumor burden was estimated via TFI, which reflects fluorescent reporter output and translational activity rather than absolute cell number and is further affected by optical attenuation in 3D matrices. Finally, therapy was initiated shortly after embedding, potentially imposing early stress on sensitive cells and transiently weakening competitive suppression. Together, these factors limit confidence in the exact magnitude and mechanistic interpretation of the observed schedule-associated differences.

Future work should focus on resolving internal 3D dynamics and improving quantitative interpretability without sacrificing long-term stability. Three immediate directions appear most informative. First, 3D-resolved imaging (confocal z-stacks, light-sheet microscopy, clearing, or cryosectioning) should be used to quantify the spatial distribution of sensitive and resistant cells throughout the spheroid volume and to separate optical attenuation from true population loss. Second, schedule comparisons should be complemented by dose-matched or exposure-matched controls and by delayed treatment initiation to reduce early embedding-related stress and better preserve competition during the initial phase. Third, tracking resistant lineages using genetic barcodes or mixtures of distinguishable resistant clones would help determine whether the dispersed patterns observed under AT arise from the persistence of multiple clones, their spatial separation, or processes such as migration and passive displacement.

More broadly, extending the assay across spheroid sizes, matrix stiffness, and nutrient conditions will help determine which physical constraints most strongly shape resistance evolution under spatial confinement. Finally, integrating these experiments with the modelling and control frameworks developed in this thesis provides a natural path toward quantitatively linking treatment schedules to ecological state variables and, ultimately, to predictive control of resistance dynamics in 3D environments.

5

Exploratory Extensions of the Experimental Framework

While the main body of this thesis focuses on the development and validation of controlled 2D and 3D model systems to study resistance evolution and AT, several complementary research directions were explored in parallel. These projects extended the established platform toward additional material, mechanical, immunological, *in vivo*, and measurement-related questions. Although they remained at an early stage, they provided practical proof-of-concept data and identified concrete routes for further development.

The bead–gel hybrid approach explores how composite matrices could introduce tunable mechanical and optical properties into spheroid cultures. Zebrafish xenografts test whether the fluorescently barcoded melanoma populations can be transferred into an optically accessible *in vivo* setting. Complementary mechanical and immune-cell co-culture experiments examine how local physical properties and immune–tumor interactions can be incorporated into the system, while S100 measurements explore whether soluble medium-based signals could complement imaging-derived estimates of population burden.

5.1 Bead–Gel Hybrid Embedding

As part of the exploratory work, I evaluated hybrid matrices combining standard hydrogels with fluorescently labelled polyacrylamide beads provided by my colleague Dr. Ruchi Goswami. The aim was to create a composite material that could provide optical landmarks within the matrix while offering a route towards tuning matrix properties such as bead stiffness, porosity, and surface functionalization.

In a first proof-of-concept experiment, pre-mixed collagen was combined at different ratios with bead suspension and subsequently mixed with 2 µl of spheroid suspension before dispensing into 96-well plates. Among the tested conditions, a final composition of approximately 40% hydrogel and 60% beads produced the most homogeneous consistency and remained compatible with fluorescence imaging.

However, bead incorporation did not resolve the inconsistent positioning of spheroids within the gel, which mirrored the variability observed in single-layer collagen and Mebiol conditions. Because vertical embedding height could not yet be controlled reliably, the approach was not pursued further within this project. Nevertheless, these trials demonstrate that fluorescent bead–gel composites can be prepared and imaged together with spheroids. Polyacrylamide beads can be engineered to differ in stiffness, elasticity, or porosity and may be functionalized with cross-linking moieties, adhesion peptides, or other biochemical cues [239]. Such modifications could enable future composite matrices with more defined mechanical and biochemical properties for 3D tumor models.

5.2 *In Vivo* Proof-of-Concept Xenografts

As a further extension of the experimental framework, I explored whether the fluorescently barcoded melanoma cells could be transferred from the *in vitro* setting into a living organism. Zebrafish larvae were chosen as a host system due to their optical transparency, rapid development, and well-established use in tumor xenograft models. In collaboration with the group of Prof. Felix Engel, mixed populations of sensitive and resistant melanoma cells were injected into the hindbrain region of zebrafish larvae. Subsequent fluorescence imaging detected both fluorescent cell populations at the injection site within the host tissue (figure 5.1). The experiments were performed under the animal-welfare procedures established at the host institution.

These proof-of-principle experiments demonstrated that the engineered melanoma populations can be distinguished by fluorescence shortly after transplantation into a living host. At the same time, several inherent limitations of the zebrafish larval model should be acknowledged. First, the spatial composition and initial arrangement of the co-injected cell populations cannot be precisely controlled during transplantation, limiting reproducibility of early spatial arrangements. Second, the physical size constraints of zebrafish larvae restrict the maximum tumor volume and experimental duration, thereby constraining long-term evolutionary dynamics. Third, ethical considerations impose strict limits on tumor burden and experimental endpoints.

Despite these constraints, the zebrafish system provides a technically feasible and biologically versatile platform for future studies that integrate therapy dynamics with vascularization, nutrient gradients, and host–tumor interactions. Its optical accessibility and whole-organism context make it particularly well suited for extending AT research beyond *in vitro* models toward physiologically integrated settings with spatially resolved readouts.



Figure 5.1: Proof-of-concept implantation of engineered cells into zebrafish larvae A mixed population of $\alpha 2$ (blue, sensitive) and $\alpha 4$ (yellow, resistant) melanoma cells was co-injected into zebrafish larvae to assess whether both fluorescently labelled populations could be detected *in vivo*. Fluorescence imaging shows melanoma cells at the injection site within the larva and as a zoom-in. Scale bar: 50 µm.

5.3 Mechanical Interactions in Immune–Tumor Spheroids

Given the barriers faced by adoptive immune-cell therapies in solid tumors, understanding the physical interactions between immune cells and tumor cells in 3D environments has become crucial [240]. While co-culture models for investigating immune–tumor dynamics exist, the mechanical dimension of these interactions remains largely unexplored. In particular, the interplay between resistance evolution or AT concepts, immune-cell

infiltration, and tumor–immune mechanical crosstalk remains insufficiently understood. Brillouin microscopy offers a unique opportunity in this regard: unlike conventional mechanical assays, it provides label-free and non-invasive optical contrast that is sensitive to local material and cellular state in living systems [241]. This allows one to probe how immune cells mechanically engage with tumor structures in real time.

To evaluate whether such an approach is technically feasible, I performed a proof-of-concept experiment in collaboration with Dr. Kyoohyun Kim, in which free-floating $\alpha 2$ spheroids were co-cultured with red-fluorescent NK-92MI cells and subsequently imaged using a combined fluorescence–Brillouin setup (see figure 5.2). Short-term co-culture proceeded without issues: NK-92MI cells remained viable throughout the imaging session, and the medium composition was fully compatible with both cell lines. Fluorescence microscopy readily revealed NK-92MI cells at the spheroid surface, demonstrating that putative immune–tumor contact regions could be visualized during imaging. Experimental conditions were not optimized to quantify immune-mediated killing, and the observation window was short. No spheroid disruption or measurable changes in Brillouin shift were observed, although these findings do not exclude cytotoxic effects of NK-92MI cells on individual melanoma cells. No obvious phototoxicity or excessive heating occurred during imaging.

Brillouin imaging demonstrated that mechanical mapping of these interactions is possible, albeit constrained by current technological limitations. Brillouin microscopy typically involves point-by-point scanning which limits acquisition speed relative to wide-field imaging approaches. As a result, high-resolution maps require considerable acquisition times, making it impractical to monitor large volumes or rapid cellular dynamics. These limitations strongly motivate the development of fluorescence-guided adaptive Brillouin imaging, in which fast fluorescence snapshots identify immune–tumor contact zones and only these spatially restricted regions are subsequently scanned in Brillouin mode. Such an approach could substantially reduce imaging time while preserving Brillouin contrast at selected immune–tumor contact regions.

Future studies should combine functional assessment of NK-92MI cytotoxicity against the melanoma cells, longer observation periods, collagen-based spheroid stabilization as established in section 4.3.1, and adaptive fluorescence-guided scanning. This would provide a focused 3D system for investigating how physical confinement, tumor mechanics, and local microenvironmental conditions shape immune-cell infiltration and killing, helping to clarify barriers to immune-cell infiltration and killing in solid tumors.

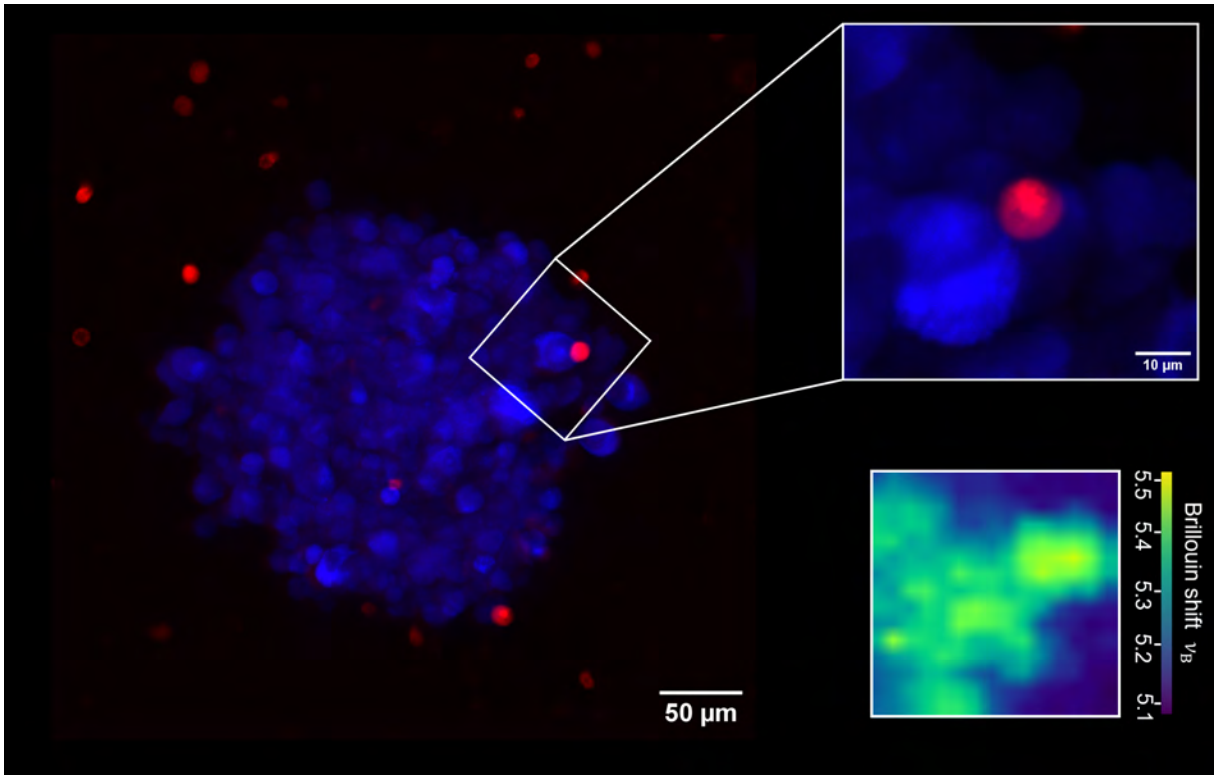


Figure 5.2: Fluorescence and Brillouin imaging of NK-92MI–α2 interaction An α2 melanoma spheroid (blue) co-cultured with NK-92MI immune cells (red). The left panel shows the full spheroid under fluorescence microscopy (scale bar: 50 µm). The magnified region shows an individual NK-92MI cell adjacent to outer α2 cells (scale bar: 10 µm). The corresponding Brillouin map (bottom right) displays the local Brillouin frequency shift within the same region of interest.

5.4 2D and 3D Mechanical Phenotyping via RT-DC and Brillouin Microscopy

To complement the spatially resolved growth analyses described above, I assessed the mechanical phenotype of the α2 cell line using two complementary techniques: RT-DC, performed in collaboration with Dr. Benedikt Hartmann, and Brillouin microscopy, conducted together with Dr. Kyoohyun Kim [76, 241]. Both methods probe cell mechanics at different time and length scales and under distinct physical conditions. RT-DC measures whole-cell deformability at high throughput by flowing suspended cells through a microfluidic constriction, capturing their deformation under hydrodynamic stress. As introduced above, Brillouin microscopy is a label-free optical technique that measures

frequency shifts from spontaneous Brillouin scattering in intact, adherent, or embedded samples, thereby providing spatially resolved optical-mechanical contrast.

To compare mechanical properties in 2D versus 3D growth conditions, $\alpha 2$ spheroids were cultured for two weeks before measurements. For Brillouin imaging, intact spheroids were first transferred into 96-well flat-bottom plates. Individual spheroids ($n = 20$) were then collected, pooled, and enzymatically dissociated for RT-DC measurements. Dissociation was performed by adding 200 µl Trypsin and applying five cycles of 3-minute incubation followed by vortexing. For comparison of outer versus inner spheroid regions, single spheroids were also dissociated independently by adding 200 µl dissociation solution directly to the medium drop containing the spheroid, followed by three cycles of 5-minute incubation and short vortexing.

Brillouin microscopy revealed higher frequency shifts in intact 3D spheroids than in 2D-grown $\alpha 2$ cells (see Supplementary figure A.5). Assuming comparable refractive indices and mass densities, this is consistent with a higher longitudinal modulus [242]. Increased mechanical constraint or compressive stress could contribute to this contrast [243], but distinguishing these mechanisms requires independent measurements.

In contrast, RT-DC measurements on dissociated spheroids revealed substantial heterogeneity between individual spheroids, making single-spheroid analyses noisy due to low cell counts. After pooling, however, 3D-derived cells showed larger cross-sectional areas and higher deformation than 2D-cultured $\alpha 2$ cells (see supplementary figure A.4), consistent with a softer apparent mechanical phenotype. Dissociation experiments designed to compare early-released and fully dissociated cells did not reveal an obvious separation between the resulting RT-DC distributions. Because deformation in RT-DC is size-dependent and Young's modulus was not calculated, this interpretation remains approximate. Nevertheless, the 3D-derived population clustered along lower iso-elasticity contours in supplementary figure A.4, supporting an overall softer apparent phenotype.

Taken together, the two methods yielded different readouts: Brillouin microscopy showed higher frequency shifts in intact 3D spheroids than in 2D cultures, whereas RT-DC showed increased deformation after spheroid dissociation. Reconciling these findings requires consideration of the different physical contexts and methodological assumptions inherent to each technique. Brillouin microscopy probes cells in their native, intact microenvironment without mechanical or enzymatic perturbation and therefore captures the physical context in which the cells reside. RT-DC, on the other hand, requires cells to be fully dissociated

into a single-cell suspension. Previous work shows that enzymatic detachment from a substrate can remodel the cytoskeleton and alter the mechanical state measured in suspension [244, 245]. Whether the present spheroid-dissociation protocol caused softening remains to be tested. Accordingly, these measurements should be interpreted as exploratory evidence that the apparent mechanical phenotype depends strongly on whether cells are measured within an intact spheroid or after dissociation.

Future work could clarify these discrepancies by combining minimally perturbative, spatially resolved mechanical assays. Atomic-force microscopy, a widely used reference method for single-cell mechanical measurements, could directly assess stiffness in intact spheroids or freshly isolated cells. In parallel, gentler dissociation protocols, such as collagenase-based digestion with reduced vortexing, may better preserve the native mechanical phenotype of 3D-grown cells. These refinements would help resolve how cell mechanics change during transitions between 2D culture, spheroid confinement, and spatially structured growth.

5.5 Evaluating S100 as a Medium-Based Readout of Total Cell Burden

In a clinical setting, obtaining high-frequency measurements of total tumor burden is difficult because imaging cannot be performed often enough to capture short-term fluctuations which might be relevant for evolutionary therapies. For this reason, circulating tumor markers are commonly used to complement imaging and provide more frequent information on disease dynamics. In melanoma, S100 is one such established biomarker [246]. Although absolute S100 concentrations measured *in serum* and *in vitro* are not directly comparable—owing to differences in volume, clearance, and physiological context—the use of a soluble marker as a proxy for tumor burden motivated exploration of whether S100 could serve a similar role in this model system.

To test this, a proof-of-concept analysis was performed using the AT100 assay described in section 3.2. Supernatant collected at each medium change was analyzed for S100 at Universitätsklinikum Erlangen, while total cell numbers were obtained using the established segmentation pipeline. S100 concentrations are reported in µg/L. Cell counts on day t were paired with S100 measurements collected three days later to account for possible accumulation during partial medium exchange (approximately 59% at each change).

The lagged measurements showed a positive descriptive association with total cell number (censored Kendall's $\tau_b = 0.78$, $n = 8$; two S100 measurements were at the assay limit of 30 µg/L; figure 5.3). The result is consistent with S100 providing a delayed readout of population-level dynamics, but requires confirmation in additional replicates.

Under the current experimental conditions, the three-day delay and the limited number of measurements at the assay ceiling mean that S100 has not yet been established as a real-time control signal for adaptive dosing. Its utility could be assessed using an assay with a higher measurement range and additional replicates. Algorithmically, accumulation effects could be modelled to estimate underlying production dynamics from the measured signal. Experimentally, increasing the effective medium-exchange fraction—for example through perfusion-based approaches—could reduce signal accumulation and bring measured S100 dynamics closer to the underlying cell-number trajectory. Overall, these preliminary results suggest that soluble tumor markers may reflect population-level changes in this system and could complement direct imaging-based measurements in control strategies that require higher-frequency information.

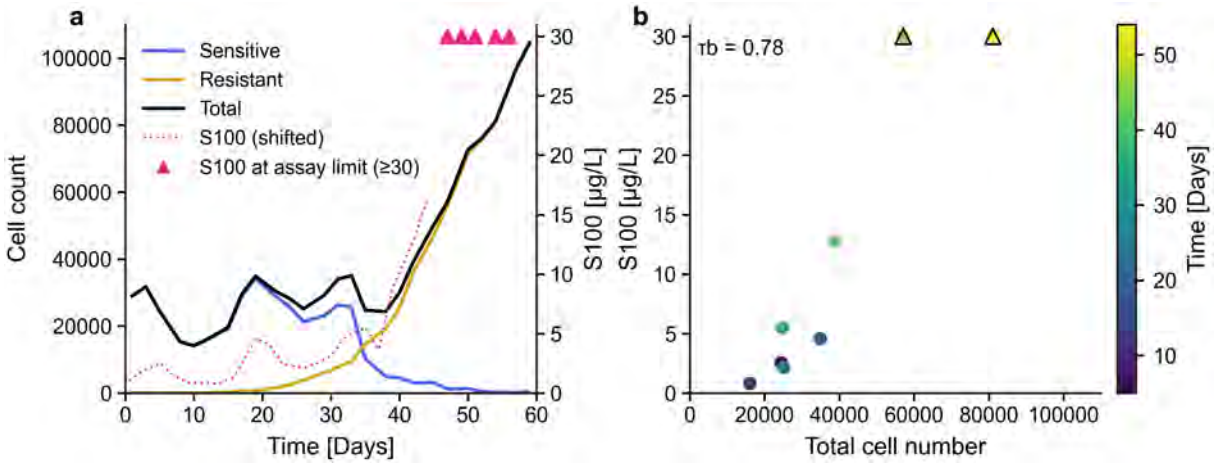


Figure 5.3: S100 tracks total cell burden with a delay in a representative 2D replicate Temporal population dynamics and S100 release in a representative AT100-treated replicate from the standardized 2D assay described in Section 3.2. Blue, yellow, and black curves show sensitive, resistant, and total cell numbers, respectively. The dotted red trace indicates S100 concentration in the culture supernatant (µg/L). In panels (a) and (b), S100 measurements are shifted by three days so that a measurement collected at $t + 3$ is displayed and paired with the cell count at t. Triangle markers indicate measurements at the assay's upper limit (30 µg/L). In panel (b), point colour indicates cell-count day. The lagged measurements showed a positive descriptive association (censored Kendall's $\tau_b = 0.78$; eight pairs, two at the assay limit).

6

Discussion and Outlook

This thesis demonstrates how spatial organization and local crowding constrain resistant expansion and influence adaptive-therapy outcomes, using experimentally tractable tumor models studied quantitatively *in vitro* over extended timescales. It combines a modular genetic-engineering toolbox for generating traceable melanoma populations with structured 2D and 3D assays and complementary *in silico* models to link local ecological interactions with treatment-level outcomes. Although developed in the context of adaptive therapy, the framework addresses the broader question of how selective treatment pressure interacts with spatial organization to shape resistance evolution.

These systems reveal how local crowding, confinement, and spatial escape govern resistant expansion. In the controlled 2D system, these mechanisms can be quantified directly and linked to effective population-level descriptions, enabling model-guided treatment strategies that delay spatial escape and treatment failure while reducing treatment exposure. In stable 3D spheroids, preserved architecture allows related ecological interactions to persist despite greater structural and diffusional complexity, with trends toward more persistent suppression of resistant growth under adaptive therapy. Together, these approaches make

spatial ecological dynamics observable, mechanistically interpretable, and, under defined conditions, actionable for AT design.

These findings support a bottom-up view of tumors as ensembles of interacting microregions with locally defined ecological states. The work therefore establishes a foundation for linking spatial ecology, experimental models, and tumour therapy optimization.

6.1 Eco-Evolutionary Dynamics Can Be Resolved in Meso-Scale Systems

A central contribution of this thesis is the establishment of spatially structured tumor model systems that enable resistance evolution and adaptive therapy to be studied at an intermediate level of complexity between mathematical models and clinical tumors, and at a mesoscopic scale between individual cancer cells and whole tumors. The systems are sufficiently controlled to isolate mechanistic hypotheses, while retaining key features relevant to tumor ecology, including spatial organization, local competition, heterogeneous treatment response, and—in the 3D setting—structural and diffusional constraints.

The framework does not seek to reproduce full tumor complexity but provides tractable microregional systems in which spatial structure, competition, and treatment response can be studied under defined conditions. The modular genetic toolbox established in chapter 2 enables fluorescently labelled populations with defined drug-response phenotypes to be tracked across 2D and 3D assays, while the broader construct library and planned IRM extend the approach toward alternative and dynamically emerging resistance mechanisms.

By linking image-derived spatial measurements with mechanistic and effective *in silico* descriptions, these systems provide a controlled testbed for treatment design. They allow us to determine which spatial mechanisms must be represented explicitly, which can be captured by effective models, and where such abstraction fails as biological complexity increases.

6.2 Spatial Architecture, Local Crowding, and Escape Govern Resistance Control

Across the experimental systems developed in this thesis, spatial organization strongly influences resistant expansion. In the controlled 2D colony system, local crowding was resolved as a proximal constraint on resistant proliferation: resistant growth was reduced in dense local neighbourhoods and increased when treatment-induced loss of sensitive cells created lower-density conditions. This interpretation is consistent with controlled epithelial model systems in which local cell density and mechanical feedback regulate the probability of cell division [133]. AT could preserve or re-establish these crowded conditions by limiting excessive depletion of the sensitive population, thereby maintaining competitive suppression for longer than CT.

Continued resistant expansion progressively eroded confinement, ultimately limiting sensitive regrowth. Escape marked a qualitative transition in this system: Before escape, global sensitive-cell abundance remained closely coupled to local crowding around resistant clones, thereby constraining their growth. After escape, this coupling progressively weakened, allowing resistant cells to proliferate with increasing independence from the sensitive population. Escape therefore emerged as a central failure mode in the 2D system, linking the breakdown of local spatial competition to treatment-level progression.

The 3D systems extend this interpretation by suggesting that the architectural integrity of spheroids determines whether spatial competition can be maintained in three dimensions. In free-floating spheroids, treatment-associated collapse disrupted coherent organization and was accompanied by rapid resistant expansion. In contrast, stable collagen-embedded spheroids maintained their architecture over extended periods, with AT showing trends toward more sustained suppression of resistant growth and distinct resistant spatial distributions compared with CT. These 3D observations do not directly resolve local crowding or confinement at the same level as the 2D assay, but support the broader conclusion that structural organization provides the conditions under which spatial ecological competition can persist.

Together, these findings show that resistant suppression in spatial systems depends on whether treatment preserves the spatial conditions under which resistant cells remain locally constrained [12, 19]. Local crowding, confinement, escape, and structural integrity therefore describe linked but distinct levels of this ecological control process: crowding

regulates resistant proliferation locally, confinement and escape define the spatial state of resistant clones relative to the colony boundary and structural integrity determines whether these interactions can remain coherently coupled in more complex 3D systems.

6.3 Multi-Scale Mechanistic Models Capture the Main Resistance Dynamics

The experimental identification of crowding and escape provided the basis for modelling the 2D system at complementary levels of abstraction:

The spatial DPD model reproduced the characteristic resistant expansion and escape dynamics of the 2D assay using minimal physical assumptions [130, 132]. This supports the sufficiency of these mechanisms in generating the observed behavior and enables their individual contributions to be explored at single-cell resolution, over extended timescales, and under controlled spatial configurations. The force-based formulation in Eq. (1.9) is consistent with the crowding-dependent growth response described by Eq. (1.7), although the underlying pressure–growth relationship was not directly tested experimentally.

By capturing pre-escape dynamics and the advantage of AT100 over CT and AT50, the effective GLV-like model further demonstrated that key aspects of spatial competition can be represented through aggregate population variables. This reduction sacrifices spatial detail but provides the computational efficiency needed for treatment optimization.

Within this framework, Reinforcement Learning served to identify effective scheduling principles rather than directly control treatment [2, 15]. Translating the learned preference for avoiding consecutive treatment during delayed responses while preserving sensitive competitors into the interpretable AT^{+} strategy yielded outcomes comparable to those of the learned policies.

Together, these modelling approaches serve complementary purposes: the spatial model reveals how local ecological interactions shape resistance dynamics, whereas the reduced model identifies when those effects can be captured in practical control rules, with RL testing the utility of the resulting effective representation.

6.4 Population-Level Observables Can Capture Local Ecological Dynamics

The results of this thesis highlight an important possibility for AT in spatial systems. Local crowding, confinement, and escape are fundamentally spatial processes, yet direct access to these variables is unlikely once model systems become more complex and is largely unavailable in clinical settings [15, 179]. The 2D experiments show that this does not necessarily preclude effective control. During the pre-escape regime, global tumor burden and recent treatment history retained sufficient information about the local ecological state to support treatment strategies that delayed progression and reduced drug exposure relative to CT.

This finding does not imply that spatial structure becomes irrelevant. Rather, it identifies a regime in which the consequences of spatial structure remain encoded in higher-order observables. In the radially organized 2D system, aggregate burden was coupled to the local density surrounding resistant clones, allowing global measurements to act as functional proxies for the competitive conditions that governed resistant growth. Once confinement was lost through escape, this correspondence weakened, defining a boundary for effective abstraction.

These results connect mechanistic understanding to practical treatment rules. Knowledge of delayed treatment responses and the ecological cost of excessive depletion enabled the learned policy to be formalized as the transparent AT^{+} rule. A central finding is therefore that spatial organization may be mechanistically essential yet partially abstractable for control, provided that aggregate measurements remain reliably coupled to local competitive conditions. Determining when this coupling persists, and when additional spatial information becomes indispensable, is a central requirement for extending these treatment principles toward more heterogeneous 3D and clinical settings.

6.5 Toward a Microregional Description of Tumor Dynamics

Extending this concept beyond individual experimental systems suggests a possible framework for describing more heterogeneous tumors through interacting local domains. The

experimental systems developed in this thesis can be viewed as controlled tumor microregions in which local ecological interactions give rise to emergent population-level behavior [10, 227]. Such a perspective could allow tumors to be represented as coupled regions with distinct local states, rather than requiring every cell to be described individually. Whether effective descriptions derived from isolated assays remain valid when regions exchange cells, nutrients, and drugs or interact mechanically remains to be tested.

Viewed through this lens, the 2D colonies and collagen-embedded spheroids represent experimentally controlled examples of distinct microregional states. In the 2D system, detailed spatial variables such as local crowding, confinement, clone position, and escape can, before escape, be summarized by effective descriptors that capture the relevant system dynamics in a simplified form. In the collagen-embedded system, the corresponding effective description remains to be established, particularly which coarse variables are sufficient to capture its three-dimensional spatial and architectural dynamics. The present systems therefore provide a first step toward identifying a limited set of effective variables through which interacting microregions could eventually be described without explicitly representing every cell.

This perspective does not assume that all tumor microregions behave identically or that a single effective description will remain valid across an entire tumor. Rather, it provides a framework for organizing increasing complexity: local states may differ in crowding, architecture, transport, mechanics, or treatment exposure, while their coupled dynamics may nevertheless generate an emergent higher-level system [247, 248]. The central challenge is therefore to determine which local variables must be resolved explicitly, which can be inferred from coarser measurements, and under which conditions microregional dynamics can be abstracted without losing information relevant to treatment control.

6.6 Limitations

The insights derived in this thesis arise from a deliberately simplified experimental and modelling framework designed to isolate specific ecological mechanisms. Accordingly, they define the scope of inference rather than providing comprehensive descriptions of tumor behavior under physiological conditions. The core experiments used predefined sensitive and resistant populations in a single melanoma background, with G418 resistance serving as a controlled experimental phenotype. This enabled direct analysis of

ecological competition and density-dependent constraints, but it does not reproduce the molecular diversity of clinical resistance mechanisms, variable fitness costs, pharmacokinetic effects, or the continual emergence of new resistant variants through mutation and phenotypic switching. This simplification was intentional, reflecting a bottom-up strategy aimed at isolating and understanding individual ecological mechanisms before introducing additional layers of biological complexity.

Beyond these biological simplifications, the 2D assay relies on a highly controlled spatial organization. Its core–rim geometry and relatively uniform nutrient and drug exposure enabled local crowding, confinement, and escape to be quantified with high resolution, while also supporting the observed coupling between aggregate population burden and local competitive state before escape. This relationship may not persist in asymmetric, heterogeneous, gradient-rich, or highly migratory tumors, but the controlled system provides a foundation for testing how it changes with increasing complexity. The collagen-embedded 3D system introduces additional structural and diffusional complexity, although current measurements do not directly resolve full clone topology, viable resistant-cell number, local density, or a three-dimensional analogue of confinement. Its findings should therefore be interpreted as evidence that architectural stability may preserve ecological coupling, rather than as a complete mechanistic extension of the 2D system.

Finally, the models presented here are intended as mechanistic and effective descriptions of the experimental systems rather than as quantitative predictors of clinical tumors. The spatial model omits many relevant processes, including vascular delivery, nutrient limitation, matrix remodelling, immune interactions, phenotypic plasticity, and active migration. Likewise, the reduced GLV-like model remains valid only insofar as spatial ecological states are sufficiently encoded in coarse observables. How broadly this condition holds, and which prerequisites enable such effective abstraction, remain open questions for which this thesis provides a foundation. Mechanical properties of cells and matrices were explored in parallel extensions, but mechanical feedback, migration, and matrix remodelling were not yet integrated into closed-loop AT decisions. These limitations constrain direct clinical translation while defining the next experimental and modelling steps required to test the generality of the framework.

6.7 Outlook

The microregion perspective outlined in this dissertation provides a structured route for increasing model complexity without sacrificing mechanistic interpretability. Rather than attempting to reproduce an entire tumor within a single experimental or computational system, future work can build increasingly realistic behavior from spatially defined local domains with distinct ecological states, ultimately enabling a higher-order effective understanding of tumors as evolutionary ensembles. The immediate task is to determine which of the interactions identified here remain relevant when such microregions differ in architecture, transport, mechanics, and treatment exposure.

Stable 3D systems provide the next setting in which to address this question. Volumetric imaging using multiphoton microscopy or optically cleared samples could resolve the three-dimensional organization of resistant cells, including their spatial distribution, local density, spheroid architecture, migratory behavior, and treatment-induced changes in clonal topology over time. This would allow the relationship between local crowding, structural integrity, and resistant expansion to be tested directly in diffusion-limited environments, while establishing whether meaningful three-dimensional analogues of confinement and escape can be defined. Recent 3D bioprinting studies in melanoma have also demonstrated that matrix composition can influence cellular proliferation and phenotype, suggesting that matrix-dependent changes in cell behavior should be considered when interpreting spatial competition in future AT studies [249].

Additional biological and physical complexity can then be introduced systematically. Phenotypic plasticity and *de novo* resistance would allow ecological and evolutionary dynamics to be studied jointly, testing whether crowding remains an exploitable mode of competition once resistant states arise dynamically [53, 102]. Functional establishment of the IRM described in chapter 2 would provide one route toward this goal. In parallel, the $BRAF^{V600E}$ donor constructs developed in Chapter 2 could enable future studies of spatial competition under BRAF inhibitor treatment. However, their effects on drug sensitivity must first be validated in the respective melanoma cell lines. Importantly, immunotherapies, particularly immune checkpoint inhibitors, have driven some of the most substantial advances in cancer treatment over the past decade, especially in melanoma [87, 106]. Incorporating immune components and immunotherapeutic interventions into these experimental systems is therefore an important step toward greater clinical relevance. Together with nutrient and drug gradients, these extensions could reveal how heterogeneous

selective pressures and interactions between tumor, immune, and stromal cells reshape ecological control [65, 247]. Matrix architecture, stiffness, viscoelasticity, and cell mechanical state represent a complementary physical axis [29, 37, 79]. The bead–gel systems, Brillouin microscopy, and RT-DC work presented in chapter 5 provide initial tools for investigating how mechanics influences local crowding, migration, and escape, and for integrating these effects into adaptive treatment assays.

Observability becomes increasingly important as system complexity grows. The effective GLV-like model showed that, before escape and under the defined conditions of the 2D assay, spatial ecological mechanisms remained sufficiently reflected in coarse population-level measurements to guide meaningful treatment strategies. Whether, for how long, and under which conditions this remains possible in asymmetric, gradient-rich, or heterogeneous systems must be determined experimentally. Repeated non-destructive burden measurements, soluble S100, and imaging-derived spatial features offer potential routes for testing when hidden ecological states can be inferred from more clinically accessible observables and when direct spatial information becomes indispensable [15, 183].

The continued development of the mechanistic *in silico* model provides the natural integration layer for this expanding framework. As additional processes are introduced experimentally, including gradients, mutation, phenotypic plasticity, mechanics, migration, and immune interactions, spatial models can test how these local processes combine to generate emergent resistance dynamics at the microregion scale. Their role is not to reproduce every aspect of tumor biology in isolation, but to identify which local variables, interactions, and timescales must be retained to capture control-relevant behavior. In this way, the model can guide experiments toward the most informative measurements and help distinguish mechanisms that remain essential from details that can be safely abstracted.

At the same time, increasingly detailed microregion-resolved models can provide a principled basis for constructing higher-level effective descriptions. Rather than representing a tumor as a single homogeneous population, a more advanced framework could treat it as an interacting collection of microregions with distinct local architecture, transport conditions, mechanical states, clonal composition, and treatment exposure. The key task would then be to determine how information flows between these scales. Which microregional states can be inferred from aggregate measurements, how local escape or structural failure propagates to tumor-scale progression, and under which conditions reduced models remain sufficient for control. RL and related approaches could then be used not as

autonomous decision-makers, but as tools for exploring transparent and robust treatment principles within these experimentally grounded multiscale models.

For clinical translation, the microregion-based model–experiment framework provides a preclinical approach to investigating how treatment decisions could be guided by evolving ecological conditions rather than fixed treatment schedules. Although it does not yet constitute a clinical digital twin, it allows systematic investigation of which measurements, model components, and levels of spatial resolution are needed to support patient-specific predictions and adaptive treatment decisions. In the longer term, clinically accessible measurements could be combined with calibrated models to infer hidden spatial constraints, compare candidate interventions, and support transparent closed-loop treatment decisions. Translating these approaches into clinical practice will require further experimental validation, careful consideration of uncertainty, and appropriate clinical oversight.

Overall, this dissertation links spatial competition to adaptive treatment design. Crowding and confinement support ecological suppression, while escape limits its persistence. The experimental and modelling framework developed here provides a basis for translating these principles toward more durable tumor control with lower treatment burden, with the ultimate aim of improving quality of life and survival for cancer patients.

7 Methods

7.1 Molecular Biology and Cell-Line Engineering

7.1.1 Polymerase Chain Reaction

PCR was used to amplify plasmid fragments, donor cassettes, and control regions. Reactions were performed with Q5® Hot Start High-Fidelity 2X Master Mix (NEB) using manufacturer-recommended buffer systems. Typical reactions contained 10–50 ng of template DNA and 0.5 µM of each primer. Cycling conditions consisted of an initial denaturation (95°C, 30 s), 30–35 cycles of denaturation (95°C, 10–30 s), annealing (60–72°C, 20–60 s depending on primer T_m), and extension at 72°C for a duration adjusted to the expected fragment length according to the manufacturer's recommendations, followed by a final extension at 72°C. Products were verified by agarose gel electrophoresis, purified with silica spin columns or gel extraction (NucleoSpin® Gel and PCR Clean-up, Macherey-Nagel), and sequenced when required. PCR was applied both for cloning

and for diagnostic assays such as colony PCR or gDNA CRISPR-Knock-In verification. Primers were ordered and synthesized by Integrated DNA Technologies.

7.1.2 Gibson Assembly

Plasmid constructs were generated using Gibson isothermal DNA assembly (NEBuilder® HiFi DNA Assembly Cloning Kit, NEB) following the manufacturer's recommendations [250]. Assembly design and fragment overlap planning were carried out using the NEBuilder Assembly Tool (https://nebuilder.neb.com), which automatically optimized overlap lengths and melting temperatures for efficient cloning. Individual fragments—typically vector backbones, resistance cassettes, or regulatory elements—were amplified by high-fidelity PCR using fragment-specific primers containing 20–30 bp homologous overlaps. Amplified products were verified by agarose gel electrophoresis and purified using silica-membrane cleanup. For each assembly reaction, vector and insert fragments were combined at equimolar ratios (typically 15–30 fmol each) in a 20 µl total reaction volume containing 10 µl of NEBuilder® HiFi DNA Assembly Master Mix (NEB). Reactions were incubated at 50°C for 60 min to allow exonuclease-mediated end processing, annealing of complementary overlaps, and polymerase-driven gap filling. Assembled plasmids were subsequently transformed into chemically competent *E. coli* and screened for correct assembly by colony PCR, restriction digest, and Sanger sequencing. Verified constructs were propagated for downstream applications and long-term storage.

7.1.3 Golden Gate Assembly

Golden Gate Assembly was used for modular and scarless cloning of multiple DNA fragments using Type IIS restriction enzymes and T4 DNA ligase. In contrast to Gibson Assembly, which relies on homologous overlaps and exonuclease activity, Golden Gate Assembly exploits the ability of Type IIS enzymes (e.g., BsaI, BsmBI, BbsI) to cut outside of their recognition sites, generating user-defined overhangs that enable directional and seamless ligation of multiple fragments in a single reaction. Correctly assembled constructs are resistant to further digestion because the Type IIS recognition sites are removed during ligation, making the process effectively irreversible [251]. Assembly reactions were performed using the NEB Golden Gate Assembly Kit (BsaI-HF® v2, NEB) according to the manufacturer's instructions. Fragments were designed to include outward-facing BsaI

sites and specific 4-bp overhangs, ensuring unique and non-repetitive junctions. Primer design and sequence verification were performed in Benchling. Typical reactions (20 µl total volume) contained equimolar amounts of vector and insert DNA, 1 µl of NEBridge® Golden Gate Enzyme Mix (BsaI-HFv2), 10x T4 Ligase buffer and nuclease-free water. Thermocycling was carried out in a standard PCR thermocycler with 30–50 alternating cycles between 37°C (digestion) and 16°C (ligation), followed by a final incubation at 50°C for 5 min and 80°C for 10 min to inactivate enzymes. Assembled products were subsequently transformed into NEB 5-alpha Competent *E. coli* and plated on selective LB agar plates. Positive clones were screened by colony PCR or diagnostic restriction digest, and sequence integrity was confirmed by Sanger sequencing.

7.1.4 Restriction Digest and Ligation

Restriction enzyme digestion was used for analytical and preparative cloning steps, such as vector linearization, insert release, and construct validation. All digests were performed using standard restriction endonucleases from New England Biolabs (NEB) under the conditions recommended by the manufacturer.

For multi-enzyme digests, compatibility of enzymes, buffers, and incubation temperatures was verified using the NEBcloner tool (https://nebcloner.neb.com/#!/redigest), which provides buffer compatibility and activity information for enzyme combinations. Unless specified otherwise, reactions were carried out in 1× NEBuffer, using 1 U enzyme per µg DNA in a total volume of 20 µl, and incubated for 1 h at the recommended temperature (typically 37°C). Ligation of digested fragments was performed on ice using T4 DNA Ligase (New England Biolabs) according to the manufacturer's protocol. Ligation mixtures were subsequently transformed into NEB 5-alpha Competent *E. coli* (NEB) and plated on selective LB agar. Positive colonies were screened by PCR with primers spanning the ligation junction to confirm correct construct assembly.

7.1.5 Gel Electrophoresis

Agarose gel electrophoresis was used to assess the size and integrity of DNA fragments following restriction digestion, PCR amplification, or cloning. Unless otherwise specified, standard analytical gels were prepared at 1% (w/v) agarose in 1× Tris-acetate-EDTA (TAE) buffer. This concentration allows reliable resolution of DNA fragments in the

range of approximately 300 bp to 10 kb. For separation of larger fragments (> 2 kb), lower agarose concentrations (0.7–0.8%) were used to improve migration and band sharpness, whereas smaller fragments (< 300 bp) were analyzed on denser gels (1.5–2%).

Once polymerized, the gel was placed in an electrophoresis chamber filled with 1× TAE buffer, ensuring complete coverage of the gel surface. DNA samples were mixed with 6× loading dye and loaded into the wells alongside a suitable DNA ladder for size reference. Electrophoresis was typically carried out at a constant voltage of 100 V for approximately 30 min.

After electrophoresis, gels were imaged directly under UV illumination using a transilluminator. Band patterns were analyzed to verify expected fragment sizes, assess DNA quality, and confirm successful cloning or restriction reactions.

7.1.6 Transformation

NEB 5-alpha Competent *E. coli* cells were used for transformation of Gibson assembly products, restriction–ligation constructs, and control plasmids. The transformation procedure followed the manufacturer's protocol for Gibson assembly products, with the modifications described below. NEB 5-alpha competent *E. coli* (stored at −70°C) and LB agar plates containing Ampicillin were used.

Those plates were prepared by dissolving 35 g of LB Broth + Agar per liter of distilled water. After autoclaving, bottles were allowed to cool to a temperature at which they could be safely held by hand (∼55°C). Ampicillin was then added to a final concentration of 100 µg/ml (1.6 ml of a 50 mg/ml stock per 800 ml agar), and the solution was gently mixed by swirling. Plates were poured immediately, allowed to solidify and dry for 1–2 days at room temperature, and then stored at 4°C.

Prior to transformation, heat blocks and selection plates were prewarmed to 37°C. In parallel, one positive control transformation and one negative control using the complete Gibson reaction without assembly mix were prepared. For each transformation, 50 µl aliquots of chemically competent *E. coli* were thawed on ice and transferred to pre-chilled 1.5 ml microcentrifuge tubes. 2 µl of chilled Gibson or Goden Gate assembly product (or 5 µl for the positive control) were gently added to the cells and mixed by flicking the tube 4–5 times without pipetting or vortexing.

The mixtures were incubated on ice for 30 min without disturbance, followed by heat shock at 42°C for 30 s in a water-filled heat block. Tubes were immediately returned to ice for 2 min. Subsequently, 950 µl of room-temperature SOC outgrowth medium (New England Biolabs) was added, and cells were incubated for 60 min at 37°C with shaking or rotation to allow recovery and expression of antibiotic resistance.

After recovery, 100 µl of the transformation mixture was spread on pre-warmed selective plates using sterile glass beads, while the remaining 900 µl was pelleted (1 min, 3000 × g), resuspended in 100 µl SOC, and plated on a second plate to maximize yield in case of low transformation efficiency. Plates were incubated overnight at 37°C or, alternatively, at room temperature for 48–72 h. Individual colonies were subsequently picked and re-streaked onto fresh LB + Amp plates for isolation.

7.1.7 Colony PCR

Single colonies were picked with sterile pipette tips and resuspended in 10 µl of LB medium. Of this suspension, 1 µl served as the template for colony PCR using plasmid-specific or insert-spanning primers. Q5® Hot Start High-Fidelity 2X Master Mix (NEB) was used with an initial denaturation at 95°C for 5 min to lyse cells, followed by 30–35 amplification cycles. PCR products were analyzed by agarose gel electrophoresis to confirm insert presence and size.

Colonies yielding positive PCR results were restreaked for isolation by transferring 2 µl of the LB suspension onto fresh LB + Amp plates and streaking once completely down to the plate corner. Up to eight confirmed clones were streaked per plate. From these single colonies, overnight cultures were initiated for subsequent plasmid preparation.

7.1.8 Plasmid Preparation and Cryostorage

Plasmid DNA was purified from overnight *E. coli* cultures using commercial silica column–based miniprep (GeneJET™ Plasmid-Miniprep-Kit, Thermo Scientific) and midiprep (NucleoBond Xtra Midi Kit für Plasmid DNA, Macherey-Nagel) kits according to the manufacturer's instructions. For routine screening and analytical digests, plasmids were isolated using the Miniprep Kit. Single colonies confirmed by colony PCR were inoculated into 3–5 ml LB medium supplemented with the appropriate antibiotic and incubated overnight

at 37°C with shaking. Bacterial cells were harvested by centrifugation, and plasmid DNA was extracted following the standard alkaline lysis and silica membrane purification procedure provided with the kit. DNA was eluted in 30–50 µl of nuclease-free water. DNA concentration and purity were determined spectrophotometrically.

For preparative purposes or downstream applications requiring higher plasmid yield and purity, plasmids were purified using the NucleoBond Xtra Midi Kit für Plasmid DNA kit. Midipreps were performed from 100 ml overnight cultures grown at 37°C and under antibiotic selection, and plasmid DNA was eluted in 200–500 µl of nuclease-free water. All purified plasmids were stored at −20°C until further use.

For long-term storage, verified *E. coli* clones were preserved as glycerol stocks by mixing 500 µl of fresh overnight culture with 500 µl of sterile 50% (v/v) glycerol. After brief vortexing, samples were frozen at −70°C using a controlled-rate freezing container (Mr. Frosty, Thermo Fisher Scientific) to minimize ice-crystal damage.

For recovery, frozen stocks were retrieved directly from −70°C and kept on ice while handling. A sterile pipette tip or inoculation loop was used to scrape a small amount of the frozen material from the surface of the glycerol block. This was streaked directly onto a selective LB agar plate without thawing the entire sample. Plates were incubated overnight at 37°C, and the resulting colonies were used to re-establish working cultures.

7.1.9 Sequencing

Plasmid constructs and PCR products were verified by Sanger sequencing (TubeSeq Supreme, Eurofins Genomics). Each sample contained 5–10 ng/µl of purified DNA and 0.5 µM of the corresponding primer, submitted in separate tubes according to provider specifications. Primers covered relevant inserts and junctions to confirm correct assembly, sequence integrity, and reading frame.

Initially, plasmids were validated by conventional Sanger sequencing using overlapping primers covering functionally relevant regions. Midway through the project, Eurofins Genomics introduced a full-plasmid sequencing service, enabling complete sequence verification without individual primers. This approach was adopted for subsequent constructs, while earlier plasmids had already been validated by targeted sequencing.

Sequencing data were analyzed using the chromatogram visualization and alignment tool Benchling.com to confirm sequence identity and detect potential mutations or frame shifts.

7.1.10 Construction of AAVS1 Donors and the IRM

Reporter–Resistance Donors

Unless otherwise stated, custom donor constructs were generated by PCR amplification of backbone and insert fragments followed by high-fidelity DNA assembly. Candidate clones were screened by colony PCR, and the relevant sequences were checked by Sanger or whole-plasmid sequencing using the procedures described above. Plasmid identities, parent–derivative relationships, and component sources are listed in tables 7.10–7.12.

To generate pME_018, the pME_012 backbone and a hygromycin B resistance cassette from pNA7 (provided by Nico Appold) were amplified by PCR and joined using NEBuilder HiFi DNA Assembly. Colony-PCR screening and sequencing identified a fully sequence-verified parental clone for the pME_020, pME_021, and pME_022 derivatives. pME_019 was derived from pME_015 by replacing the reporter-associated puromycin resistance cassette with hygromycin B resistance; subsequent removal of the NLS from pME_019 yielded pME_019b. pME_029 was generated by adding an NLS to pME_016.

The construction relationships among the custom donors are summarized in figure 7.1.

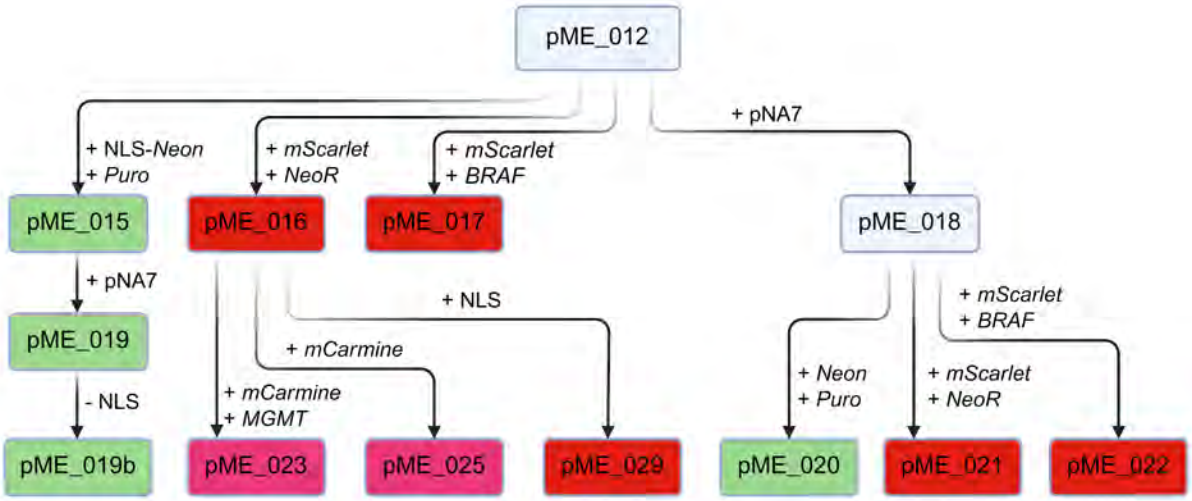


Figure 7.1: Construction relationships in the AAVS1 donor library. Construction routes from the empty AAVS1 donor backbone pME_012 to custom reporter–resistance donors. Internal pME identifiers correspond to the functional names listed in tables 7.10 and 7.11. Green, red, and pink indicate *Neon*, *mScarlet*, and *mCarmine*, respectively. NLS denotes nuclear localization, and pNA7 supplied the hygromycin B resistance cassette. Created with BioRender.com [20]

Inducible Resistance Module

For IRM construction, pME_012 was modified by removing two internal BsaI sites through silent mutations and adding BsaI sites at defined positions for modular type IIS assembly. Three overlapping PCR fragments were joined by Gibson Assembly, and candidate clones were checked by colony PCR and sequencing. The resulting backbone was designated pME_012b.

The IRM cassette was ordered as six double-stranded synthetic DNA modules (gBlocks, Integrated DNA Technologies). The supplier's synthesis-length limit at the time of ordering was 3 kb per fragment. Alternative versions of selected promoter and resistance modules were obtained, with one version of each element used in a given assembly. The module options are listed in table 2.2. Complete ordered-fragment sequences and construction records are archived and available on request.

Initial approaches included one-step Golden Gate assembly and Gibson Assembly of the six modules together with the backbone, using 20–30 bp overlaps for the latter. Two intermediate plasmids were subsequently generated by Gibson Assembly: Part I (pME_026) comprised the modified backbone, SV40 promoter, *CreERT2*, and EF1α–*mKate2* modules; Part II (pME_027) contained the CMV, *NeoR–EGFP*–T2A, and hygromycin B resistance modules. Both intermediates were screened by colony PCR and sequencing, and AfeI/FseI restriction analysis yielded the expected fragment sizes.

Joining of the intermediates was tested by restriction–ligation using AfeI/FseI and subsequently NotI/FseI, with Part I as the receiving backbone and gel-extracted Part II material as the insert. Gibson-based joining approaches included assembly across the AfeI/FseI junction, PCR-mediated joining with extended overlaps, long-overlap assembly, and full-backbone PCR. Junction-spanning colony PCR, restriction analysis, and sequencing were used to assess recovered constructs; these early approaches did not yield a correct full-length donor.

For the final four-fragment assembly, Part II was amplified as three fragments of approximately 5.5, 1.4, and 2.5 kb, and the Part I insert was amplified as an approximately 4.7 kb fragment. Adjacent fragments carried 20–30 bp overlaps. Fragments were combined in equimolar ratios using Gibson Assembly Master Mix and incubated at 50°C for 60 min. Following transformation, colonies were screened by PCR across backbone regions, junctions, and internal features. Approximately 50 colonies were screened, and

13 plasmids were miniprepped and sequenced. Two full-length clones, pME_028a and pME_028b, were purified by MidiPrep and assessed by whole-plasmid sequencing. Both contained the nonsynonymous puromycin-resistance sequence variant reported in Chapter 2. An additional substitution in the *CreERT2* gene was recorded under the laboratory designation Clone 30.

7.1.11 CRISPR/Cas9-Mediated Genetic Modification

Guide RNA Design and Validation

The primary sgRNA sequence (5'-GGGGCCACTAGGGACAGGAT-3') was encoded directly in AAVS1_Cas9 (pME_001; table 7.10). This plasmid expresses the AAVS1-T2-targeting sgRNA from a U6 promoter and eSpCas9(1.1) from a chicken β-actin hybrid (cBh) promoter. As a safeguard, additional candidate sgRNAs were identified using CRISPOR and Benchling, though these were not required in practice [252]. Prior to proceeding with transfections, I confirmed the accessibility and integrity of the AAVS1 locus in our chosen cell lines (Mel Juso and SK-Mel-28) through PCR amplification, verifying that no pre-existing mutations or rearrangements would compromise targeting efficiency.

Transfection Procedure

For all knock-in experiments, I established a standardized workflow that was reused throughout the project. Cells were thawed (day 0), split to ensure healthy growth (day 1), and seeded at defined density ($\sim$2–2.5 $\times$ 10^5 cells per well in a 6-well plate) on day 3. Transfection was performed on day 4 using Lipofectamine™ 3000 Transfection Reagent (Thermo Fisher Scientific) together with the P3000 reagent in Opti-MEM using their standard protocol. For each well, I typically used 0.75 µg of the Cas9/gRNA plasmid and 1.75 µg of donor plasmid, diluted in 125 µl Opti-MEM. This was combined with 125 µl Opti-MEM containing 3.75 µl Lipofectamine 3000, incubated for 10–15 minutes at room temperature, and then added dropwise to the cells.

Antibiotic Selection and Controls

Forty-eight hours post-transfection, cells were exposed to antibiotic selection corresponding to the donor cassette. Kill-curve experiments were performed to determine effective concentrations in both melanoma lines. SK-Mel-28 cells were relatively sensitive to puromycin (Thermo Fisher Scientific), with complete elimination of unedited cells at 5–7 µg/ml. In contrast, Mel Juso cells tolerated higher levels, requiring up to 10 µg/ml for complete clearance. To preserve edited cells during early outgrowth, I adopted a stepwise escalation strategy, starting at 0.2–2 µg/ml and gradually increasing to the full concentration over several days. Hygromycin B (AppliChem) and G418 (Corning) were used analogously for constructs carrying these resistance markers, based on pre-established kill curves. Hygromycin B inhibits ribosomal translation [253], whereas hygromycin B phosphotransferase confers resistance by phosphorylating and inactivating the antibiotic [254]. In every experiment, negative controls (Cas9-only, donor-only, and no-DNA conditions) were included to detect random integration, leaky expression, or antibiotic batch issues.

7.1.12 Clonal Isolation and Confirmation

Following one to two weeks of selection, resistant populations were enriched and positive clones became apparent. To obtain monoclonal lines, I used fluorescence-activated cell sorting (FACS) at the Core Unit für Zellsortierung und Immunomonitoring in Erlangen to deposit single cells into individual wells based on their reporter signal. Colonies were then expanded stepwise, first into 12-well and 24-well plates, and subsequently into T25/T75 flasks. At this stage, validation focused on expansion and cryopreservation, while molecular confirmation of integration was carried out as a separate step.

Genomic DNA (gDNA) was isolated from transfected cell populations to confirm successful CRISPR/Cas9-mediated integration at the AAVS1 locus. Approximately 50,000 cells were collected per sample, pelleted at 300 × g for 5 min, and resuspended in 200 µl of QuickExtract™ DNA Extraction Solution (Lucigen). The extraction procedure followed the manufacturer's protocol. The suspension was vortexed briefly and incubated at 65°C for 6 min to lyse the cells, followed by vortexing for 15 s and a final incubation at 98°C for 2 min to inactivate nucleases. The resulting lysate was mixed by brief vortexing, and 1–2 µl of the crude extract (corresponding to approximately 20 ng DNA per 25 µl reaction) was used directly as PCR template for integration analysis. PCR primers were

designed to span the genomic–insert junctions, thereby confirming site-specific knock-in at the AAVS1 locus.

While this approach reliably verified correct genomic insertion, it did not permit discrimination between heterozygous and homozygous integration events. As allele-specific dosage effects were not expected to influence the experimental outcomes, additional copy-number analysis was not pursued.

Due to inherent heterogeneity, some of the generated cell lines required a second round of FACS to obtain populations with uniform fluorescence. In addition, certain clones exhibited only moderate resistance following initial selection, particularly among clones selected with G418 or vemurafenib. To consolidate resistance, I therefore applied a dose-escalation regime, gradually increasing the concentration of the respective drug until stable growth was achieved under high selection pressure. From these cultures, priority and back-up clones were chosen based on growth kinetics and fluorescence homogeneity. Selected clones were subsequently expanded and cryopreserved for long-term use.

7.2 Cell Culture and Phenotypic Characterisation

7.2.1 Cell Culture Maintenance

Human melanoma cell lines Mel Juso and SK-Mel-28 were used as parental lines throughout this study. Both were originally obtained from the laboratories of Prof. Anja Boßerhoff and Prof. Silke Kuphal (Friedrich-Alexander-Universität Erlangen-Nürnberg). Mel Juso cells originate from a primary melanoma, whereas SK-Mel-28 cells derive from a metastatic lesion. The parental lines are designated $\alpha 1$ and $\beta 1$, respectively. Their engineered derivatives and corresponding transfection plasmids are listed in table 7.13.

Cells were maintained under standard culture conditions at 37°C in a humidified atmosphere with 8% CO_2 using a Heracell VIOS 160i CO_2 incubator (Thermo Scientific). Both cell lines were cultured in RPMI 1640 medium (PAN-Biotech) supplemented with 10% fetal bovine serum (FBS, Thermo Fisher Scientific). For resistant derivatives, the corresponding selection antibiotics (e.g., G418 at 0.5 g/l) were included during routine maintenance to preserve fluorescence and resistance phenotypes. For mixed-population

2D and 3D experiments, routine selection antibiotics were omitted after seeding. G418 was added only according to the assigned experimental treatment schedule.

Cells were subcultured two to three times per week depending on confluency, typically when reaching 60–80%. After aspirating the medium, cells were rinsed once with Dulbecco's phosphate-buffered saline (DPBS, Gibco), detached using 1 ml Trypsin–EDTA (Thermo-Fisher Scientific) for approximately 5 min, and resuspended in fresh RPMI + 10% FBS medium. Mel Juso cultures were usually split 1:4, and SK-Mel-28 cultures 1:3, into new flasks containing 10 ml of complete medium per T-25 flask. To ensure culture integrity, all cell lines were tested for mycoplasma contamination prior to freezing and periodically during continuous culture.

Cryogenic vials were retrieved from liquid nitrogen and thawed rapidly in a 37°C water bath until only a small ice crystal remained. The thawed suspension was transferred into 8 ml of pre-warmed RPMI medium in a 50 ml Falcon tube, mixed gently by pipetting, and centrifuged for 5 min at 1200 rpm to remove residual DMSO. The resulting pellet was resuspended in 10 ml of complete medium (5 ml for T-25 flasks) and seeded into new culture flasks. Medium was replaced the following day to remove dead cells and debris.

For cryopreservation, exponentially growing cells were trypsinized and pelleted as described above. The freezing medium was freshly prepared and consisted of 60% RPMI, 30% FBS, and 10% DMSO. Cell pellets were resuspended in 1 ml of freezing medium per cryovial (approximately 10^6 cells/vial). Vials were labeled, placed in a Mr. Frosty (Thermo Fisher Scientific) freezing container, and cooled gradually overnight at −70°C before being transferred to liquid nitrogen for long-term storage.

7.2.2 AlamarBlue Assay

To quantify cell viability and determine drug resistance profiles of the engineered melanoma cell lines, I employed the AlamarBlue Cell Viability Reagent (Thermo Scientific), following the general protocol described by the manufacturer. The AlamarBlue assay is a resazurin-based, non-toxic method that measures metabolic activity through the reduction of resazurin—a non-fluorescent, cell-permeable compound—into resorufin, a strongly red-fluorescent molecule. The fluorescence intensity of resorufin, detected at 554 nm excitation and 593 nm emission, correlates with the number of metabolically active, viable cells.

For calibration, 0, 50, 100, 200, 400, 800, 1600, 3200, and 6400 cells per well were seeded in standard 96-well plates (outer wells excluded to minimize edge effects), with six technical replicates per density and per cell line. Each plate additionally contained blank wells (medium + AlamarBlue only) for background correction and RhodamineB calibration wells (1:100 dilution of a 1.8 mg/ml stock in Dimethyl sulfoxide (DMSO), diluted in PBS) to standardize plate-reader sensitivity. After 24h of adhesion, 10% (v/v) AlamarBlue reagent was added to the culture medium, followed by a 2 h incubation at 37°C before fluorescence measurement at 554 nm excitation and 593 nm emission.

For the drug-response experiments, $\alpha 2$, $\alpha 3$, and $\alpha 4$ cell lines were seeded at 1,000 cells per well in 100 µl medium. A pretreatment measurement was performed two days after seeding, after which drug treatment was initiated. Further measurements were performed 2, 5, and 7 days after treatment initiation, corresponding to days 4, 7, and 9 after seeding. Fluorescence values were normalized to the pretreatment measurement, which defines treatment day 0. Serial dilutions of G418 (0–5 mg/ml) and hygromycin B (0–64 µg/ml) were tested in independent plates, with 8 technical replicates per concentration. In parallel, maximum growth rates were determined under drug-free conditions in separate plates containing 16 replicates per cell line

Control experiments confirmed that neither G418 nor hygromycin B interfered with the redox reaction underlying the assay. Growth rates were derived from nonlinear regression of normalized fluorescence over time. These quantitative measurements defined the effective concentration ranges used in subsequent AT assays.

7.3 2D Adaptive-Therapy Protocol

7.3.1 Colony Seeding

Initial Mixed-Droplet Assay

Early assay-development experiments used circular colonies generated by mixing sensitive and resistant cells at defined initial ratios, typically 0.1% or 0.025% resistant cells. Cryopreserved cells were thawed and allowed to recover for approximately two weeks before experimental use. After washing with DPBS and trypsinization, cells were counted (LUNA-II™ automated counter), pelleted ($300 \times$ g, 3 min), and resuspended in complete

medium at 3×10^6 cells/ml. To limit evaporation during droplet attachment, warm DPBS was placed in outer wells and plates were maintained in humidified incubation boxes. A 3 µl droplet containing approximately 9,000 cells was pipetted into the center of a 48-well plate and incubated overnight at 37°C and 8% CO_2. The following day, 1 ml RPMI was added carefully along the well wall. This workflow was used for the assay-development experiments described in Section 3.1.

Final Sequential Core–Rim Assay

The standardized 2D therapy experiments described in Section 3.2 used a sequential core–rim seeding workflow in 24-well plates. Resistant $\alpha 4$ cells were first seeded centrally as a 3 µl droplet at 1,333 cells/ml and allowed to adhere for 3 h. Sensitive $\alpha 2$ cells were then added to the same position as a 20 µl droplet at 1×10^6 cells/ml and allowed to adhere for a further 18 h before medium was added carefully along the well wall. This procedure produced colonies with a sparse centrally positioned resistant population surrounded by a dense sensitive-cell rim. Cells were used after approximately two weeks of post-thaw recovery. For the main CT, AT50, and AT100 experiments, treatment was initiated three days after addition of the sensitive-cell droplet. The later AT^+ experiment used the same sequential seeding workflow but was allowed to grow for two additional days before treatment initiation. Cell counts remained normalized to the day-1 total cell count, while the total cell count on day 3 defined the treatment-decision threshold. Further procedural and analytical details are provided in [1].

7.3.2 Maintenance, Imaging, and Treatment Workflow

Experiments alternated between treatment-decision and non-decision days. Imaging and treatment decisions were initially performed daily or every second day and were later standardized to Monday/Wednesday/Friday. Each decision day comprised imaging, segmentation, treatment assignment, and medium exchange.

Plates were wiped with ethanol and imaged on a Zeiss LSM900 in widefield mode using a 5× objective in brightfield, EGFP, and mScarlet channels. Stage positions were re-established and focus adjusted only along the Z-axis. Channel acquisition and tiling were configured in ZEN, after which images were merged, stitched in ZEN 3.11, and visually

inspected in Fiji [255]. Tile dimensions were increased when colonies outgrew the field of view. Detailed imaging parameters are provided below.

Channel	LED module	LED power [%]	Exposure time [ms]
Coherence contrast	Focus Finder LED	80	1
EGFP	475 nm	60	80
mScarlet	555 nm	60	1,000
EGFP (low)	475 nm	10	80
mScarlet (low)	555 nm	10	1,000

Table 7.1: Imaging-channel acquisition settings. LED modules, illumination power, and exposure times used for bright-field and fluorescence imaging in the 2D colony experiments.

Fluorescence images were segmented using two separately trained CellPose models [154]: a cyto3-based model for EGFP-positive sensitive cells and a cyto2-based model for mScarlet-positive resistant cells. The models were trained from pretrained CellPose models using manually annotated images, as summarized in the table below. The `flow_threshold` was set to 0.0 for all segmentations, thereby disabling flow-based exclusion of irregularly shaped objects.

Parameter	α**2_cyto3_Epi_1000**	α**4b4_cyto2**
Used initial model	cyto3	cyto2
Learning rate	0.1	0.1
Weight decay	0.0001	0.0001
Number of training epochs	1,000	100
Segmented cells	Green fluorescent sensitive cells	Red fluorescent resistant cells
Model trained on	Green fluorescent sensitive cells	Green fluorescent sensitive and red fluorescent resistant cells

Table 7.2: Cellpose model training settings. Training configuration and training data used to establish the segmentation models for sensitive and resistant melanoma cells.

For most 2D experiments, segmentation was performed using a single imaging round with identical LED power and exposure times across colony samples. The only exception was the experiment with 0.2 mg/ml G418, in which some colonies exhibited three-dimensional

growth, resulting in overexposed regions where CellPose could not reliably detect cells. Reducing illumination or exposure time would have compromised detection in regions with weaker fluorescence.

Therefore, a second imaging round with reduced LED power in the fluorescence channels was introduced for this experiment. These images were used to generate CellPose segmentation masks in highly fluorescent regions. All other experiments in Section 3.1 and all experiments in Section 3.2 relied on a single imaging round. The remaining channel-specific segmentation parameters are listed below.

Channel	**Segmentation model**	**cellprob_threshold**	**min. size**
EGFP	α2_cyto3_Epi_1000	-1	250 px
mScarlet	α4b4_cyto2	-1	250 px
EGFP (low)	α2_cyto3_Epi_1000	-3	150 px
mScarlet (low)	α4b4_cyto2	-1	250 px

Table 7.3: Cellpose inference parameters. Segmentation models and channel-specific threshold settings used for image analysis.

For the assay-development experiments in Section 3.1, crosstalk was reduced by excluding green objects whose centre of mass fell within red masks. For dual-exposure acquisitions, saturated-region masks were taken from the low-power images. For the final experiments in Section 3.2, an additional channel-overlap filter was applied to reduce residual fluorescence crosstalk between segmentation masks.

Channel-specific and total cell counts were extracted for each colony and time point. Counts were normalized relative to the assay-specific baseline, as defined below for the respective experimental schedules. For spatial analyses, cell-centre coordinates extracted from the segmentation masks were used to estimate local cell-number densities with an Epanechnikov kernel of 120 µm radius. Density changes between successive observations were divided by the elapsed time to obtain local density-change rates.

Treatment decisions were performed individually for each colony based on normalized total cell count. The following schedules were implemented:

- **No Treatment (NT):** Continuous culture without G418 exposure.
- **Continuous Therapy (CT):** Continuous G418 exposure.

- **Intermittent Therapy (IT):** G418 exposure alternated with drug-free intervals at successive medium exchanges.
- **Adaptive Therapy 50 (AT50):** Treatment was applied while total cell count remained at or above 50% of baseline and was paused below this threshold. Treatment resumed after regrowth to at least 100% of baseline.
- **Adaptive Therapy 80 (AT80):** Treatment was applied while total cell count remained at or above 80% of baseline and was paused below this threshold. Treatment resumed after regrowth to at least 100% of baseline.
- **Adaptive Therapy 100 (AT100):** Treatment was applied while total cell count remained at or above baseline and was paused below baseline. Treatment resumed after regrowth to baseline.
- **Effective Adaptive Therapy 100 (eAT100):** An exploratory adaptation of AT100 intended to account for delayed treatment effects. Treatment was initiated above baseline and withdrawn at a higher burden threshold (1.37) to avoid delayed overshoot.
- **Adaptive Therapy agent (ATagent):** A binary treatment schedule selected at each treatment-decision point by the reinforcement-learning policy described in section 7.5 and [1].
- **Adaptive Therapy$^+$ (AT$^+$):** a constrained version of AT100 in which each treatment episode was limited to a single decision interval and was obligatorily followed by one drug-free interval. Treatment could resume at the subsequent decision point only if total cell count remained at or above its day-3 value, corresponding to a normalized treatment threshold of $\eta_{\mathrm{tot}}(\text{day 3})$ relative to the day-1 normalization reference.

For CT, AT50, and AT100, cell counts were normalized to the total cell count on the third day after sequential seeding, which marked treatment initiation and was designated as day 1 in Section 3. The AT$^+$ experiment retained this day-1 normalization reference, but treatment was initiated two days later, on day 3. Its treatment-decision threshold was defined by the total cell count on day 3, corresponding to $\eta_{\mathrm{tot}}(\text{day 3})$ in the reported day-1-normalized coordinates. Thus, the normalization reference and treatment-decision threshold were distinct for AT$^+$. Imaging continued until $\eta_{\mathrm{tot}}>3$, which served as the termination criterion.

TTF, TTE, spatial confinement metrics, including confinement margin and the partitioning of resistant growth into escaped and confined regions, were derived from segmented

population masks as described in [1].

On treatment-decision days, medium handling followed a standardized plan:

1. For each well, the day's decision (no drug / continue / start drug) was compiled into a treatment map. Fresh medium was prepared per well at 1.3 ml volume, color-coded for handling: green = no drug, orange = continue, red = start. For the core CT, AT50, and AT100 experiments, drug-on intervals were maintained at a final in-well G418 concentration of 0.4 mg/ml. Because medium exchange was partial, fresh medium was prepared at 0.6769 mg/ml when treatment was initiated, yielding the intended final in-well concentration of 0.4 mg/ml after mixing with residual drug-free medium. For continued treatment intervals, fresh medium was prepared at 0.4 mg/ml.

2. Aspirate spent medium (multi-channel, gentle). Using an Xplorer Plus MoveIt multi-channel pipette (Eppendorf) in fixed-volume presets, medium was removed into a labeled collection plate (plate ID, date, operator, plate code). To equalize residual films without disturbing colonies, removal volumes were slightly adjusted by well position: center: 2×630 µl; edge: 2×615 µl; corner: 2×600 µl. Collected medium was archived by freezing for potential downstream analysis.

3. 1.3 ml per well was added with a 5 ml Combitip on a Multipipette E3 (Eppendorf) at the lowest speed, dispensing along the wall to avoid shear on colonies. This explicit per-well preparation allowed colony-specific dosing in lockstep with the decision logic, while the position-dependent aspiration minimized mechanical disturbance and helped maintain consistent hydration across the plate.

7.4 Three-Dimensional Spheroid Experiments

7.4.1 Preliminary Embedded and Free-Floating Spheroids

The preliminary 3D experiments described in Section 4.2 were performed during iterative development of the spheroid platforms and comprised embedded and free-floating formats. These experiments were intended to assess the feasibility of longitudinal adaptive treatment and to identify technical determinants of spheroid stability, imaging, and treatment response.

Embedded Spheroids

Early embedded spheroid experiments used Mebiol-based and later collagen-based workflows in which mixed $\alpha 2/\alpha 4$ spheroids were cultured under daily confocal imaging. Adaptive schedules (AT50, AT80, and AT100) were compared with continuous and intermittent treatment controls using G418 concentrations between 0.1 and 0.15 mg/ml. For the preliminary embedded-spheroid analysis, fluorescence was normalized to total fluorescence on workbook day 2, corresponding to treatment initiation. Relative resistant growth was calculated as percentage fluorescence change between successive retained observations divided by elapsed days.

Free-Floating Spheroids

For treatment-fraction analysis, spheroids with destruction annotations in the retained data were excluded. Observations with missing required measurements and designated artifact days (15, 27, and 29) were omitted. Treatment fraction was calculated as the percentage of retained post-baseline observations labelled treatment-on, excluding the two untreated baseline observations on days -2 and -1. Individual observation windows were used without elapsed-time weighting or truncation at treatment failure. Free-floating spheroids were maintained in low-adhesion plates and imaged daily by confocal microscopy. Subsequent experiments were typically initiated with approximately 3,000 cells per spheroid and included continuous CollagenR supplementation (5 µg/ml) to improve structural integrity. Adaptive schedules (AT50, AT80, and AT100) were compared with CT, intermittent treatment, and pulse schedules using 0.2 mg/ml G418. Treatment failure was dated at the second of two successive retained observations with relative total fluorescence > 5, or relative resistant-channel fluorescence > 0.1. Spheroids annotated as lost or destroyed were right-censored unless threshold failure had already occurred. Remaining non-failing spheroids were censored at their last valid observation.

For preliminary free-floating spheroids, time to failure was analysed using Kaplan–Meier curves with pairwise log-rank testing, and treatment burden was compared using Kruskal–Wallis tests. Longitudinal resistant-cell growth was analysed using GEEs clustered by spheroid. For the global competition analysis, an exploratory GEE separated normalized sensitive fluorescence into each spheroid's mean and its observation-level deviation from that mean, adjusting for treatment strategy, centred time, and treatment during the

preceding interval. The outcome was percentage resistant-fluorescence change divided by elapsed days. Sensitive fluorescence was measured at the interval endpoint. Treatment-associated growth differences were assessed by comparing treatment-on and treatment-off intervals, with treatment annotations aligned to the ensuing interval because imaging preceded treatment changes. Intervals with unknown or changing treatment states were excluded from this comparison, which was complemented by a paired within-spheroid analysis. Collapse-associated changes were assessed by comparing percentage resistant-fluorescence change per day in intervals ending before versus on or after the earliest observed collapse, identified from the original annotation sheets before imaging or growth filtering.

7.4.2 Collagen Preparation and Two-Layer Embedding

Collagen I gels were prepared on ice from acidic Collagen R and Collagen G stock solutions (Matrix Biosciences), which were mixed at a fixed 1:2 mass ratio. Gel composition was calculated individually from the desired final collagen concentration, total preparation volume, and lot-specific stock concentrations. The mixture was supplemented with 10× RPMI, $NaHCO_3$, and 1 M NaOH immediately before plating. Gels were polymerized at 37 °C until visibly gelled, typically for 5–10 min.

For each preparation, the total collagen mass was calculated from the target final collagen concentration and gel volume. One third of this mass was provided by CollagenR and two thirds by CollagenG. The required volumes of both collagen stocks were calculated from their respective lot-specific concentrations. Equal volumes of 10×RPMI and 23 mg/ml $NaHCO_3$ were included, and the remaining volume was supplied by dilution medium or cell suspension in 1×RPMI.

Because the required NaOH volume depended on the calculated gel composition and collagen-lot properties, it was calibrated separately for each collagen batch. CollagenR, CollagenG, 10×RPMI, and $NaHCO_3$ were mixed on ice according to the calculated formulation, and 1 M NaOH was added dropwise until a pre-polymerization pH of approximately 9 was reached. The resulting NaOH volume was then used for subsequent preparations with the same collagen lot and calculated gel formulation.

Tenfold concentrated RPMI was prepared by dissolving RPMI powder in fetal bovine serum (FBS; 16.4 mg/ml final concentration). Sodium bicarbonate was prepared as a

23 mg/ml stock solution, and sodium hydroxide was used at 1 M. Dilution medium consisted of one part $NaHCO_3$ stock, one part 10×RPMI stock, and eight parts ultrapure water.

For collagen-embedded spheroid experiments, the total gel volume was calculated based on the number of wells to be seeded. All collagen components except NaOH and spheroids were mixed on ice in the calculated proportions. Spheroids were then mixed with the collagen-mix. Immediately before plating, the lot- and formulation-specific volume of NaOH determined during batch calibration was added while keeping all solutions on ice. The collagen–spheroid mixture was dispensed into the required well format and transferred directly to a 37 °C incubator for polymerization.

7.4.3 Collagen-Embedded Adaptive-Therapy Experiment

For the final collagen-embedded experiments described in Section 4.3, the workflow established by Isabel Schneider was used. Spheroids were generated in ultra-low-attachment 96-well plates by seeding 500 α2 cells together with two α4 cells per well in 100 µl medium. Following overnight incubation, 100 µl medium supplemented with CollagenR (4 µg/ml) was added. After eight days of maturation, spheroids with visibly detectable mScarlet-positive cells were selected for embedding.

Spheroids were embedded in a two-layer collagen-I configuration at a final collagen concentration of 2.4 mg/ml. For this formulation, the gel contained 0.807 mg/ml CollagenR, 1.615 mg/ml CollagenG, 1×RPMI prepared from the 10×stock, 27 mM $NaHCO_3$, and 11 mM NaOH, as described by Isabel Schneider in her master's thesis. A 50 µl collagen base layer was polymerized in each well before a second 50 µl collagen layer containing one spheroid was added. After polymerization, 185 µl culture medium was added. Spheroids were allowed to equilibrate for three days. This first treatment day was defined as day 0.

Imaging, treatment decisions, and complete medium exchange were performed on a Monday, Wednesday, and Friday schedule. Whole-spheroid epifluorescence images were acquired using the Zeiss LSM900 without z-stacking. TFI was calculated as the EGFP fluorescence intensity plus calibrated mScarlet fluorescence intensity and normalized to the initial TFI of the respective spheroid.

CT spheroids received 0.15 mg/ml G418 at every decision point. In the AT group, G418 was withheld while TFI remained below baseline, applied at 0.15 mg/ml when TFI reached

or exceeded baseline, and increased to 0.30 mg/ml when TFI reached or exceeded 150% of baseline. Treatment was withdrawn once TFI fell below baseline. Experiments were followed for up to 72 days. Therapy failure was defined as the first measured time point at which background-corrected calibrated mScarlet fluorescence reached or exceeded 200% of the background-corrected initial total fluorescence. Imaging continued after this analysis endpoint. Only spheroids that showed mScarlet fluorescence above background during the experiment were included.

7.5 Statistical Analysis and Computational Modelling

Unless stated otherwise, analyses and visualizations were performed in Python. Colonies and spheroids were treated as experimental units for treatment-response and time-to-event analyses. Clone- and sector-level summaries were used for spatial analyses. Exact sample sizes, exclusions, and statistical tests are reported in the respective figure captions.

7.5.1 Statistical Analysis

Longitudinal 2D resistant-growth analyses used GEEs with colony identity as the clustering variable, including sensitive-cell abundance, treatment strategy, and time as predictors. TTE and TTF were evaluated by Kaplan–Meier analysis with pairwise unadjusted log-rank testing. Treatment effects were expressed as differences in Kaplan–Meier median event times relative to CT. Colonies without failure were right-censored at their final valid observation time or predefined truncation time. Non-parametric comparisons used Kruskal–Wallis, Mann–Whitney U, or Wilcoxon signed-rank tests as appropriate.

Spatial associations between density, resistant growth, confinement, and escape were assessed using Pearson or Spearman correlations on quality-filtered sector- or clone-level summaries and interpreted as descriptive associations. Full image-processing, spatial-analysis, statistics and filtering details are reported in [1].

For preliminary free-floating spheroids, treatment failure was analysed by Kaplan–Meier curves with log-rank testing, and treatment burden was compared using Kruskal–Wallis tests. For the final collagen-embedded spheroid experiment, spheroids were treated as

experimental units. Group comparisons used Mann–Whitney U tests and time to treatment failure curves were compared by log-rank testing, as described in Isabel Schneider's master's thesis. For each spheroid, the spatial distribution of mScarlet fluorescence was visually assessed across the complete image series by a single independent rater blinded to treatment assignment. A semi-quantitative score from 1 (dispersed, scattered distribution) to 10 (compact, bulk-like growth) was assigned. Group differences were tested by exhaustive two-sided permutation of treatment labels using pooled midranks.

7.5.2 Computational Modelling and Reinforcement Learning

Computational modelling and reinforcement-learning-based treatment optimization followed the framework described in Eiche *et al.* [1]. This included the DPD model, a reduced two-population GLV-type model, and RL for treatment schedule optimization. Details of model construction, calibration, RL training, and *in silico* policy evaluation are provided in the original publication.

7.6 Materials

General Materials

Material	Manufacturer / supplier	Catalog No.
RPMI 1640 medium	PAN-Biotech	P04-18047
RPMI 1640 powder, HEPES-modified, with L-glutamine and without $NaHCO_3$	Sigma-Aldrich / Merck	R4130
Fetal bovine serum (FBS)	Thermo Fisher Scientific	A5256701
Opti-MEM™ I Reduced Serum Medium	Gibco / Thermo Fisher Scientific	31985062
Dulbecco's phosphate-buffered saline (DPBS)	Gibco / Thermo Fisher Scientific	14190250
Trypsin–EDTA	Thermo Fisher Scientific	25200072
Dimethyl sulfoxide (DMSO), ≥99.9%, Ultra Pure Grade	VWR Chemicals / Avantor	N182-5X10ML
Puromycin	Thermo Fisher Scientific	A1113803
Hygromycin B	AppliChem	A2175,0005
G418	Corning	30-234-CR
Vemurafenib	Cell Signaling Technology	17531
Ampicillin sodium salt, ≥97%	Carl Roth	K029.1
SOC Outgrowth Medium	New England Biolabs	B9020S
LB Broth with agar (Lennox)	Sigma-Aldrich	L2897
Glycerol, molecular biology grade, ≥99.0%	Sigma-Aldrich	G5516
CollagenR	Matrix Biosciences	50301
CollagenG	Matrix Biosciences	50104
Mebiol® Gel	Cosmo Bio (via Hölzel Biotech)	MBG-PMW20-5001
Gly-Arg-Gly-Asp-Ser (GRGDS; RGD-containing peptide), ≥97% (HPLC)	Sigma-Aldrich	G4391
Hyaluronic acid sodium salt from *Streptococcus equi*, 91%, ≥1 MDa	Thermo Scientific Chemicals / Alfa Aesar (via Fisher Scientific)	J66993.03; Fisher: 15480687
Sodium bicarbonate, BioReagent	Sigma-Aldrich	S5761
Sodium hydroxide solution, 1 mol/l, Titripur®	Supelco / Merck	1.09137

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Table 7.4 (continued)

Material	Manufacturer / supplier	Catalog No.
1H,1H,2H,2H-Perfluorodecyl triethoxysilane	Sigma-Aldrich	658758
Agarose, low gelling temperature, BioReagent	Sigma-Aldrich	A9414
TAE buffer, 10×, molecular biology grade	Calbiochem / Merck	574797
GeneRuler 1 kb Plus DNA Ladder (75–20,000 bp)	Thermo Scientific	SM1331
Rhodamine B, ≥95% (HPLC)	Sigma-Aldrich	R6626
Hoechst 33342 (bisBenzimide H 33342)	Sigma-Aldrich	14533
QuickExtract™ DNA Extraction Solution	Lucigen	QE09050

Table 7.4: Consumables, chemicals, and media. Reagents used in the experimental work, including materials tested during assay development.

Kits and Enzymes

Kit or enzyme	Manufacturer / supplier	Catalog No.
AlamarBlue Cell Viability Reagent	Thermo Scientific	DAL 1025
Q5® Hot Start High-Fidelity 2X Master Mix	New England Biolabs	M0494L
NEBuilder® HiFi DNA Assembly Cloning Kit	New England Biolabs	E5520S
NEB Golden Gate Assembly Kit BsaI-HF® v2	New England Biolabs	E1601S
NucleoSpin® Gel and PCR Clean-up	Macherey-Nagel	740609.250
GeneJET™ Plasmid Miniprep Kit	Thermo Scientific	K0503
NucleoBond Xtra Midi Kit	Macherey-Nagel	740410,1
Lipofectamine™ 3000 Transfection Reagent	Thermo Fisher Scientific	L3000001
NEB 5-alpha Competent *E. coli*	New England Biolabs	C2987H
LookOut® Mycoplasma PCR Detection Kit	Sigma-Aldrich	MP0035
FastDigest NotI	Thermo Scientific	FD0595
AfeI / Eco47III; FseI	Thermo Scientific	–
T4 DNA Ligase	New England Biolabs	M0202

Table 7.5: Assay and molecular-biology kits and enzymes. Commercial assay, cloning, purification, transfection, and bacterial-transformation reagents.

Culture Plates

Plate	Manufacturer / supplier	Catalog No.
24-well tissue-culture plate, flat bottom, TC-treated	VWR / Avantor	734-2325
96-well Ultra-Low Attachment plate, clear, round bottom, with lid	Corning / Costar	7007
96-well SCREENSTAR microplate, black, flat bottom, 190 µm clear cycloolefin film bottom, TC-treated, with lid	Greiner Bio-One	655866

Table 7.6: Culture plates. Identified culture plates used for the 2D colony assay, spheroid generation, and AlamarBlue measurements. Other culture formats used standard well plates and flasks.

Laboratory Devices

Device	Manufacturer / source
Mr. Frosty freezing container	Thermo Fisher Scientific
Heracell VIOS 160i CO_2 incubator	Thermo Scientific
LSM900 confocal microscope	Zeiss
Synergy H1 multimode microplate reader	BioTek Instruments
LUNA-II™ automated cell counter	Logos Biosystems
Xplorer Plus MoveIt multi-channel pipette	Eppendorf
Multipipette E3	Eppendorf
Custom-built confocal Brillouin microscope with a two-stage VIPA spectrometer	Guck laboratory
Real-time deformability cytometry (RT-DC) system	Zellmechanik Dresden

Table 7.7: Laboratory devices. Core equipment used for cell culture, imaging, mechanical phenotyping, and liquid handling. VIPA denotes virtually imaged phased array. Single-cell sorting was provided by the core facility listed in table 7.9.

Software

Software	Producer / reference
NEBuilder Assembly Tool (https://nebuilder.neb.com)	New England Biolabs
Benchling (https://benchling.com)	Benchling
NEBcloner tool (https://nebcloner.neb.com/#!/redigest)	New England Biolabs
CRISPOR (http://crispor.org/)	Concordet et al., 2018
ZEN 3.11	Zeiss
Fiji (ImageJ)	Schindelin et al., 2012
Cellpose	Stringer et al., 2021
Python	Python Software Foundation
BioRender	BioRender.com

Table 7.8: Software used in this work. Software used for experimental design, image acquisition and analysis, statistical analysis, and figure preparation.

Services

Service	Type	Provider
TubeSeq Supreme	Sanger sequencing	Eurofins Genomics
Oligonucleotide synthesis	Primer synthesis	Integrated DNA Technologies
gBlocks	Synthetic DNA fragments	Integrated DNA Technologies
Whole Plasmid Sequencing	Whole-plasmid sequencing	Eurofins Genomics
Single-cell sorting	Fluorescence-activated cell sorting	Core Unit für Zellsortierung und Immunomonitoring, FAU Erlangen-Nürnberg
S100 measurement	Analysis of culture supernatant	Universitätsklinikum Erlangen

Table 7.9: External services used in this work. Commercial sequencing and DNA-synthesis services, core-facility cell sorting, and S100 analysis.

Plasmids and Construct Sources

The Roman identifiers and functional names used in Chapter 2 correspond to the internal pME identifiers listed below. The internal identifiers are retained in laboratory records, plasmid maps, and the cell-line inventory.

Internal ID	Functional name	Source and role
pME_001	I. AAVS1_Cas9	eSpCas9(1.1)_No_FLAG_AAVS1_T2; Addgene #79888. Cas9/sgRNA targeting plasmid.
pME_002	II. AAVS1_EGFP	pAAVS1-P-CAG-GFP; Addgene #80491. EGFP donor used for $\alpha 2$ [145].
pME_012	III. AAVS1_empty	pAAVS1-P-MCS; Addgene #80488. Empty backbone with AAVS1 homology arms [145].

Table 7.10: Source plasmids for AAVS1 targeting.

Internal ID	Functional name	Parent or composition
pME_015	AAVS1_NLS-Neon_PuroR	pME_012 backbone; NLS-Neon and puromycin-resistance payload.
pME_016	IV. AAVS1_mScarlet_NeoR	pME_012 backbone; mScarlet and NeoR. Used to generate $\alpha 4$.
pME_017	V. AAVS1_mScarlet_BRAF	pME_012 backbone; mScarlet and $BRAF^{V600E}$.
pME_018	Hygromycin B-selectable empty donor	pME_012 modified with a hygromycin B selection cassette; parent of pME_020–pME_022.
pME_019	VIII. AAVS1_NLS-Neon_HygroR	pME_015 derivative; puromycin resistance replaced by hygromycin B resistance.
pME_019b	AAVS1_Neon_HygroR	pME_019 derivative without the NLS.
pME_020	Hygromycin-based Neon–PuroR donor	pME_018 backbone; Neon and puromycin-resistance payload.
pME_021	Hygromycin-based mScarlet–NeoR donor	pME_018 backbone; mScarlet and NeoR.
pME_022	Hygromycin-based mScarlet–$BRAF^{V600E}$ donor	pME_018 backbone; mScarlet and $BRAF^{V600E}$.

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Table 7.11 (continued)

Internal ID	Functional name	Parent or composition
pME_023	VI. AAVS1_mCarmine_MGMT	AAVS1 donor carrying mCarmine and MGMT.
pME_025	VII. AAVS1_mCarmine_NeoR	AAVS1 donor carrying mCarmine and NeoR.
pME_029	IX. AAVS1_NLS-mScarlet_NeoR	pME_016 derivative with an NLS added to mScarlet.
pME_012b	Modified IRM assembly backbone	pME_012 derivative with BsaI-site modifications.
pME_026	IRM intermediate I	Modified backbone, SV40 promoter, CreERT2, and EF1α–mKate2 modules.
pME_027	IRM intermediate II	CMV, NeoR–EGFP–T2A, and hygromycin B resistance modules.
pME_028a, pME_028b	First-generation IRM donor clones	Full-length four-fragment assembly products; sequence deviations described in Chapter 2.

Table 7.11: Engineered donors and IRM intermediates.

Source plasmid	Component	Recipient constructs
3xnls-Neon	Neon/NLS-Neon [206]	pME_015, pME_020
pTRE-T2-ΔmIR-IRES-Puromycin	Puromycin resistance; SV40 terminator	Puromycin: pME_015, pME_020. SV40 terminator: pME_022.
pmScarlet-H_C1	mScarlet [143]	pME_016, pME_017, pME_021, pME_022
pTS2404-Tier1-PhCMV-NeoR	NeoR [202]; SV40 terminator	NeoR: pME_016, pME_021, pME_025. SV40 terminator: pME_017.
pHIV-BRAFV600E-mOrange2	$BRAF^{V600E}$ [203]	pME_017, pME_022
mCarmine_in_pcDNA3	mCarmine [201]	pME_023
pIRES-Puro-MGMT	MGMT [209]	pME_023
pNA7 (Nico Appold)	Hygromycin B resistance	pME_018, pME_019

Table 7.12: Source plasmids for donor components.

Complete plasmid sequences, annotated maps, primer lists, and detailed assembly records are retained by the author and are available on request.

Cancer Cells

ID	Parental line	Fluorophore	Selection marker(s)	Resistance	Constructs	Origin / notes
α1	Mel Juso	–	–	–	–	Obtained 11/2021, Prof. Anja Boßerhoff
α2	Mel Juso	EGFP	Puromycin	–	pME_001 + pME_002	Built 03/2022 from α1
α3	Mel Juso	NLS-Neon	Puromycin, Hygromycin	–	pME_001 + pME_019	Built 05/2023 from α1
α4	Mel Juso	mScarlet	Puromycin	G418	pME_001 + pME_016	Built 04/2023 from α1
α5	Mel Juso	mScarlet	Puromycin	Vemurafenib	pME_001 + pME_017	Built 05/2023 from α1
β1	SK-Mel-28	–	–	–	–	Obtained 11/2021, Prof. Anja Boßerhoff
β2	SK-Mel-28	EGFP	Puromycin	–	pME_001 + pME_002	Built 03/2022 from β1
β3	SK-Mel-28	NLS-Neon	Puromycin, Hygromycin B	–	pME_001 + pME_019	Built 05/2023 from β1
β4	SK-Mel-28	mScarlet	Puromycin	G418	pME_001 + pME_016	Built 04/2023 from β1
β5	SK-Mel-28	mScarlet	Puromycin	Vemurafenib	pME_001 + pME_017	Built 05/2023 from β1

Table 7.13: Cell lines used and generated in this work. Overview of parental melanoma lines and their engineered derivatives, including fluorescent reporter, selection markers, resistance phenotype, transfection plasmids, and origin. For engineered lines, the Constructs column lists the Cas9/sgRNA plasmid followed by the donor plasmid; their corresponding functional names are given in tables 7.10 and 7.11.

A

Supplementary Figures

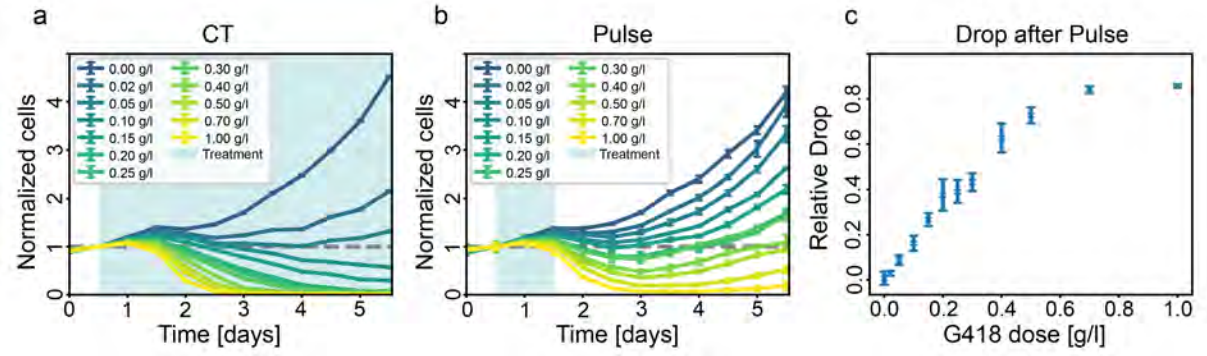


Figure A.1: Stronger treatment pulses cause larger cell count drops. Shown are cell-count dynamics under continuous and pulsed G418 treatment. Under CT (a), counts declined over time and are normalized to day 1; error bars indicate Poisson error ($\sqrt{\text{cell count}}$). In the pulse setting (b), cells were exposed to G418 for two days before recovery; error bars indicate the SD of three replicates. Panel (c) quantifies the delayed post-pulse reduction as the relative decrease from pulse end (day 3) to the subsequent minimum. Counts are normalized to day 1 (gray dashed line); error bars indicate the SD of three biological replicates. Turquoise shading marks treatment periods. Figure and text adapted from Elias Fischer's Master's thesis.

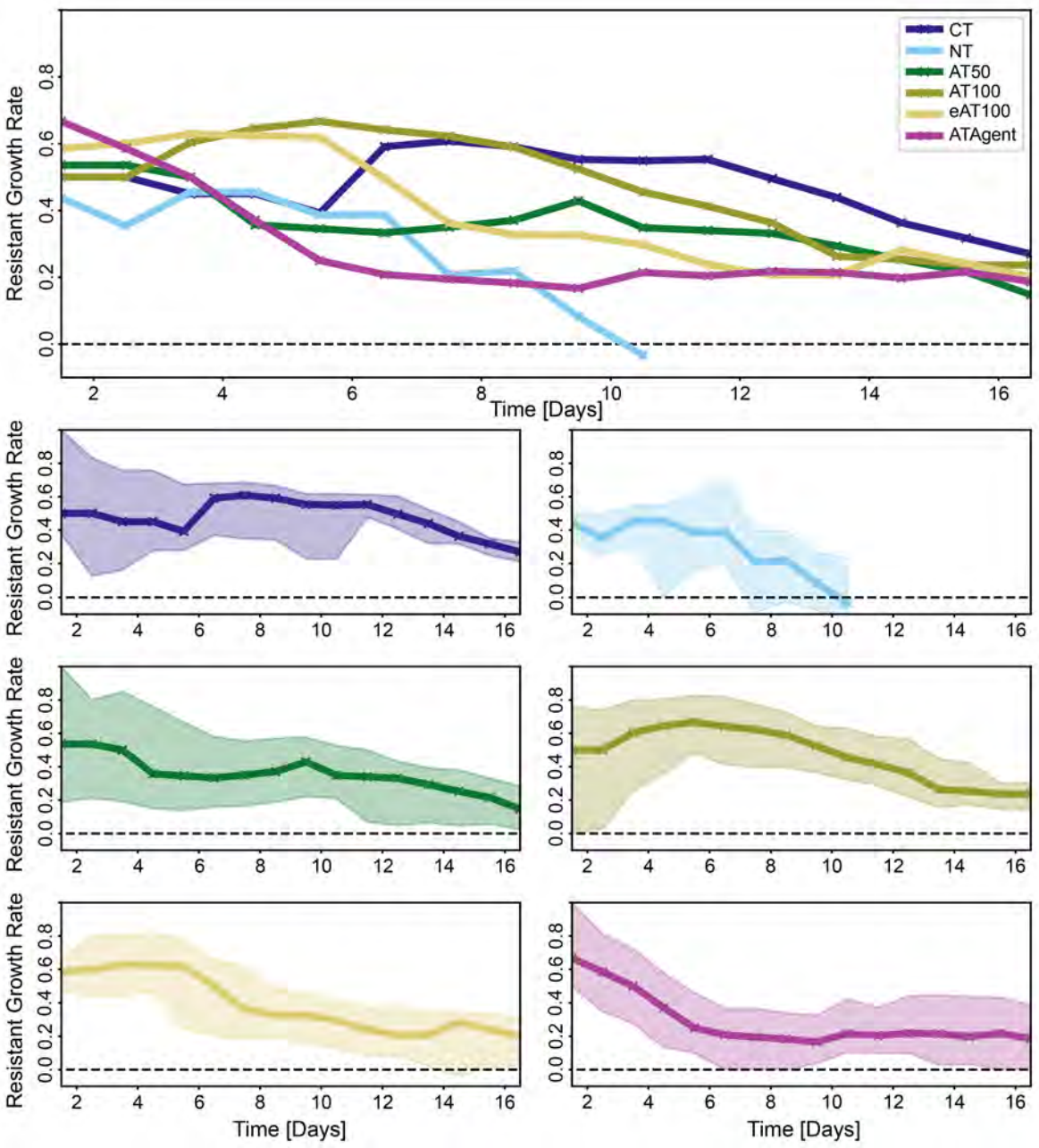


Figure A.2: Resistant growth trajectories across preliminary treatment strategies. Resistant growth rate was calculated as the day-to-day change in resistant cell number, normalized to the resistant-cell count at the start of each interval. The upper panel compares trajectories across treatment strategies, while lower panels show the corresponding strategy-specific trajectories over the full observation period. Shaded regions indicate SD where shown. ATagent denotes a preliminary RL-derived adaptive strategy. Figure and text adapted from Elias Fischer's Master's thesis.

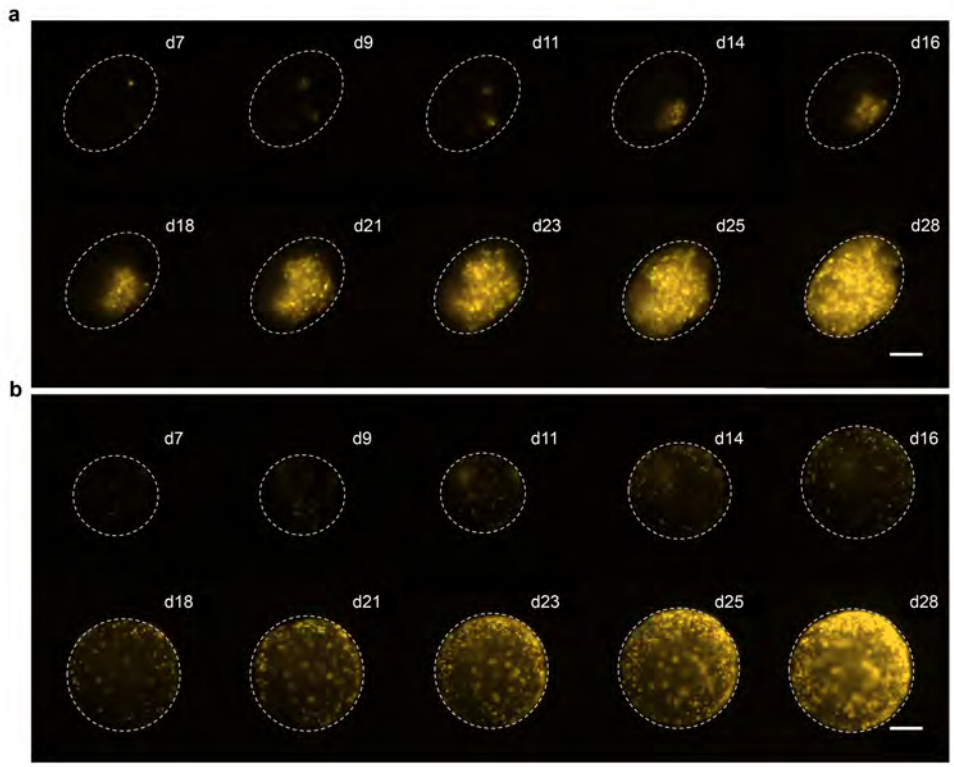


Figure A.3: Guide to the eye visualization for dispersed growth behaviour under AT. Representative epifluorescence time-lapse images of mScarlet-positive resistant cells (yellow) within the spheroid boundary (dashed white line) under CT (a) and AT (b), respectively. Scale bars: 200 µm. Adapted from Isabel Schneider's master's thesis.

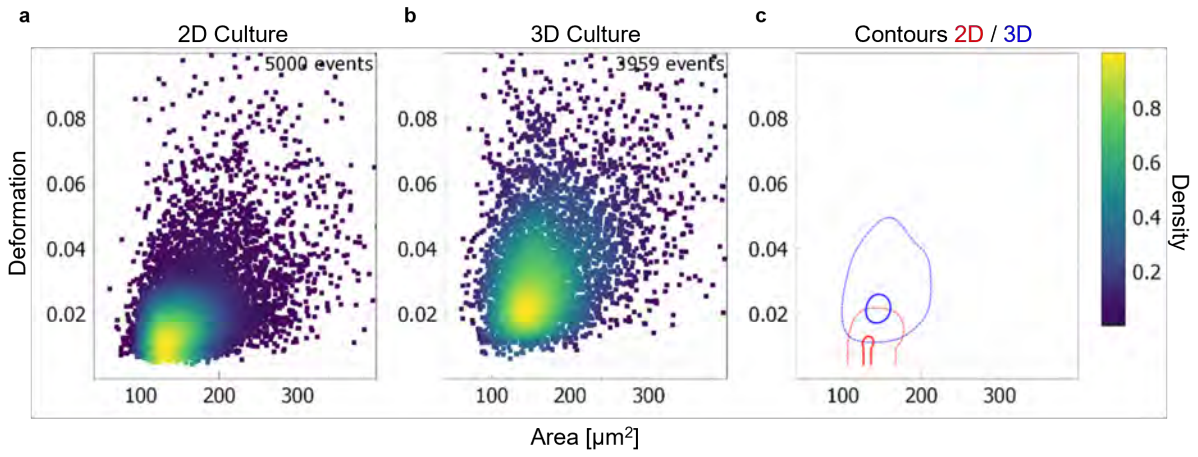


Figure A.4: Mechanical phenotypes differ between 2D and 3D samples. (a, b) Individual $\alpha 2$ cell events from 2D culture and dissociated 3D spheroids, respectively, plotted by cross-sectional area and deformation. The 3D dataset pools cells from $n = 20$ spheroids. (c) Density contours of the 2D (red) and 3D (blue) populations are overlaid on RT-DC iso-elasticity contours.

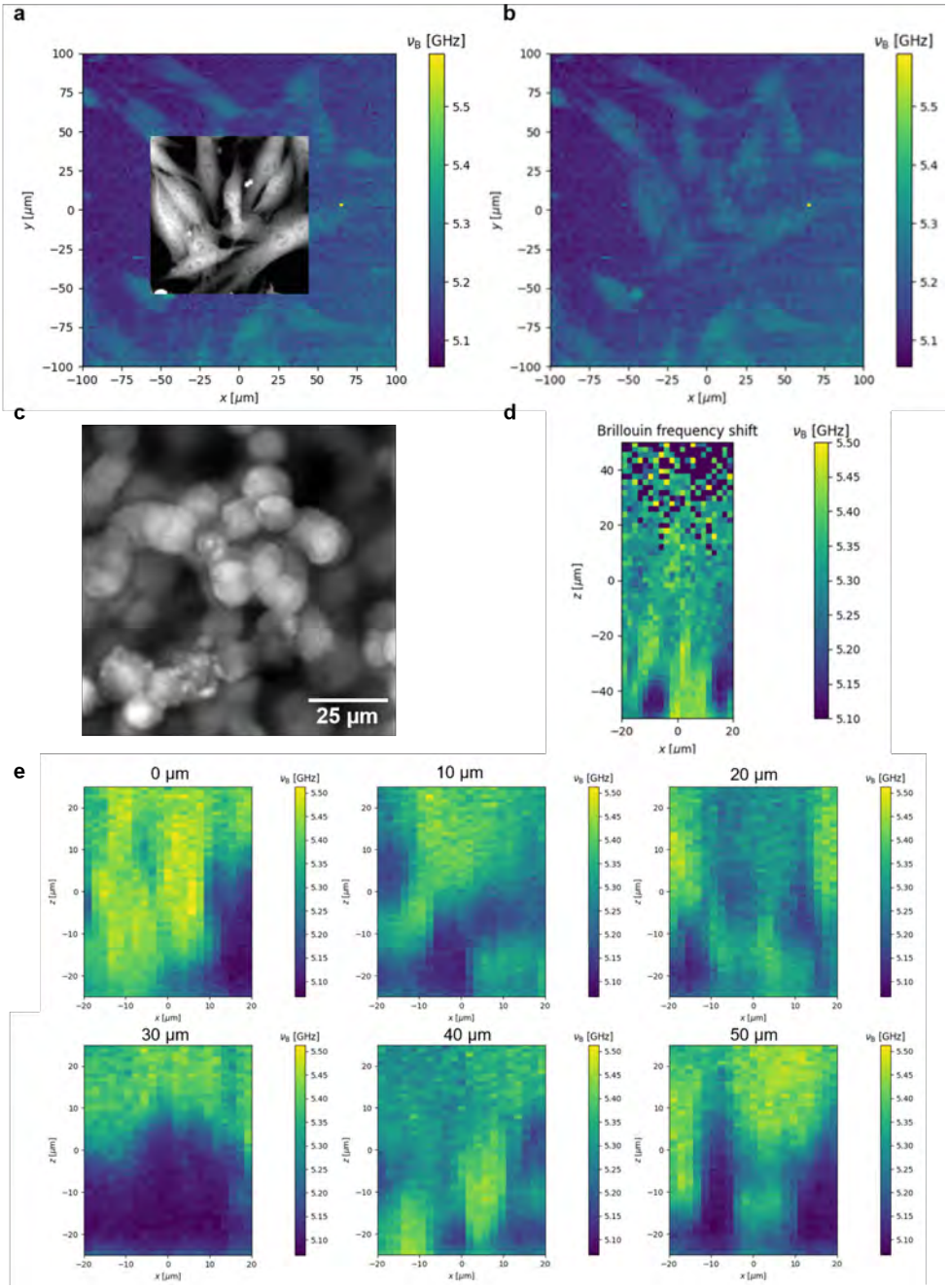


Figure A.5: Brillouin shifts differ between α2 cells in 2D monolayers and 3D spheroids. (a, b) Brillouin and corresponding fluorescence image (inset) of α2 cells in 2D culture, with Brillouin shifts below approximately 5.4 GHz. (c) Fluorescence image of a compact α2 spheroid. (d) High-resolution Brillouin cross-section of the spheroid periphery. Signal quality decreases beyond approximately 50 µm depth because of light scattering and absorption. (e) Multiple x–z cross-sections acquired at 10 µm intervals along the y-axis, showing higher Brillouin shifts in 3D than in 2D, with local maxima exceeding approximately 5.45 GHz.

B

Plasmid Constructs and Resource Availability

This appendix documents the three plasmids directly used to establish the validated AAVS1-targeted melanoma reporter system. These comprise the CRISPR/Cas9 targeting plasmid and the two donor plasmids used to generate the $\alpha 2$ and $\alpha 4$ cell lines.

The annotated maps below show the principal functional elements of these constructs. Complete nucleotide sequences, full annotated plasmid maps, primer lists, gBlock designs, detailed Gibson and Golden Gate assembly records, PCR conditions, screening records, and documentation for additional toolbox constructs and IRM intermediates are retained by the author and are available upon reasonable request.

B.1 Core Constructs for Cell-Line Generation

The validated $\alpha 2$–$\alpha 4$ reporter-cell pair was generated through CRISPR/Cas9-mediated homology-directed integration at the AAVS1 locus. The targeting plasmid pME_001 provided both the eSpCas9(1.1) nuclease and the AAVS1-T2-targeting single-guide RNA, thereby directing the genomic double-strand break required for donor integration. It

was used together with either pME_002 or pME_016, which contain the AAVS1 donor architecture and the respective reporter and resistance cassettes.

pME_002 was used to establish the EGFP-labelled, G418-sensitive $\alpha 2$ population. In addition to the fluorescent reporter cassette, this donor retains the promoterless Puromycin-selection design of the AAVS1 targeting system, linking antibiotic selection to correct integration at the target locus. pME_016 was used to establish the mScarlet-labelled, G418-resistant $\alpha 4$ population. Its *NeoR* cassette provides the defined Geneticin-resistance phenotype used as the selective difference in the 2D and 3D competition experiments.

Together, these three plasmids generated the traceable sensitive–resistant cell-line pair used throughout the central experimental chapters. This appendix focuses on the constructs directly relevant to the validated 2– 4 experiments, while the broader plasmid toolbox, including additional fluorophores, resistance cassettes, and inducible resistance-module intermediates, is described in Chapter 2.

Plasmid	**Function in the workflow**	**Experimental output**
pME_001	AAVS1-targeting CRISPR/Cas9 plasmid	Provided eSpCas9(1.1) and the AAVS1-T2 sgRNA for targeted donor integration at the AAVS1 locus.
pME_002	AAVS1 donor plasmid carrying *EGFP*	Used to generate the EGFP-labelled, G418-sensitive $\alpha 2$ cell line.
pME_016	AAVS1 donor plasmid carrying *mScarlet* and *NeoR*	Used to generate the mScarlet-labelled, G418-resistant $\alpha 4$ cell line.

Table B.1: Core plasmids used to establish the $\alpha 2$–$\alpha 4$ cell-line pair. The AAVS1-targeting CRISPR/Cas9 plasmid and donor plasmids used to generate the validated reporter-cell pair underlying the core 2D and 3D experiments.

B.2 Annotated Plasmid Maps

B.2.1 pME_001: AAVS1-Targeting CRISPR/Cas9 Plasmid

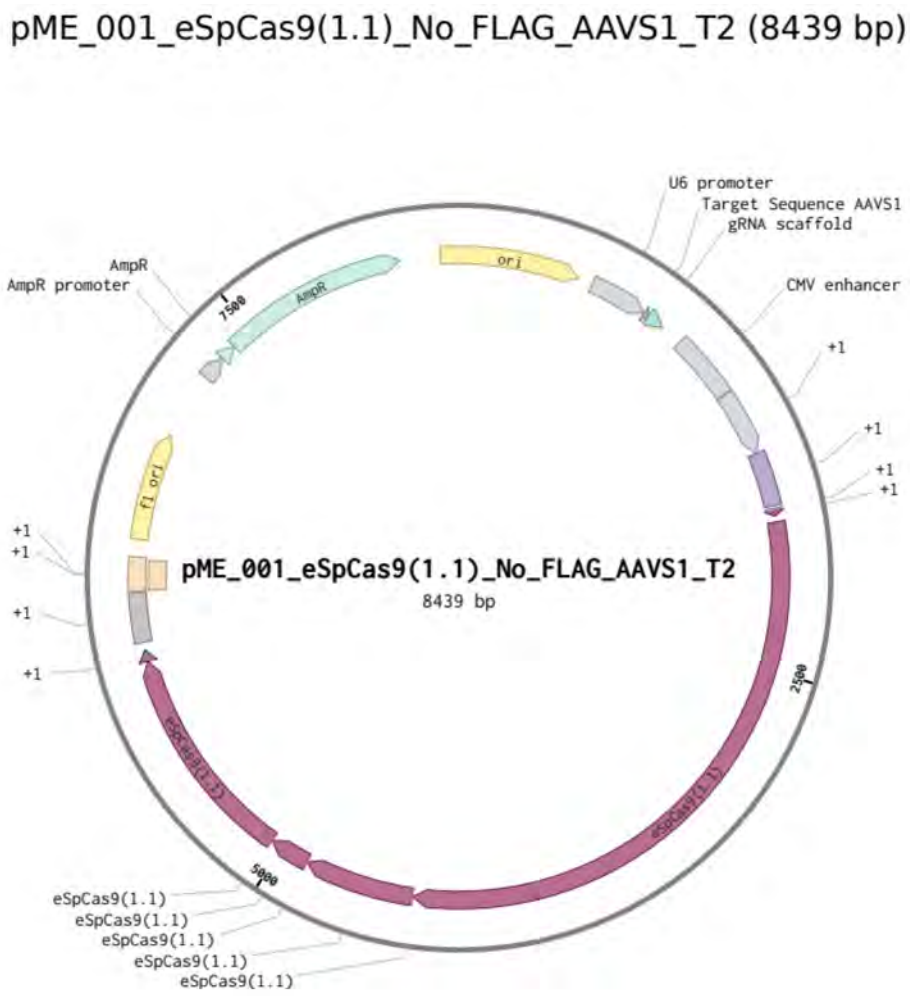


Figure B.1: Annotated plasmid map of pME_001 The eSpCas9(1.1)_No_FLAG_AAVS1_T2 plasmid was used for AAVS1-directed CRISPR/Cas9-mediated knock-in. Major functional elements are indicated.

B.2.2 pME_002: EGFP Donor Plasmid for the α2 Cell Line

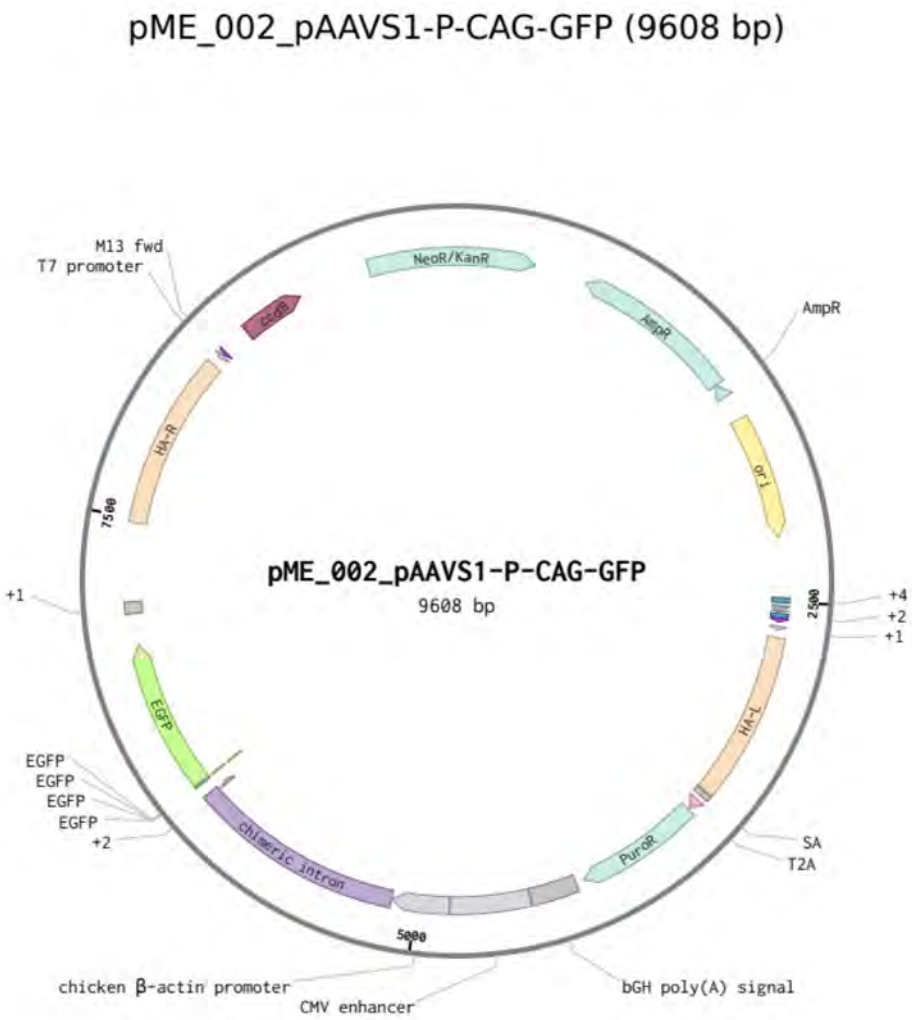


Figure B.2: Annotated plasmid map of pME_002 This AAVS1 donor plasmid was used to establish the EGFP-labelled, G418-sensitive α2 cell line. Major donor and expression-cassette elements are indicated.

B.2.3 pME_016: mScarlet–NeoR Donor Plasmid for the α4 Cell Line

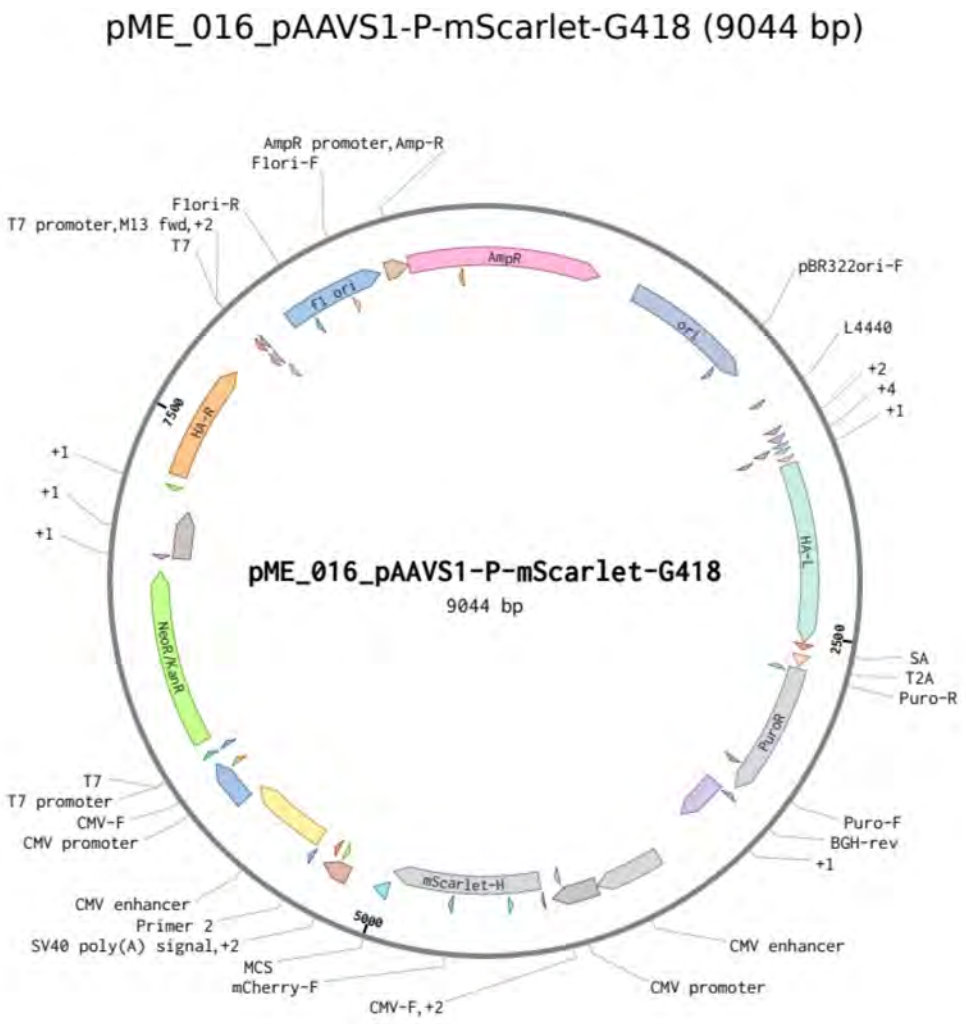


Figure B.3: Annotated plasmid map of pME_016 This AAVS1 donor plasmid was used to establish the mScarlet-labelled, G418-resistant α4 cell line. Major donor, reporter, and resistance-cassette elements are indicated.

List of Acronyms and Abbreviations

2D	Two-Dimensional
3D	Three-Dimensional
3R	Replace, Reduce, Refine
4-OHT	4-hydroxytamoxifen
AT	Adaptive Therapy
AAVS1	Adeno-associated virus integration site 1
CAG	Cytomegalovirus early enhancer/chicken β-actin
CBh	chicken β-actin hybrid
CI	Confidence interval
ctDNA	circulating tumor DNA
CT	Continuous Therapy
CMV	Cytomegalovirus
CreERT2	Cre recombinase fused to a tamoxifen-inducible estrogen receptor mutant
CRISPR	Clustered regularly interspaced short palindromic repeats
CTLA-4	Cytotoxic T-lymphocyte–associated protein 4
DNA	Deoxyribonucleic acid
DMSO	Dimethyl sulfoxide
DPBS	Dulbecco's phosphate-buffered saline
DPD	Dissipative particle dynamics
ECM	Extracellular matrix
EGFP	Enhanced green fluorescent protein
eSpCas9(1.1)	Enhanced specificity Cas9
FACS	Fluorescence-activated cell sorting
FAK	Focal adhesion kinase
FBS	Fetal bovine serum
G418	Geneticin
GEE	Generalized estimating equation
GLV	Generalized Lotka-Volterra
HA	Hyaluronic acid
HDR	Homology-directed repair
IQR	Interquartile range

IRM	Inducible Resistance Module
IT	Intermittent Therapy
KPZ	Kardar–Parisi–Zhang
MAPK	Mitogen-activated protein kinase
MEK	Mitogen-activated protein kinase kinase
MRI	Magnetic resonance imaging
MIP	Maximum intensity projection
Neon	mNeonGreen
NeoR	Neomycin phosphotransferase gene
NLS	Nuclear localization signal
NT	No treatment
PAM	Protospacer adjacent motif
PBS	phosphate-buffered saline
PCR	Polymerase chain reaction
PD-1	Programmed cell death protein 1
PD-L1	Programmed death-ligand 1
PI3K	Phosphoinositide 3-kinase
PSA	Prostate-specific antigen
RL	Reinforcement Learning
RPMI	Roswell Park Memorial Institute
RT-DC	Real-time deformability cytometry
RTK	Receptor tyrosine kinase
SD	Standard deviation
sgRNA	Single-guide RNA
Silane	1H,1H,2H,2H-Perfluorodecyl-triethoxysilane
SV40	Simian Virus 40
TAE	Tris-acetate-EDTA
TFI	Total fluorescence intensity
TTE	Time to escape
TTF	Time to treatment failure
UV	Ultraviolet

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References

[1] M. Eiche, E. F. Fischer, M. Milas, T. Citak, S. Aif, A.-S. Smith, and J. Kayser, *Crowding governs resistance confinement and guides the improvement of adaptive therapy in a 2D melanoma model system*, 2026.

[2] S. Aif, M. Eiche, N. Appold, E. Fischer, T. Citak, and J. Kayser, "Reinforcement Failing guides the discovery of emergent physical dynamics in adaptive tumor therapy," npj Artificial Intelligence **2**, 76 (2026).

[3] M. Eiche and J. Kayser, "Wie mikrobielle Modellsysteme helfen, Tumorevolution zu entschlüsseln," BIOspektrum **28**, 250–252 (2022).

[4] N. Bauer, B. Oswald, M. Eiche, L. Schiller, E. Langguth, C. Schantz, A. Osterlehner, A. Shen, S. Misaghi, J. Stingele, and S. Ausländer, "An arrayed CRISPR screen reveals Myc depletion to increase productivity of difficult-to-express complex antibodies in CHO cells," Synthetic Biology **7**, ysac026 (2022).

[5] H. Sung, J. Ferlay, R. L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, and F. Bray, "Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries," CA: a cancer journal for clinicians **71**, 209–249 (2021).

[6] C. Holohan, S. Van Schaeybroeck, D. B. Longley, and P. G. Johnston, "Cancer drug resistance: an evolving paradigm," Nature Reviews Cancer **13**, 714–726 (2013).

[7] R. J. Gillies, D. Verduzco, and R. A. Gatenby, "Evolutionary dynamics of carcinogenesis and why targeted therapy does not work," Nature Reviews Cancer **12**, 487–493 (2012).

[8] A. Marusyk and K. Polyak, "Tumor heterogeneity: causes and consequences," Biochimica Et Biophysica Acta **1805**, 105–117 (2010).

[9] J.-C. Marine, S.-J. Dawson, and M. A. Dawson, "Non-genetic mechanisms of therapeutic resistance in cancer," Nature Reviews Cancer **20**, 743–756 (2020).

[10] S. M. Mumenthaler, J. Foo, N. C. Choi, N. Heise, K. Leder, D. B. Agus, W. Pao, F. Michor, and P. Mallick, "The Impact of Microenvironmental Heterogeneity on the Evolution of Drug Resistance in Cancer Cells," Cancer Informatics **14**, 19–31 (2015).

[11] U. Ledzewicz, H. Schättler, M. R. Gahrooi, and S. M. Dehkordi, "On the MTD paradigm and optimal control for multi-drug cancer chemotherapy," Mathematical biosciences and engineering: MBE **10**, 803–819 (2013).

[12] R. A. Gatenby, A. S. Silva, R. J. Gillies, and B. R. Frieden, "Adaptive therapy," Cancer Research **69**, 4894–4903 (2009).

[13] P. M. Enriquez-Navas, Y. Kam, T. Das, S. Hassan, A. Silva, P. Foroutan, E. Ruiz, G. Martinez, S. Minton, R. J. Gillies, and R. A. Gatenby, "Exploiting evolutionary principles to prolong tumor control in preclinical models of breast cancer," Science Translational Medicine **8**, 327ra24 (2016).

[14] J. Zhang, J. J. Cunningham, J. S. Brown, and R. A. Gatenby, "Integrating evolutionary dynamics into treatment of metastatic castrate-resistant prostate cancer," Nature Communications **8**, 1816 (2017).

[15] J. West, F. Adler, J. Gallaher, M. Strobl, R. Brady-Nicholls, J. Brown, M. Roberson-Tessi, E. Kim, R. Noble, Y. Viossat, D. Basanta, and A. R. A. Anderson, "A survey of open questions in adaptive therapy: Bridging mathematics and clinical translation," eLife **12**, e84263 (2023).

[16] M. a. R. Strobl, J. Gallaher, M. Robertson-Tessi, J. West, and A. R. A. Anderson, "Treatment of evolving cancers will require dynamic decision support," Annals of Oncology: Official Journal of the European Society for Medical Oncology **34**, 867–884 (2023).

[17] G. Szakács, M. D. Hall, M. M. Gottesman, A. Boumendjel, R. Kachadourian, B. J. Day, H. Baubichon-Cortay, and A. Di Pietro, "Targeting the Achilles heel of multidrug-resistant cancer by exploiting the fitness cost of resistance," Chemical Reviews **114**, 5753–5774 (2014).

[18] M. Kapałczyńska, T. Kolenda, W. Przybyła, M. Zajączkowska, A. Teresiak, V. Filas, M. Ibbs, R. Bliźniak, Ł. Łuczewski, and K. Lamperska, "2D and 3D cell cultures - a comparison of different types of cancer cell cultures," Archives of medical science: AMS **14**, 910–919 (2018).

[19] M. A. R. Strobl, J. Gallaher, J. West, M. Robertson-Tessi, P. K. Maini, and A. R. A. Anderson, "Spatial structure impacts adaptive therapy by shaping intra-tumoral competition," Communications Medicine **2**, 46 (2022).

[20] M. Eiche, *Biorender Image*, 2026, https://BioRender.com/k01kptf.

[21] P. C. Nowell, "The Clonal Evolution of Tumor Cell Populations: Acquired genetic lability permits stepwise selection of variant sublines and underlies tumor progression.," Science **194**, 23–28 (1976).

[22] W. Zhou, X. Liu, Z. Li, B. Jia, X. Lei, K. Sun, P. Yang, S. He, D. Wang, H. Zhang, and S. Wang, "Epigenetic crosstalk between stem cells and tumors: mechanisms and emerging perspectives," American Journal of Stem Cells **14**, 98–133 (2025).

[23] G. Ciriello, L. Magnani, S. J. Aitken, L. Akkari, S. Behjati, D. Hanahan, D. A. Landau, N. Lopez-Bigas, D. G. Lupiáñez, J.-C. Marine, A. Martin-Villalba, G. Natoli, A. C. Obenauf, E. Oricchio, P. Scaffidi, A. Sottoriva, A. Swarbrick, G. Tonon, S. Vanharanta, and J. Zuber, "Cancer Evolution: A Multifaceted Affair," Cancer Discovery **14**, 36–48 (2024).

[24] C. C. Bell and O. Gilan, "Principles and mechanisms of non-genetic resistance in cancer," British Journal of Cancer **122**, 465–472 (2020).

[25] D. F. Quail and J. A. Joyce, "Microenvironmental regulation of tumor progression and metastasis," Nature Medicine **19**, 1423–1437 (2013).

[26] D. Hanahan, "Hallmarks of Cancer: New Dimensions," Cancer Discovery **12**, 31–46 (2022).

[27] N. N. Pavlova and C. B. Thompson, "The Emerging Hallmarks of Cancer Metabolism," Cell Metabolism **23**, 27–47 (2016).

[28] F. Rambow, J.-C. Marine, and C. R. Goding, "Melanoma plasticity and phenotypic diversity: therapeutic barriers and opportunities," Genes & Development **33**, 1295–1318 (2019).

[29] H. T. Nia, L. L. Munn, and R. K. Jain, "Physical traits of cancer," Science **370**, eaaz0868 (2020).

[30] M. Gerlinger, A. J. Rowan, S. Horswell, J. Larkin, D. Endesfelder, E. Gronroos, P. Martinez, N. Matthews, A. Stewart, P. Tarpey, I. Varela, B. Phillimore, S. Begum, N. Q. McDonald, A. Butler, D. Jones, K. Raine, C. Latimer, C. R. Santos, M. Nohadani, A. C. Eklund, B. Spencer-Dene, G. Clark, L. Pickering, G. Stamp, M. Gore, Z. Szallasi, J. Downward, P. A. Futreal, and C. Swanton, "Intratumor Heterogeneity and Branched Evolution Revealed by Multiregion Sequencing," New England Journal of Medicine **366**, 883–892 (2012).

[31] A. Sottoriva, H. Kang, Z. Ma, T. A. Graham, M. P. Salomon, J. Zhao, P. Marjoram, K. Siegmund, M. F. Press, D. Shibata, and C. Curtis, "A Big Bang model of human colorectal tumor growth," Nature Genetics **47**, 209–216 (2015).

[32] M. J. Williams, B. Werner, C. P. Barnes, T. A. Graham, and A. Sottoriva, "Identification of neutral tumor evolution across cancer types," Nature Genetics **48**, 238–244 (2016).

[33] M. R. Sullivan and M. G. Vander Heiden, "Determinants of nutrient limitation in cancer," Critical Reviews in Biochemistry and Molecular Biology **54**, 193–207 (2019).

[34] M. G. Vander Heiden and R. J. DeBerardinis, "Understanding the Intersections between Metabolism and Cancer Biology," Cell **168**, 657–669 (2017).

[35] M. W. Dewhirst and T. W. Secomb, "Transport of drugs from blood vessels to tumour tissue," Nature Reviews. Cancer **17**, 738–750 (2017).

[36] D. Hanahan and R. A. Weinberg, "The hallmarks of cancer," Cell **100**, 57–70 (2000).

[37] R. K. Jain, J. D. Martin, and T. Stylianopoulos, "The role of mechanical forces in tumor growth and therapy," Annual Review of Biomedical Engineering **16**, 321–346 (2014).

[38] H. T. Nia, H. Liu, G. Seano, M. Datta, D. Jones, N. Rahbari, J. Incio, V. P. Chauhan, K. Jung, J. D. Martin, V. Askoxylakis, T. P. Padera, D. Fukumura, Y. Boucher, F. J. Hornicek, A. J. Grodzinsky, J. W. Baish, L. L. Munn, and R. K. Jain, "Solid stress and elastic energy as measures of tumour mechanopathology," Nature Biomedical Engineering **1**, 0004 (2016).

[39] F. Moradi Kashkooli, F. Mirala, M. H H Tehrani, M. Alirahimi, M. Souri, A. Golzaryan, S. Kar, and M. Soltani, "Mechanical Forces in Tumor Growth and Treatment: Perspectives From Biology, Physics, Engineering, and Mathematical Modeling," WIREs mechanisms of disease **17**, e70000 (2025).

[40] E. Cambria, M. F. Coughlin, M. A. Floryan, G. S. Offeddu, S. E. Shelton, and R. D. Kamm, "Linking cell mechanical memory and cancer metastasis," Nature Reviews. Cancer **24**, 216–228 (2024).

[41] M. Delarue, F. Montel, D. Vignjevic, J. Prost, J.-F. Joanny, and G. Cappello, "Compressive stress inhibits proliferation in tumor spheroids through a volume limitation," Biophysical Journal **107**, 1821–1828 (2014).

[42] S. Dupont, L. Morsut, M. Aragona, E. Enzo, S. Giulitti, M. Cordenonsi, F. Zanconato, J. Le Digabel, M. Forcato, S. Bicciato, N. Elvassore, and S. Piccolo, "Role of YAP/TAZ in mechanotransduction," Nature **474**, 179–183 (2011).

[43] M. H. Kim, J. Kim, H. Hong, S.-H. Lee, J.-K. Lee, E. Jung, and J. Kim, "Actin remodeling confers BRAF inhibitor resistance to melanoma cells through YAP/TAZ activation," The EMBO journal **35**, 462–478 (2016).

[44] R. Noble, D. Burri, C. Le Sueur, J. Lemant, Y. Viossat, J. N. Kather, and N. Beerenwinkel, "Spatial structure governs the mode of tumour evolution," Nature Ecology & Evolution **6**, 207–217 (2021).

[45] O. Hallatschek, P. Hersen, S. Ramanathan, and D. R. Nelson, "Genetic drift at expanding frontiers promotes gene segregation," Proceedings of the National Academy of Sciences **104**, 19926–19930 (2007).

[46] A. I. Minchinton and I. F. Tannock, "Drug penetration in solid tumours," Nature Reviews. Cancer **6**, 583–592 (2006).

[47] C. F. Schreck, D. Fusco, Y. Karita, S. Martis, J. Kayser, M.-C. Duvernoy, and O. Hallatschek, "Impact of crowding on the diversity of expanding populations," Proceedings of the National Academy of Sciences of the United States of America **120**, e2208361120 (2023).

[48] O. Hallatschek and D. R. Nelson, "Life at the front of an expanding population," Evolution **64**, 193–206 (2010).

[49] D. Fusco, M. Gralka, J. Kayser, A. Anderson, and O. Hallatschek, "Excess of mutational jackpot events in expanding populations revealed by spatial Luria–Delbrück experiments," Nature Communications **7**, 12760 (2016).

[50] K. S. Korolev, M. J. I. Müller, N. Karahan, A. W. Murray, O. Hallatschek, and D. R. Nelson, "Selective sweeps in growing microbial colonies," Physical Biology **9**, 026008 (2012).

[51] J. Kayser, C. F. Schreck, M. Gralka, D. Fusco, and O. Hallatschek, "Collective motion conceals fitness differences in crowded cellular populations," Nature Ecology & Evolution **3**, 125–134 (2018).

[52] A. Giometto, D. R. Nelson, and A. W. Murray, "Physical interactions reduce the power of natural selection in growing yeast colonies," Proceedings of the National Academy of Sciences **115**, 11448–11453 (2018).

[53] S. Aif, N. Appold, L. Kampman, O. Hallatschek, and J. Kayser, "Evolutionary rescue of resistant mutants is governed by a balance between radial expansion and selection in compact populations," Nature Communications **13**, 7916 (2022).

[54] R. Wittmann, G. H. P. Nguyen, H. Löwen, F. J. Schwarzendahl, and A. Sengupta, "Collective mechano-response dynamically tunes cell-size distributions in growing bacterial colonies," Communications Physics **6**, 331 (2023).

[55] M. Gralka, F. Stiewe, F. Farrell, W. Möbius, B. Waclaw, and O. Hallatschek, "Allele surfing promotes microbial adaptation from standing variation," Ecology Letters **19**, edited by B. Blasius, 889–898 (2016).

[56] B. Waclaw, I. Bozic, M. E. Pittman, R. H. Hruban, B. Vogelstein, and M. A. Nowak, "A spatial model predicts that dispersal and cell turnover limit intratumour heterogeneity," Nature **525**, 261–264 (2015).

[57] A. Stein, K. Bostock, R. Kizhuttil, M. Bak, and R. Noble, "Selective sweep probabilities in spatially expanding populations," Nature Communications **17**, 2181 (2026).

[58] M. A. Ahmed and A. Nagelkerke, "Current developments in modelling the tumour microenvironment in vitro: Incorporation of biochemical and physical gradients," Organs-on-a-Chip **3**, 100012 (2021).

[59] K. M. Almazrouei, V. Mishra, H. Pandya, K. Sambhav, and S. N. Bhavsar, "Tumor Microenvironment and Its Role in Cancer Progression: An Integrative Review," Cureus, 10.7759/cureus.92707 (2025).

[60] T. Alarcón, H. M. Byrne, and P. K. Maini, "A Multiple Scale Model for Tumor Growth," Multiscale Modeling & Simulation **3**, 440–475 (2005).

[61] Z. Chen, F. Han, Y. Du, H. Shi, and W. Zhou, "Hypoxic microenvironment in cancer: molecular mechanisms and therapeutic interventions," Signal Transduction and Targeted Therapy **8**, 70 (2023).

[62] A. Ray, O. Lee, Z. Win, R. M. Edwards, P. W. Alford, D.-H. Kim, and P. P. Provenzano, "Anisotropic forces from spatially constrained focal adhesions mediate contact guidance directed cell migration," Nature Communications **8**, 14923 (2017).

[63] T. Stylianopoulos, J. D. Martin, V. P. Chauhan, S. R. Jain, B. Diop-Frimpong, N. Bardeesy, B. L. Smith, C. R. Ferrone, F. J. Hornicek, Y. Boucher, L. L. Munn, and R. K. Jain, "Causes, consequences, and remedies for growth-induced solid stress in murine and human tumors," Proceedings of the National Academy of Sciences of the United States of America **109**, 15101–15108 (2012).

[64] T. P. Padera, B. R. Stoll, J. B. Tooredman, D. Capen, E. D. Tomaso, and R. K. Jain, "Cancer cells compress intratumour vessels," Nature **427**, 695–695 (2004).

[65] J. A. Joyce and D. T. Fearon, "T cell exclusion, immune privilege, and the tumor microenvironment," Science (New York, N.Y.) **348**, 74–80 (2015).

[66] N. Appold, T. Citak, A. A. Palm, and J. Kayser, "A microbial therapy-mimicry assay shows how spatial resource dynamics control resistance escape," Nature Ecology & Evolution, 10.1038/s41559-026-03178-z (2026).

[67] S. Goel, D. G. Duda, L. Xu, L. L. Munn, Y. Boucher, D. Fukumura, and R. K. Jain, "Normalization of the vasculature for treatment of cancer and other diseases," Physiological Reviews **91**, 1071–1121 (2011).

[68] F. Mpekris, J. W. Baish, T. Stylianopoulos, and R. K. Jain, "Role of vascular normalization in benefit from metronomic chemotherapy," Proceedings of the National Academy of Sciences of the United States of America **114**, 1994–1999 (2017).

[69] S. Jafari Nivlouei, M. Soltani, E. Shirani, M. R. Salimpour, R. Travasso, and J. Carvalho, "A multiscale cell-based model of tumor growth for chemotherapy assessment and tumor-targeted therapy through a 3D computational approach," Cell Proliferation **55**, e13187 (2022).

[70] G. Helmlinger, P. A. Netti, H. C. Lichtenbeld, R. J. Melder, and R. K. Jain, "Solid stress inhibits the growth of multicellular tumor spheroids," Nature Biotechnology **15**, 778–783 (1997).

[71] B. Ladoux and R.-M. Mège, "Mechanobiology of collective cell behaviours," Nature Reviews. Molecular Cell Biology **18**, 743–757 (2017).

[72] J.-A. Park, J. H. Kim, D. Bi, J. A. Mitchel, N. T. Qazvini, K. Tantisira, C. Y. Park, M. McGill, S.-H. Kim, B. Gweon, J. Notbohm, R. Steward Jr, S. Burger, S. H. Randell, A. T. Kho, D. T. Tambe, C. Hardin, S. A. Shore, E. Israel, D. A. Weitz, D. J. Tschumperlin, E. P. Henske, S. T. Weiss, M. L. Manning, J. P. Butler, J. M. Drazen, and J. J. Fredberg, "Unjamming and cell shape in the asthmatic airway epithelium," Nature Materials **14**, 1040–1048 (2015).

[73] A. Puliafito, L. Hufnagel, P. Neveu, S. Streichan, A. Sigal, D. K. Fygenson, and B. I. Shraiman, "Collective and single cell behavior in epithelial contact inhibition," Proceedings of the National Academy of Sciences **109**, 739–744 (2012).

[74] T. B. Saw, A. Doostmohammadi, V. Nier, L. Kocgozlu, S. Thampi, Y. Toyama, P. Marcq, C. T. Lim, J. M. Yeomans, and B. Ladoux, "Topological defects in epithelia govern cell death and extrusion," Nature **544**, 212–216 (2017).

[75] M. Basan, T. Risler, J.-F. Joanny, X. Sastre-Garau, and J. Prost, "Homeostatic competition drives tumor growth and metastasis nucleation," HFSP Journal **3**, 265–272 (2009).

[76] O. Otto, P. Rosendahl, A. Mietke, S. Golfier, C. Herold, D. Klaue, S. Girardo, S. Pagliara, A. Ekpenyong, A. Jacobi, M. Wobus, N. Töpfner, U. F. Keyser, J. Mansfeld, E. Fischer-Friedrich, and J. Guck, "Real-time deformability cytometry: on-the-fly cell mechanical phenotyping," Nature Methods **12**, 199–202 (2015).

[77] J. Guck, S. Schinkinger, B. Lincoln, F. Wottawah, S. Ebert, M. Romeyke, D. Lenz, H. M. Erickson, R. Ananthakrishnan, D. Mitchell, J. Käs, S. Ulvick, and C. Bilby, "Optical Deformability as an Inherent Cell Marker for Testing Malignant Transformation and Metastatic Competence," Biophysical Journal **88**, 3689–3698 (2005).

[78] S. E. Cross, Y.-S. Jin, J. Rao, and J. K. Gimzewski, "Nanomechanical analysis of cells from cancer patients," Nature Nanotechnology **2**, 780–783 (2007).

[79] O. Ilina, P. G. Gritsenko, S. Syga, J. Lippoldt, C. A. M. La Porta, O. Chepizhko, S. Grosser, M. Vullings, G.-J. Bakker, J. Starruß, P. Bult, S. Zapperi, J. A. Käs, A. Deutsch, and P. Friedl, "Cell-cell adhesion and 3D matrix confinement determine jamming transitions in breast cancer invasion," Nature Cell Biology **22**, 1103–1115 (2020).

[80] A. Palamidessi, C. Malinverno, E. Frittoli, S. Corallino, E. Barbieri, S. Sigismund, G. V. Beznoussenko, E. Martini, M. Garre, I. Ferrara, C. Tripodo, F. Ascione, E. A. Cavalcanti-Adam, Q. Li, P. P. Di Fiore, D. Parazzoli, F. Giavazzi, R. Cerbino, and G. Scita, "Unjamming overcomes kinetic and proliferation arrest in terminally differentiated cells and promotes collective motility of carcinoma," Nature Materials **18**, 1252–1263 (2019).

[81] D. E. Discher, P. Janmey, and Y.-L. Wang, "Tissue cells feel and respond to the stiffness of their substrate," Science (New York, N.Y.) **310**, 1139–1143 (2005).

[82] Z. Sun, S. S. Guo, and R. Fässler, "Integrin-mediated mechanotransduction," The Journal of Cell Biology **215**, 445–456 (2016).

[83] A. J. Rice, E. Cortes, D. Lachowski, B. C. H. Cheung, S. A. Karim, J. P. Morton, and A. Del Río Hernández, "Matrix stiffness induces epithelial–mesenchymal transition and promotes chemoresistance in pancreatic cancer cells," Oncogenesis **6**, e352–e352 (2017).

[84] M. J. Paszek, N. Zahir, K. R. Johnson, J. N. Lakins, G. I. Rozenberg, A. Gefen, C. A. Reinhart-King, S. S. Margulies, M. Dembo, D. Boettiger, D. A. Hammer, and V. M. Weaver, "Tensional homeostasis and the malignant phenotype," Cancer Cell **8**, 241–254 (2005).

[85] J. Schrader, T. T. Gordon-Walker, R. L. Aucott, M. van Deemter, A. Quaas, S. Walsh, D. Benten, S. J. Forbes, R. G. Wells, and J. P. Iredale, "Matrix stiffness modulates proliferation, chemotherapeutic response, and dormancy in hepatocellular carcinoma cells," Hepatology (Baltimore, Md.) **53**, 1192–1205 (2011).

[86] C. M. Nelson, R. P. Jean, J. L. Tan, W. F. Liu, N. J. Sniadecki, A. A. Spector, and C. S. Chen, "Emergent patterns of growth controlled by multicellular form and mechanics," Proceedings of the National Academy of Sciences of the United States of America **102**, 11594–11599 (2005).

[87] B. Switzer, I. Puzanov, J. J. Skitzki, L. Hamad, and M. S. Ernstoff, "Managing Metastatic Melanoma in 2022: A Clinical Review," JCO oncology practice **18**, 335–351 (2022).

[88] M. Das Thakur, F. Salangsang, A. S. Landman, W. R. Sellers, N. K. Pryer, M. P. Levesque, R. Dummer, M. McMahon, and D. D. Stuart, "Modelling vemurafenib resistance in melanoma reveals a strategy to forestall drug resistance," Nature **494**, 251–255 (2013).

[89] X. Fu, Y. Zhao, J. I. Lopez, A. Rowan, L. Au, A. Fendler, S. Hazell, H. Xu, S. Horswell, S. T. C. Shepherd, C. E. Spencer, L. Spain, F. Byrne, G. Stamp, T. O'Brien, D. Nicol, M. Augustine, A. Chandra, S. Rudman, A. Toncheva, A. J. S. Furness, L. Pickering, S. Kumar, D.-M. Koh, C. Messiou, D. A. Dafydd, M. R. Orton, S. J. Doran, J. Larkin, C. Swanton, E. Sahai, K. Litchfield, S. Turajlic, on behalf of the TRACERx Renal Consortium, Ben Challacombe, S. Chowdhury, W. Drake, A. Fernando, N. Fotiadis, E. Hatipoglu, K. Harrison-Phipps, P. Hill, C. Horsfield, T. Marafioti, J. Olsburgh, A. Polson, S. Quezada, M. Varia, H. Verma, and P. A. Bates, "Spatial patterns of tumour growth impact clonal diversification in a computational model and the TRACERx Renal study," Nature Ecology & Evolution **6**, 88–102 (2021).

[90] R. Nazarian, H. Shi, Q. Wang, X. Kong, R. C. Koya, H. Lee, Z. Chen, M.-K. Lee, N. Attar, H. Sazegar, T. Chodon, S. F. Nelson, G. McArthur, J. A. Sosman, A. Ribas, and R. S. Lo, "Melanomas acquire resistance to B-RAF(V600E) inhibition by RTK or N-RAS upregulation," Nature **468**, 973–977 (2010).

[91] P. I. Poulikakos, Y. Persaud, M. Janakiraman, X. Kong, C. Ng, G. Moriceau, H. Shi, M. Atefi, B. Titz, M. T. Gabay, M. Salton, K. B. Dahlman, M. Tadi, J. A. Wargo, K. T. Flaherty, M. C. Kelley, T. Misteli, P. B. Chapman, J. A. Sosman, T. G. Graeber, A. Ribas, R. S. Lo, N. Rosen, and D. B. Solit, "RAF inhibitor resistance is mediated by dimerization of aberrantly spliced BRAF(V600E)," Nature **480**, 387–390 (2011).

[92] J. Villanueva, A. Vultur, J. T. Lee, R. Somasundaram, M. Fukunaga-Kalabis, A. K. Cipolla, B. Wubbenhorst, X. Xu, P. A. Gimotty, D. Kee, A. E. Santiago-Walker, R. Letrero, K. D'Andrea, A. Pushparajan, J. E. Hayden, K. D. Brown, S. Laquerre, G. A. McArthur, J. A. Sosman, K. L. Nathanson, and M. Herlyn, "Acquired resistance to BRAF inhibitors mediated by a RAF kinase switch in melanoma can be overcome by cotargeting MEK and IGF-1R/PI3K," Cancer Cell **18**, 683–695 (2010).

[93] R. Haq, J. Shoag, P. Andreu-Perez, S. Yokoyama, H. Edelman, G. C. Rowe, D. T. Frederick, A. D. Hurley, A. Nellore, A. L. Kung, J. A. Wargo, J. S. Song, D. E. Fisher, Z. Arany, and H. R. Widlund, "Oncogenic BRAF regulates oxidative metabolism via PGC1α and MITF," Cancer Cell **23**, 302–315 (2013).

[94] H. Shi, G. Moriceau, X. Kong, M.-K. Lee, H. Lee, R. C. Koya, C. Ng, T. Chodon, R. A. Scolyer, K. B. Dahlman, J. A. Sosman, R. F. Kefford, G. V. Long, S. F. Nelson, A. Ribas, and R. S. Lo, "Melanoma whole-exome sequencing identifies V600EB-RAF amplification-mediated acquired B-RAF inhibitor resistance," Nature Communications **3**, 724 (2012).

[95] I. Tirosh, B. Izar, S. M. Prakadan, M. H. Wadsworth, D. Treacy, J. J. Trombetta, A. Rotem, C. Rodman, C. Lian, G. Murphy, M. Fallahi-Sichani, K. Dutton-Regester, J.-R. Lin, O. Cohen, P. Shah, D. Lu, A. S. Genshaft, T. K. Hughes, C. G. K. Ziegler, S. W. Kazer, A. Gaillard, K. E. Kolb, A.-C. Villani, C. M. Johannessen, A. Y. Andreev, E. M. Van Allen, M. Bertagnolli, P. K. Sorger, R. J. Sullivan, K. T. Flaherty, D. T. Frederick, J. Jané-Valbuena, C. H. Yoon, O. Rozenblatt-Rosen, A. K. Shalek, A. Regev, and L. A. Garraway, "Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq," Science **352**, 189–196 (2016).

[96] H. Shi, W. Hugo, X. Kong, A. Hong, R. C. Koya, G. Moriceau, T. Chodon, R. Guo, D. B. Johnson, K. B. Dahlman, M. C. Kelley, R. F. Kefford, B. Chmielowski, J. A. Glaspy, J. A. Sosman, N. Van Baren, G. V. Long, A. Ribas, and R. S. Lo, "Acquired Resistance and Clonal Evolution in Melanoma during BRAF Inhibitor Therapy," Cancer Discovery **4**, 80–93 (2014).

[97] G. V. Long, D. Stroyakovskiy, H. Gogas, E. Levchenko, F. De Braud, J. Larkin, C. Garbe, T. Jouary, A. Hauschild, J. J. Grob, V. Chiarion Sileni, C. Lebbe, M. Mandalà, M. Millward, A. Arance, I. Bondarenko, J. B. Haanen, J. Hansson, J. Utikal, V. Ferraresi, N. Kovalenko, P. Mohr, V. Probachai, D. Schadendorf, P. Nathan, C. Robert, A. Ribas, D. J. DeMarini, J. G. Irani, M. Casey, D. Ouellet, A.-M. Martin, N. Le, K. Patel, and K. Flaherty, "Combined BRAF and MEK Inhibition versus BRAF Inhibition Alone in Melanoma," New England Journal of Medicine **371**, 1877–1888 (2014).

[98] J. Gallaher, M. Strobl, J. West, R. Gatenby, J. Zhang, M. Robertson-Tessi, and A. R. A. Anderson, "Intermetastatic and Intrametastatic Heterogeneity Shapes Adaptive Therapy Cycling Dynamics," Cancer Research **83**, 2775–2789 (2023).

[99] C. R. Goding, "A picture of Mitf in melanoma immortality," Oncogene **30**, 2304–2306 (2011).

[100] K. S. Hoek and C. R. Goding, "Cancer stem cells versus phenotype-switching in melanoma," Pigment Cell & Melanoma Research **23**, 746–759 (2010).

[101] H. E. Seberg, E. Van Otterloo, and R. A. Cornell, "Beyond MITF: Multiple transcription factors directly regulate the cellular phenotype in melanocytes and melanoma," Pigment Cell & Melanoma Research **30**, 454–466 (2017).

[102] S. M. Hossain and M. R. Eccles, "Phenotype Switching and the Melanoma Microenvironment; Impact on Immunotherapy and Drug Resistance," International Journal of Molecular Sciences **24**, 1601 (2023).

[103] M. Willemsen, J. Bulgarelli, S. Chauhan, R. Lereim, D. Angeli, G. Grisendi, G. Krebbers, I. Davidson, J. Kyte, M. Guidoboni, R. Luiten, and W. Bakker, "Changes in AXL and/or MITF melanoma subpopulations in patients receiving immunotherapy," Immuno-Oncology and Technology **24**, 101009 (2024).

[104] F. S. Hodi, S. J. O'Day, D. F. McDermott, R. W. Weber, J. A. Sosman, J. B. Haanen, R. Gonzalez, C. Robert, D. Schadendorf, J. C. Hassel, W. Akerley, A. J. Van Den Eertwegh, J. Lutzky, P. Lorigan, J. M. Vaubel, G. P. Linette, D. Hogg, C. H. Ottensmeier, C. Lebbé, C. Peschel, I. Quirt, J. I. Clark, J. D. Wolchok, J. S. Weber, J. Tian, M. J. Yellin, G. M. Nichol, A. Hoos, and W. J. Urba, "Improved Survival with Ipilimumab in Patients with Metastatic Melanoma," New England Journal of Medicine **363**, 711–723 (2010).

[105] C. Robert, J. Schachter, G. V. Long, A. Arance, J. J. Grob, L. Mortier, A. Daud, M. S. Carlino, C. McNeil, M. Lotem, J. Larkin, P. Lorigan, B. Neyns, C. U. Blank, O. Hamid, C. Mateus, R. Shapira-Frommer, M. Kosh, H. Zhou, N. Ibrahim, S. Ebbinghaus, and A. Ribas, "Pembrolizumab versus Ipilimumab in Advanced Melanoma," New England Journal of Medicine **372**, 2521–2532 (2015).

[106] A. Ribas and J. D. Wolchok, "Cancer immunotherapy using checkpoint blockade," Science (New York, N.Y.) **359**, 1350–1355 (2018).

[107] M. B. Atkins, S. J. Lee, B. Chmielowski, A. A. Tarhini, G. I. Cohen, T.-G. Truong, H. H. Moon, D. Davar, M. O'Rourke, J. J. Stephenson, B. D. Curti, W. J. Urba, J. M. Brell, P. Funchain, K. L. Kendra, A. P. Ikeguchi, A. Jaslowski, C. L. Bane, M. A. Taylor, M. Bajaj, R. M. Conry, R. J. Ellis, T. F. Logan, N. Laudi, J. A. Sosman, D. G. Crockett, A. L. Pecora, I. J. Okazaki, S. Reganti, S. Chandra, S. Guild, H. X. Chen, H. Z. Streicher, J. D. Wolchok, A. Ribas, and J. M. Kirkwood, "Combination Dabrafenib and Trametinib Versus Combination Nivolumab and Ipilimumab for Patients With Advanced BRAF-Mutant Melanoma: The DREAMseq Trial-ECOG-ACRIN EA6134," Journal of Clinical Oncology: Official Journal of the American Society of Clinical Oncology **41**, 186–197 (2023).

[108] R. Gutzmer, D. Stroyakovskiy, H. Gogas, C. Robert, K. Lewis, S. Protsenko, R. P. Pereira, T. Eigentler, P. Rutkowski, L. Demidov, G. M. Manikhas, Y. Yan, K.-C. Huang, A. Uyei, V. McNally, G. A. McArthur, and P. A. Ascierto, "Atezolizumab, vemurafenib, and cobimetinib as first-line treatment for unresectable advanced BRAFV600 mutation-positive melanoma (IMspire150): primary analysis of the randomised, double-blind, placebo-controlled, phase 3 trial," The Lancet **395**, 1835–1844 (2020).

[109] P. A. Ascierto, M. Casula, J. Bulgarelli, M. Pisano, C. Piccinini, L. Piccin, A. Cossu, M. Mandalà, P. F. Ferrucci, M. Guidoboni, P. Rutkowski, V. Ferraresi, A. Arance, M. Guida, E. Maiello, H. Gogas, E. Richtig, M. T. Fierro, C. Lebbe, H. Helgadottir, P. Queirolo, F. Spagnolo, M. Tucci, M. Del Vecchio, M. G. Cao, A. M. Minisini, S. De Placido, M. F. Sanmamed, D. Mallardo, M. Paone, M. G. Vitale, I. Melero, A. M. Grimaldi, D. Giannarelli, R. Dummer, V. C. Sileni, and G. Palmieri, "Sequential immunotherapy and targeted therapy for metastatic BRAF V600 mutated melanoma: 4-year survival and biomarkers evaluation from the phase II SECOMBIT trial," Nature Communications **15**, 146 (2024).

[110] L. Haas, A. Elewaut, C. L. Gerard, C. Umkehrer, L. Leiendecker, M. Pedersen, I. Krecioch, D. Hoffmann, M. Novatchkova, M. Kuttke, T. Neumann, I. P. da Silva, H. Witthock, M. A. Cuendet, S. Carotta, K. J. Harrington, J. Zuber, R. A. Scolyer, G. V. Long, J. S. Wilmott, O. Michielin, S. Vanharanta, T. Wiesner, and A. C. Obenauf, "Acquired resistance to anti-MAPK targeted therapy confers an immune-evasive tumor microenvironment and cross-resistance to immunotherapy in melanoma," Nature Cancer **2**, 693–708 (2021).

[111] S. Belkhir, F. Thomas, and B. Roche, "Darwinian Approaches for Cancer Treatment: Benefits of Mathematical Modeling," Cancers **13**, 4448 (2021).

[112] A. P. Algazi, M. Othus, A. I. Daud, R. S. Lo, J. M. Mehnert, T.-G. Truong, R. Conry, K. Kendra, G. C. Doolittle, J. I. Clark, M. J. Messino, D. F. Moore, C. Lao, B. A. Faller, R. Govindarajan, A. Harker-Murray, L. Dreisbach, J. Moon, K. F. Grossmann, and A. Ribas, "Continuous versus intermittent BRAF and MEK inhibition in patients with BRAF-mutated melanoma: a randomized phase 2 trial," Nature Medicine **26**, 1564–1568 (2020).

[113] J. Chahoud, A. R. A. Anderson, J. Zhang, J. Brown, and R. A. Gatenby, "Evolutionary Dynamics and Intermittent Therapy for Metastatic Cancers," Journal of Clinical Oncology: Official Journal of the American Society of Clinical Oncology **41**, 4469–4471 (2023).

[114] M. Hussain, C. M. Tangen, D. L. Berry, C. S. Higano, E. D. Crawford, G. Liu, G. Wilding, S. Prescott, S. Kanaga Sundaram, E. J. Small, N. A. Dawson, B. J. Donnelly, P. M. Venner, U. N. Vaishampayan, P. F. Schellhammer, D. I. Quinn, D. Raghavan, B. Ely, C. M. Moinpour, N. J. Vogelzang, and I. M. Thompson, "Intermittent versus Continuous Androgen Deprivation in Prostate Cancer," New England Journal of Medicine **368**, 1314–1325 (2013).

[115] M. A. Swartz and A. W. Lund, "Lymphatic and interstitial flow in the tumour microenvironment: linking mechanobiology with immunity," Nature Reviews. Cancer **12**, 210–219 (2012).

[116] K. Bacevic, R. Noble, A. Soffar, O. Wael Ammar, B. Boszonyik, S. Prieto, C. Vincent, M. E. Hochberg, L. Krasinska, and D. Fisher, "Spatial competition constrains resistance to targeted cancer therapy," Nature Communications **8**, 1995 (2017).

[117] J. A. Gallaher, J. S. Brown, and A. R. A. Anderson, "The impact of proliferation-migration tradeoffs on phenotypic evolution in cancer," Scientific Reports **9**, 2425 (2019).

[118] L. Imamovic and M. O. A. Sommer, "Use of Collateral Sensitivity Networks to Design Drug Cycling Protocols That Avoid Resistance Development," Science Translational Medicine **5**, 10.1126/scitranslmed.3006609 (2013).

[119] D. I. Andersson and D. Hughes, "Antibiotic resistance and its cost: is it possible to reverse resistance?" Nature Reviews Microbiology **8**, 260–271 (2010).

[120] B. E. Tabashnik and B. A. Croft, "Managing Pesticide Resistance in Crop-Arthropod Complexes: Interactions Between Biological and Operational Factors," Environmental Entomology **11**, 1137–1144 (1982).

[121] Y. Carrière, C. Ellers-Kirk, K. Hartfield, G. Larocque, B. Degain, P. Dutilleul, T. J. Dennehy, S. E. Marsh, D. W. Crowder, X. Li, P. C. Ellsworth, S. E. Naranjo, J. C. Palumbo, A. Fournier, L. Antilla, and B. E. Tabashnik, "Large-scale, spatially-explicit test of the refuge strategy for delaying insecticide resistance," Proceedings of the National Academy of Sciences **109**, 775–780 (2012).

[122] C. Danel, R. Moh, A. Minga, A. Anzian, O. Ba-Gomis, C. Kanga, G. Nzunetu, D. Gabillard, F. Rouet, S. Sorho, M.-L. Chaix, S. Eholié, H. Menan, D. Sauvageot, E. Bissagnene, R. Salamon, X. Anglaret, and Trivacan ANRS 1269 trial group, "CD4-guided structured antiretroviral treatment interruption strategy in HIV-infected adults in west Africa (Trivacan ANRS 1269 trial): a randomised trial," Lancet (London, England) **367**, 1981–1989 (2006).

[123] H. P. Greenspan, "Models for the Growth of a Solid Tumor by Diffusion," Studies in Applied Mathematics **51**, 317–340 (1972).

[124] J. A. Sherratt and M. A. Nowak, "Oncogenes, anti-oncogenes and the immune response to cancer: a mathematical model," Proceedings. Biological Sciences **248**, 261–271 (1992).

[125] P. Gerlee, "The Model Muddle: In Search of Tumor Growth Laws," Cancer Research **73**, 2407–2411 (2013).

[126] Z. Wang, J. D. Butner, R. Kerketta, V. Cristini, and T. S. Deisboeck, "Simulating cancer growth with multiscale agent-based modeling," Seminars in Cancer Biology **30**, 70–78 (2015).

[127] K. Gallagher, M. A. R. Strobl, D. S. Park, F. C. Spoendlin, R. A. Gatenby, P. K. Maini, and A. R. A. Anderson, "Mathematical Model-Driven Deep Learning Enables Personalized Adaptive Therapy," Cancer Research **84**, 1929–1941 (2024).

[128] J. West, M. Robertson-Tessi, and A. R. A. Anderson, "Agent-based methods facilitate integrative science in cancer," Trends in Cell Biology **33**, 300–311 (2023).

[129] B. Colyer, M. Bak, D. Basanta, and R. Noble, "A seven-step guide to spatial, agent-based modelling of tumour evolution," Evolutionary Applications **17**, e13687 (2024).

[130] M. Basan, J. Prost, J.-F. Joanny, and J. Elgeti, "Dissipative particle dynamics simulations for biological tissues: rheology and competition," Physical Biology **8**, 026014 (2011).

[131] P. Español and P. B. Warren, "Perspective: Dissipative particle dynamics," The Journal of Chemical Physics **146**, 150901 (2017).

[132] S. Kaliman, M. Hubert, C. Wollnik, L. Nuić, D. Vurnek, S. Gehrer, J. Lovrić, D. Dudziak, F. Rehfeldt, and A.-S. Smith, "Mechanical Regulation of Epithelial Tissue Homeostasis," Physical Review X **11**, 031029 (2021).

[133] K. Höllring, L. Nuić, L. Rogić, S. Kaliman, S. Gehrer, C. Wollnik, F. Rehfeldt, M. Hubert, and A.-S. Smith, "Capturing the mechanosensitivity of cell proliferation in models of epithelium," Proceedings of the National Academy of Sciences of the United States of America **121**, e2308126121 (2024).

[134] A. J. Lotka, *Elements of Physical Biology* (Williams & Wilkins, Baltimore, 1925).

[135] V. Volterra, "Variations and Fluctuations of the Number of Individuals in Animal Species living together," ICES Journal of Marine Science **3**, 3–51 (1928).

[136] L.-B. Weiswald, D. Bellet, and V. Dangles-Marie, "Spherical cancer models in tumor biology," Neoplasia (New York, N.Y.) **17**, 1–15 (2015).

[137] M. Hidalgo, F. Amant, A. V. Biankin, E. Budinská, A. T. Byrne, C. Caldas, R. B. Clarke, S. de Jong, J. Jonkers, G. M. Mælandsmo, S. Roman-Roman, J. Seoane, L. Trusolino, and A. Villanueva, "Patient-derived xenograft models: an emerging platform for translational cancer research," Cancer Discovery **4**, 998–1013 (2014).

[138] R. W. Jenkins, A. R. Aref, P. H. Lizotte, E. Ivanova, S. Stinson, C. W. Zhou, M. Bowden, J. Deng, H. Liu, D. Miao, M. X. He, W. Walker, G. Zhang, T. Tian, C. Cheng, Z. Wei, S. Palakurthi, M. Bittinger, H. Vitzthum, J. W. Kim, A. Merlino, M. Quinn, C. Venkataramani, J. A. Kaplan, A. Portell, P. C. Gokhale, B. Phillips, A. Smart, A. Rotem, R. E. Jones, L. Keogh, M. Anguiano, L. Stapleton, Z. Jia, M. Barzily-Rokni, I. Cañadas, T. C. Thai, M. R. Hammond, R. Vlahos, E. S. Wang, H. Zhang, S. Li, G. J. Hanna, W. Huang, M. P. Hoang, A. Piris, J.-P. Eliane, A. O. Stemmer-Rachamimov, L. Cameron, M.-J. Su, P. Shah, B. Izar, M. Thakuria, N. R. LeBoeuf, G. Rabinowits, V. Gunda, S. Parangi, J. M. Cleary, B. C. Miller, S. Kitajima, R. Thummalapalli, B. Miao, T. U. Barbie, V. Sivathanu, J. Wong, W. G.

Richards, R. Bueno, C. H. Yoon, J. Miret, M. Herlyn, L. A. Garraway, E. M. Van Allen, G. J. Freeman, P. T. Kirschmeier, J. H. Lorch, P. A. Ott, F. S. Hodi, K. T. Flaherty, R. D. Kamm, G. M. Boland, K.-K. Wong, D. Dornan, C. P. Paweletz, and D. A. Barbie, "*Ex Vivo* Profiling of PD-1 Blockade Using Organotypic Tumor Spheroids," Cancer Discovery **8**, 196–215 (2018).

[139] M. Vinci, S. Gowan, F. Boxall, L. Patterson, M. Zimmermann, W. Court, C. Lomas, M. Mendiola, D. Hardisson, and S. A. Eccles, "Advances in establishment and analysis of three-dimensional tumor spheroid-based functional assays for target validation and drug evaluation," BMC biology **10**, 29 (2012).

[140] D. Antoni, H. Burckel, E. Josset, and G. Noel, "Three-Dimensional Cell Culture: A Breakthrough in Vivo," International Journal of Molecular Sciences **16**, 5517–5527 (2015).

[141] H.-e. C. Bhang, D. A. Ruddy, V. Krishnamurthy Radhakrishna, J. X. Caushi, R. Zhao, M. M. Hims, A. P. Singh, I. Kao, D. Rakiec, P. Shaw, M. Balak, A. Raza, E. Ackley, N. Keen, M. R. Schlabach, M. Palmer, R. J. Leary, D. Y. Chiang, W. R. Sellers, F. Michor, V. G. Cooke, J. M. Korn, and F. Stegmeier, "Studying clonal dynamics in response to cancer therapy using high-complexity barcoding," Nature Medicine **21**, 440–448 (2015).

[142] J. Livet, T. A. Weissman, H. Kang, R. W. Draft, J. Lu, R. A. Bennis, J. R. Sanes, and J. W. Lichtman, "Transgenic strategies for combinatorial expression of fluorescent proteins in the nervous system," Nature **450**, 56–62 (2007).

[143] D. S. Bindels, L. Haarbosch, L. Van Weeren, M. Postma, K. E. Wiese, M. Mastop, S. Aumonier, G. Gotthard, A. Royant, M. A. Hink, and T. W. J. Gadella, "mScarlet: a bright monomeric red fluorescent protein for cellular imaging," Nature Methods **14**, 53–56 (2017).

[144] M. Dalvai, J. Loehr, K. Jacquet, C. C. Huard, C. Roques, P. Herst, J. Côté, and Y. Doyon, "A Scalable Genome-Editing-Based Approach for Mapping Multiprotein Complexes in Human Cells," Cell Reports **13**, 621–633 (2015).

[145] F. Oceguera-Yanez, S.-I. Kim, T. Matsumoto, G. W. Tan, L. Xiang, T. Hatani, T. Kondo, M. Ikeya, Y. Yoshida, H. Inoue, and K. Woltjen, "Engineering the AAVS1 locus for consistent and scalable transgene expression in human iPSCs and their differentiated derivatives," Methods (San Diego, Calif.) **101**, 43–55 (2016).

[146] R. C. DeKelver, V. M. Choi, E. A. Moehle, D. E. Paschon, D. Hockemeyer, S. H. Meijsing, Y. Sancak, X. Cui, E. J. Steine, J. C. Miller, P. Tam, V. V. Bartsevich, X. Meng, I. Rupniewski, S. M. Gopalan, H. C. Sun, K. J. Pitz, J. M. Rock, L. Zhang, G. D. Davis, E. J. Rebar, I. M. Cheeseman, K. R. Yamamoto, D. M. Sabatini, R. Jaenisch, P. D. Gregory, and F. D. Urnov, "Functional genomics, proteomics, and regulatory DNA analysis in isogenic settings using zinc finger nuclease-driven transgenesis into a safe harbor locus in the human genome," Genome Research **20**, 1133–1142 (2010).

[147] R. Feil, J. Brocard, B. Mascrez, M. LeMeur, D. Metzger, and P. Chambon, "Ligand-activated site-specific recombination in mice," Proceedings of the National Academy of Sciences of the United States of America **93**, 10887–10890 (1996).

[148] L. A. Gilbert, M. H. Larson, L. Morsut, Z. Liu, G. A. Brar, S. E. Torres, N. Stern-Ginossar, O. Brandman, E. H. Whitehead, J. A. Doudna, W. A. Lim, J. S. Weissman, and L. S. Qi, "CRISPR-mediated modular RNA-guided regulation of transcription in eukaryotes," Cell **154**, 442–451 (2013).

[149] A. Chavez, J. Scheiman, S. Vora, B. W. Pruitt, M. Tuttle, E. P R Iyer, S. Lin, S. Kiani, C. D. Guzman, D. J. Wiegand, D. Ter-Ovanesyan, J. L. Braff, N. Davidsohn, B. E. Housden, N. Perrimon, R. Weiss, J. Aach, J. J. Collins, and G. M. Church, "Highly efficient Cas9-mediated transcriptional programming," Nature Methods **12**, 326–328 (2015).

[150] K. Paludan, M. Duch, P. Jørgensen, N. O. Kjeldgaard, and F. S. Pedersen, "Graduated resistance to G418 leads to differential selection of cultured mammalian cells expressing the neo gene," Gene **85**, 421–426 (1989).

[151] S. Bar-Nun, Y. Shneyour, and J. S. Beckmann, "G-418, an elongation inhibitor of 80 S ribosomes," Biochimica Et Biophysica Acta **741**, 123–127 (1983).

[152] M. Poujade, E. Grasland-Mongrain, A. Hertzog, J. Jouanneau, P. Chavrier, B. Ladoux, A. Buguin, and P. Silberzan, "Collective migration of an epithelial monolayer in response to a model wound," Proceedings of the National Academy of Sciences **104**, 15988–15993 (2007).

[153] W. Xi, S. Sonam, T. Beng Saw, B. Ladoux, and C. Teck Lim, "Emergent patterns of collective cell migration under tubular confinement," Nature Communications **8**, 1517 (2017).

[154] C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu, "Cellpose: a generalist algorithm for cellular segmentation," Nature Methods **18**, 100–106 (2021).

[155] D. R. Grimes, C. Kelly, K. Bloch, and M. Partridge, "A method for estimating the oxygen consumption rate in multicellular tumour spheroids," Journal of the Royal Society, Interface **11**, 20131124 (2014).

[156] L. Wullkopf, A.-K. V. West, N. Leijnse, T. R. Cox, C. D. Madsen, L. B. Oddershede, and J. T. Erler, "Cancer cells' ability to mechanically adjust to extracellular matrix stiffness correlates with their invasive potential," Molecular Biology of the Cell **29**, edited by D. Discher, 2378–2385 (2018).

[157] J. Sievers, V. Mahajan, P. B. Welzel, C. Werner, and A. Taubenberger, "Precision Hydrogels for the Study of Cancer Cell Mechanobiology," Advanced Healthcare Materials **12**, e2202514 (2023).

[158] B. Olson, Y. Li, Y. Lin, E. T. Liu, and A. Patnaik, "Mouse Models for Cancer Immunotherapy Research," Cancer Discovery **8**, 1358–1365 (2018).

[159] H. Hockings, E. Lakatos, W. Huang, M. Mossner, M. A. Khan, S. Metcalf, F. Nicolini, K. Smith, A.-M. Baker, T. A. Graham, and M. Lockley, "Adaptive therapy achieves long-term control of chemotherapy resistance in high grade ovarian cancer," bioRxiv: The Preprint Server for Biology, 2023.07.21.549688 (2023).

[160] D. Deb, S. Zhu, M. J. LeBlanc, and T. Danino, "Assessing chemotherapy dosing strategies in a spatial cell culture model," Frontiers in Oncology **12**, 980770 (2022).

[161] *Adaptive BRAF-MEK Inhibitor Therapy for Advanced BRAF Mutant Melanoma (NCT03543969).*

[162] U. A. Mukherjee, H. Hockings, N. Counsell, A. Patel, P. Narayanan, K. Wilkinson, H. Dhanda, K. Robinson, I. McNeish, A. R. A. Anderson, R. Miller, C. Gourley, T. Graham, and M. Lockley, "Study protocol for Adaptive ChemoTherapy for Ovarian cancer (ACTOv): a multicentre phase II randomised controlled trial to evaluate the efficacy of adaptive therapy (AT) with carboplatin, based on changes in CA125, in patients with relapsed platinum-sensitive high-grade serous or high-grade endometrioid ovarian cancer," BMJ Open **14**, e091262 (2024).

[163] T. O. McDonald, Y.-C. Cheng, C. Graser, P. B. Nicol, D. Temko, and F. Michor, "Computational approaches to modelling and optimizing cancer treatment," Nature Reviews Bioengineering **1**, 695–711 (2023).

[164] M. A. R. Strobl, J. West, Y. Viossat, M. Damaghi, M. Robertson-Tessi, J. S. Brown, R. A. Gatenby, P. K. Maini, and A. R. A. Anderson, "Turnover Modulates the Need for a Cost of Resistance in Adaptive Therapy," Cancer Research **81**, 1135–1147 (2021).

[165] K. S. Korolev, J. B. Xavier, and J. Gore, "Turning ecology and evolution against cancer," Nature Reviews Cancer **14**, 371–380 (2014).

[166] J. West, L. You, J. Zhang, R. A. Gatenby, J. S. Brown, P. K. Newton, and A. R. A. Anderson, "Towards Multidrug Adaptive Therapy," Cancer Research **80**, 1578–1589 (2020).

[167] F. Rambow, A. Rogiers, O. Marin-Bejar, S. Aibar, J. Femel, M. Dewaele, P. Karras, D. Brown, Y. H. Chang, M. Debiec-Rychter, C. Adriaens, E. Radaelli, P. Wolter, O. Bechter, R. Dummer, M. Levesque, A. Piris, D. T. Frederick, G. Boland, K. T. Flaherty, J. Van Den Oord, T. Voet, S. Aerts, A. W. Lund, and J.-C. Marine, "Toward Minimal Residual Disease-Directed Therapy in Melanoma," Cell **174**, 843–855.e19 (2018).

[168] I. Smalley, E. Kim, J. Li, P. Spence, C. J. Wyatt, Z. Eroglu, V. K. Sondak, J. L. Messina, N. A. Babacan, S. S. Maria-Engler, L. De Armas, S. L. Williams, R. A. Gatenby, Y. A. Chen, A. R. A. Anderson, and K. S. M. Smalley, "Leveraging transcriptional dynamics to improve BRAF inhibitor responses in melanoma," EBioMedicine **48**, 178–190 (2019).

[169] D. Lau, P. G. Corrie, and F. A. Gallagher, "MRI techniques for immunotherapy monitoring," Journal for Immunotherapy of Cancer **10**, e004708 (2022).

[170] T. Persigehl, S. Lennartz, and L. H. Schwartz, "iRECIST: how to do it," Cancer Imaging: The Official Publication of the International Cancer Imaging Society **20**, 2 (2020).

[171] C. Parisi, M. Tagliamento, L. Belcaid, M. Aldea, A. Bayle, J. Remon-Masip, A. Italiano, D. Planchard, B. Besse, and F. Barlesi, "Circulating tumor DNA in clinical trials for solid tumors: Challenges and current applications," The Journal of Liquid Biopsy **1**, 100007 (2023).

[172] S. Ahmed and C. G. Loiselle, "Patient Adherence to Oral Anticancer Agents: A Mapping Review of Supportive Interventions," Current Oncology **30**, 10224–10236 (2023).

[173] J.-N. Eckardt, K. Wendt, M. Bornhäuser, and J. M. Middeke, "Reinforcement Learning for Precision Oncology," Cancers **13**, 4624 (2021).

[174] M. Murtaza, S.-J. Dawson, D. W. Y. Tsui, D. Gale, T. Forshew, A. M. Piskorz, C. Parkinson, S.-F. Chin, Z. Kingsbury, A. S. C. Wong, F. Marass, S. Humphray, J. Hadfield, D. Bentley, T. M. Chin, J. D. Brenton, C. Caldas, and N. Rosenfeld, "Non-invasive analysis of acquired resistance to cancer therapy by sequencing of plasma DNA," Nature **497**, 108–112 (2013).

[175] L. Kolla, F. K. Gruber, O. Khalid, C. Hill, and R. B. Parikh, "The case for AI-driven cancer clinical trials - The efficacy arm in silico," Biochimica Et Biophysica Acta. Reviews on Cancer **1876**, 188572 (2021).

[176] D. Barbolosi, J. Ciccolini, B. Lacarelle, F. Barlési, and N. André, "Computational oncology–mathematical modelling of drug regimens for precision medicine," Nature Reviews. Clinical Oncology **13**, 242–254 (2016).

[177] D. Zheng, K. Preuss, M. T. Milano, X. He, L. Gou, Y. Shi, B. Marples, R. Wan, H. Yu, H. Du, and C. Zhang, "Mathematical modeling in radiotherapy for cancer: a comprehensive narrative review," Radiation Oncology **20**, 49 (2025).

[178] A. S. Darwich, K. Ogungbenr, O. J. Hatley, and A. Rostami-Hodjegan, "Role of pharmacokinetic modeling and simulation in precision dosing of anticancer drugs," Translational Cancer Research **6**, S1512–S1529 (2017).

[179] A. Soboleva, I. Grossmann, A.-M. C. Dingemans, J. Rezaei, and K. Staňková, "Bringing evolutionary cancer therapy to the clinic: a systems approach," npj Systems Biology and Applications **11**, 56 (2025).

[180] P. M. Altrock, L. L. Liu, and F. Michor, "The mathematics of cancer: integrating quantitative models," Nature Reviews. Cancer **15**, 730–745 (2015).

[181] A. S. Kazerouni, M. Gadde, A. Gardner, D. A. Hormuth, A. M. Jarrett, K. E. Johnson, E. A. B. F. Lima, G. Lorenzo, C. Phillips, A. Brock, and T. E. Yankeelov, "Integrating Quantitative Assays with Biologically Based Mathematical Modeling for Predictive Oncology," iScience **23**, 101807 (2020).

[182] A. Silva and N. Vale, "Digital Twins in Personalized Medicine: Bridging Innovation and Clinical Reality," Journal of Personalized Medicine **15**, 503 (2025).

[183] C. Wu, G. Lorenzo, D. A. Hormuth, E. A. B. F. Lima, K. P. Slavkova, J. C. DiCarlo, J. Virostko, C. M. Phillips, D. Patt, C. Chung, and T. E. Yankeelov, "Integrating mechanism-based modeling with biomedical imaging to build practical digital twins for clinical oncology," Biophysics Reviews **3**, 021304 (2022).

[184] D. A. Hormuth, J. A. Weis, S. L. Barnes, M. I. Miga, E. C. Rericha, V. Quaranta, and T. E. Yankeelov, "Predicting in vivo glioma growth with the reaction diffusion equation constrained by quantitative magnetic resonance imaging data," Physical Biology **12**, 046006 (2015).

[185] A. S. Darwich, T. M. Polasek, J. K. Aronson, K. Ogungbenro, D. F. B. Wright, B. Achour, J.-L. Reny, Y. Daali, B. Eiermann, J. Cook, L. Lesko, A. J. McLachlan, and A. Rostami-Hodjegan, "Model-Informed Precision Dosing: Background, Requirements, Validation, Implementation, and Forward Trajectory of Individualizing Drug Therapy," Annual Review of Pharmacology and Toxicology **61**, 225–245 (2021).

[186] B. C. Agema, B. C. P. Koch, R. H. J. Mathijssen, and S. L. W. Koolen, "From Prospective Evaluation to Practice: Model-Informed Dose Optimization in Oncology," Drugs **85**, 487–503 (2025).

[187] C.-Y. Yang, C. Shiranthika, C.-Y. Wang, K.-W. Chen, and S. Sumathipala, "Reinforcement learning strategies in cancer chemotherapy treatments: A review," Computer Methods and Programs in Biomedicine **229**, 107280 (2023).

[188] B. Eastman, M. Przedborski, and M. Kohandel, "Reinforcement learning derived chemotherapeutic schedules for robust patient-specific therapy," Scientific Reports **11**, 17882 (2021).

[189] A. N. Hata, M. J. Niederst, H. L. Archibald, M. Gomez-Caraballo, F. M. Siddiqui, H. E. Mulvey, Y. E. Maruvka, F. Ji, H.-e. C. Bhang, V. Krishnamurthy Radhakrishna, G. Siravegna, H. Hu, S. Raoof, E. Lockerman, A. Kalsy, D. Lee, C. L. Keating, D. A. Ruddy, L. J. Damon, A. S. Crystal, C. Costa, Z. Piotrowska, A. Bardelli, A. J. Iafrate, R. I. Sadreyev, F. Stegmeier, G. Getz, L. V. Sequist, A. C. Faber, and J. A. Engelman, "Tumor cells can follow distinct evolutionary paths to become resistant to epidermal growth factor receptor inhibition," Nature Medicine **22**, 262–269 (2016).

[190] Strategies for Management of Antiretroviral Therapy (SMART) Study Group, W. M. El-Sadr, J. D. Lundgren, J. D. Neaton, F. Gordin, D. Abrams, R. C. Arduino, A. Babiker, W. Burman, N. Clumeck, C. J. Cohen, D. Cohn, D. Cooper, J. Darbyshire, S. Emery, G. Fätkenheuer, B. Gazzard, B. Grund, J. Hoy, K. Klingman, M. Losso, N. Markowitz, J. Neuhaus, A. Phillips, and C. Rappoport, "CD4+ count-guided interruption of antiretroviral treatment," The New England Journal of Medicine **355**, 2283–2296 (2006).

[191] Y. Viossat and R. Noble, "A theoretical analysis of tumour containment," Nature Ecology & Evolution **5**, 826–835 (2021).

[192] F. Markowetz, "All models are wrong and yours are useless: making clinical prediction models impactful for patients," npj Precision Oncology **8**, 54 (2024).

[193] P. Bayer, J. S. Brown, J. Dubbeldam, and M. Broom, "A Markovian decision model of adaptive cancer treatment and quality of life," Journal of Theoretical Biology **551–552**, 111237 (2022).

[194] M. M. T. Kuijpers, H. van Veenendaal, V. Engelen, E. Visserman, E. A. Noteboom, A. M. Stiggelbout, A. M. May, N. de Wit, E. van der Wall, and C. W. Helsper, "Shared decision making in cancer treatment: A Dutch national survey on patients' preferences and perceptions," European Journal of Cancer Care **31**, e13534 (2022).

[195] M. Sadelain, E. P. Papapetrou, and F. D. Bushman, "Safe harbours for the integration of new DNA in the human genome," Nature Reviews Cancer **12**, 51–58 (2012).

[196] X. Chen, J. Du, S. Yun, C. Xue, Y. Yao, and S. Rao, "Recent advances in CRISPR-Cas9-based genome insertion technologies," Molecular Therapy. Nucleic Acids **35**, 102138 (2024).

[197] C. J. A. Ramachandra, M. Shahbazi, T. W. X. Kwang, Y. Choudhury, X. Y. Bak, J. Yang, and S. Wang, "Efficient recombinase-mediated cassette exchange at the AAVS1 locus in human embryonic stem cells using baculoviral vectors," Nucleic Acids Research **39**, e107 (2011).

[198] I. M. Slaymaker, L. Gao, B. Zetsche, D. A. Scott, W. X. Yan, and F. Zhang, "Rationally engineered Cas9 nucleases with improved specificity," Science **351**, 84–88 (2016).

[199] B. P. Cormack, R. H. Valdivia, and S. Falkow, "FACS-optimized mutants of the green fluorescent protein (GFP)," Gene **173**, 33–38 (1996).

[200] N. C. Shaner, G. G. Lambert, A. Chammas, Y. Ni, P. J. Cranfill, M. A. Baird, B. R. Sell, J. R. Allen, R. N. Day, M. Israelsson, M. W. Davidson, and J. Wang, "A bright monomeric green fluorescent protein derived from Branchiostoma lanceolatum," Nature Methods **10**, 407–409 (2013).

[201] A. Fabritius, D. Ng, A. M. Kist, M. Erdogan, R. Portugues, and O. Griesbeck, "Imaging-Based Screening Platform Assists Protein Engineering," Cell Chemical Biology **25**, 1554–1561.e8 (2018).

[202] V. Haellman, T. Strittmatter, A. Bertschi, P. Stücheli, and M. Fussenegger, "A versatile plasmid architecture for mammalian synthetic biology (VAMSyB)," Metabolic Engineering **66**, 41–50 (2021).

[203] H. Zeng, A. Jorapur, A. H. Shain, U. E. Lang, R. Torres, Y. Zhang, A. S. McNeal, T. Botton, J. Lin, M. Donne, I. N. Bastian, R. Yu, J. P. North, L. Pincus, B. S. Ruben, N. M. Joseph, I. Yeh, B. C. Bastian, and R. L. Judson, "Bi-allelic Loss of CDKN2A Initiates Melanoma Invasion via BRN2 Activation," Cancer Cell **34**, 56–68.e9 (2018).

[204] G. V. Long, C. Fung, A. M. Menzies, G. M. Pupo, M. S. Carlino, J. Hyman, H. Shahheydari, V. Tembe, J. F. Thompson, R. P. Saw, J. Howle, N. K. Hayward, P. Johansson, R. A. Scolyer, R. F. Kefford, and H. Rizos, "Increased MAPK reactivation in early resistance to dabrafenib/trametinib combination therapy of BRAF-mutant metastatic melanoma," Nature Communications **5**, 5694 (2014).

[205] P. I. Poulikakos, C. Zhang, G. Bollag, K. M. Shokat, and N. Rosen, "RAF inhibitors transactivate RAF dimers and ERK signalling in cells with wild-type BRAF," Nature **464**, 427–430 (2010).

[206] A. O. Chertkova, M. Mastop, M. Postma, N. Van Bommel, S. Van Der Niet, K. L. Batenburg, L. Joosen, T. W. Gadella, Y. Okada, and J. Goedhart, *Robust and Bright Genetically Encoded Fluorescent Markers for Highlighting Structures and Compartments in Mammalian Cells*, 2017.

[207] A. Shang, S. Shao, L. Zhao, and B. Liu, "Far-Red Fluorescent Proteins: Tools for Advancing In Vivo Imaging," Biosensors **14**, 359 (2024).

[208] Y. W. Jun, H. R. Kim, Y. J. Reo, M. Dai, and K. H. Ahn, "Addressing the autofluorescence issue in deep tissue imaging by two-photon microscopy: the significance of far-red emitting dyes," Chem. Sci. **8**, 7696–7704 (2017).

[209] J. Li, C. A. Koczor, K. M. Saville, F. Hayat, A. Beiser, S. McClellan, M. E. Migaud, and R. W. Sobol, "Overcoming Temozolomide Resistance in Glioblastoma via Enhanced NAD+ Bioavailability and Inhibition of Poly-ADP-Ribose Glycohydrolase," Cancers **14**, 3572 (2022).

[210] H. A. Tawbi, L. Villaruz, A. Tarhini, S. Moschos, M. Sulecki, F. Viverette, J. Shipe-Spotloe, R. Radkowski, and J. M. Kirkwood, "Inhibition of DNA repair with MGMT pseudosubstrates: phase I study of lomeguatrib in combination with dacarbazine in patients with advanced melanoma and other solid tumours," British Journal of Cancer **105**, 773–777 (2011).

[211] M. Esteller, J. Garcia-Foncillas, E. Andion, S. N. Goodman, O. F. Hidalgo, V. Vanaclocha, S. B. Baylin, and J. G. Herman, "Inactivation of the DNA-Repair Gene *MGMT* and the Clinical Response of Gliomas to Alkylating Agents," New England Journal of Medicine **343**, 1350–1354 (2000).

[212] E. R. Szelenyi, J. S. Navarrete, A. D. Murry, Y. Zhang, K. S. Girven, L. Kuo, M. M. Cline, M. X. Bernstein, M. Burdyniuk, B. Bowler, N. L. Goodwin, B. Juarez, L. S. Zweifel, and S. A. Golden, "An arginine-rich nuclear localization signal (ArgiNLS) strategy for streamlined image segmentation of single-cells," bioRxiv: The Preprint Server for Biology, 2023.11.22.568319 (2023).

[213] A. M. Haugh, A. K. S. Salama, and D. B. Johnson, "Advanced Melanoma: Resistance Mechanisms to Current Therapies," Hematology/Oncology Clinics of North America **35**, 111–128 (2021).

[214] S. Liu, P. Dharanipragada, S. H. Lomeli, Y. Wang, X. Zhang, Z. Yang, R. J. Lim, C. Dumitras, P. O. Scumpia, S. M. Dubinett, G. Moriceau, D. B. Johnson, S. J. Moschos, and R. S. Lo, "Multi-organ landscape of therapy-resistant melanoma," Nature Medicine **29**, 1123–1134 (2023).

[215] Y. Iwasa, M. A. Nowak, and F. Michor, "Evolution of resistance during clonal expansion," Genetics **172**, 2557–2566 (2006).

[216] A. Acar, D. Nichol, J. Fernandez-Mateos, G. D. Cresswell, I. Barozzi, S. P. Hong, N. Trahearn, I. Spiteri, M. Stubbs, R. Burke, A. Stewart, G. Caravagna, B. Werner, G. Vlachogiannis, C. C. Maley, L. Magnani, N. Valeri, U. Banerji, and A. Sottoriva, "Exploiting evolutionary steering to induce collateral drug sensitivity in cancer," Nature Communications **11**, 1923 (2020).

[217] X. Tian and B. Zhou, "Strategies for site-specific recombination with high efficiency and precise spatiotemporal resolution," The Journal of Biological Chemistry **296**, 100509 (2021).

[218] K. Liu, H. Jin, and B. Zhou, "Genetic lineage tracing with multiple DNA recombinases: A user's guide for conducting more precise cell fate mapping studies," The Journal of Biological Chemistry **295**, 6413–6424 (2020).

[219] D. Shcherbo, C. S. Murphy, G. V. Ermakova, E. A. Solovieva, T. V. Chepurnykh, A. S. Shcheglov, V. V. Verkhusha, V. Z. Pletnev, K. L. Hazelwood, P. M. Roche, S. Lukyanov, A. G. Zaraisky, M. W. Davidson, and D. M. Chudakov, "Far-red fluorescent tags for protein imaging in living tissues," Biochemical Journal **418**, 567–574 (2009).

[220] K. Zhou, Y. Ding, I. Vuletic, Y. Tian, J. Li, J. Liu, Y. Huang, H. Sun, C. Li, Q. Ren, and Y. Lu, "In vivo long-term investigation of tumor bearing mKate2 by an in-house fluorescence molecular imaging system," BioMedical Engineering OnLine **17**, 187 (2018).

[221] A. Fus-Kujawa, P. Prus, K. Bajdak-Rusinek, P. Teper, K. Gawron, A. Kowalczuk, and A. L. Sieron, "An Overview of Methods and Tools for Transfection of Eukaryotic Cells in vitro," Frontiers in Bioengineering and Biotechnology **9**, 701031 (2021).

[222] M. Pachitariu and C. Stringer, "Cellpose 2.0: how to train your own model," Nature Methods **19**, 1634–1641 (2022).

[223] C. Stringer and M. Pachitariu, "Cellpose3: one-click image restoration for improved cellular segmentation," Nature Methods **22**, 592–599 (2025).

[224] J. Devany, M. J. Falk, L. J. Holt, A. Murugan, and M. L. Gardel, "Epithelial tissue confinement inhibits cell growth and leads to volume-reducing divisions," Developmental Cell **58**, 1462–1476.e8 (2023).

[225] N. Podewitz, F. Jülicher, G. Gompper, and J. Elgeti, "Interface dynamics of competing tissues," New Journal of Physics **18**, 083020 (2016).

[226] M. Bondarenko, M. Le Grand, Y. Shaked, Z. Raviv, G. Chapuisat, C. Carrère, M.-P. Montero, M. Rossi, E. Pasquier, M. Carré, and N. André, "Metronomic Chemotherapy Modulates Clonal Interactions to Prevent Drug Resistance in Non-Small Cell Lung Cancer," Cancers **13**, 2239 (2021).

[227] R. M. Sutherland, "Cell and environment interactions in tumor microregions: the multicell spheroid model," Science (New York, N.Y.) **240**, 177–184 (1988).

[228] J. P. Freyer and R. M. Sutherland, "Regulation of growth saturation and development of necrosis in EMT6/Ro multicellular spheroids by the glucose and oxygen supply," Cancer Research **46**, 3504–3512 (1986).

[229] M. J. Lerman, J. Lembong, S. Muramoto, G. Gillen, and J. P. Fisher, "The Evolution of Polystyrene as a Cell Culture Material," Tissue Engineering Part B: Reviews **24**, 359–372 (2018).

[230] K. M. Lonkwic, R. Zajdel, and K. Kaczka, "Unlocking the Potential of Spheroids in Personalized Medicine: A Systematic Review of Seeding Methodologies," International Journal of Molecular Sciences **26**, 6478 (2025).

[231] C. Xing, A. Kemas, E. Mickols, K. Klein, P. Artursson, and V. M. Lauschke, "The choice of ultra-low attachment plates impacts primary human and primary canine hepatocyte spheroid formation, phenotypes, and function," Biotechnology Journal **19**, 2300587 (2024).

[232] D. Khaitan, S. Chandna, M. B. Arya, and B. S. Dwarakanath, "Establishment and characterization of multicellular spheroids from a human glioma cell line; Implications for tumor therapy," Journal of Translational Medicine **4**, 12 (2006).

[233] H. Karimi, P. Moskal, A. Żak, and E. Ł. Stępień, "3D melanoma spheroid model for the development of positronium biomarkers," Scientific Reports **13**, 7648 (2023).

[234] H. Ohguro, M. Watanabe, T. Sato, F. Hikage, M. Furuhashi, M. Okura, T. Hida, and H. Uhara, "3D Spheroid Configurations Are Possible Indictors for Evaluating the Pathophysiology of Melanoma Cell Lines," Cells **12**, 759 (2023).

[235] D. A. Fletcher, "Bottom-Up Biology: Harnessing Engineering to Understand Nature," Developmental Cell **38**, 587–589 (2016).

[236] K. Kataoka and N. Huh, "Application of a thermo-reversible gelation polymer, mebiol gel, for stem cell culture and regenerative medicine," Journal of Stem Cells & Regenerative Medicine **6**, 10–14 (2010).

[237] K. M. Charoen, B. Fallica, Y. L. Colson, M. H. Zaman, and M. W. Grinstaff, "Embedded multicellular spheroids as a biomimetic 3D cancer model for evaluating drug and drug-device combinations," Biomaterials **35**, 2264–2271 (2014).

[238] U. Cheema, Z. Rong, O. Kirresh, A. J. MacRobert, P. Vadgama, and R. A. Brown, "Oxygen diffusion through collagen scaffolds at defined densities: implications for cell survival in tissue models," Journal of Tissue Engineering and Regenerative Medicine **6**, 77–84 (2012).

[239] R. Goswami, K. Kim, A. R. Boccaccini, J. Guck, and S. Girardo, "Fine-tuning cell-mimicking polyacrylamide microgels: Sensitivity to microscale reaction conditions in droplet microfluidics," Materials & Design **262**, 115450 (2026).

[240] H. M. Knochelmann, A. S. Smith, C. J. Dwyer, M. M. Wyatt, S. Mehrotra, and C. M. Paulos, "CAR T Cells in Solid Tumors: Blueprints for Building Effective Therapies," Frontiers in Immunology **9**, 1740 (2018).

[241] R. Prevedel, A. Diz-Muñoz, G. Ruocco, and G. Antonacci, "Brillouin microscopy: an emerging tool for mechanobiology," Nature Methods **16**, 969–977 (2019).

[242] Z. N. Coker, M. Troyanova-Wood, Z. A. Steelman, B. L. Ibey, J. N. Bixler, M. O. Scully, and V. V. Yakovlev, "Brillouin microscopy monitors rapid responses in subcellular compartments," PhotoniX **5**, 9 (2024).

[243] V. Mahajan, T. Beck, P. Gregorczyk, A. Ruland, S. Alberti, J. Guck, C. Werner, R. Schlüßler, and A. V. Taubenberger, "Mapping Tumor Spheroid Mechanics in Dependence of 3D Microenvironment Stiffness and Degradability by Brillouin Microscopy," Cancers **13**, 5549 (2021).

[244] A. Nyga, K. Plak, M. Kräter, M. Urbanska, K. Kim, J. Guck, and B. Baum, "Dynamics of cell rounding during detachment," iScience **26**, 106696 (2023).

[245] R. A. Badley, A. Woods, L. Carruthers, and D. A. Rees, "Cytoskeleton changes in fibroblast adhesion and detachment," Journal of Cell Science **43**, 379–390 (1980).

[246] E. A. Janka, T. Várvölgyi, Z. Sipos, A. Soós, P. Hegyi, S. Kiss, F. Dembrovszky, D. Csupor, P. Kéringer, D. Pécsi, M. Solymár, and G. Emri, "Predictive Performance of Serum S100B Versus LDH in Melanoma Patients: A Systematic Review and Meta-Analysis," Frontiers in Oncology **11**, 772165 (2021).

[247] P. Vaupel and A. Mayer, "Tumor Hypoxia: Causative Mechanisms, Microregional Heterogeneities, and the Role of Tissue-Based Hypoxia Markers," in *Oxygen Transport to Tissue XXXVIII*, Vol. 923, edited by Q. Luo, L. Z. Li, D. K. Harrison, H. Shi, and D. F. Bruley (Springer International Publishing, Cham, 2016), pp. 77–86.

[248] M. D. Nieskoski, K. Marra, J. R. Gunn, P. J. Hoopes, M. M. Doyley, T. Hasan, B. S. Trembly, and B. W. Pogue, "Collagen Complexity Spatially Defines Microregions of Total Tissue Pressure in Pancreatic Cancer," Scientific Reports **7**, 10093 (2017).

[249] S. K. Schmidt, S. Fischer, Z. El Ahmad, R. Schmid, E. Metzger, R. Schüle, C. Hellerbrand, A. Arkudas, A. Kengelbach-Weigand, M. Kappelmann-Fenzl, and A. K. Bosserhoff, "Modeling a mesenchymal cell state by bioprinting for the molecular analysis of dormancy in melanoma," Materials Today Bio **32**, 101674 (2025).

[250] D. G. Gibson, L. Young, R.-Y. Chuang, J. C. Venter, C. A. Hutchison, and H. O. Smith, "Enzymatic assembly of DNA molecules up to several hundred kilobases," Nature Methods **6**, 343–345 (2009).

[251] J. E. Bird, J. Marles-Wright, and A. Giachino, "A User's Guide to Golden Gate Cloning Methods and Standards," ACS synthetic biology **11**, 3551–3563 (2022).

[252] J.-P. Concordet and M. Haeussler, "CRISPOR: intuitive guide selection for CRISPR/-Cas9 genome editing experiments and screens," Nucleic Acids Research **46**, W242–W245 (2018).

[253] M. A. Borovinskaya, S. Shoji, K. Fredrick, and J. H. D. Cate, "Structural basis for hygromycin B inhibition of protein biosynthesis," RNA (New York, N.Y.) **14**, 1590–1599 (2008).

[254] R. N. Rao, N. E. Allen, J. N. Hobbs, W. E. Alborn, H. A. Kirst, and J. W. Paschal, "Genetic and enzymatic basis of hygromycin B resistance in Escherichia coli," Antimicrobial Agents and Chemotherapy **24**, 689–695 (1983).

[255] J. Schindelin, I. Arganda-Carreras, E. Frise, V. Kaynig, M. Longair, T. Pietzsch, S. Preibisch, C. Rueden, S. Saalfeld, B. Schmid, J.-Y. Tinevez, D. J. White, V. Hartenstein, K. Eliceiri, P. Tomancak, and A. Cardona, "Fiji: an open-source platform for biological-image analysis," Nature Methods **9**, 676–682 (2012).

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Statement on Generative AI Use

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Unless otherwise indicated, the scientific questions, experimental work, analytical design and decisions, data processing, validation, interpretation of results, and conclusions presented in this dissertation were developed and carried out by the author. Contributions from collaborators and other researchers are explicitly acknowledged and attributed in the relevant sections. The author assumes full responsibility for the selection, presentation, and interpretation of the material and for the overall conclusions of this dissertation.

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Maximilian Eiche